I

ONE COMPLETE NOTEBOOK

Notes In Dental Sleep Medicine

An IFDSM examination revision companion · Integrated edition

Ten IFDSM examination competencies, with the full explanations

Condensed high-yield summaries, definitions and key numbers

Clinical illustrations, reference tables and examination traps

Self-tests and one 60-question mock paper with answers

Compiled from the August 2026 Condensed Revision Notes and IFDSM Examination Revision Companion; with integrated topic navigation and consolidated examination resources.

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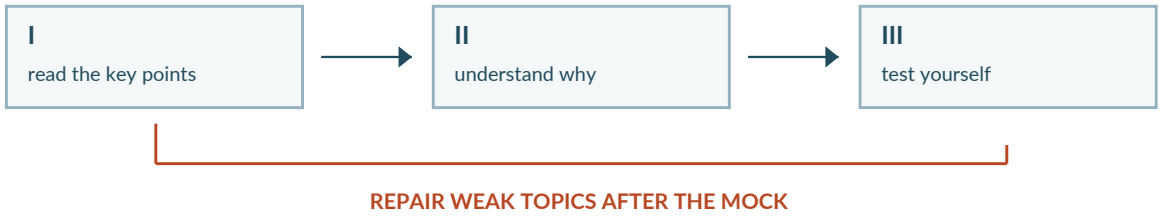
NOVEMBER 2026

The notebook

One integrated book, three ways to use it. The same ten IFDSM competencies are revisited at two levels of detail, with a mock paper and references to test and consolidate learning.

I	**High-yield recall** Condensed revision	The rapid-review layer: essential definitions, numerical thresholds, examination traps, selected illustrations and short recall checks.
II	**Detailed understanding** Full companion	The teaching layer: mechanisms, clinical decision-making, treatment pathways, tables, diagrams, longer self-tests and supporting references.
III	**Examination practice** Mock and reference material	A single 60-question examination with model answers, followed by abbreviations and the consolidated bibliography.

Which way to move



Read to learn; then test to retrieve. Start with the high-yield layer for a familiar topic. If an answer is unclear, continue directly into the full companion section for the same numbered competency. After the mock, return to the topic rather than simply memorising the answer key.

WHEN	USE
Final-day memorisation	**High-yield notes + key numbers**
An unfamiliar mechanism or clinical question	**Detailed companion of that topic**
Checking readiness under examination conditions	**60-question mock and answer key**

INTERACTIVE QUICK LINKS

Contents & page map

Every topic begins with high-yield notes, followed immediately by the full companion.

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Page numbers refer to this integrated notebook; original source footer numbers are retained.

Master numbers

If you learn nothing else, learn these two pages.

Diagnosis and severity

– **Apnoea:** ≥90% airflow reduction, ≥10 s. Obstructive = effort persists; central = effort absent; mixed = starts central, ends obstructive.
– **Hypopnoea (AASM recommended):** ≥30% reduction, ≥10 s, with ≥3% desaturation *and/or* arousal.
– **RERA:** increasing effort or flattened nasal pressure waveform → arousal, not meeting apnoea/hypopnoea criteria.
– **AHI** = apnoeas + hypopnoeas per hour. **RDI** = AHI + RERAs, so RDI is always ≥ AHI.
– **OSA diagnosis:** RDI >15/h, *or* >5/h with symptoms. ICSD-3-TR: AHI ≥5 with symptoms/comorbidity, or ≥15 regardless.
– **Adult severity:** normal <5 · mild 5–<15 · moderate 15–<30 · severe ≥30.
– **Paediatric severity:** normal <1 · mild 1–<5 · moderate 5–<10 · severe ≥10. Obstructive AHI ≥1 is abnormal.
– **Childhood SDB cut-off age:** 12 years.
– **AHI ≥15** = material rise in insulin resistance, dyslipidaemia, vascular disease, death.
– **Primary snoring:** habitual snoring with AHI <5 (some sources <15); OSA must be excluded by a physician first.
– **SaO_2** normal 95–100%, ≥90% acceptable. Minimum SaO_2 is an independent severity marker.
– **Sleep efficiency:** ≥85% normal (>90% young adults); 75–84% suboptimal; <75% suggests insomnia.
– **SOL and WASO** <30 min. **REM latency** ~90 min.

Sleep architecture

– NREM 75–80% : REM 20–25%.
– Cycle 70–100 min (first), then 90–120 min; 4–6 cycles/night.
– N1 2–5% · N2 45–55% · N3 ~13–23%.
– Alpha 8–13 Hz; sleep spindles 11–16 Hz; sawtooth waves in REM; ~80% dream recall from REM.
– REM raises BP by up to 30%; **thermoregulation absent in REM**; muscle atonia in REM.
– Growth hormone in early slow-wave sleep; thyroid hormone late evening.
– Adults 12–20 breaths/min asleep; infants/young children up to 30–60.

Screening thresholds

– **ESS** >10 of 24.
– **STOP-BANG** ≥3 (higher cut-off = better specificity). BMI >35, age >50, neck >40 cm.
– **OSA50** ≥5 of 10: waist 3 (>102 cm M / >88 cm F), snoring 3, apnoeas 2, age ≥50 y 2. **94% sensitivity, 31% specificity** for moderate–severe OSA.
– **Berlin:** high risk = positive in ≥2 of 3 categories.
– **NoSAS** ≥8 of 17: neck >40 cm 4, BMI ≥30 5, snoring 2, age >55 y 4, male 2.
– **PSQ** >0.33 (>8 of 22 items positive).
– **FAIREST-6:** 2 mild · 4 moderate · 6 severe.
– **Paediatric ESS** ≥10.
– **Neck circumference:** >43 cm (17 in) male, >40 cm (16 in) female.
– **Uvula** >15 mm long, >10 mm wide → linked to SDB.
– **Tonsils:** obstruction >50% warrants evaluation. Brodsky 0–4: within pillars · <25% · 25–50% · 50–75% · >75%.
– **Australian direct-referral:** positive questionnaire (Berlin positive / OSA50 ≥5 / STOP-BANG ≥4) **plus** ESS ≥8.

Oral appliance therapy

– **Effective appliance:** increments ≤1 mm · range ≥5 mm · reversible · verifiable setting · patient-removable · stable retention.
– **Initial protrusion** 50–70% of maximum (AADSM 2025: 25–75% acceptable). Start 50%, do not exceed 75%.
– **Vertical:** ~2 mm interincisal (grey George Gauge fork; white 5 mm fork usually too much). Delivery check 2–4 mm over habitual. Posterior clearance: nylon ~3 mm, acrylic ~5 mm, some designs 6 mm.
– **Increasing vertical opening reduces efficacy** and promotes mouth breathing.
– **Titration increments** 0.1–0.4 mm/turn; 10 turns = +1 mm on interlocking dorsal designs.
– **Protrusive capacity** ≥5–8 mm needed (normal 10–12 mm); <6 mm is a contraindication. Max opening 25–40 mm (normal 40–60 mm).
– **Dentition:** ≥6 teeth/arch (stricter sources 8–10 sound teeth/arch with adequate bone).
– **BMI ≥35** predicts failure.
– **Response definitions:** complete = symptoms resolved + AHI <5; partial = symptoms improved + ≥50% AHI reduction but AHI ≥5; failure = ongoing symptoms ± <50% reduction.
– **Efficacy:** AHI <5 ~37% · AHI <10 ~52% · ≥50% reduction ~64%. ~⅔ meaningful benefit, ~⅓ none. Up to ~80% success in mild, ~60% in severe.
– **Indirect AHI reduction:** OA ~42% · CPAP ~75% · UPPP ~30%.

– Snoring reduced ~45%; ESS 11.2 → 7.8; BP ~2–4.2 mmHg reduction; 90–95% report better sleep quality.
– **MDA:** OA ~51% vs CPAP ~50%.
– **AADSM compliance:** ≥80% of the night, ≥5 nights/week, with ≥7 h sleep. (CPAP convention ≥4 h/night.)
– **Adherence:** ~80% at 1 mo → ~67% at 6 mo, then stable; 6.1–6.7 h/night; MAD 6.5 vs CPAP 5.2 h.
– **Titration timeline:** reassess ≤30 days; 2-week acclimatisation then 6-week self-adjustment; total ~2–5 months; monthly for first 6 months.
– **Follow-up:** 6-monthly in year 1, then at least annually; physician reassessment ~every 5 years. Appliance life 2–3 years.

Side effects

– Discontinuation from side effects ~7.5% (commonest cause **TMJ pain**); side effects cause 20–30% of all discontinuations.
– Short term: salivation ~27% · dry mouth ~38% · tooth pain ~36% · allergy <1%.
– Long term: jaw pain ~34% · bite change 14–26%.
– Occlusal change: OJ ~1.9 mm, OB ~2.3 mm (average ~2.5 mm described); ~⅓ exceed 1 mm; at 17 y median −1.1 mm OJ, −1.6 mm OB.
– ~80% of nightly users at 7 years show change; 85% detectable on models/cephalograms.
– 51% posterior open bite; 62% anterior crossbite of ≥1 tooth.
– Facial height and jaw relation change as early as 6 months.
– Greatest vulnerability in first 2 years; resolves within ~2 weeks of cessation with exercises; leave alone 12–18 months as most self-correct.
– Protrusion >50% of max = significantly higher risk; >80% = vulnerability factor.
– Morning aligner worn 15–20 min on waking, custom, recorded in centric occlusion.

PAP, other therapies, epidemiology

– **CPAP:** start 4 cmH_2O, max ~20; >20 → BPAP; 4–6 cmH_2O abolishes simple snoring.
– CPAP compliance 40–60% at 12 months; threshold ≥4 h/night; failure rates 29–83%.
– **Weight:** 10% gain → ~32% AHI rise; 10% loss → ~26% fall. BMI target <30. Bariatric threshold BMI ≥35 with comorbidity.
– **Hypoglossal nerve stimulation:** BMI <35, AHI 20–50, non-concentric retropalatal collapse; exclusions = complete concentric collapse, grade 3–4 tonsils; adherence ~86%.
– **Positional OSA** = non-supine AHI <5.
– **EPAP:** Provent ~8 cmH_2O; Theravent 2–6 cmH_2O.
– **UPPP success** = >50% AHI reduction and AHI <20; Friedman stage I best, III mostly unsuccessful; ~4% surgical adverse event rate.
– **P_{crit}:** healthy negative, OSA more positive. Mild <−2, moderate −2 to +2, severe >+2 cmH_2O; distribution ~19% / 58% / 23%.
– **Non-anatomical traits:** ~70% have ≥1. High loop gain ~36% · low arousal threshold ~37% · poor muscle responsiveness ~36%.
– **Adult prevalence:** AHI>5 — 9% F / 24% M (Wisconsin 1993); moderate–severe 6% F / 13% M (2007–2010); global 936 million any, 425 million moderate–severe (aged 30–69).
– **Paediatric:** OSA 1–5% (typically 2–3%); snoring 3–12%; Down syndrome ~50% at AHI ≥5.
– **Sleep duration:** 4–12 mo 12–16 h · 1–2 y 11–14 h · 3–5 y 10–13 h · 6–12 y 9–12 h · 13–18 y 8–10 h. <8 h in adolescence → ~3× suicide-attempt risk.
– **Insomnia:** symptoms 33–50% · disorder 10–15% · chronic 5–15% · with comorbidity 50–75%. Chronic = ≥3 nights/week for ≥3 months.
– **Sleep bruxism:** reported 6–91%; self-reported adults ~8%; children/adolescents 14–46%; 30–50% of OSA patients.
– **TMD in OSA** 2–52%; **OSA in TMD** ~28%; ~2% cannot tolerate a MAD.
– Type 2 study failure ~6.6–7%; home study underestimates AHI ~10%.
– Sensor counts: type 1 = 15–20 · type 2 = 10–15 · type 3 = 4–7 · type 4 = 1–2.

Master table of numbers

Everything numerical in one place. If you learn nothing else, learn this page.

A. Diagnostic and severity thresholds

Parameter	Value	Note
Apnoea	≥90% airflow reduction, ≥10 s	Obstructive if respiratory effort persists; central if effort absent
Hypopnoea (AASM recommended)	≥30% airflow reduction, ≥10 s, with ≥3% desaturation *or* arousal	The ≥4% desaturation rule became *optional* in Scoring Manual v3 (2023); some payers still require 4%
Normal oxygen saturation in sleep	≥90% (typically 95–100%)	Minimum SaO_2 is an independent severity marker
Adult OSA — normal	AHI < 5	ICSD-3-TR: AHI ≥ 5 with symptoms/comorbidity, *or* AHI ≥ 15 regardless of symptoms
Adult OSA — mild	AHI 5 to < 15	
Adult OSA — moderate	AHI 15 to < 30	
Adult OSA — severe	AHI ≥ 30	
Paediatric OSA — normal	AHI < 1	Paediatric thresholds are approximately one third of adult values. A child with an AHI of 6 has *moderate* disease, not mild
Paediatric OSA — mild	AHI 1 to < 5	
Paediatric OSA — moderate	AHI 5 to < 10	
Paediatric OSA — severe	AHI ≥ 10	
Age cut-off for "childhood" SDB	12 years	Above this, adult scoring criteria are generally applied
Primary snoring	Habitual snoring with AHI < 5 (some sources < 15)	Requires exclusion of OSA by a physician before it can be called primary
Sleep efficiency — normal	≥85% (often >90% in young adults)	75–84% suboptimal; <75% suggests insomnia
Excessive daytime sleepiness (ESS)	>10 of 24	ESS correlates poorly with AHI — never use alone

B. Screening questionnaires

Tool	Positive threshold	Performance and use
STOP-BANG (8 items)	≥3 (higher cut-off = better specificity)	High sensitivity for moderate–severe OSA; the workhorse adult dental screening tool
OSA50 (4 items, /10)	≥5	Waist circumference 3, snoring 3, apnoeas 2, age ≥50 y 2. Validation: 94% sensitivity, 31% specificity for moderate–severe OSA
Berlin (11 items, 3 categories)	High risk = positive in ≥2 categories	Snoring severity, daytime sleepiness, hypertension/obesity
NoSAS (/17)	≥8	Neck >40 cm 4, BMI ≥30 5, snoring 2, age >55 y 4, male 2
Epworth Sleepiness Scale	>10 of 24	Measures sleep propensity, not disease severity
Paediatric Sleep Questionnaire	>0.33 (i.e. >8 of 22 positive)	Most sensitive paediatric screening questionnaire; four subscales
FAIREST-6		Six dental "red flags" for paediatric SDB

Tool	Positive threshold	Performance and use
	2 mild · 4 moderate · 6 severe	
Neck circumference (adult risk)	>43 cm (17 in) male; >40 cm (16 in) female	Recorded by the dentist at screening

C. Oral appliance therapy

Parameter	Value	Note
Initial protrusive setting	50–70% of maximum protrusion (AADSM 2025: 25–75% acceptable)	Do not start at maximum; >75% predisposes to TMD
Vertical opening at construction bite	~2 mm interincisal (grey George Gauge fork)	Increasing vertical opening *reduces* efficacy and promotes mouth breathing
Minimum titration increment	≤1 mm (commonly 0.1–0.4 mm per turn)	Definition of an effective appliance requires ≥5 mm adjustment range and reversibility
Minimum protrusive capacity for a MAD	≥5–6 mm (normal 10–12 mm)	<6 mm is a relative contraindication
Minimum dentition	6–10 sound teeth per arch with adequate bone support	Sources vary; state the principle (adequate retention and anchorage)
Titration period	~2–5 months (longer than CPAP)	Reassess within 30 days of insertion
Complete response	Symptom resolution + AHI < 5	Learn all three definitions — examiners ask candidates to distinguish them
Partial response	Improved symptoms + ≥50% AHI reduction but AHI ≥ 5	
Treatment failure	Ongoing symptoms ± <50% AHI reduction	
Overall MAD efficacy	AHI <5 in ~37%; AHI <10 in ~52%; ≥50% reduction in ~64%	Roughly two thirds derive meaningful benefit; one third do not
Success rate by severity	Up to ~80% in mild OSA; ~60% in more severe disease	
AHI reduction, indirect comparison	OA ~42% · CPAP ~75% · UPPP ~30%	
Blood pressure effect	~2–4 mmHg mean reduction	Comparable to CPAP in several trials
Mean disease alleviation	OA ~51% vs CPAP ~50%	Efficacy × adherence — the reconciliation of the CPAP/OA paradox
AADSM compliance definition	Worn ≥80% of the night, ≥5 nights per week	CPAP research convention is ≥4 h per night
Objective OA adherence	~80% at 1 month falling to ~67% at 6 months, then stable; ~6.1–6.7 h per night	MAD nightly use exceeds CPAP (~6.5 vs ~5.2 h)
Discontinuation from side effects	~7.5% (most often TMJ pain)	
Occlusal change — overjet/overbite	~1.9 mm OJ and ~2.3 mm OB at 5 y; median −1.1 mm OJ and −1.6 mm OB at 17 y	~⅓ of patients exceed 1 mm; ~80% show change by 7 years
Appliance lifespan	~2–3 years	Assess yearly for wear, fracture and biofilm
Follow-up schedule	Every 6 months in year 1, then at least annually	Refer back to the physician approximately every 5 years for reassessment

D. Sleep architecture, PAP and other therapies

Parameter	Value	Note
NREM : REM proportion of sleep	75–80% : 20–25%	
Sleep cycle length	70–100 min (first), then 90–120 min	4–6 cycles per night
Stage proportions (adult)	N1 2–5% · N2 45–55% · N3 ~13–23%	REM latency ~90 min from sleep onset
CPAP pressure range	Start 4 cmH_2O; maximum ~20 cmH_2O	>20 cmH_2O required → consider BPAP; 4–6 cmH_2O abolishes simple snoring
CPAP adherence	40–60% at 12 months; threshold ≥4 h per night	Adherence in the first days-to-weeks predicts long-term use
Weight and AHI	10% weight gain → ~32% AHI increase; 10% loss → ~26% AHI decrease	Obesity is the major reversible risk factor; BMI target <30
Bariatric surgery threshold	BMI ≥35 with comorbidity, after failed non-surgical weight loss	Reduces AHI substantially but frequently does not cure OSA
Hypoglossal nerve stimulation selection	BMI <35, AHI 20–50, non-concentric retropalatal collapse	Complete concentric collapse and grade 3–4 tonsils are exclusions; adherence ~86%
Positional OSA definition	Non-supine AHI < 5	A strict definition — supine-predominant is not the same thing
Critical closing pressure (P_{crit})	Healthy negative; OSA more positive. Mild <−2, moderate −2 to +2, severe >+2 cmH_2O	Anatomical impairment: ~19% mild, ~58% moderate, ~23% severe
Non-anatomical trait prevalence in OSA	~70% have ≥1; high loop gain ~36%, low arousal threshold ~37%, poor muscle responsiveness ~36%	The PALM construct: P_{crit}, Arousal threshold, Loop gain, Muscle responsiveness
Adult OSA prevalence	Moderate–severe ~6% women, ~13% men (US 2007–2010); global 936 million mild–severe, 425 million moderate–severe (aged 30–69)	Wisconsin cohort 1993: AHI>5 in 9% of women, 24% of men
Paediatric prevalence	OSA 1–5% (typically 2–3%); habitual snoring 3–12%	Down syndrome: OSA in ~50% at AHI ≥5
Recommended sleep duration	4–12 mo 12–16 h · 1–2 y 11–14 h · 3–5 y 10–13 h · 6–12 y 9–12 h · 13–18 y 8–10 h	<8 h in adolescence: ~3-fold increase in suicide attempt risk

TOPIC 1

Normal sleep, sleep disorders and the pathophysiology of SDB

BLUEPRINT

1. Health benefits of normal sleep · 2. Physiology of normal sleep · 3. Pathophysiology of SDB: anatomical and non-anatomical factors · 4. Prevalence and comorbidities · 5. Risk factors including age and gender · 6. Signs and symptoms · 7. Impact on the individual · 8. Impact on society · 9. Insomnia · 10. Other respiratory sleep disorders · 11. Other non-respiratory sleep disorders · 12. Effects of pharmacotherapeutics on sleep

1.1 Benefits and functions of sleep

- **Physical restoration and repair** — protein synthesis, cell and tissue repair, anabolic dominance.
- **Biochemical refreshment** — synaptic and neuronal function; glial-mediated **glymphatic clearance** of metabolites; glycogen restoration; neurogenesis.
- **Memory consolidation** — declarative in NREM (sleep spindles), procedural and emotional in REM.
- **Emotional regulation** — REM-dependent processing of affect.
- **Hormone regulation** — growth hormone in early SWS; prolactin, testosterone, PTH; thyroid hormone late evening; melatonin under light–dark control.
- **Immune function** and **metabolic regulation** — insulin sensitivity, leptin/ghrelin balance.

1.2 Physiology of normal sleep

DEFINITION — VERBATIM

Sleep is a reversible behavioural state characterised by decreased responsiveness to arousing stimuli and disengagement from environmental surroundings. It is a complex physiological and behavioural process orchestrated by distinct neuronal networks, and is qualitatively distinct from rest. Assessed by polysomnographic recording of brain waves, eye movements and muscle tone.

Two-process regulation

- **Process S (homeostatic drive)** — sleep pressure accumulates with time awake; adenosine is the principal candidate mediator.
- **Process C (circadian)** — endogenous ~24 h oscillator in the **suprachiasmatic nucleus**, entrained by light via the retinohypothalamic tract. **Melatonin** (pineal) induces sleepiness partly by opposing the SCN alerting signal, and is suppressed by light.
- Their interaction explains the 04:00 shift-worker trough and the adolescent phase delay (Topic 8).

Neurochemistry

- **Wake-promoting:** orexin/hypocretin (lateral hypothalamus), noradrenaline (locus coeruleus), serotonin (raphe), histamine (tuberomammillary), acetylcholine, dopamine.
- **NREM-promoting:** GABA and galanin from the ventrolateral preoptic area.
- **REM:** cholinergic "REM-on" neurons (pedunculopontine/laterodorsal tegmentum) active while monoaminergic and orexinergic "REM-off" activity reaches its nadir. Explains why REM-suppressing drugs are serotonergic/noradrenergic, and why orexin loss causes narcolepsy.

Stages

- **N1** — 2–5%, 1–7 min first cycle. Alpha (8–13 Hz) → low-voltage mixed frequency; vertex sharp waves; easily interrupted. Transitional role in cycling.
- **N2** — 45–55%, 10–25 min lengthening each cycle. **Sleep spindles** (11–16 Hz) and **K-complexes**. Spindle density rises after learning a new task.
- **N3** (old stages 3+4) — ~13–23%, 20–40 min first cycle. High-voltage slow delta; **highest arousal threshold of all NREM stages**; concentrated in the first third of the night; GH pulse here; disorders of arousal arise here.
- **REM** — 20–25%, 1–5 min first cycle, progressively prolonged. Desynchronised EEG, **sawtooth waves**, theta 3–7 Hz, rapid eye movement bursts, **skeletal muscle atonia**. ~80% vivid dream recall.

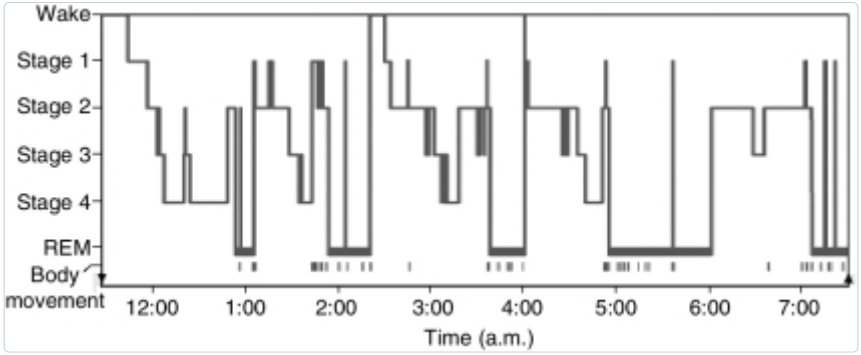


Fig 1.1 — Hypnogram, normal adult night. Slow-wave sleep concentrated in the first third; REM periods lengthen and dominate the final third. **Examinable because:** explains why REM-predominant OSA is worst in the early hours, why a split-night study under-samples REM, and why successful treatment shows **REM rebound** rather than more total sleep.

Physiological changes during sleep

- **NREM:** brain activity, HR, BP and sympathetic activity all fall; muscle tone similar to wake; respiration slower, deeper, very regular; minute ventilation falls; chemosensitivity blunted; cerebral blood flow and metabolism significantly reduced; temperature regulated at a lower set point.
- **REM:** BP rises up to 30% and fluctuates; HR rises and varies; sympathetic activity markedly above waking; **muscle tone absent**; respiration irregular and faster with brief central pauses; cough suppressed; total cerebral blood flow ≈ wake, regionally increased in limbic and visual association areas; **thermoregulation absent**.
- **Both:** upper airway resistance increases from wake. Renal — reduced sodium, potassium, chloride and calcium excretion, concentrated urine, reduced flow (hence nocturia is *abnormal*).
- Sympathetic bursts occur in NREM with **K-complexes, arousals and large body movements**. The morning HR/BP surge on awakening explains the **morning peak in myocardial infarction** — repeated dozens of times a night in OSA.

TRAP — NOMENCLATURE

Older texts (Rechtschaffen & Kales) describe **four** NREM stages. Since AASM 2007, stages 3 and 4 are merged as **N3** and the scheme is N1/N2/N3/R. Signal that you know both: "slow-wave sleep, stage N3, previously stages 3 and 4."

1.3 Pathophysiology of SDB

DEFINITIONS — VERBATIM

SDB — a group of disorders characterised by abnormalities of breathing, respiratory pattern or quantity of ventilation during sleep; spanning increased airflow resistance, heavy snoring, hypopnoea and apnoea.

OSA — a form of SDB involving snoring but caused by more significant upper airway obstruction, with consequent sleep fragmentation, hypoxaemia, or both.

OSA vs OSA syndrome — objective PSG evidence *without* symptoms = OSA alone; with symptoms = OSA *syndrome*.

Core mechanism

- Wake: tonic neural drive to pharyngeal dilators (chiefly **genioglossus**) resists inspiratory negative pressure. Sleep onset withdraws this compensatory input.
- Therefore **airway narrowing during sleep is a normal physiological event**; in OSA the normal response is *exaggerated* by combined abnormal anatomy and physiology, so narrowing becomes repetitive collapse.
- Obstruction may occur anywhere nasopharynx → larynx; commonest sites are **retropalatal** and **retroglossal**.
- Mechanically a **Starling resistor**; collapsibility quantified by $\mathbf{P_{crit}}$, the minimum pressure preventing complete collapse. Healthy = negative; OSA shifted positive; more positive = more collapsible.

Anatomical factors

- **Craniofacial:** retrognathia, micrognathia, maxillary constriction with high-arched/narrow palate, steep mandibular plane angle, excessive vertical facial growth, inferiorly positioned hyoid, reduced maxillary/mandibular length.
- **Soft tissue:** macroglossia, elongated soft palate, enlarged uvula, adenotonsillar hypertrophy (dominant in children), parapharyngeal fat pads, large mandibular tori, lateral pharyngeal wall thickening.
- **Nasal:** septal deviation, turbinate hypertrophy, polyps, allergic rhinitis — increase resistance, promote mouth breathing, reduce PAP tolerance.

- **Obesity:** peripharyngeal fat mechanically loads the airway, offsetting dilator activity most markedly in sleep; reduces **functional residual capacity** and chest wall compliance, so less caudal tracheal traction on the pharynx; disturbs the relationship between respiratory drive and load compensation.
- **Other:** rostral fluid shift to the neck on recumbency, pharyngeal sensory neuropathy from chronic vibration, reduced lung volumes in neuromuscular disease.

Non-anatomical factors — PALM

PALM — THE FRAMEWORK THAT CARRIES THIS TOPIC

- **P — P_{crit} / passive anatomy.** Impaired in essentially all patients, severity varying: mild <−2, moderate −2 to +2, severe >+2 cmH_2O; distribution ~19% / 58% / 23%. → CPAP, oral appliance, weight loss, positional therapy, surgery.
- **A — Arousal threshold, low in ~37%.** Wakes prematurely, *before* dilators recruit → fragmentation and ventilatory instability. → hypnotic (eszopiclone) to raise threshold.
- **L — Loop gain, high in ~36%.** Oversensitive ventilatory control: excessive CO_2 response → overshoot → hypocapnic undershoot → unstable periodic breathing, promoting obstructive *and* central events. → acetazolamide, supplemental oxygen.
- **M — Muscle responsiveness, poor in ~36%.** Dilators fail to recruit to negative pressure. → hypoglossal nerve stimulation, myofunctional therapy, atomoxetine + oxybutynin.
- **~70% of patients have ≥1 non-anatomical impairment.** Of those with moderate anatomical impairment (the largest group), ~64% have ≥1 non-anatomical impairment (→ combination therapy) and ~36% do not (→ anatomical intervention alone). Severe anatomical impairment requires major anatomical intervention regardless.

WHY THIS MATTERS TO A DENTIST

An appliance is an **anatomical** intervention: it reduces collapsibility dose-dependently *without* systematically changing genioglossus function, loop gain or arousal threshold. So anatomical, mild-to-moderate-collapsibility patients respond best; a highly collapsible airway plus unstable ventilatory control is a predictable non-responder, and no amount of further protrusion fixes a loop-gain problem. This is the mechanistic case for **combination therapy** and the best available answer to "why does OAT fail in a third of patients?"

Estimating phenotype without a research lab

- **CPAP dial-down:** CPAP withdrawn for 5 breaths, repeated 5–10×. Peak inspiratory flow as % of CPAP-supported baseline; **PIF ≥30% = only mildly collapsible airway**; higher PIF = more negative P_{crit}.
- **Awake upper airway collapsibility index:** choanal and epiglottic pressure catheters during negative-pressure inspiration; % collapsibility = (choanal − epiglottic)/choanal. Higher = more collapsible, correlating with higher sleeping P_{crit}. Takes ~10 min; easier and faster.
- Both comparable to gold-standard P_{crit}; clinical/PSG prediction models emerging.

1.4–1.5 Prevalence, comorbidities, risk factors

- **Prevalence:** AHI>5 — 9% women / 24% men; AHI>5 with sleepiness — 2% / 4%; AHI>15 — 4% / 9% (Wisconsin 1993). Moderate–severe 6% / 13% (US 2007–2010). ~17% of US adults, up to ~38% in some estimates. Globally 936 million any, 425 million moderate–severe, aged 30–69.
- **Variation between studies** depends on hypopnoea definition (3% vs 4%), AHI threshold, symptom requirement, and population age/obesity. Always quote a figure *with* its definition.
- **Predictive comorbidities:** obesity, hypertension, cardiovascular disease (especially ischaemic heart disease), type 2 diabetes. Long-term: neurocognitive impairment, metabolic syndrome, systemic hypertension, cardiovascular disease.
- **Obesity is the most important and the only readily reversible risk factor**, and the relationship is **bidirectional**: fragmentation lowers leptin and raises ghrelin → increased hunger and appetite (especially calorie-dense carbohydrate) → reduced activity → weight gain → worse OSA. 10% gain → ~32% AHI rise; 10% loss → ~26% fall.
- **Age:** prevalence rises with age, peaking around 60, possibly plateauing or declining in very advanced age. In children the dominant factor shifts from adenotonsillar hypertrophy (peak 2–8 y) to overweight in adolescence.
- **Gender:** ~2:1–3:1 male predominance; the gap **narrows with age and after the menopause** — loss of oestrogen's protective effect plus post-menopausal weight gain and central fat redistribution. Women present atypically (insomnia, fatigue, mood) and are under-diagnosed.
- **Others:** narrowed airway, craniofacial abnormality, large neck circumference, large tongue, family history, smoking, alcohol/sedatives/hypnotics/opioids, nasal congestion, rhinitis, sinusitis, inflammation, supine posture, hypothyroidism, acromegaly, PCOS, heart failure, type 2 diabetes.

1.6 Signs and symptoms

- **Nocturnal:** snoring (loud, habitual, socially disruptive); witnessed apnoeas; waking choking or gasping; restless unrefreshing sleep; frequent arousals; insomnia; **nocturia**; night sweats; vivid dreams; salivation and tooth grinding.
- **Daytime:** excessive sleepiness; tiredness and fatigue; impaired concentration and cognition; **morning headache**; mood change and irritability; **dry mouth** on waking; reduced libido and impotence; nasal congestion; restless legs.
- May be entirely asymptomatic. Sympathetic activity and arterial pressure are commonly elevated even without symptoms — the basis for treating asymptomatic severe disease.

THREE REASONS AN ADULT ATTENDS A SLEEP CLINIC

1. **Own daytime symptoms** — unrefreshing sleep, tiredness, fatigue, poor concentration and focus, low mood, excessive sleepiness.
2. **Suspicion of OSA and concern about consequences** — often prompted by partner-witnessed apnoeas; concerns range from immediate (driving, workplace safety) to long-term (cardiovascular disease, stroke, cognitive decline).
3. **Effect on the bed partner** — socially disruptive snoring ± witnessed apnoeas is a very common trigger.

Children differ, and this contrast is frequently examined: **inattention, irritability, behavioural and learning problems** rather than sleepiness.

1.7–1.8 Impact on the individual and on society

- **Individual:** impaired alertness and daytime sleepiness; hypertension including treatment-resistant; stroke, MI, congestive heart failure, coronary artery disease; atrial fibrillation difficult to control; increased mortality and premature death; type 2 diabetes and glucose intolerance; metabolic syndrome; dementia and neurocognitive decline; reduced quality of life; pain; social disharmony and relationship difficulty from snoring, which also has an independent adverse effect on the carotid arteries.
- **Public safety:** impaired reflexes and judgement → higher risk of serious and fatal crashes; workplace accidents. Suspected OSA in a **commercial driver or safety-critical worker**, or sleepiness while driving, needs **urgent** assessment and explicit driving advice.
- **Economic:** reduced productivity, lost work hours; disproportionately affects the middle-aged workforce (more sick leave, more work disability); greater healthcare resource use.
- **Mental health and social:** worsens depression and anxiety; social withdrawal and reduced desire for interaction, contributing to loneliness. A public health issue, not only an individual one.

Sleep, pain and TMD — the dentally relevant impact

- Experimentally reducing total sleep time and disrupting slow-wave and REM sleep **evokes musculoskeletal pain, lowers pain thresholds, increases pain sensitivity, and reduces the antinociceptive potency of morphine**.
- In a population-based cohort free of TMD at baseline, ≥2 OSA signs/symptoms conferred a **73% greater incidence of first-onset TMD**, independent of age, gender, ethnicity, obesity, smoking and autonomic parameters — and OSA symptoms *preceded* TMD onset.
- In the case-control arm, chronic TMD was **>3× more frequent** in adults with low versus high likelihood of OSA.
- **Three proposed mechanisms:** (i) central sensitisation with impaired pain inhibition — experimental inspiratory resistance reduces baroreflex sensitivity, and impaired baroreflex sensitivity is implicated in TMD; (ii) sympathetic activation, with catecholamine-sensitive individuals at higher risk of first-onset TMD; (iii) sleep bruxism, commoner in OSA, as a possible contributor.

1.9 Insomnia

DEFINITION — VERBATIM

Insomnia — a subjective perception of difficulty with sleep initiation, duration, consolidation or quality that occurs *despite adequate opportunity for sleep*, and that results in some form of daytime impairment. **Both** nocturnal symptomatology and daytime dysfunction are required; a person who sleeps five hours and functions well is a short sleeper, not an insomniac.

- **Chronic insomnia disorder** — sleep-onset and/or maintenance complaints with daytime impairment, exceeding the minimal frequency and duration thresholds associated with clinically significant morbidity: **≥3 nights/week for ≥3 months**.
- **Short-term insomnia disorder** — clinically significant dissatisfaction or impairment failing those frequency/duration criteria. **Other insomnia disorder** — rare cases not meeting short-term criteria but warranting attention.

- **Prevalence:** symptoms 33–50%; ~⅓ report dissatisfaction with sleep; with distress/impairment 10–15%; chronic 5–15%; with medical/psychiatric comorbidity 50–75%. Risk factors: increasing age, female sex, comorbidity, shift work, unemployment, lower socioeconomic status.
- **Pathophysiology — hyperarousal** across both sleep and wake, with heightened sympathetic and HPA-axis activity perpetuating dysfunction. Correlates: increased HR, altered HR variability, increased whole-body metabolic rate, elevated cortisol/ACTH/CRF (particularly near sleep onset), increased body temperature, increased high-frequency (beta/gamma) EEG in NREM, increased cerebral glucose metabolism on PET. **No discrete structural brain pathology** except where comorbid with stroke, brain trauma or multiple sclerosis.
- **Clinical features:** adults — difficulty initiating or maintaining sleep, early awakening, non-restorative sleep. Children (caregiver-reported) — bedtime resistance, frequent night awakenings, inability to sleep independently. **≥1 daytime impairment required:** fatigue/malaise; attention, concentration or memory impairment; social, vocational or school dysfunction; mood disturbance or irritability; daytime sleepiness; reduced motivation, energy or initiative; proneness to errors or accidents; tension, headaches or GI symptoms in response to sleep loss; concerns or worries about sleep.
- **Insomnia is best viewed as a comorbid disorder warranting separate treatment attention** — not a symptom that resolves when the "primary" condition is treated.
- **Treatment: CBT-I is first line**, without tolerance or the adverse effects of pharmacotherapy. Five components, more effective combined: **cognitive therapy, stimulus control, sleep restriction, sleep hygiene, relaxation training** (stimulus control and sleep restriction have the strongest evidence).
- **Effect on outcomes:** marked, durable improvement in **SOL, WASO and sleep efficiency**; only a **small** improvement in total sleep time, augmenting over time. Sleep-diary measures are the most consistently used outcome because of ease of assessment.
- Pharmacotherapy is second line despite hypnotics being the most commonly used treatment; second-generation antipsychotics (quetiapine) are increasingly used out of dissatisfaction rather than evidence of superiority.

> **TRAP — DURATION THRESHOLD**
>
> Older material says chronic insomnia requires "at least a month", with acute/transient lasting days to weeks — that was ICSD-2/ DSM-IV. **ICSD-3 and ICSD-3-TR require ≥3 nights/week for ≥3 months**; DSM-5 aligns. Quote three months.

1.10 Other respiratory sleep disorders

Central sleep apnoea

- **Definition:** a lack of drive to breathe during sleep → repetitive insufficient ventilation and compromised gas exchange. Central apnoea = >10 s cessation of airflow **in the absence of respiratory effort**. CSA is the primary diagnosis when **≥50% of apnoeas are central**. The discriminator is **respiratory effort**, detected by thoracic and abdominal RIP belts.
- **Aetiologies:** high-altitude periodic breathing (which most healthy people develop given sufficient alveolar hypoxia); treatment-emergent CSA; opioid-induced; Cheyne–Stokes breathing (heart failure, stroke); obesity hypoventilation syndrome; idiopathic CSA — uncommon, possibly <5% of patients.
- **Classified by awake CO_2:** *hypercapnic* — abnormal central pattern generator output ("won't breathe") or impaired motor output caudal to the generator ("can't breathe"); *non-hypercapnic* — typically high loop gain.
- **Complications:** frequent awakenings, excessive sleepiness, adverse cardiovascular outcomes. **Bidirectional with OSA** — central events provoke obstructive events in vulnerable pharyngeal anatomy and vice versa; this mixed picture is present in the majority of sleep apnoea patients. **ASV is contraindicated in symptomatic heart failure with reduced ejection fraction.**

Hypoventilation, hypoxaemia, catathrenia

- **Hypoventilation syndromes** — insufficient alveolar ventilation causing CO_2 retention and hypoxaemia. Includes **obesity hypoventilation syndrome** (obesity with awake hypercapnia not otherwise explained), medication/substance-induced, and disorder-related (Duchenne muscular dystrophy, kyphoscoliosis, COPD).
- **Sleep-related hypoxaemia disorder** — sustained desaturation without comparable CO_2 rise.
- **Catathrenia** — slow, monotone, groaning expiratory vocalisation; benign, often mistaken for a parasomnia.
- **UARS** — increased resistance producing repetitive RERAs and fragmentation without qualifying events; often young, non-obese women with fatigue and insomnia rather than sleepiness. **ICSD-3 does not recognise UARS as a separate entity**, regarding it as part of the OSA spectrum captured by the RDI.

Primary snoring

DEFINITIONS

Snoring — audible vibration of the upper airway during respiration in sleep, on a continuum: primary snoring → upper airway resistance syndrome → symptomatic SDB. **Primary snoring** — habitual snoring in the absence of significant sleep apnoea or oxygen desaturation.

- Affects up to **50%** of the population; primary snoring prevalence is hard to establish because OSA must first be excluded.
- **~⅓ of OSA patients do not snore; ~⅓ of snorers have no OSA** (AHI <5). Snoring neither rules OSA in nor out.
- **Not benign:** heavy snoring (>50% of sleep time) may induce endothelial injury at the carotid bifurcation given its proximity to the hypopharyngeal walls → increased intima-media thickness, carotid atherosclerosis, metabolic syndrome. Also bed-partner sleep deprivation, depression and fatigue.

Graded management of primary snoring

- **Weight loss and reduced alcohol** (alcohol reduces dilator tone); **benzodiazepine and opioid reduction** (benzodiazepines increase NREM snoring and upper airway resistance even in healthy people).
- **Positional therapy** in supine-dominant snorers — physical barrier or a position-sensing vibratory device. Reduces snoring frequency and rate in most non-apnoeic snorers (AHI<5) but **not** in those with OSA; older devices had poor long-term adherence.
- **MAD is first-line in dentate patients**, after assessment by both an appropriately trained dentist and a sleep physician. Mechanism: protrusion pulls the tongue forward and widens the oropharynx, secondarily drawing the soft palate anteriorly and inferiorly via palatoglossus, displacing the velopharynx forward and controlling soft-tissue vibration. Occlusal assessment required first. Custom titratable > thermoplastic "boil-and-bite"; a partially titratable thermoplastic device reduces snoring but with more interim side effects; **a sham placebo exacerbated snoring**.
- **CPAP** in those already committed or willing — effective and safe, compliance highly variable; **4–6 cmH_2O** eliminates snoring.
- **Nasal surgery** — septoplasty with turbinate reduction only where there are *both* symptoms *and* objective nasendoscopic evidence of obstruction; benefit as a primary anti-snoring procedure is uncertain and inconsistent, but useful as a preliminary or adjunctive step.
- **Palatal surgery** — radiofrequency (submucosal volume reduction and stiffening; safe and effective short-term, minimal morbidity, may need additional therapy); modified UPPP (reconstructive; appropriate anatomy on nasendoscopy, after considering CPAP and MAD; main concern unpredictable scarring); palatal implants (three midline implants inducing fibrosis without volume reduction — **unacceptably high extrusion rate**, limited durability, potential to worsen OSA); injection snoreplasty (submucosal sclerosant for palatal flutter — reduces snoring at 6–12 months, minimal side effects, but efficacy declines and scarring is unpredictable); LAUP and Er:YAG laser.
- **Complementary:** nasal valve dilators, lubricating sprays, herbal remedies. Dilator therapy is effective and safe in mild–moderate OSA but without long-term data; theoretical lipoid pneumonia concern with oil-based products. Not advised, given uncertain long-term outcome and cost.
- **Nasal EPAP** — one-way resistor valves creating positive end-expiratory pressure to stabilise the airway; overall improvement but less than CPAP; minimally harmful, costly long term.
- **Myofunctional therapy** — speech-pathologist-delivered over ~3 months, isotonic and isometric exercises for tongue, soft palate and lateral pharyngeal walls. Reduces snoring severity, generally in the presence of OSA. A 15-min daily smartphone regimen over 8 weeks reduced acoustically measured snoring rate by **22%**. Low risk but significant commitment for small gain.
- **Combination therapy** — pre-phase + adjunctive, adjunctive + mainstream, or multiple mainstream. Higher efficacy, limited evidence.

1.11 Other non-respiratory sleep disorders

- **Parasomnias.** *NREM (disorders of arousal, from N3):* confusional arousals, sleepwalking, sleep terrors, sleep-related eating disorder — first third of the night, amnesia for the event, commoner in children. *REM:* **REM sleep behaviour disorder** — loss of REM atonia with dream enactment, typically older men, an important **prodrome of the alpha-synucleinopathies** (Parkinson's, dementia with Lewy bodies, multiple system atrophy); recurrent isolated sleep paralysis; nightmare disorder.
- **Circadian rhythm sleep–wake disorders.** Delayed phase (adolescents), advanced phase (older adults), irregular and non-24-hour rhythms, shift work disorder, jet lag.
- **Sleep-related movement disorders. Restless legs syndrome** (Willis–Ekbom) — urge to move with creeping/crawling sensations, worse in the evening and at rest, relieved by movement; associated with iron deficiency and dopaminergic dysfunction. **PLMD** — PLM index >15/h adults (>5/h children) is abnormal; the *disorder* requires associated sleep disturbance. Also sleep bruxism and rhythmic movement disorder.
- **Central disorders of hypersomnolence. Narcolepsy type 1** — hypocretin/orexin deficiency with cataplexy; type 2 without. MSLT: mean sleep latency ≤8 min with ≥2 SOREMPs; patients may enter sleep directly into REM. Also idiopathic hypersomnia, Kleine–Levin syndrome, insufficient sleep syndrome.

1.12 Pharmacotherapeutics and sleep

Drug Class	Main Sleep Effects	Notable Risks/Considerations
SSRIs/SNRIs	↓ REM, ↑ REM latency, possible insomnia	RLS, bruxism, nightmares
Sedating Antidepressants	↓ Sleep latency, ↑ SWS, oversedation	Long-term sedation
Benzodiazepines	↑ N2, ↓ REM/N3, improved sleep onset	Dependence, cognitive effects
Antipsychotics	Variable, often sedating	Daytime sleepiness
Melatonin/Orexin Antagonists	↑ Sleep time/efficiency (orexin)	Mixed evidence (melatonin)
Dexmedetomidine	↑ Sleep quality (ICU)	Bradycardia, hypotension

FIGURE 1 Table summarizing drug class effects on sleep

Fig 1.2 — Drug class effects on sleep. SSRIs/SNRIs (↓REM, ↑REM latency, possible insomnia; RLS, **bruxism**, nightmares) · sedating antidepressants (↓sleep latency, ↑SWS; long-term oversedation) · benzodiazepines (↑N2, ↓REM and N3, improved sleep onset; dependence, cognitive effects) · antipsychotics (variable, often sedating; daytime sleepiness) · melatonin and orexin antagonists (↑sleep time and efficiency; mixed evidence for melatonin) · dexmedetomidine (↑sleep quality in ICU; bradycardia, hypotension). **Examinable because:** learn the direction of the arrows plus the one signature adverse effect per class. The SSRI row is doubly relevant to a dentist because of **drug-induced sleep bruxism**.

- **Alcohol** — reduces sleep latency but suppresses REM in the first half with rebound in the second; fragments sleep; reduces pharyngeal dilator tone and blunts arousal, worsening OSA and snoring.
- **Caffeine** — adenosine receptor antagonist (blocking the mediator of homeostatic sleep pressure), half-life ~5 h. Shortens duration, prolongs SOL, increases WASO and daytime sleepiness, reduces slow-wave sleep dose-dependently.
- **Nicotine** — stimulant plus nocturnal withdrawal arousals; smoking also causes upper airway inflammation.
- **Opioids** — dose-dependent respiratory depression with central apnoeas and ataxic breathing; a recognised cause of CSA. Nuance: they may improve *subjective* sleep quality where pain disturbs sleep, but impair architecture and respiratory control, so the net effect in SDB is adverse.
- **Suggested for chronic insomnia:** short/intermediate-acting benzodiazepine receptor agonists (zolpidem, temazepam, eszopiclone, zaleplon, triazolam); ramelteon; suvorexant; sedating antidepressants (low-dose doxepin, trazodone, amitriptyline); other sedating agents (gabapentin, olanzapine, quetiapine). Reduce sleep latency, improve maintenance, may cause residual sedation.
- **Recommended against** for chronic insomnia: OTC antihistamines (diphenhydramine), herbal and nutritional substances (melatonin as hypnotic, valerian, tryptophan), barbiturates.

DEFINITION

Sleep hygiene — the practices and habits that affect the quality of sleep, including use of stimulants, alcohol, exercise, napping, time spent in bed, and quality of the sleep environment. Avoid late-afternoon and evening naps, heavy meals, caffeine and tobacco near bedtime; exercise earlier; establish a relaxing routine and a regular schedule; limit distractions, light and devices. **Insufficient as monotherapy for chronic insomnia** — it is one of five CBT-I components and the weakest.

TRAP — THE SEDATIVE–OSA INTERACTION

A benzodiazepine or hypnotic in **untreated** OSA can worsen it: reduced muscle tone, blunted arousal response to hypoxia and hypercapnia, raised arousal threshold → more and longer apnoeas. The sophisticated addition: raising the arousal threshold is precisely the *intended* action of eszopiclone in the low-arousal-threshold endotype. The same drug is harmful in one endotype and therapeutic in another — specify which you are treating, and confirm the OSA is being treated concurrently.

NUMBERS — TOPIC 1

- NREM 75–80% / REM 20–25% · cycle 70–100 then 90–120 min · REM latency ~90 min · N1 2–5% / N2 45–55% / N3 ~13–23%
- Alpha 8–13 Hz · spindles 11–16 Hz · theta 3–7 Hz · REM raises BP up to 30% · ~80% dream recall from REM
- Prevalence: 9% F / 24% M at AHI>5; 6% / 13% moderate–severe; 936 M / 425 M globally
- Weight: 10% gain → +32% AHI; 10% loss → −26% AHI
- PALM: ~70% ≥1 non-anatomical trait; loop gain ~36%, arousal threshold ~37%, muscle responsiveness ~36%; P_{crit} ~19/58/23%
- OSA–TMD: ≥2 signs/symptoms → 73% greater first-onset TMD incidence; chronic TMD >3× more frequent
- Snoring affects up to 50%; ⅓ of OSA don't snore; ⅓ of snorers have no OSA; heavy = >50% of sleep time; smartphone myofunctional programme −22% snoring rate
- CSA when ≥50% of apnoeas central; idiopathic CSA <5% of CSA
- Insomnia: symptoms 33–50% · disorder 10–15% · chronic 5–15% · comorbid 50–75% · chronic = ≥3 nights/week × ≥3 months
- CPAP 4–6 cmH_2O for snoring; PLM index >15/h adults, >5/h children; MSLT ≤8 min + ≥2 SOREMPs

SELF-TEST — TOPIC 1

1. Is upper airway narrowing during sleep abnormal?

No — narrowing, and even brief complete occlusion, is a normal physiological event caused by withdrawal of tonic neural drive to the pharyngeal dilators. In OSA this normal response is *exaggerated* by combined abnormal anatomy and physiology, so narrowing becomes repetitive collapse with fragmentation, hypoxaemia or both.

2. Name the four PALM traits with prevalences and one targeted therapy each.

P_{crit} / passive anatomy — impaired in essentially all, severity varying → CPAP, appliance, weight loss, positional therapy, surgery. Arousal threshold, low in ~37% → eszopiclone. Loop gain, high in ~36% → acetazolamide or oxygen. Muscle responsiveness, poor in ~36% → hypoglossal nerve stimulation, myofunctional therapy, atomoxetine + oxybutynin. ~70% have ≥1 non-anatomical impairment.

3. Why does an appliance fail in a third of patients despite correct fitting and full titration?

Because it is an anatomical intervention: it reduces collapsibility dose-dependently but does not systematically change genioglossus function, loop gain or arousal threshold. A patient driven substantially by a non-anatomical trait, or with P_{crit} >+2 cmH_2O, will not be corrected by protrusion alone — further advancement adds side effects without benefit. The rational response is combination or alternative therapy.

4. How would you phrase the OSA–TMD relationship?

A robust association with plausible mechanisms, **not** an established causal relationship. Evidence: ≥2 OSA signs/symptoms conferred 73% greater incidence of first-onset TMD independent of confounders, with OSA symptoms preceding onset; chronic TMD was >3× more frequent in low- versus high-OSA-likelihood adults. Mechanisms: central sensitisation with impaired pain inhibition via reduced baroreflex sensitivity; sympathetic activation in catecholamine-sensitive individuals; and sleep bruxism. Either way the prevalence justifies screening for TMD before starting appliance therapy.

TOPIC 1

Normal sleep, sleep disorders and the pathophysiology of sleep-disordered breathing

EXAMINER'S BLUEPRINT

1. Health benefits of normal sleep
2. Physiology of normal sleep
3. Pathophysiology of SDB including anatomical and non-anatomical factors
4. Prevalence and comorbidities of SDB
5. Risk factors for SDB including age and gender
6. Signs and symptoms of SDB
7. Impact of sleep disorders on the individual
8. Impact of sleep disorders on society
9. Prevalence, pathophysiology and clinical features of insomnia
10. Other respiratory sleep disorders
11. Other non-respiratory sleep disorders
12. The effects of pharmacotherapeutics on sleep

This topic is the foundation on which the other nine rest. Examiners use it to establish whether a candidate is a dentist who has learned some appliance technique, or a clinician who understands the disease being treated. The single most valuable investment here is the **anatomical/non-anatomical framework** in section 1.3: it explains why oral appliance therapy works in some patients and not others, why the appliance is not simply "a weaker CPAP", and it is the intellectual scaffolding for Topics 3, 5 and 6.

1.1 Health benefits of normal sleep

Sleep is not passive rest. It is an actively generated, reversible behavioural state with distinct restorative functions, and the examinable point is that each function maps onto a recognisable clinical consequence when sleep is lost.

Function of sleep	Mechanism	Consequence when disrupted
Physical restoration and repair	Protein synthesis, cell and tissue repair, anabolic dominance	Impaired recovery, injury susceptibility
Biochemical refreshment	Restoration of synaptic and neuronal function; glial-mediated glymphatic clearance of metabolites; glycogen restoration; neurogenesis	Cognitive slowing; accumulation of neurotoxic metabolites implicated in neurodegeneration
Memory consolidation	Declarative memory in NREM (sleep spindles); procedural and emotional memory in REM	Impaired learning, poor academic and occupational performance
Emotional regulation	REM-dependent processing of affective experience; limbic reactivation	Irritability, mood lability, amplification of depression and anxiety
Hormone regulation	Growth hormone pulse in early slow-wave sleep; prolactin, testosterone, parathyroid hormone; thyroid hormone in late evening; melatonin under light–dark control	Growth failure in children; metabolic dysregulation; reduced libido
Immune function	Cytokine-mediated host defence and reset	Increased infection susceptibility, systemic inflammation
Metabolic regulation	Insulin sensitivity; leptin and ghrelin balance	Increased appetite, insulin resistance, weight gain

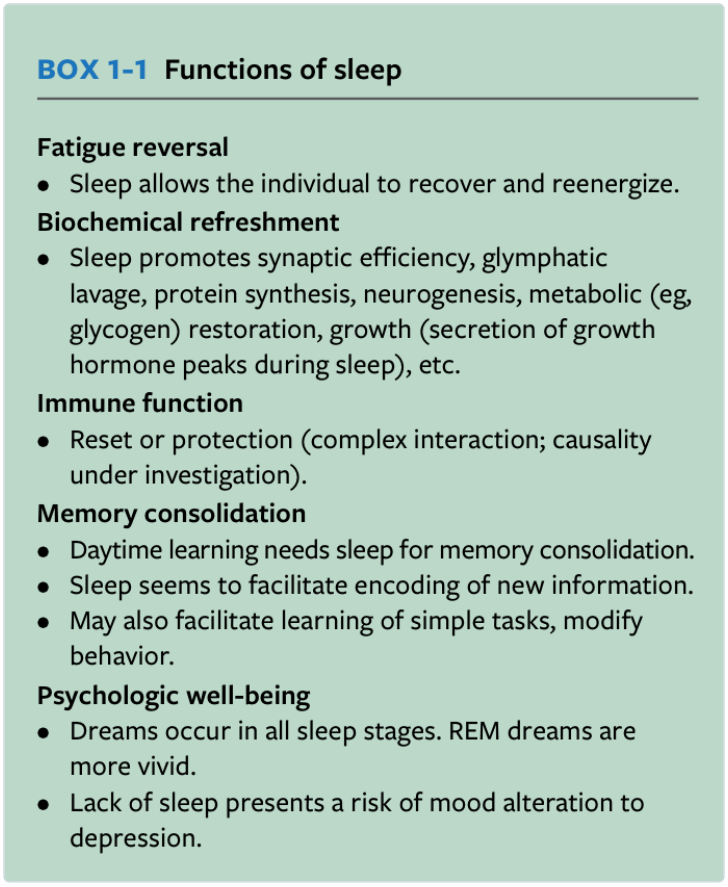


Figure 1.1 — Functions of sleep. Fatigue reversal, biochemical refreshment (synaptic efficiency, glymphatic lavage, protein synthesis, neurogenesis, growth hormone peak), immune function, memory consolidation and psychological well-being.

Why it is examinable: a favourite opening viva question is simply "why do we sleep?". A structured five-heading answer with one mechanism each scores far better than a list of adjectives.

1.2 Physiology of normal sleep

> **DEFINITION TO KNOW VERBATIM**
>
> **Sleep** is a reversible behavioural state characterised by decreased responsiveness to arousing stimuli and disengagement from environmental surroundings. It is a complex physiological and behavioural process orchestrated by distinct neuronal networks, and is qualitatively distinct from rest. Human sleep physiology is assessed by polysomnographic recording of brain waves, eye movements and muscle tone.

Two-process regulation

- **Process S — homeostatic sleep drive.** Sleep pressure accumulates with time awake (adenosine accumulation is the principal candidate mediator) and dissipates during sleep.
- **Process C — circadian rhythm.** An endogenous ~24-hour oscillator in the **suprachiasmatic nucleus** of the hypothalamus, entrained principally by light via the retinohypothalamic tract, promoting wake during the day and sleep at night. **Melatonin**, secreted by the pineal gland, induces sleepiness partly by opposing the SCN alerting signal and is suppressed by light — the mechanistic basis for the effect of evening screen use.

The interaction of the two explains clinical phenomena examiners like: why a night-shift worker is sleepy at 04:00 despite adequate prior sleep (circadian trough), and why an adolescent with a delayed circadian phase cannot fall asleep at 22:00 no matter how tired (Topic 8).

Neurochemistry

- **Wake-promoting:** orexin/hypocretin (lateral hypothalamus), noradrenaline (locus coeruleus), serotonin (raphe), histamine (tuberomammillary nucleus), acetylcholine, dopamine.
- **NREM-promoting:** GABA and galanin from the ventrolateral preoptic area; overall reduction in neuronal firing.
- **REM generation:** cholinergic ("REM-on") neurons of the pedunculopontine and laterodorsal tegmentum become active while monoaminergic ("REM-off") serotonergic, noradrenergic and orexinergic activity falls to its nadir. This reciprocal switch explains why REM-suppressing drugs are typically serotonergic or noradrenergic (1.12), and why loss of orexin neurons produces narcolepsy with intrusion of REM phenomena into wakefulness (1.11).

Sleep architecture and cycling

Sleep alternates cyclically between NREM and REM. A normal episode begins in NREM and progresses through deepening stages before the first REM period.

Stage	% of total sleep	Duration (first cycle)	EEG and clinical features
N1	2–5%	1–7 min	Transition from wakefulness; alpha (8–13 Hz) replaced by low-voltage mixed-frequency activity; vertex sharp waves; easily interrupted by noise. Transitional role in cycling
N2	45–55%	10–25 min, lengthening each cycle	Low-voltage mixed frequency with **sleep spindles** (11–16 Hz) and **K-complexes**. Spindle density rises after learning a new task — the electrophysiological correlate of memory consolidation. Requires a more intense stimulus to wake than N1
N3 (old stages 3 + 4)	~13–23%	20–40 min	**Slow-wave sleep:** high-voltage (>75 µV) slow (0.5–2 Hz) delta activity in ≥20% of the epoch. Highest arousal threshold of all NREM stages. Concentrated in the first third of the night; the growth hormone pulse occurs here. Disorders of arousal (sleepwalking, sleep terrors) arise from N3
REM	20–25%	1–5 min, progressively prolonged	Desynchronised low-voltage mixed-frequency EEG, **saw-tooth waves**, theta (3–7 Hz), bursts of rapid eye movements, and **skeletal muscle atonia** which prevents dream enactment. ~80% report vivid dreaming on arousal from REM. Longest in the final third of the night

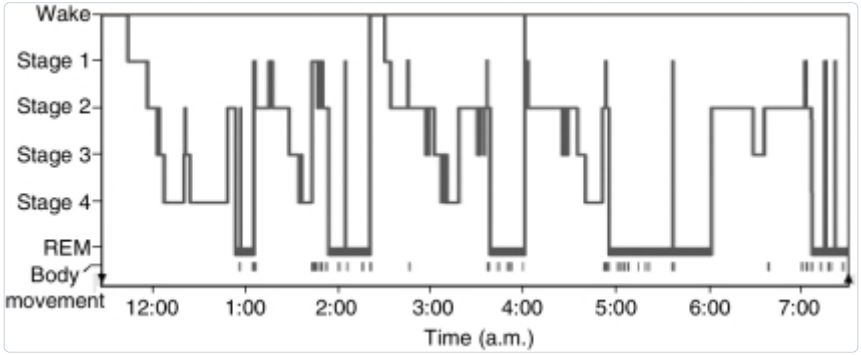


Figure 1.2 — Hypnogram of a normal adult night. Note the two features examiners look for: slow-wave sleep (stages 3–4) is concentrated in the first third of the night and largely disappears later, while REM periods lengthen progressively and dominate the final third.

Why it is examinable: this pattern underpins several downstream facts — why REM-predominant OSA is worst in the early hours, why a split-night study can under-sample REM, and why a successfully treated patient shows REM rebound with an increase in REM percentage rather than an increase in total sleep time.

Physiological changes during sleep

These changes are well tolerated by healthy individuals but may destabilise a patient whose cardiovascular or respiratory reserve is already marginal — the reason OSA is dangerous in the comorbid patient rather than merely inconvenient.

System	NREM	REM
Brain activity	Decreases from wakefulness	Increases in motor and sensory areas; other regions similar to NREM
Heart rate	Slows	Increases and becomes variable
Blood pressure	Falls (physiological nocturnal "dip")	Rises, by up to ~30%, and fluctuates
Sympathetic nerve activity	Decreases as NREM deepens, but bursts occur with K-complexes, arousals and large body movements	Increases markedly above waking levels
Muscle tone	Similar to wakefulness	**Absent** (atonia) — including the pharyngeal dilators
Respiration	Slower, deeper, very regular; minute ventilation falls; chemosensitivity to hypoxia and hypercapnia is blunted	Irregular, faster, less predictable; brief central pauses may occur; cough is suppressed
Upper airway resistance	Increases from wakefulness	Increases further and varies
Cerebral blood flow	Significant reduction in flow and metabolism	Total flow comparable to wakefulness; regionally increased in limbic and visual association areas
Thermoregulation	Regulated at a lower set point; shivering threshold lowered	**Not regulated** — no shivering or sweating; temperature drifts toward ambient
Renal	Reduced excretion of sodium, potassium, chloride and calcium; concentrated urine and reduced flow — hence nocturia is *abnormal* and a useful OSA symptom	
Endocrine	Growth hormone in early SWS; thyroid hormone in the late evening; melatonin under light–dark control	

HIGH-YIELD

- The sharp rise in heart rate and blood pressure on morning awakening explains the **morning peak in myocardial infarction**. In OSA this surge is repeated dozens of times a night.
- REM atonia applies to the pharyngeal dilators too. This is why REM-predominant OSA exists and why it responds relatively poorly to mandibular advancement (Topic 7).
- Thermoregulation is **suspended** in REM. A memorable single fact that is easy to ask and easy to mark.

EXAM TRAP — NOMENCLATURE

Older texts (Rechtschaffen & Kales) describe **four** NREM stages. Since the AASM 2007 scoring manual, stages 3 and 4 have been merged into a single stage **N3**, and stage nomenclature is N1/N2/N3/R. Your source notes use the four-stage terminology. Both are acceptable if you signal that you know the modern convention: say "slow-wave sleep, stage N3, previously stages 3 and 4". Using the old scheme without acknowledgement can read as out-of-date reading.

1.3 Pathophysiology of SDB: anatomical and non-anatomical factors

DEFINITIONS TO KNOW VERBATIM

Sleep-disordered breathing (SDB) — a group of disorders characterised by abnormalities of breathing, respiratory pattern or quantity of ventilation during sleep. It spans a wide spectrum: increased upper airway airflow resistance, heavy snoring, marked reduction in airflow (hypopnoea) and complete cessation of breathing (apnoea).

Obstructive sleep apnoea (OSA) — a form of SDB involving snoring but caused by more significant upper airway obstruction, with consequent sleep fragmentation, hypoxaemia, or both.

OSA versus OSA syndrome — where objective polysomnographic evidence exists *without* symptoms, the correct term is OSA alone; when accompanied by symptoms it is OSA *syndrome*. Examiners do ask for this distinction.

The core mechanism

During wakefulness, tonic neural activation of the pharyngeal dilator muscles (principally **genioglossus**) maintains airway patency against the negative intraluminal pressure of inspiration. With sleep onset this compensatory tonic input is withdrawn. Some degree of airway narrowing during sleep is therefore a **normal physiological event**. In the patient with OSA the normal response is exaggerated by the combination of abnormal anatomy and abnormal physiology, so that narrowing becomes repetitive collapse. Obstruction may occur anywhere between the nasopharynx and larynx; the commonest sites are **retropalatal** (behind the soft palate) and **retroglossal** (behind the tongue base).

Mechanically the pharynx behaves as a collapsible tube — a **Starling resistor**. Its collapsibility is quantified by the **critical closing pressure (P_{crit})**: the minimum airway pressure required to prevent complete collapse. A healthy airway has a negative P_{crit}; the OSA airway is shifted towards positive values, and the more positive the value, the more collapsible the airway.

Anatomical factors

Category	Contributors
Craniofacial skeletal	Retrognathia and micrognathia; maxillary constriction with a high-arched or narrow palate; steep mandibular plane angle; excessive vertical facial growth; inferiorly positioned hyoid bone; reduced maxillary and mandibular length
Soft tissue volume	Macroglossia and increased tongue volume; elongated soft palate and enlarged uvula; adenotonsillar hypertrophy (dominant in children); parapharyngeal fat pads; large mandibular tori reducing oral volume; lateral pharyngeal wall thickening
Nasal	Septal deviation, turbinate hypertrophy, polyps, allergic rhinitis — increasing resistance, promoting mouth breathing and reducing tolerance of nasal PAP
Obesity	Peripharyngeal fat deposition mechanically loads the airway and offsets dilator activity, an effect most marked in sleep when neuromuscular activity falls. Obesity also reduces functional residual capacity and chest wall compliance, which reduces caudal tracheal traction on the pharynx and further increases collapsibility; it disturbs the relationship between respiratory drive and load compensation
Other	Rostral fluid shift to the neck on recumbency; pharyngeal sensory neuropathy from chronic vibration; reduced lung volumes in neuromuscular disease

Non-anatomical factors — the PALM construct

Approximately **all** patients with OSA have some degree of anatomical impairment, but the magnitude varies widely, and around **70%** have impairment of one or more non-anatomical traits. Four traits are conventionally described, summarised by the mnemonic **PALM**.

PALM	Trait	Mechanism and consequence	Targeted intervention
P	P_{crit} — passive anatomy/collapsibility	The anatomical substrate. Mild <−2, moderate −2 to +2, severe >+2 cmH_2O. Distribution in OSA: ~19% mild, ~58% moderate, ~23% severe	CPAP, oral appliance, weight loss, positional therapy, upper airway surgery
A	Arousal threshold (low in ~37%)	The patient wakes prematurely in response to a respiratory disturbance — *before* the dilator muscles have had time to recruit — producing sleep fragmentation and ventilatory instability. Paradoxically a low threshold can also limit apnoea duration	Hypnotic agents (e.g. eszopiclone) to raise the threshold
L	Loop gain (high in ~36%)	An oversensitive ventilatory control loop: excessive ventilatory response to CO_2 causes overshoot, subsequent hypocapnic undershoot and unstable periodic breathing, promoting both obstructive and central events	Acetazolamide, supplemental oxygen
M	Muscle responsiveness (poor in ~36%)	Failure of the dilators to recruit in response to negative pressure sensed at the pharynx and epiglottis	Hypoglossal nerve stimulation, myofunctional therapy, noradrenergic–antimuscarinic agents (atomoxetine + oxybutynin)

Clinically the traits combine. Of patients with **moderate** anatomical impairment — the largest group — about 64% have at least one significant non-anatomical impairment and are candidates for combined anatomical and non-anatomical therapy, while about 36% have no major non-anatomical impairment and can be managed with anatomical intervention alone. Patients with **severe** anatomical impairment require major anatomical intervention regardless.

HIGH-YIELD — WHY THIS FRAMEWORK MATTERS TO A DENTIST

An oral appliance is an **anatomical** intervention. It reduces pharyngeal collapsibility in a dose-dependent manner without systematically altering genioglossus function, loop gain or arousal threshold. It therefore follows that:

- Patients whose disease is predominantly anatomical and mild-to-moderate in collapsibility respond best.
- A patient with a highly collapsible airway plus unstable ventilatory control is a predictable non-responder — and no amount of further protrusion will fix a loop-gain problem.
- This is the mechanistic justification for **combination therapy**, and it is the most sophisticated answer available to the question "why does oral appliance therapy fail in a third of patients?"

Estimating phenotype without a research laboratory. Two practical surrogates have been described. A **CPAP dial-down manoeuvre** during a titration night (CPAP briefly withdrawn for five breaths, repeated five to ten times) allows calculation of peak inspiratory flow as a percentage of the CPAP-supported baseline; a value ≥30% indicates a only mildly collapsible airway and correlates with a more negative P_{crit}. The awake **upper airway collapsibility index** uses choanal and epiglottic pressure catheters during negative-pressure inspiration, takes about ten minutes, and correlates with sleeping P_{crit}. Both are comparable to the gold-standard P_{crit} measurement, and clinical/polysomnographic prediction models are emerging.

1.4 Prevalence and comorbidities of SDB

Population and definition	Women	Men	Source
AHI > 5, aged 30–60	9%	24%	Wisconsin Sleep Cohort, 1993
AHI > 5 *plus* excessive sleepiness (i.e. OSA syndrome)	2%	4%	
AHI > 15 (moderate–severe)	4%	9%	
Moderate–severe, aged 30–70, USA 2007–2010	6%	13%	Peppard *et al.* 2013
Any OSA, US adults	~17%; up to ~38% in some adult estimates		Varies with definition
Global, aged 30–69, AHI ≥5	936 million (mild–severe); 425 million moderate–severe		Benjafield *et al.* 2019

The wide variation between studies is itself examinable: prevalence depends on the **hypopnoea definition** used (3% versus 4% desaturation), the AHI threshold, whether symptoms are required, and the age and obesity profile of the population. If asked "how common is OSA?", the correct answer names a figure *and* its definition.

Comorbidity	Association with SDB	Clinical Impact
Hypertension	Strong, independent	Increased risk, dose-response effect
Cardiovascular disease	Strong, bidirectional	Higher morbidity/mortality, arrhythmias
Metabolic syndrome/diabetes	Strong, independent	Insulin resistance, worsened outcomes
GORD	Frequent coexistence	Mutual exacerbation
Depression/anxiety	Increased prevalence	Lower quality of life, more severe SDB
Neurocognitive decline	Increased risk	Impaired function, even in mild SDB

FIGURE 1 Summary of major comorbidities associated with sleep-disordered breathing

Figure 1.3 — Major comorbidities of sleep-disordered breathing, with the strength and direction of association and the clinical impact of each: hypertension (strong, independent, dose-response), cardiovascular disease (strong, bidirectional), metabolic syndrome and diabetes (strong, independent), GORD (frequent coexistence, mutual exacerbation), depression and anxiety (increased prevalence), and neurocognitive decline.

Why it is examinable: note the column headed "association" — examiners reward candidates who distinguish independent from bidirectional associations, and who identify hypertension as the one with a genuine dose–response relationship.

Predictive comorbidities — the conditions whose presence should raise suspicion of OSA in a dental patient — are obesity, hypertension, cardiovascular disease (particularly ischaemic heart disease), and type 2 diabetes. Long-term sequelae include neurocognitive impairment, metabolic syndrome, systemic hypertension and cardiovascular disease.

1.5 Risk factors for SDB including age and gender

Obesity — the single most important, and reversible, risk factor

The relationship is **bidirectional**, which is the point examiners probe. Obesity increases OSA severity by the anatomical and mechanical routes described in 1.3. Conversely, sleep fragmentation lowers leptin and raises ghrelin, increasing hunger and appetite — particularly for calorie-dense carbohydrate — while daytime sleepiness reduces physical activity. A self-perpetuating cycle results: disrupted sleep → increased appetite and reduced activity → weight gain → worse OSA → more disrupted sleep. Quantitatively, a 10% weight gain predicts a ~32% increase in AHI, and a 10% weight loss a ~26% decrease.

Age

- Prevalence rises with age, with a peak in the sixties, and may plateau or decline in very advanced age.
- In children the dominant risk factor shifts with age: adenotonsillar hypertrophy predominates in younger children (peak 2–8 years, when tonsillar tissue is proportionally largest), while overweight becomes the major factor in adolescence.

Gender

- SDB is substantially more prevalent in men — roughly a 2:1 to 3:1 ratio at any given threshold.
- The gender gap **narrows with advancing age and after the menopause**, implicating sex hormones. Loss of the protective effect of oestrogen, together with post-menopausal weight gain and central fat redistribution, increases both SDB and cardiovascular risk in women.
- Women more often present atypically — insomnia, fatigue, mood disturbance rather than classic witnessed apnoea — and are therefore under-diagnosed. This is a good point to raise in a viva about screening.

Other risk factors

- Anatomical: narrowed airway, craniofacial abnormality, large neck circumference
- Large tongue
- Family history of SDB (genetic predisposition to craniofacial and body-fat distribution phenotypes)
- Smoking — chronic upper airway inflammation and oedema
- Alcohol, sedatives, hypnotics and opioids — reduced dilator tone and blunted arousal
- Nasal congestion, rhinitis, sinusitis
- Systemic and local inflammatory mechanisms
- Comorbid disease: hypertension, congestive heart failure, type 2 diabetes, hypothyroidism, acromegaly, PCOS, neurological disease
- Supine sleep posture

1.6 Signs and symptoms of SDB

Nocturnal	Daytime
• Snoring (loud, habitual, socially disruptive) • Witnessed apnoeas • Waking with choking or gasping • Restless, unrefreshing sleep • Frequent arousals; insomnia • **Nocturia** • Night sweats • Vivid dreams • Salivation and tooth grinding	• Excessive daytime sleepiness • Tiredness and fatigue • Impaired concentration and cognition • **Morning headache** • Mood change, irritability, low mood • **Dry mouth** on waking • Reduced libido, impotence • Nasal congestion • Restless legs

OSA may be entirely asymptomatic. Sympathetic nerve activity and arterial pressure are commonly elevated in people with OSA even in the absence of symptoms — the basis for treating asymptomatic severe disease.

HIGH-YIELD — THE THREE REASONS AN ADULT ATTENDS A SLEEP CLINIC

1. **Their own daytime symptoms** — unrefreshing sleep, tiredness and fatigue, poor concentration and focus, low mood, or excessive daytime sleepiness.
2. **Suspicion of OSA and concern about consequences** — often prompted by apnoeas witnessed by a partner; concerns range from immediate (driving and workplace safety) to long-term (cardiovascular disease, stroke, cognitive decline).
3. **The effect on their bed partner** — socially disruptive snoring with or without witnessed apnoeic episodes is a very common trigger symptom.

Children present differently and this contrast is frequently examined: **inattention, irritability, behavioural and learning problems** rather than sleepiness (Topic 8).

1.7 Impact of sleep disorders on the individual

- Impaired alertness and excessive daytime sleepiness
- Hypertension, including treatment-resistant hypertension
- Cardiovascular disease: stroke, myocardial infarction, congestive heart failure, coronary artery disease
- Atrial fibrillation that is difficult to control (and recurs after ablation)
- Increased mortality and premature death
- Type 2 diabetes and glucose intolerance; metabolic syndrome
- Dementia and neurocognitive decline
- Reduced quality of life
- Pain — see below
- Social disharmony and relationship difficulty from snoring; snoring also has an independent adverse effect on the carotid arteries

Sleep, pain and temporomandibular disorder — the dentally relevant impact

Experimentally reducing total sleep time and selectively disrupting slow-wave and REM sleep evokes musculoskeletal pain, lowers pain thresholds, increases pain sensitivity, and reduces the antinociceptive potency of morphine. This is the mechanistic backdrop to the observed association between OSA and TMD.

- In a population-based cohort free of TMD at baseline, OSA signs and symptoms were associated with an increased incidence of **first-onset TMD**. Men and women with two or more signs or symptoms of OSA had a **73% greater incidence** of first-onset TMD, independently of age, gender, race/ethnicity, obesity, smoking history and autonomic parameters. OSA symptoms *preceded* first-onset TMD.
- In the parallel case-control analysis, chronic TMD was more than **three times** as frequent among adults with a low relative to a high likelihood of OSA, independently of the same factors.

Three mechanisms are proposed: (i) central sensitisation and pain amplification with impaired pain-inhibitory function — experimental inspiratory resistance reduces baroreceptor reflex sensitivity, and impaired baroreflex sensitivity is implicated in TMD; (ii) increased sympathetic activation, with individuals genetically predisposed to catecholamine sensitivity at higher risk of first-onset TMD; and (iii) sleep bruxism, which is more common in people with OSA, as a possible contributor to TMD development.

1.8 Impact of sleep disorders on society

Domain	Impact
Public safety	Sleepy drivers have impaired reflexes and judgement; driver sleepiness contributes to a significant proportion of serious and fatal crashes. Workplace accidents increase through slowed reaction time and impaired cognition, endangering workers and the public. This is why suspected OSA in a **commercial driver or other safety-critical worker**, or reported sleepiness while driving, requires *urgent* assessment and explicit advice about driving risk
Economic	Reduced productivity and lost work hours; disproportionately affects the middle-aged workforce, so more sick leave and more work disability; greater consumption of healthcare resources through consultations, prescriptions and hospital services
Health system	Direct and indirect costs of untreated disease place a substantial burden on health systems; OSA is a public health issue, not only an individual one
Mental health and social function	Worsening of depression and anxiety; social withdrawal and reduced desire for social interaction, contributing to loneliness
Physical health at population level	Higher risk of chronic illness, compromised immune function and increased infection susceptibility

1.9 Insomnia: prevalence, pathophysiology and clinical features

DEFINITION TO KNOW VERBATIM

Insomnia is a subjective perception of difficulty with sleep initiation, duration, consolidation or quality that occurs *despite adequate opportunity for sleep*, and that results in some form of daytime impairment.

Both elements are essential to the diagnosis: **nocturnal symptomatology and daytime dysfunction**. A patient who sleeps five hours, feels well and functions normally is a short sleeper, not an insomniac.

Classification and duration criteria

- **Chronic insomnia disorder** — sleep-onset and/or sleep-maintenance complaints with associated daytime impairment, exceeding minimal frequency and duration thresholds shown to be associated with clinically significant morbidity. Under ICSD-3/ICSD-3-TR this means symptoms at least **three nights per week for at least three months**.
- **Short-term insomnia disorder** — sleep–wake difficulties with clinically significant sleep dissatisfaction or waking impairment that fail to meet the frequency and duration criteria of chronic insomnia disorder.
- **Other insomnia disorder** — rare cases that do not meet criteria for short-term insomnia disorder yet have sufficient symptoms to warrant clinical attention.

EXAM TRAP — THE DURATION THRESHOLD

Older material (and some revision notes) state that chronic insomnia requires symptoms "for at least a month", with acute or transient insomnia lasting days to weeks. That was the ICSD-2/DSM-IV framing. **ICSD-3 and ICSD-3-TR require ≥3 nights per week for ≥3 months**, and DSM-5 aligns with this. Quote the three-month figure.

Prevalence

- Insomnia *symptoms*: 33–50% of the adult population; about one third report dissatisfaction with sleep.
- Insomnia symptoms *plus* distress or impairment (insomnia disorder): 10–15%.
- Meeting diagnostic criteria for **chronic** insomnia: 5–15%.
- In patients with medical and psychiatric comorbidity, prevalence rises to 50–75%.
- Risk factors: increasing age, female sex, comorbid disorders, shift work, unemployment, lower socioeconomic status.

Pathophysiology — hyperarousal

The unifying concept is **physiological, cognitive and cortical hyperarousal** present during both sleep and wakefulness, with heightened activity of the sympathetic nervous system and the hypothalamic–pituitary–adrenal axis thought to perpetuate sleep–wake dysfunction. Objective correlates include:

- Increased heart rate and altered heart rate variability
- Increased whole-body metabolic rate
- Elevated cortisol, ACTH and corticotropin-releasing factor, particularly near sleep onset
- Increased body temperature
- Increased high-frequency (beta and gamma) EEG activity during NREM sleep
- Increased cerebral glucose metabolism on PET

No discrete structural brain pathology is identifiable, except where insomnia is comorbid with a neurological disorder such as stroke, brain trauma or multiple sclerosis. Genetics, environment, behavioural patterns and psychological stress all contribute to development and perpetuation.

Clinical features

Adults typically report difficulty initiating or maintaining sleep, early-morning awakening, and non-restorative sleep. In children, the caregiver reports bedtime resistance, frequent night awakenings and inability to sleep independently. At least one of the following forms of related daytime impairment must be present:

- Fatigue or malaise
- Attention, concentration or memory impairment
- Social or vocational dysfunction, or poor school performance
- Mood disturbance or irritability
- Daytime sleepiness
- Reduced motivation, energy or initiative
- Proneness for errors or accidents at work or while driving
- Tension, headaches or gastrointestinal symptoms in response to sleep loss
- Concerns or worries about sleep

Insomnia often accompanies comorbid medical illness (hypertension, type 2 diabetes), mental disorders (anxiety, depression), other sleep disorders, substance use and exposure (alcohol, medication), and pain. It is associated with increased risk of motor vehicle and worksite accidents and of incident psychiatric and cardiovascular disorders. **Insomnia is best viewed as a comorbid disorder that warrants separate treatment attention** rather than a symptom that will resolve when the "primary" condition is treated.

Treatment

Cognitive behavioural therapy for insomnia (CBT-I) is recommended as first-line treatment, without the risk of tolerance or the adverse effects of pharmacological approaches. Its five components are more effective in combination than alone.

Table 1. Components of CBT-i

Component	Description
Cognitive therapy	Aims to identify, challenge, and replace dysfunctional beliefs and attitudes about sleep and insomnia. Such misconceptions may include unrealistic expectations of sleep, fear of missing out on sleep, and overestimation of the consequences of poor sleep.
Stimulus control	Behavioral instructions aimed at strengthening the association between bed and sleep and preventing conditioning of the patient to associate bed with other stimulating activities. Such instructions include avoiding nonsleep activities in the bedroom; going to bed only when sleepy; and leaving the bedroom when unable to sleep for 15–20 min, returning to bed only when sleepy.
Sleep restriction	Behavioral instructions to limit time in bed to match perceived sleep duration in order to increase sleep drive and further reduce time awake in bed. Time allowed in bed is initially restricted to the average time perceived as sleep per night and then adjusted to ensure sleep efficiency remains >85%.
Sleep hygiene	General recommendations relating to environmental factors, physiologic factors, behavior, and habits that promote sound sleep. Specific instructions include advice on control of the bedroom environment, including avoiding visual access to a clock; regular sleep scheduling and avoiding long daytime naps; and limiting alcohol, caffeine, and nicotine intake, especially before bed.
Relaxation	Any relaxation technique that the patient finds effective can be used to limit cognitive arousal and reduce muscular tension to facilitate sleep. Specific techniques that may be used include meditation, mindfulness, progressive muscle relaxation, guided imagery, and breathing techniques.

Figure 1.4 — The components of CBT-I: cognitive therapy (identifying and challenging dysfunctional beliefs and attitudes about sleep and about the consequences of poor sleep), stimulus control (re-associating bed and bedroom with sleep, and re-establishing a consistent sleep–wake schedule), sleep restriction (limiting time in bed to match perceived sleep duration to increase sleep drive, then extending it as sleep efficiency improves), sleep hygiene, and relaxation training.

Why it is examinable: candidates who can name all five components — and identify sleep restriction and stimulus control as the two with the strongest evidence — clearly distinguish themselves from those who answer only "sleep hygiene".

Table 2. Glossary

Outcome Measure	Definition	Abbreviation	Unit
Sleep onset latency	Average time to enter sleep after lights out, over the diary period	SOL	Minutes
Wake after sleep onset	Average time spent awake during the night after first entering sleep, over the diary period	WASO	Minutes
Total sleep time	Average total nighttime sleep, over the diary period	TST	Minutes
Sleep efficiency	Total sleep time divided by average time spent in bed, over the diary period	SE%	Percentage

Figure 1.5 — The four sleep-continuity outcome measures used to judge response in insomnia: sleep onset latency (SOL), wake after sleep onset (WASO), total sleep time (TST) and sleep efficiency (SE).

*Why it is examinable: CBT-I produces a marked and durable effect on **SOL, WASO and sleep efficiency**, but only a small improvement in **total sleep time** which augments over time. Sleep-diary measures of these variables are the most consistently used outcome in chronic insomnia because of ease of assessment. Being able to say which variable improves, and by how much, is a high-scoring detail.*

Pharmacological options are second line. Hypnotics — benzodiazepines and benzodiazepine receptor agonists ("Z-drugs") — remain the most commonly used treatment despite this. Second-generation antipsychotics such as quetiapine are increasingly used, driven by dissatisfaction with available treatments rather than by evidence of superiority.

1.10 Other respiratory sleep disorders

Central sleep apnoea (CSA)

DEFINITION

CSA reflects a **lack of drive to breathe** during sleep, producing repetitive periods of insufficient ventilation and compromised gas exchange. A central apnoea is a cessation of airflow of >10 s **in the absence of respiratory effort**. CSA is conventionally considered the primary diagnosis when **≥50% of apnoeas are central in origin**.

The discriminating feature against OSA is therefore **respiratory effort**, detected by thoracic and abdominal respiratory inductance plethysmography belts: present in obstructive events, absent in central events.

- **Aetiologies:** high-altitude periodic breathing (which most healthy people develop if the ascent is sufficient to cause substantial alveolar hypoxia); treatment-emergent CSA; opioid/narcotic-induced central apnoea; Cheyne–Stokes breathing (classically with congestive heart failure and stroke); obesity hypoventilation syndrome; idiopathic CSA, which is uncommon and may account for <5% of patients.
- **Classification by awake CO_2:** *hypercapnic* CSA, reflecting either abnormal central pattern generator output ("won't breathe") or impairment of respiratory motor output caudal to the pattern generator ("can't breathe"); and *non-hypercapnic* CSA, typically a high-loop-gain disorder.
- **Complications:** frequent nocturnal awakenings, excessive daytime sleepiness and increased risk of adverse cardiovascular outcomes.

- **Bidirectional relationship with OSA:** central events provoke obstructive events in patients with vulnerable pharyngeal anatomy and vice versa; this mixed picture is present in the majority of sleep apnoea patients. Treatment of choice for CSA with Cheyne–Stokes breathing depends on ejection fraction — adaptive servo-ventilation is contraindicated in symptomatic heart failure with reduced ejection fraction.

Sleep-related hypoventilation and hypoxaemia

- **Hypoventilation syndromes** — insufficient alveolar ventilation during sleep causing CO_2 retention and hypoxaemia. Includes **obesity hypoventilation syndrome** (obesity with awake hypercapnia not otherwise explained), medication- or substance-induced hypoventilation, and hypoventilation due to a medical or neurological disorder (e.g. Duchenne muscular dystrophy, kyphoscoliosis, COPD).
- **Sleep-related hypoxaemia disorder** — sustained desaturation during sleep without a comparable rise in CO_2.
- **Catathrenia** — slow, monotone, groaning expiratory vocalisation during sleep; benign but often mistaken for a parasomnia by the bed partner.

Primary snoring

DEFINITIONS

Snoring — audible vibration of the upper airway during respiration in sleep. It exists on a continuum: primary snoring → upper airway resistance syndrome → symptomatic SDB.

Primary snoring — habitual snoring in the absence of significant sleep apnoea or oxygen desaturation.

- **Prevalence:** snoring affects up to 50% of the population; the prevalence of *primary* snoring is hard to establish because OSA must first be excluded.
- **Relationship to OSA:** approximately one third of patients with OSA do not snore, and about one third of snorers have no OSA (AHI <5). Snoring therefore neither rules OSA in nor out — the single most useful sentence on this subject.
- **It is not benign.** Vibration may injure adjacent vasculature: heavy snoring (>50% of sleep time) may induce endothelial injury at the carotid bifurcation because of its proximity to the hypopharyngeal walls, with increased carotid intima-media thickness, carotid atherosclerosis and metabolic syndrome. There is also the potential for chronic sleep deprivation, depression and fatigue in the bed partner.

Management of primary snoring — the graded approach

Intervention	Evidence and comment
Weight loss and reduced alcohol	Recommended where appropriate; alcohol reduces pharyngeal dilator muscle tone
Benzodiazepine and opioid reduction	Advise where clinical judgement permits; benzodiazepines increase NREM snoring and upper airway resistance even in healthy people
Positional therapy	Consider in supine-dominant snorers; snoring severity is greater supine. Either a physical barrier or a device that detects position and delivers a vibratory stimulus. Reduces snoring frequency and rate in most non-apnoeic snorers (AHI <5) but not in those with OSA. Older devices had poor long-term adherence due to discomfort and ineffectiveness
Mandibular advancement device	**First-line in dentate patients**, following assessment by both an appropriately trained dentist and a sleep physician. Mechanism: mandibular protrusion pulls the tongue forward and widens the oropharynx, secondarily drawing the soft palate anteriorly and inferiorly via the palatoglossus, displacing the velopharynx forward and controlling airway soft-tissue vibration. Significantly reduces snoring outcome scores. Limited evidence shows a custom titratable device is more efficacious than a thermoplastic "boil-and-bite" device; a partially titratable thermoplastic device reduces snoring but with more interim side effects. A sham placebo *exacerbated* snoring. Occlusal assessment is required before treatment

Intervention	Evidence and comment
CPAP	May be recommended in those already committed to its use or willing to try; effective and safe but with highly variable compliance. A low pressure of 4–6 cmH_2O is sufficient to eliminate snoring
Nasal surgery	Septoplasty with concurrent turbinate reduction is recommended **only** in patients with both symptoms *and* objective nasendoscopic evidence of nasal obstruction. Benefit as a primary anti-snoring procedure is uncertain and inconsistent, but it may be a useful preliminary or adjunctive step before non-surgical therapy
Palatal surgery	*Radiofrequency palatal surgery* — temperature-controlled submucosal volume reduction and stiffening; safe and effective in the short term with minimal morbidity, but patients may require additional therapy. *Modified UPPP* — a reconstructive procedure replacing the original ablative operations; can be offered where snoring is the primary symptom, anatomy is favourable on nasendoscopy, and CPAP and MAD have been considered; principal concern is unpredictable scarring. *Palatal implants* — three permanent implants into the midline soft palate musculature initiating fibrosis without volume reduction; effective short-term but with an unacceptably high extrusion rate, limited durability and potential to worsen OSA. *Injection snoreplasty* — submucosal sclerosant (sodium tetradecyl sulfate, or lidocaine with ethanol) for palatal flutter under local anaesthesia; reduces snoring at 6–12 months with minimal side effects but efficacy declines over time and scarring is unpredictable. *LAUP and Er:YAG laser (Nightlase)* — also described
Complementary and alternative	External and internal nasal valve dilators, lubricating sprays, herbal remedies. Nasal dilator therapy is effective and safe for subjective and objective snoring severity in mild–moderate OSA but without long-term data; theoretical concern regarding lipoid pneumonia with oil-based products. Not advised, given uncertain long-term outcome and cost
Nasal EPAP	Disposable one-way resistor valves over the nostrils creating positive end-expiratory pressure and stabilising the airway. Overall improvement but less than CPAP; minimally harmful with some treatment effect, but costly in the long run
Myofunctional therapy	Delivered by a speech pathologist over ~3 months using isotonic and isometric exercises targeting tongue, soft palate and lateral pharyngeal walls. Reduces snoring severity, generally in the presence of OSA. A 15-minute daily smartphone-delivered regimen over 8 weeks decreased acoustically measured mean snoring rate by 22% and reduced snoring frequency and volume. Low risk, but requires significant commitment for a small reduction
Combination therapy	Pre-phase plus adjunctive, adjunctive plus mainstream, or more than one mainstream treatment. Higher therapeutic efficacy but limited evidence

Upper airway resistance syndrome (UARS)

Increased upper airway resistance producing repetitive respiratory effort-related arousals (RERAs) and fragmented sleep, without qualifying apnoeas, hypopnoeas or desaturation. Patients are often young, non-obese women presenting with fatigue and insomnia rather than sleepiness. Note that **ICSD-3 does not recognise UARS as a separate diagnostic entity**, regarding it as part of the OSA spectrum captured by the respiratory disturbance index; saying so demonstrates current reading.

1.11 Other non-respiratory sleep disorders

Category	Examples and examinable detail
Parasomnias Abnormal behaviours or experiences during sleep	*NREM parasomnias (disorders of arousal, from N3):* confusional arousals, sleepwalking, sleep terrors, sleep-related eating disorder. Typically in the first third of the night, with amnesia for the event; commoner in children. *REM parasomnias:* **REM sleep behaviour disorder** — loss of REM atonia with dream enactment, typically in older men, and an important prodrome of the alpha-synucleinopathies (Parkinson's disease, dementia with Lewy bodies, multiple system atrophy); recurrent isolated sleep paralysis; nightmare disorder
Circadian rhythm sleep–wake disorders	Misalignment of the internal clock with the external environment: delayed sleep–wake phase disorder (typical of adolescents), advanced sleep–wake phase disorder (older adults), irregular and non-24-hour rhythms, shift work disorder, jet lag
Sleep-related movement disorders	**Restless legs syndrome** (Willis–Ekbom disease) — an urge to move the legs with uncomfortable creeping or crawling sensations, worse in the evening and at rest, relieved by movement; associated with iron deficiency and dopaminergic dysfunction. **Periodic limb movement disorder** — repetitive limb twitching or jerking during sleep; a PLM index >15/h in adults (>5/h in children) is abnormal, and the *disorder* requires associated sleep disturbance. Also sleep bruxism (Topic 4) and rhythmic movement disorder
Central disorders of hypersomnolence	**Narcolepsy type 1** — hypocretin/orexin deficiency with cataplexy; type 2 without cataplexy. Diagnostic MSLT findings: mean sleep latency ≤8 min with ≥2 sleep-onset REM periods. Patients may enter sleep directly into REM. Also idiopathic hypersomnia, Kleine–Levin syndrome, and insufficient sleep syndrome
Insomnia disorders	As in 1.9

1.12 The effects of pharmacotherapeutics on sleep

Drug Class	Main Sleep Effects	Notable Risks/Considerations
SSRIs/SNRIs	↓ REM, ↑ REM latency, possible insomnia	RLS, bruxism, nightmares
Sedating Antidepressants	↓ Sleep latency, ↑ SWS, oversedation	Long-term sedation
Benzodiazepines	↑ N2, ↓ REM/N3, improved sleep onset	Dependence, cognitive effects
Antipsychotics	Variable, often sedating	Daytime sleepiness
Melatonin/Orexin Antagonists	↑ Sleep time/efficiency (orexin)	Mixed evidence (melatonin)
Dexmedetomidine	↑ Sleep quality (ICU)	Bradycardia, hypotension

FIGURE 1 Table summarizing drug class effects on sleep

Figure 1.6 — Drug class effects on sleep, with main effects and notable risks: SSRIs/SNRIs (↓REM, ↑REM latency, possible insomnia; RLS, bruxism, nightmares), sedating antidepressants (↓sleep latency, ↑SWS; long-term oversedation), benzodiazepines (↑N2, ↓REM and N3, improved sleep onset; dependence and cognitive effects), antipsychotics (variable, often sedating; daytime sleepiness), melatonin and orexin antagonists (↑sleep time and efficiency; mixed evidence for melatonin), and dexmedetomidine (↑sleep quality in ICU; bradycardia and hypotension).

Why it is examinable: this is the single most efficient revision object for 1.12. Learn the direction of the arrows for each class, and the one signature adverse effect. The SSRI row is doubly relevant to a dentist because of ***drug-induced sleep bruxism****.*

Substances

- **Alcohol** — reduces sleep latency but suppresses REM in the first half of the night with rebound in the second; fragments sleep; reduces pharyngeal dilator muscle tone and blunts arousal responses, worsening OSA and snoring. Advise avoidance in the hours before bed.
- **Caffeine** — an adenosine receptor antagonist (blocking the mediator of homeostatic sleep pressure) with a half-life of roughly five hours. Shortens sleep duration, prolongs sleep onset latency, increases WASO and daytime sleepiness, and reduces slow-wave sleep in a dose-related manner (see Topic 8 for the adolescent vicious cycle).
- **Nicotine** — stimulant with additional nocturnal withdrawal-related arousals; smoking also causes upper airway inflammation.
- **Opioids** — dose-dependent respiratory depression with central apnoeas and ataxic breathing patterns; chronic use is a recognised cause of CSA. Note carefully: while opioids may improve *subjective* sleep quality in patients whose sleep is disturbed by pain, they impair sleep architecture and respiratory control, so the net effect in a patient with SDB is adverse.

Pharmacological treatment of chronic insomnia

Suggested agents include short- to intermediate-acting benzodiazepine receptor agonists (e.g. zolpidem, temazepam, eszopiclone, zaleplon, triazolam), the melatonin receptor agonist ramelteon, the orexin receptor antagonist suvorexant, sedating antidepressants (low-dose doxepin, trazodone, amitriptyline), and other sedating agents such as gabapentin or atypical antipsychotics (olanzapine, quetiapine). These reduce sleep latency, improve sleep maintenance, and may cause residual sedation. Guidelines recommend **against** over-the-counter antihistamines (diphenhydramine), herbal and nutritional substances (melatonin as a hypnotic, valerian, tryptophan) and barbiturates for chronic insomnia.

EXAM TRAP — THE SEDATIVE–OSA INTERACTION

Treating comorbid insomnia or anxiety with a benzodiazepine or hypnotic in a patient with untreated OSA can **worsen the OSA**: reduced muscle tone, blunted arousal response to hypoxia and hypercapnia, and a raised arousal threshold together increase the number and duration of apnoeas. The nuance worth adding is that *raising* the arousal threshold is precisely the intended action of eszopiclone in the low-arousal-threshold phenotype — so the same drug is harmful in one endotype and therapeutic in another. That is a very strong answer.

Sleep hygiene

DEFINITION

Sleep hygiene — the practices and habits that affect the quality of sleep, including use of stimulants, alcohol, exercise, napping, time spent in bed, and quality of the sleep environment.

Good practice includes avoiding naps in the late afternoon and evening; avoiding heavy meals, caffeine and tobacco close to bedtime; exercising earlier in the day; establishing a relaxing bedtime routine; maintaining a regular sleep schedule; and limiting distractions, light and electronic device use in the sleep environment. Sleep hygiene alone is **insufficient** as monotherapy for chronic insomnia — it is one of five CBT-I components, and the weakest of them.

NUMBERS TO MEMORISE — TOPIC 1

- NREM 75–80% / REM 20–25% of total sleep; cycle 70–100 min then 90–120 min; REM latency ~90 min
- N1 2–5% · N2 45–55% · N3 ~13–23%; alpha 8–13 Hz; spindles 11–16 Hz; sawtooth waves in REM; ~80% dream recall from REM
- REM raises blood pressure by up to 30%; thermoregulation absent in REM
- OSA prevalence: AHI>5 — 9% F / 24% M (Wisconsin 1993); moderate–severe — 6% F / 13% M (2007–2010); global 936 million / 425 million moderate–severe
- 10% weight gain → ~32% AHI increase; 10% loss → ~26% AHI decrease
- Non-anatomical traits: ~70% have ≥1; loop gain ~36%, low arousal threshold ~37%, poor muscle responsiveness ~36%; P_{crit} distribution ~19% / 58% / 23%
- OSA and TMD: ≥2 OSA signs/symptoms → 73% greater incidence of first-onset TMD; chronic TMD >3× more frequent in high-OSA-likelihood adults
- Snoring affects up to 50% of the population; ⅓ of OSA patients do not snore; ⅓ of snorers have no OSA
- CSA when ≥50% of apnoeas are central; idiopathic CSA <5% of CSA patients
- Insomnia: symptoms 33–50%; disorder 10–15%; chronic 5–15%; with comorbidity 50–75%. Chronic = ≥3 nights/week for ≥3 months
- CPAP 4–6 cmH_2O abolishes simple snoring; heavy snoring = >50% of sleep time; smartphone myofunctional programme reduced snoring rate by 22%

SELF-TEST — TOPIC 1

1. A candidate is asked: "Is upper airway narrowing during sleep abnormal?" What is the best answer?

> No — airway narrowing, and even brief complete occlusion, is a normal physiological event during sleep, caused by withdrawal of the tonic neural drive to the pharyngeal dilators. In OSA this normal response is *exaggerated* by the combination of abnormal anatomy and abnormal physiology, so that narrowing becomes repetitive collapse with sleep fragmentation, hypoxaemia or both.

2. Name the four traits of the PALM construct, give the approximate prevalence of impairment in each non-anatomical trait, and name one targeted therapy for each.

> P_{crit} (passive anatomy — impaired in essentially all patients, severity varying) → CPAP, oral appliance, weight loss, positional therapy, surgery. Arousal threshold (low in ~37%) → hypnotic such as eszopiclone. Loop gain (high in ~36%) → acetazolamide or supplemental oxygen. Muscle responsiveness (poor in ~36%) → hypoglossal nerve stimulation, myofunctional therapy, atomoxetine with oxybutynin. Around 70% of patients have at least one non-anatomical impairment.

3. Why does an oral appliance fail in roughly one third of patients, despite correct fitting and full titration?

> Because a mandibular advancement device is an *anatomical* intervention: it reduces pharyngeal collapsibility in a dose-dependent manner but does not systematically change genioglossus function, loop gain or arousal threshold. A patient whose disease is driven substantially by a non-anatomical trait, or whose airway is severely collapsible (P_{crit} >+2 cmH_2O), will not be corrected by protrusion alone. Further advancement will add side effects without adding benefit; the rational response is combination or alternative therapy.

4. Distinguish OSA from OSA syndrome, and central from obstructive apnoea.

> OSA describes objective polysomnographic evidence of obstructive events without symptoms; OSA *syndrome* requires accompanying symptoms. An apnoea is ≥90% airflow reduction for ≥10 s; it is obstructive when respiratory effort persists (detected by thoracoabdominal inductance plethysmography) and central when effort is absent. Central sleep apnoea is the primary diagnosis when ≥50% of apnoeas are central.

5. A patient reports that their partner does not snore, so cannot have sleep apnoea. Correct them using figures.

> Snoring is neither necessary nor sufficient. Approximately one third of people with OSA do not snore, and approximately one third of snorers have no OSA (AHI <5). Absence of snoring does not exclude OSA, and its presence does not establish it; formal assessment is required. Equally, negative screening questionnaires — particularly the ESS — do not rule out clinically significant OSA.

6. What is the evidence that OSA and TMD are related, and how would you phrase the relationship?

> In a population-based cohort free of TMD at baseline, individuals with two or more OSA signs or symptoms had a 73% greater incidence of first-onset TMD, independently of age, gender, ethnicity, obesity, smoking and autonomic parameters, and OSA symptoms preceded TMD onset. In the case-control arm, chronic TMD was more than three times as frequent in those with a low relative to a high likelihood of OSA. Proposed mechanisms are central sensitisation with impaired pain inhibition (via re-
> duced baroreflex sensitivity), sympathetic activation in catecholamine-sensitive individuals, and sleep bruxism. Phrase it as a **robust association with plausible mechanisms, not an established causal relationship** — and note that screening for TMD before commencing appliance therapy is warranted by the high prevalence either way.

7. Give the first-line treatment for chronic insomnia, its components, and its effect on outcome measures.

> CBT-I, comprising cognitive therapy, stimulus control, sleep restriction, sleep hygiene and relaxation training — more effective in combination than singly, and without the tolerance and adverse effects of hypnotics. It produces a marked and sustained improvement in sleep onset latency, wake after sleep onset and sleep efficiency, with only a small improvement in total sleep time that augments over time. Sleep-diary measures are the most consistently used outcome because of ease of assessment.

8. A patient with untreated moderate OSA and comorbid insomnia asks for "something to help me sleep". Discuss.

> Both conditions need treatment — this is comorbid insomnia and sleep apnoea (COMISA), and reducing the AHI alone will not resolve the patient's complaint. CBT-I is first line for the insomnia. Benzodiazepines and Z-drugs should be avoided or used with caution in untreated OSA because they reduce muscle tone, blunt the arousal response to hypoxia and hypercapnia and raise the arousal threshold, increasing the number and duration of apnoeic events. The apparent contradiction — that raising the arousal threshold is the therapeutic goal in the low-arousal-threshold endotype — resolves once you specify which endotype you are treating and confirm that the OSA is being treated concurrently.

REFERENCES — TOPIC 1

1. Carskadon MA, Dement WC. Normal human sleep: an overview. In: Kryger MH, Roth T, Dement WC, eds. *Principles and Practice of Sleep Medicine*. 6th ed. Elsevier; 2017:15–24.
2. Borbély AA. A two-process model of sleep regulation. *Hum Neurobiol*. 1982;1(3):195–204.
3. Xie L, Kang H, Xu Q, et al. Sleep drives metabolite clearance from the adult brain. *Science*. 2013;342(6156):373–377.
4. Berry RB, Quan SF, Abreu AR, et al. *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications*. Version 3. American Academy of Sleep Medicine; 2023.
5. American Academy of Sleep Medicine. *International Classification of Sleep Disorders*. 3rd ed, Text Revision (ICSD-3-TR). AASM; 2023.
6. Eckert DJ, White DP, Jordan AS, Malhotra A, Wellman A. Defining phenotypic causes of obstructive sleep apnea: identification of novel therapeutic targets. *Am J Respir Crit Care Med*. 2013;188(8):996–1004.
7. Eckert DJ. Phenotypic approaches to obstructive sleep apnoea — new pathways for targeted therapy. *Sleep Med Rev*. 2018;37:45–59.
8. Schwartz AR, Patil SP, Laffan AM, Polotsky V, Schneider H, Smith PL. Obesity and obstructive sleep apnea: pathogenic mechanisms and therapeutic approaches. *Proc Am Thorac Soc*. 2008;5(2):185–192.
9. Young T, Palta M, Dempsey J, Skatrud J, Weber S, Badr S. The occurrence of sleep-disordered breathing among middle-aged adults. *N Engl J Med*. 1993;328(17):1230–1235.
10. Peppard PE, Young T, Barnet JH, Palta M, Hagen EW, Hla KM. Increased prevalence of sleep-disordered breathing in adults. *Am J Epidemiol*. 2013;177(9):1006–1014.
11. Benjafield AV, Ayas NT, Eastwood PR, et al. Estimation of the global prevalence and burden of obstructive sleep apnoea: a literature-based analysis. *Lancet Respir Med*. 2019;7(8):687–698.
12. Peppard PE, Young T, Palta M, Skatrud J. Prospective study of the association between sleep-disordered breathing and hypertension. *N Engl J Med*. 2000;342(19):1378–1384.
13. Peppard PE, Young T, Palta M, Dempsey J, Skatrud J. Longitudinal study of moderate weight change and sleep-disordered breathing. *JAMA*. 2000;284(23):3015–3021.
14. Sanders AE, Essick GK, Fillingim R, et al. Sleep apnea symptoms and risk of temporomandibular disorder: OPPERA cohort. *J Dent Res*. 2013;92(7 Suppl):70S–77S.
15. Ohayon MM, Li KK, Guilleminault C. Risk factors for sleep bruxism in the general population. *Chest*. 2001;119(1):53–61.
16. Sateia MJ, Buysse DJ, Krystal AD, Neubauer DN, Heald JL. Clinical practice guideline for the pharmacologic treatment of chronic insomnia in adults: an AASM clinical practice guideline. *J Clin Sleep Med*. 2017;13(2):307–349.
17. Edinger JD, Arnedt JT, Bertisch SM, et al. Behavioral and psychological treatments for chronic insomnia disorder in adults: an AASM clinical practice guideline. *J Clin Sleep Med*. 2021;17(2):255–262.
18. Riemann D, Spiegelhalder K, Feige B, et al. The hyperarousal model of insomnia: a review of the concept and its evidence. *Sleep Med Rev*. 2010;14(1):19–31.
19. Aurora RN, Chowdhuri S, Ramar K, et al. The treatment of central sleep apnea syndromes in adults: practice parameters with an evidence-based literature review and meta-analyses. *Sleep*. 2012;35(1):17–40.
20. Australasian Sleep Association. Position statement on the assessment and management of primary snoring in adults. 2021.
21. Guilleminault C, Stoohs R, Clerk A, Cetel M, Maistros P. A cause of excessive daytime sleepiness: the upper airway resistance syndrome. *Chest*. 1993;104(3):781–787.
22. Camacho M, Certal V, Abdullatif J, et al. Myofunctional therapy to treat obstructive sleep apnea: a systematic review and meta-analysis. *Sleep*. 2015;38(5):669–675.
23. Sweetman A, Lack L, Bastien C. Co-morbid insomnia and sleep apnea (COMISA): prevalence, consequences, methodological considerations, and recent randomized controlled trials. *Brain Sci*. 2019;9(12):371.

TOPIC 2

Diagnostic sleep testing in sleep-disordered breathing

BLUEPRINT

1. Normal sleep architecture and respiratory parameters on PSG · 2. Pathological architecture and parameters · 3. Interpreting the PSG and its influence on treatment · 4. Interpreting treatment efficacy on PSG · 5. Attended vs unattended PSG · 6. Vigilance testing (MSLT, MWT) · 7. Screening for SDB

The examinable skill: read a sleep study competently, extract what changes your dental plan, and stop short of diagnosis. The event definitions in 2.2 are the highest-yield sentences in the syllabus.

2.1 Normal architecture and parameters

DEFINITION

Polysomnography — a multi-parametric test, or comprehensive recording, of the biophysiological changes that occur during sleep; used in the study of sleep and as a diagnostic tool in sleep medicine.

Channels and what each establishes

- **EEG** (scalp) — sleep staging and arousals. **EOG** (periorbital) — eye movements, identifies REM and sleep onset. **EMG** (submental, tibial) — muscle tone/REM atonia, leg movements; masseter EMG for bruxism.
- **EEG + EOG + EMG together determine sleep stage.** No staging is possible without them — the fundamental limitation of level 3 and 4 studies.
- **Airflow** — oronasal thermal sensor (apnoea) *and* nasal pressure transducer (hypopnoea); ideally both.
- **Effort** — thoracic and abdominal RIP belts, diaphragm EMG. Distinguishes obstructive from central; paradoxical movement indicates obstruction.
- **SpO_2** — pulse oximetry, finger probe, high sampling rate, fast averaging. **ECG** — rate, rhythm, arrhythmia during events.
- **Body position** — positional dependence; **directly relevant to appliance and positional therapy decisions**.
- **Snoring** — microphone/sound level meter 1 m above the head; frequency per hour and intensity in dB.
- **End-tidal or transcutaneous CO_2** — hypoventilation; a **standard measurement in paediatric PSG**. **Video/audio** — parasomnia, seizure, positional confirmation.
- **RIP mechanism:** thoracic and abdominal bands each contain a conductor forming a wire loop; respiration changes circumference, so cross-sectional area and hence inductance change; a sensor converts this to a voltage reflecting thoracoabdominal volume. Calibrated against a spirometer or pneumotachograph it gives quantitative tidal volume and flow. RIP is the recommended effort sensor for both laboratory PSG and portable monitoring.

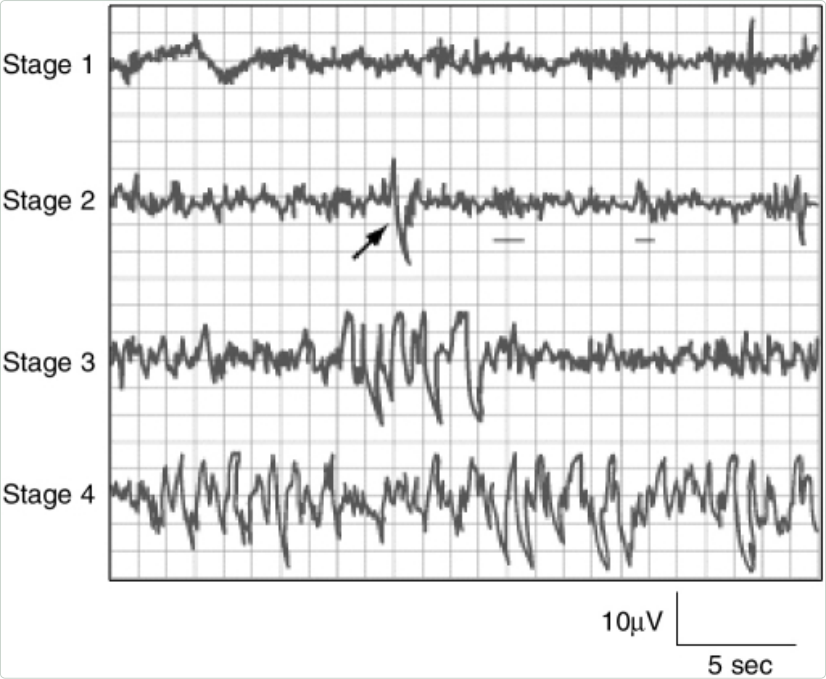


Fig 2.1 — EEG of the NREM stages. Stage 1: low-voltage mixed frequency replacing waking alpha. Stage 2: **sleep spindles** (underlined) and a **K-complex** (arrow). Stages 3–4 (now N3): progressively more high-voltage slow-wave activity. Calibration 10 µV, 5 s. **Examinable because:** you may be shown a trace and asked to stage it. Name the spindle, the K-complex and the delta of N3 confidently.

Normal values

- SOL <30 min · WASO <30 min · sleep efficiency ≥85% (>90% young adults; 75–84% suboptimal; <75% suggests insomnia) · REM latency ~90 min.
- NREM 75–80% dominating the first half, REM 20–25% dominating the second. Arousal index low, scored by the **ASDA 3-second rule**.
- AHI <5/h · ODI low (desaturations ≥3% or ≥4% per hour) · SpO_2 95–100% stable, ≥90% acceptable, minimum SaO_2 an independent severity marker.
- Respiratory rate 12–20/min in adults, up to 30–60/min in infants and young children. TST spent supine reported as % — needed to judge positional dependence.

2.2 Pathological architecture and parameters

EVENT DEFINITIONS — VERBATIM

- **Apnoea** — cessation of airflow (≥90% reduction) for ≥10 s. Obstructive if effort continues; central if effort absent; mixed if it begins central and ends obstructive.
- **Hypopnoea** — reduction in airflow and/or thoracoabdominal movement of ≥30% of baseline for ≥10 s, with ≥3% desaturation *and/or* an arousal. (Historically expressed as >50% reduction; the ≥4% desaturation alternative became **optional** in Scoring Manual v3, 2023, though some payers still require 4%.)
- **RERA** — increasing respiratory effort or flattening of the nasal pressure waveform leading to an arousal, where the event does not meet apnoea or hypopnoea criteria.
- **AHI** — apnoeas + hypopnoeas per hour of sleep. **RDI** — apnoeas + hypopnoeas **+ RERAs** per hour; therefore always ≥ AHI. **Arousal index** — arousals per hour of sleep.
- **OSA diagnosis** — obstructive events per hour of sleep (the RDI) **>15/h**, *or* **>5/h in a patient reporting any of**: unintentional sleep episodes during wakefulness; daytime sleepiness; unrefreshing sleep; fatigue; insomnia; waking with breath-holding, gasping or choking; or a bed partner describing loud snoring, breathing interruptions, or both.

- **Severity — adult:** normal <5 · mild 5–<15 · moderate 15–<30 · severe ≥30. **Paediatric:** <1 · 1–<5 · 5–<10 · ≥10.
- Oxygen desaturation is an **independent** severity measure but harder to quantify with no universally agreed gradation. **AHI ≥15** is the level at which insulin resistance, dyslipidaemia, vascular disease and death risk rise materially — hence the threshold for treating asymptomatic disease.
- **Fragmentation** — increased arousal index, increased stage shifts, increased N1, reduced slow-wave and REM sleep: the signature of untreated OSA.
- **Irregular or absent stages** point to specific disorders — narcolepsy patients may enter sleep directly into REM.
- **Chronic insomnia on PSG:** increased sleep latency and/or WASO (often >30 min) with reduced sleep efficiency for age; increased N1, decreased slow-wave sleep; greater high-frequency (beta/gamma) EEG power; and a characteristic **reverse first-night effect**

— better sleep on the first laboratory night than the second. Some patients have a normal study. **PSG is not indicated in chronic insomnia** unless a breathing or movement disorder is suspected, the diagnosis is uncertain, treatment has failed, or precipitous arousals occur with violent or injurious behaviour.

- **Variables reported.** Sleep: TST, REM min and % of TST, NREM min and %, % TST supine, arousal index, sleep efficiency. Respiratory: AHI, RDI, minimum SaO_2, ODI, snoring frequency per hour, mean and maximum snoring intensity in dB, limb movements, stage distribution.

2.3 Interpreting the PSG — the five things that change your plan

WHAT A DENTIST READS A SLEEP STUDY FOR

1. **AHI/RDI and severity band** — sets the efficacy baseline and the expectation. High AHI (≥30) often needs more aggressive treatment; both CPAP and appliances reduce AHI but CPAP achieves complete response (AHI<5) more often. Severity predicts appliance response.
2. **Desaturation and minimum SaO_2** — CPAP is superior for desaturation. Significant desaturation with cardiovascular disease argues for CPAP first or combination therapy, and lowers your threshold for insisting on objective follow-up.
3. **Positional dependence** (supine vs non-supine AHI, % TST supine) — the most actionable single item. Supine dependence favours **positional therapy** and increases appliance success. A patient supine only 20% of the night may have been under-sampled.
4. **Sleep-stage dependence** — **REM-predominant OSA responds relatively poorly to mandibular advancement** because REM atonia defeats the mechanical advantage. Warn the patient and plan objective verification.
5. **Arousal index, architecture, other diagnoses** — CPAP has a greater effect on arousals in some studies. RBD, periodic limb movements or central events change the *referral pathway*, not the appliance; RBD warrants neurological evaluation given its link to neurodegeneration.

- **CPAP** favoured for high AHI, significant desaturation, or where an appliance is less likely to control events or improve sleep quality. **OAT** for mild–moderate OSA, or severe where CPAP is refused/not tolerated/ineffective. **Behavioural** — positional therapy where supine-dependent; weight loss where obese. **Surgery** for severe disease intolerant of CPAP and appliances, particularly with a correctable anatomical obstruction. **Pharmacotherapy** for other disorders diagnosed on the study (e.g. RBD) or residual sleepiness — **not** a primary OSA treatment.

2.4 Interpreting treatment efficacy

THE THREE RESPONSE CATEGORIES — GUARANTEED EXAMINATION ITEM

- **Complete response** — resolution of OSA symptoms **and** treatment AHI <5/h.
- **Partial response** — improved symptoms **and** ≥50% AHI reduction, but AHI remains ≥5/h.
- **Treatment failure** — ongoing clinical symptoms, with or without <50% AHI reduction.
- Some literature defines *success* loosely as ≥50% reduction and reports "near-complete response" at AHI <10. Give the three-category scheme and note that definitions vary — which is why efficacy figures differ so widely between papers.

- **Signature of a good outcome.** Sleep: REM **increases**, NREM **decreases**; **TST typically unchanged** — significant redistribution to more REM (REM rebound), not more sleep; arousal index **decreases**. Respiratory: AHI reduces, minimum SaO_2 increases, snoring frequency and intensity reduce.

TABLE 1. EFFECT OF MAS ON POLYSOMNOGRAPHIC VARIABLES*

Variable	Control	MAS	p Value
Sleep variables			
TST, min	372 ± 5	363 ± 5	NS
REM sleep, min	61 ± 4	75 ± 4	< 0.05
REM, %	16 ± 1	21 ± 1	< 0.005
NREM sleep, min	311 ± 4	287 ± 4	< 0.0001
NREM, %	84 ± 1	80 ± 1	< 0.005
TST spent supine, %	46 ± 4	55 ± 4	NS
Arousal Index/h	41 ± 2	27 ± 2	< 0.0001
Sleep efficiency, %	87 ± 1	85 ± 1	NS
Respiratory variables			
AHI/h	30 ± 2	14 ± 2	< 0.0001
MinSa_{O_2}, %	87 ± 1	91 ± 1	< 0.0001
Snoring frequency/h	402 ± 29	242 ± 28	< 0.005
Mean snoring intensity, dB	52 ± 1	49 ± 1	< 0.0001
Maximum snoring intensity, dB	70 ± 1	68 ± 1	NS

Definition of abbreviations: MAS = mandibular advancement splint; NREM% = time in NREM sleep as a % of TST; REM% = time in REM sleep as a % of TST; TST = total sleep time.
* All values represent mean ± SEM.

Fig 2.2 — Effect of a mandibular advancement splint on PSG variables (control vs MAS, with p values). Sleep: TST unchanged 372→363 min (NS), REM 61→75 min and 16%→21% of TST, NREM falls, arousal index 41→27/h. Respiratory: AHI 30→14/h, minimum SaO_2 87→91%, snoring frequency 402→242/h, mean intensity 52→49 dB, **maximum intensity unchanged**. **Examinable because:** empirical proof of the pattern above. Two details worth quoting — TST does **not** change significantly, and **maximum** snoring intensity is unchanged while **mean** intensity and frequency fall.

2.5 Attended vs unattended studies

- **Type 1** — attended, in-laboratory; 15–20 sensors, ≥7 channels. Full montage: EEG, EOG, EMG, ECG, oximetry, airflow, effort, position, limb movements. Measures sleep stage and quantifies respiratory events accurately.
- **Type 2** — unattended, home (occasionally laboratory-applied then home); 10–15 sensors, ≥7 channels. Full montage without a technician. **High agreement with type 1 for diagnosing OSA**, but greater signal loss and study failure (~7%).
- **Type 3** — home; 4–7 sensors. Limited channel monitoring of breathing parameters **without sleep assessment**: airflow (≥2 channels of respiratory movement, or movement and airflow), effort, oximetry, heart rate/ECG.
- **Type 4** — home; 1–2 bioparameters, typically oximetry ± airflow or heart rate.
- Types 3 and 4 can **confirm** OSA in highly selected patients with high pre-test probability of moderate–severe disease without significant comorbidity. In Australia MBS reimbursement is not available for types 3 and 4, so use there has remained limited.

Type 1 vs type 2

- **Type 1 indications:** low probability of moderate–severe OSA; unsuitable for a type 2 study; failed type 2 study. Also the standard for narcolepsy, central sleep apnoea, parasomnias and movement disorders.
- **Type 2 indications:** high probability of moderate–severe OSA without significant medical comorbidity or other suspected sleep disorder.
- **Factors mandating type 1 instead.** *Patient:* severe intellectual disability; neuromuscular disease; major communication difficulties; severe physical disability with inadequate carer attendance; unsuitable home environment (noise, family interactions, distance from laboratory, staff safety); second opinion where the original diagnosis is uncertain; serious medicolegal consequences; former results inconsistent with clinical impression; patient preference. *Sleep disorder:* consideration of central sleep apnoea, hypoventilation, heart failure, neurological disorders, sleep-related movement disorder, parasomnia, seizure disorder or unexplained hypersomnolence; need for infrared camera or extended EEG montage; transcutaneous CO_2 monitoring; video confirmation of positional/rotational aspects or associated movements.
- **Type 1 advantages:** most accurate and reliable reference standard; low failure rate including for non-OSA SDB; accurate staging and event quantification; expert supervision permitting real-time sensor troubleshooting and immediate comfort intervention. **Disadvantages:** expensive, technically demanding, labour-intensive; unfamiliar environment and strange technician stressful (particularly for children), impairing sleep quality and cooperation; limited accessibility.
- **Type 2 advantages:** familiar home environment giving more representative sleep; cost-effective, widely available; improves access including remotely; avoids hospital-acquired infection. **Disadvantages:** failure rate ~6.6%, especially when wired up unsupervised or wired in the laboratory then sent home; **underestimates AHI by ~10%** through loss of effort signals; valid for OSA only in conjunction with comprehensive sleep evaluation and well-defined symptoms; insufficient for non-respiratory disorders or complex breathing patterns such as CSA; lost or poor-quality data may miss important measurements.

TRAP — DIRECTION OF THE ERROR

A home study **underestimates** AHI, typically by ~10%, because the denominator is recording rather than sleep time and effort signals are more often lost. Corollary: **a positive home study can be believed; a negative home study in a symptomatic patient cannot** and must be followed by in-laboratory PSG. Candidates frequently reverse this.

2.6 MSLT and MWT

ONE SENTENCE THAT CAPTURES THE PAIR

The **MSLT** asks the patient to *try to sleep* and measures how quickly they do; the **MWT** asks them to *try to stay awake* and measures how long they last. The first diagnoses hypersomnolence; the second informs safety and fitness-to-drive judgements.

- **MSLT indications:** diagnosing and quantifying physiological sleepiness in narcolepsy and idiopathic hypersomnia; evaluating persistent sleepiness after treatment; objectively confirming reported improvement after intervention; assessing the effects of hypnotics.
- **MSLT method:** follows an overnight PSG to confirm adequate prior sleep. Four to five nap opportunities at ~2 h intervals; in a dark quiet room the patient lies down and is told to *try to fall asleep*. Each trial terminates after 20 min if no sleep. The *clinical* version lets each nap continue to 15 min after sleep onset to permit REM to emerge; the *research* version terminates as soon as a sleep-onset criterion is met, limiting total sleep obtained.
- **MSLT interpretation: mean sleep latency** averaged across naps — shorter = more severe sleepiness. Also a measure of "sleepability": high sleepability gives MSL 6–8 min even after two weeks with 10 h time in bed; low sleepability gives long MSL, as in insomnia. **≥2 SOREMPs with MSL ≤8 min supports narcolepsy** — but SOREMPs are also frequent in untreated OSA. Negative correlation between prior-night TST and next-day MSL.
- **MWT method:** the patient **sits up in a chair** in a quiet dimly lit room, instructed to *stay awake*; no vocalisation or movement. Four or five trials beginning 1.5–3 h after awakening, repeating two-hourly. Each ends after 20 min if no sleep (modern protocols use 40 min) or immediately after sleep onset; patients stay awake and occupied between trials.
- **MWT interpretation:** a longer mean sleep latency indicates greater ability to maintain wakefulness; considered a more accurate reflection of the ability to function and maintain alertness in common situations of inactivity.
- **Confounders.** *Physiological:* age (longest latency in prepubescent children, shorter in young adults, longer again at 50 and 80); prior sleep quantity and quality (MSL falls after deprivation, disruption or fragmentation). *Drugs:* MSL falls after barbiturates and hypnotics, rises after stimulants. *Circadian:* in shift workers and late sleepers, timing affects latencies and SOREMP occurrence. *Psychological:* anxiety and stress increase MSL; depression may shorten it; anticipation of going home lengthens it; increased CNS arousal (psychophysiological insomnia) lengthens it. *Protocol:* floor effect on the MSLT, ceiling effect on the MWT; a five-nap protocol gives longer latency (last-nap effect).
- **Limitations:** MSL should **not** be the sole criterion for determining sleepiness or certifying a diagnosis or treatment response. It discriminates poorly between sleep-disorder patients and normals, with large standard deviation and floor/ceiling effects — though it does change appropriately from initial to subsequent testing after treatment. Neither test perfectly predicts real-world alertness or driving performance; clinical judgement alongside the result is essential for safety decisions. Normative data are limited, especially for children and shift workers.

2.7 Screening for SDB

THE DENTIST'S ROLE, STATED PRECISELY

Screen for symptoms and signs of OSA, then **refer to a sleep physician** for clinical review and diagnostic testing. Dentists are not authorised to diagnose sleep-related breathing disorders. Since 2017 the ADA has held that **all dentists should screen** — screening is not an optional sub-speciality activity.

- **Sensitivity** = proportion *with* OSA who screen positive; for a highly sensitive test a **negative** result rules disease *out*. **Specificity** = proportion *without* OSA who screen negative; for a highly specific test a **positive** result rules disease *in*.
- OSA screening questionnaires have **good but not optimal sensitivity and poor specificity** for AHI ≥15 — producing many false positives. They are screening, not diagnostic, tools, and predictive value improves with increasing severity.

TRAP — NEGATIVE QUESTIONNAIRES DO NOT EXCLUDE OSA

A negative questionnaire — **particularly the ESS** — does not rule out clinically significant OSA; these patients still need referral if clinical suspicion exists. The ESS correlates poorly with AHI because not all OSA patients are detectably sleepy and because excessive sleepiness has many other causes (other sleep disorders, depression, insufficient sleep, medication). **Never use the ESS alone.** In combination with another questionnaire it helps identify patients with high risk of *symptomatic* disease who will benefit from treatment.

- **ESS** — 8 items scored 0–3, total 0–24; >10 indicates increased daytime sleepiness. Measures the propensity for dozing, *not* OSA severity.
- **STOP-BANG** — **S**noring, **T**iredness, **O**bserved apnoea, **P**ressure (hypertension), **B**MI >35, **A**ge >50, **N**eck >40 cm, **G**ender male. ≥3 = increased risk; higher scores correlate with greater risk; high sensitivity for moderate–severe OSA; a higher cut-off reduces sensitivity and increases specificity.
- **OSA50** — 4 items /10: waist circumference 3 (>102 cm male, >88 cm female), snoring 3, witnessed apnoeas 2, age ≥50 y 2. ≥5 positive; **94% sensitivity, 31% specificity** for moderate–severe OSA — the classic "sensitive but not specific" illustration.
- **Berlin** — 11 items in 3 categories (snoring severity; daytime sleepiness; hypertension or obesity); high risk = positive in ≥2. **NoSAS** — neck >40 cm 4, BMI ≥30 5, snoring 2, age >55 y 4, male 2; ≥8 suggests high probability. **PSQ** — 22 items in 4 subscales; >0.33 (Topic 8).

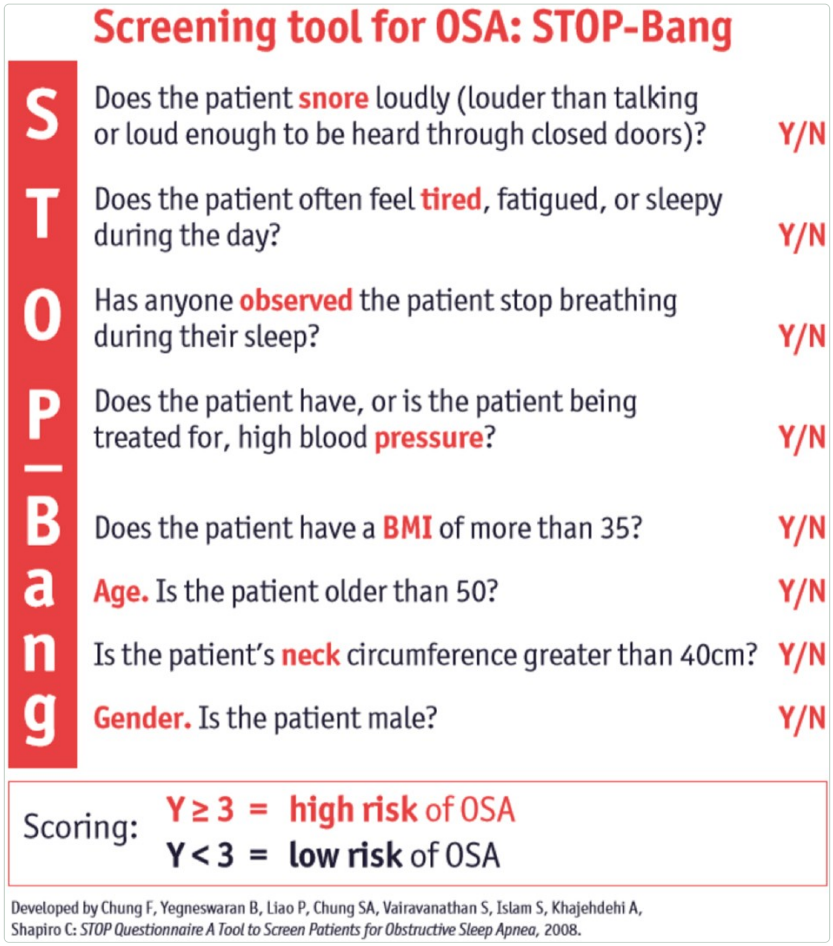


Fig 2.3 — STOP-BANG. Y ≥3 = high risk; Y <3 = low risk (Chung *et al.* 2008). **Examinable because:** recite the eight items from the acronym unprompted. The single most likely tool you will be asked to name, describe and score.

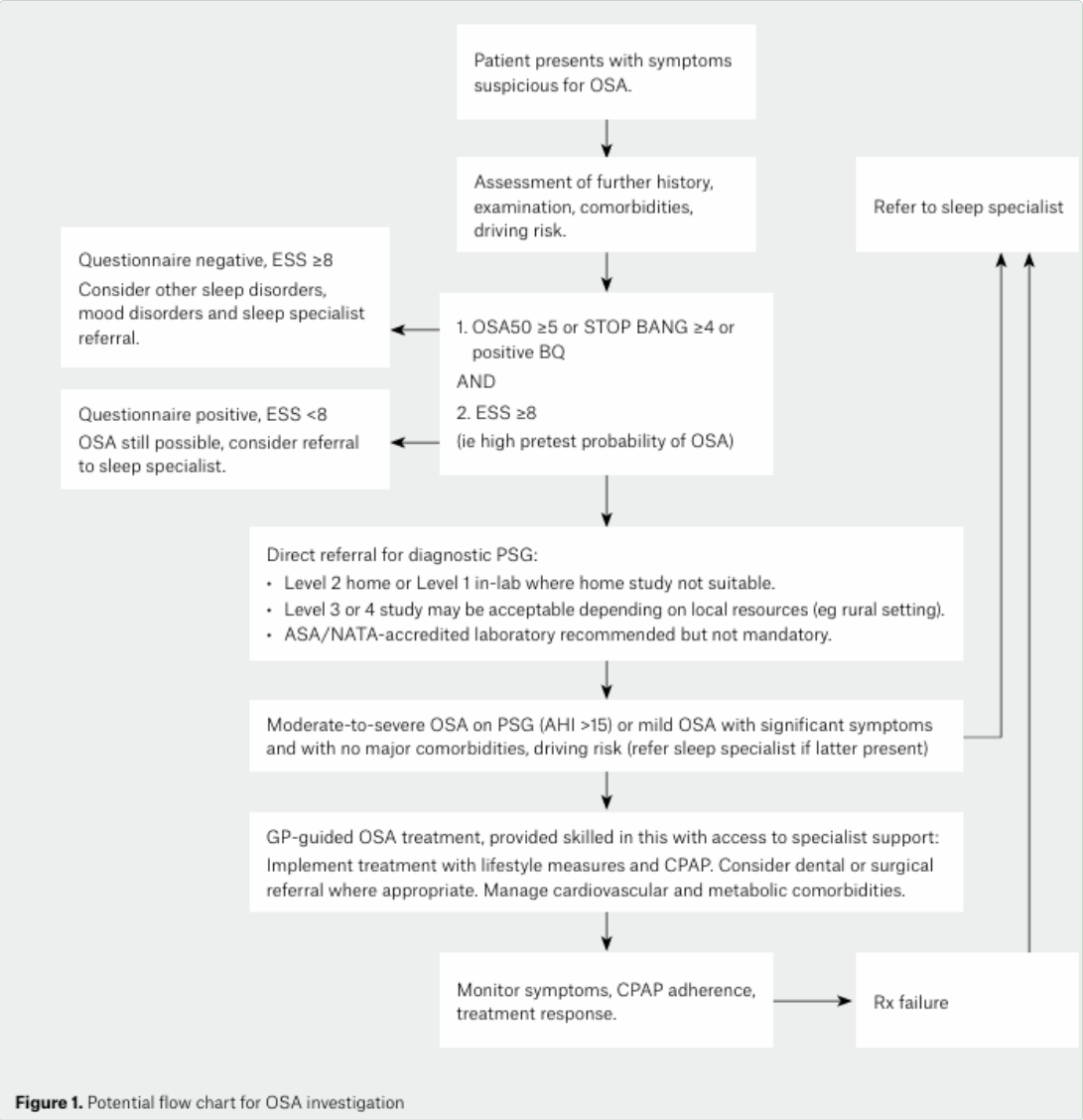


Fig 2.4 — Flow chart for OSA investigation. From symptoms → history, examination, comorbidities and driving risk → questionnaire triage → direct diagnostic study or specialist referral → monitoring of symptoms, CPAP adherence and response. **Examinable because:** this is the **Australian Medicare pathway**. Two routes: (i) GP refers to a sleep or respiratory physician who then orders testing; or (ii) GP orders a study *directly* if criteria are met — a positive questionnaire (Berlin positive, OSA50 ≥5, or STOP-BANG ≥4) **plus ESS ≥8**. The intent is to expedite diagnosis in a highly symptomatic group; the caveat you must state is that **many patients with significant OSA will not meet this combination**. Levels 3 and 4 are screening, not diagnostic. Administrative thresholds are jurisdiction-specific.

The dental physical examination as screening

- Record **height, weight and neck circumference**; evaluate history for obesity, hypertension, type 2 diabetes, cardiovascular disease, family history of sleep apnoea; review smoking, alcohol and medications affecting muscle tone. **Referral triggers:** increased BMI, witnessed apnoeas, excessive daytime sleepiness, medical comorbidity.
- **Neck circumference** >43 cm (17 in) male, >40 cm (16 in) female is strongly associated with higher risk; document central obesity and BMI.
- **Retrognathia/micrognathia** reduces airway space; a steep mandibular plane angle is linked to higher risk; lower pharyngeal changes include elongated pharyngeal structures and narrowed posterior airway space, often with an inferiorly positioned hyoid.
- **Macroglossia — scalloping** (crenations) of the lateral tongue borders suggests the tongue is too wide for the arch, causing oropharyngeal crowding; often with inferior-posterior hyoid position placing tongue muscles at mechanical disadvantage. Digital morphometry correlates increased tongue width and thickness with higher AHI.
- **High palatal vault / narrow arch** — maxillary constriction reduces oral cavity space; narrow tapered V-shaped maxillary arch, possibly with posterior crossbites or compensatory mandibular constriction; frequently associated with chronic mouth breathing.
- **Large mandibular tori** reduce oral volume, displacing the tongue. **Pharyngeal crowding** — enlarged uvula, elongated soft palate, excessive oropharyngeal soft tissue; an **enlarged uvula >15 mm long and >10 mm wide** is directly linked to SDB.
- **Mouth breathing** with dry mouth — compensation for nasal obstruction. **Bruxism** — common in disrupted sleep and airway problems. **Dental erosion** — may indicate GORD, which frequently coexists.
- **Modified Mallampati** — describes oral airway patency. Technique: patient **sits upright, opens widely, protrudes the tongue maximally without phonation** (the "without phonation" distinguishes the modified from the original anaesthetic Mallampati).

Class I: soft palate, uvula, fauces and pillars fully visible. II: soft palate and majority of uvula. III: soft palate and base of uvula only. IV: hard palate only. **Classes III–IV are the "at-risk" findings.**

NUMBERS — TOPIC 2

- Apnoea ≥90% ≥10 s · hypopnoea ≥30% ≥10 s + ≥3% desat or arousal · SaO_2 ≥90% normal
- OSA: RDI >15/h or >5/h with symptoms; AHI ≥15 = material metabolic/vascular risk. Adult 5/15/30 · paediatric 1/5/10
- Sleep efficiency ≥85% · SOL and WASO <30 min · REM latency ~90 min
- Sensors: type 1 = 15–20 · 2 = 10–15 · 3 = 4–7 · 4 = 1–2. Type 2 failure ~6.6–7%; underestimates AHI ~10%
- MSLT: MSL ≤8 min + ≥2 SOREMPs; high sleepability MSL 6–8 min; trials end at 20 min (MWT modern 40 min)
- ESS >10 · STOP-BANG ≥3 · OSA50 ≥5 (94%/31%) · Berlin ≥2 categories · NoSAS ≥8 · PSQ >0.33
- Australian direct referral: positive questionnaire (Berlin / OSA50 ≥5 / STOP-BANG ≥4) **plus** ESS ≥8
- Neck >43 cm M, >40 cm F · uvula >15 mm × >10 mm
- Good outcome: REM ↑ · NREM ↓ · TST unchanged · arousal index ↓ · AHI ↓ · min SaO_2 ↑ · snoring frequency and mean intensity ↓

SELF-TEST — TOPIC 2

1. Define apnoea, hypopnoea, AHI and RDI. Which is always larger?

Apnoea ≥90% airflow reduction ≥10 s (obstructive if effort persists, central if absent). Hypopnoea ≥30% reduction ≥10 s with ≥3% desaturation and/or arousal. AHI = apnoeas + hypopnoeas per hour of sleep; RDI = AHI + RERAs. **RDI is always ≥ AHI**, because RERAs by definition do not meet apnoea or hypopnoea criteria.

2. A home study reports AHI 4/h. The patient snores loudly, has witnessed apnoeas and an ESS of 15. What do you do?

Do not accept the negative result. A type 2 study underestimates AHI by ~10%, has ~7% failure rate, and uses recording rather than sleep time as the denominator. Refer back to the sleep physician for in-laboratory type 1 PSG. I would not start appliance therapy — there is no confirmed diagnosis and no reliable baseline for judging efficacy.

3. What does a successfully treated PSG look like?

REM increases and NREM decreases with TST essentially unchanged — redistribution, not more sleep (REM rebound). Arousal index falls; AHI falls; minimum SaO_2 rises; snoring frequency and mean intensity fall. Published MAS figures: AHI 30→14/h, min SaO_2 87→91%, arousal index 41→27/h, REM 16%→21% of TST, snoring 402→242/h, with TST (372→363 min) and maximum snoring intensity unchanged.

4. Distinguish sensitivity from specificity here, and state the practical consequence.

Sensitivity = proportion with OSA screening positive; a negative result on a sensitive test rules out. Specificity = proportion without OSA screening negative; a positive result on a specific test rules in. These questionnaires have good but suboptimal sensitivity and **poor specificity** for AHI ≥15, generating many false positives — acceptable for a tool whose job is to trigger referral, and the reason none can diagnose. Consequence: a positive screen mandates referral; a negative screen in a clinically suspicious patient does not permit reassurance.

TOPIC 2

Diagnostic sleep testing in sleep-disordered breathing

EXAMINER'S BLUEPRINT

1. Normal sleep architecture and respiratory parameters on polysomnography
2. Pathological sleep architecture and respiratory parameters on polysomnography
3. Interpretation of the polysomnogram and how it influences treatment recommendations (behavioural therapies, CPAP, oral appliance, surgery, pharmacotherapeutics)
4. Interpretation of treatment efficacy on the polysomnogram
5. Comparison of attended (laboratory) and unattended (home) PSG — indications, advantages, disadvantages
6. Other sleep testing: vigilance testing (MSLT and MWT) — indications, method, interpretation
7. Screening for sleep-disordered breathing

This is the topic where a dentist most easily betrays a superficial understanding, because it requires you to read a document you did not generate and are not permitted to interpret diagnostically. The examinable skill is precise: you must be able to **read a sleep study competently, extract what changes your dental plan, and stop short of diagnosis**. Learn the definitions in 2.2 word-perfectly — they are the highest-yield sentences in the entire syllabus.

2.1 Normal sleep architecture and respiratory parameters on polysomnography

DEFINITION

Polysomnography — a multi-parametric test, or comprehensive recording, of the biophysiological changes that occur during sleep; used both in the study of sleep and as a diagnostic tool in sleep medicine. Also called a sleep study.

What is recorded, and why

Channel	Sensor	What it establishes
Electroencephalogram (EEG)	Scalp electrodes	Sleep staging; arousals
Electro-oculogram (EOG)	Periorbital electrodes	Eye movements — identifies REM and sleep onset
Electromyogram (EMG)	Submental (chin) and tibial	Muscle tone (REM atonia); leg movements. Masseter EMG for bruxism
EEG + EOG + EMG together determine sleep stage. No sleep staging is possible without them — the fundamental limitation of level 3 and 4 studies		
Airflow	Oronasal thermal sensor (apnoea) and nasal pressure transducer (hypopnoea)	Detection and classification of respiratory events. Ideally *both* sensor types
Respiratory effort	Thoracic and abdominal respiratory inductance plethysmography (RIP) belts; diaphragm EMG	Distinguishes obstructive from central events; paradoxical movement indicates obstruction
Oxygen saturation	Pulse oximetry (finger probe), high sampling rate, fast averaging	Desaturation events; ODI; minimum SaO_2
Electrocardiogram	Chest leads	Rate, rhythm, arrhythmia during events
Body position	Position sensor	Positional dependence — **directly relevant to appliance and positional therapy decisions**
Snoring	Microphone / sound level meter	Snoring frequency (per hour) and intensity (dB), measured 1 m above the head
End-tidal or transcutaneous CO_2	Capnography	Hypoventilation. A **standard measurement in paediatric** PSG
Video and audio	Infrared camera	Parasomnia, seizure, positional and rotational confirmation

Respiratory inductance plethysmography — how it works

Two elastic bands, one thoracic and one abdominal, each containing a conductor forming a wire loop. As the chest and abdominal circumference change with respiration, the cross-sectional area of the loop changes, altering its inductance. A sensor converts the inductance change into a voltage output reflecting thoracoabdominal volume change. When calibrated against a spirometer or pneumotachograph, RIP can provide quantitative tidal volume and flow data. RIP is the recommended effort sensor for both laboratory PSG and portable monitoring.

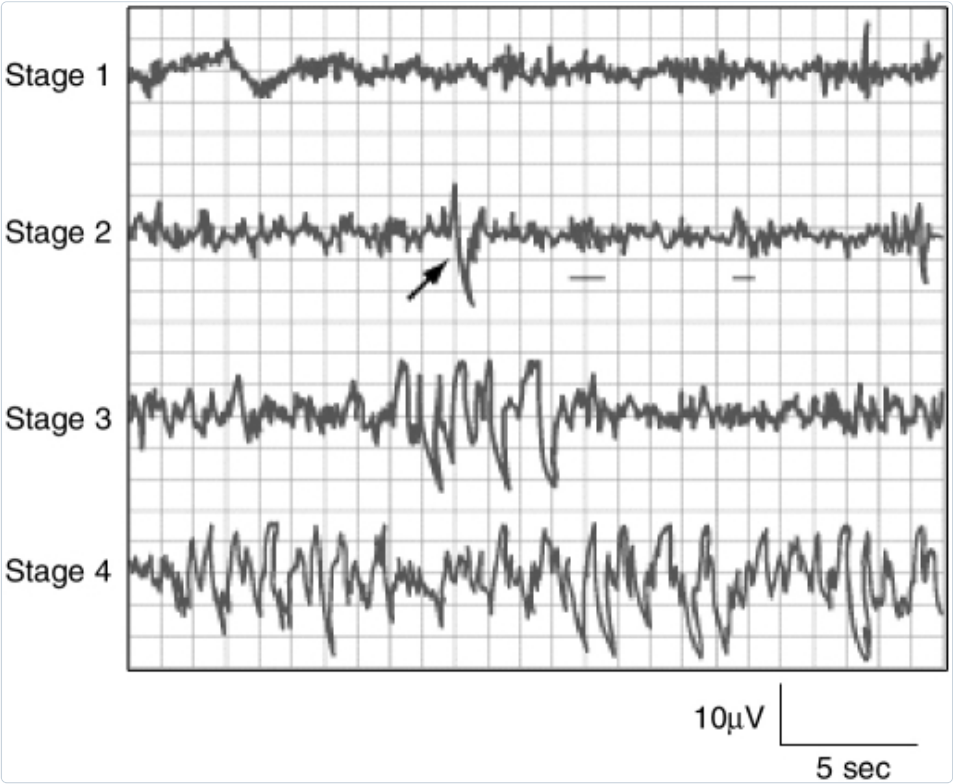


Figure 2.1 — EEG characteristics of the NREM stages. Stage 1 shows low-voltage mixed-frequency activity replacing waking alpha; stage 2 shows sleep spindles (underlined) and a K-complex (arrow); stages 3 and 4 (now combined as N3) show progressively more high-voltage slow-wave activity. Calibration: 10 µV, 5 s.

*Why it is examinable: you may be shown a trace and asked to identify the stage. The two graphoelements to name confidently are the **sleep spindle** and the **K-complex** in N2, and the high-voltage delta of N3.*

Normal values

Measure	Normal	Comment
Total sleep time (TST)	Age-dependent	Total time actually asleep
Sleep onset latency (SOL)	<30 min	Lights out to sleep onset
Wake after sleep onset (WASO)	<30 min	Wakefulness after initial sleep onset
Sleep efficiency (SE)	≥85%, often >90% in young adults	TST ÷ time in bed. 75–84% suboptimal; <75% suggests insomnia
REM latency	~90 min	Shortened, with sleep-onset REM, in narcolepsy
Stage distribution	NREM 75–80%, REM 20–25%	NREM dominates the first half, REM the second
Arousal index	Low	Arousals per hour of sleep, scored by the ASDA 3-second rule
AHI	<5/h	Apnoeas + hypopnoeas per hour of sleep
Oxygen desaturation index (ODI)	Low	Desaturations of ≥3% (or ≥4%) per hour
Oxygen saturation	95–100%, stable; ≥90% acceptable	Minimum SaO_2 is an independent severity marker
Respiratory rate in sleep	Adults 12–20/min; infants and young children up to 30–60/min	
TST spent supine	Reported as %	Needed to judge positional dependence

2.2 Pathological sleep architecture and respiratory parameters

EVENT DEFINITIONS — LEARN THESE VERBATIM

Apnoea — cessation of airflow (≥90% reduction) for ≥10 seconds. *Obstructive* if respiratory effort continues; *central* if effort is absent; *mixed* if it begins central and ends obstructive.

Hypopnoea — a reduction in airflow (and/or thoracoabdominal wall movement) of ≥30% of baseline for ≥10 seconds, accompanied by ≥3% oxygen desaturation *and/or* an arousal. (Historically expressed as a >50% reduction; the AASM-recommended rule is ≥30% with ≥3% desaturation or arousal. The ≥4% desaturation alternative became **optional** in Scoring Manual version 3 in 2023, although some payers — including Medicare in some jurisdictions — still require 4%.)

Respiratory effort-related arousal (RERA) — increasing respiratory effort or flattening of the nasal pressure waveform leading to an arousal, where the event does not meet apnoea or hypopnoea criteria.

Apnoea–hypopnoea index (AHI) — apnoeas plus hypopnoeas per hour of sleep.

Respiratory disturbance index (RDI) — apnoeas plus hypopnoeas **plus RERAs** per hour of sleep. Because it includes RERAs, RDI is always ≥ AHI.

Arousal index — the number of arousals per hour of sleep.

DIAGNOSTIC CRITERIA FOR OSA

OSA is diagnosed when the number of obstructive events per hour of sleep (apnoeas, hypopnoeas and RERAs — the RDI) on polysomnography is **>15 events/hour**, **or >5 events/hour in a patient who reports any of**: unintentional sleep episodes during wakefulness; daytime sleepiness; unrefreshing sleep; fatigue; insomnia; waking with breath-holding, gasping or choking; or a bed partner describing loud snoring, breathing interruptions, or both during sleep.

Severity grading

Severity	Adult AHI (events/hour)	Paediatric AHI (events/hour)
Normal	< 5	< 1
Mild	5 to < 15	1 to < 5
Moderate	15 to < 30	5 to < 10
Severe	≥ 30	≥ 10

The degree of oxygen desaturation during sleep is an **independent** measure of severity, but it is harder to quantify and there is no universally agreed gradation. An AHI of 15 or above is the level at which risk of insulin resistance, dyslipidaemia, vascular disease and death rises materially — which is why 15 is the threshold for treating asymptomatic disease.

Pathological architecture

- **Fragmentation:** increased arousal index, increased stage shifts, increased N1 and reduced slow-wave and REM sleep — the signature of untreated OSA.
- **Irregular or absent stages:** irregular cycling or missing stages point to specific disorders — for example, a patient with narcolepsy may enter sleep directly into REM (sleep-onset REM period).
- **Chronic insomnia disorder:** increased sleep latency and/or increased WASO (often exceeding 30 minutes) with reduced sleep efficiency relative to age norms; increased N1 and decreased slow-wave sleep; relatively greater EEG power in the high-frequency beta and gamma bands; and a characteristic **reverse first-night effect** — better sleep on the first laboratory night than the second. Some patients have an entirely normal study. **PSG is not indicated in chronic insomnia** unless a breathing or movement disorder is suspected, the initial diagnosis is uncertain, treatment has failed, or precipitous arousals occur with violent or injurious behaviour.

Variables reported

Sleep variables	Respiratory variables
• Total sleep time (min) • REM sleep (min) and REM as % of TST • NREM sleep (min) and NREM as % of TST • TST spent supine (%) • Arousal index (per hour) • Sleep efficiency (%)	• AHI (per hour); RDI • Minimum SaO_2 (%) — minimum oxyhaemoglobin saturation • ODI (per hour) • Snoring frequency (snores per hour) • Mean and maximum snoring intensity (dB) • Limb movements; sleep stage distribution

2.3 Interpreting the polysomnogram and its influence on treatment

The polysomnogram is the **gold standard for objectively assessing OSA severity**, and it is requested and reported by a sleep physician. As a dentist you read it to answer five practical questions.

What you look for	What it means for your plan
1. AHI / RDI and severity band	Establishes the baseline against which treatment success will be judged, and shapes expectation. A high AHI (≥30) suggests severe disease often requiring more aggressive treatment; both CPAP and oral appliances reduce AHI, but CPAP achieves complete response (AHI <5) more often. Severity is predictive of response to an oral appliance
2. Degree of oxygen desaturation and minimum SaO_2	CPAP is superior to oral appliances in alleviating desaturation. Significant desaturation in a patient with cardiovascular disease argues for CPAP first, or for combination therapy, and lowers your threshold for insisting on follow-up objective testing
3. Positional dependence (supine vs non-supine AHI, % TST supine)	The most actionable single item for a dentist. Supine-dependent disease favours **positional therapy** and increases the likelihood of oral appliance success. Note that a patient who spent only 20% of the night supine may have had their disease under-sampled
4. Sleep-stage dependence (REM-related events)	REM-predominant OSA responds relatively poorly to mandibular advancement, because REM atonia defeats the mechanical advantage of protrusion. Warn the patient and plan objective verification
5. Arousal index and architecture; other diagnoses	CPAP has shown a greater effect on arousals in some studies. Identification of REM sleep behaviour disorder, periodic limb movements or central events changes the referral pathway — RBD in particular warrants neurological evaluation because of its link to neurodegenerative disease

How findings map onto treatment

- **CPAP** — favoured when AHI is high, desaturation significant, or where an appliance is less likely to control events or improve sleep quality.
- **Oral appliance therapy** — for mild to moderate OSA, or severe OSA where CPAP is refused, not tolerated or ineffective. Positional dependence increases the likelihood of success.
- **Behavioural therapy** — positional therapy where the study shows supine dependence; weight loss where obesity is present.
- **Surgery** — considered for severe disease where CPAP and appliances are not tolerated or effective, particularly where a specific correctable anatomical obstruction exists.
- **Pharmacotherapy** — used for other disorders diagnosed on the study (e.g. RBD) or for residual sleepiness; it is **not** a primary treatment for OSA itself.

2.4 Interpreting treatment efficacy on the polysomnogram

THE THREE RESPONSE CATEGORIES — A GUARANTEED EXAMINATION ITEM

Category	Definition
Complete response	Resolution of OSA symptoms **and** treatment AHI < 5/h
Partial response	Improved symptoms **and** ≥50% reduction in AHI, but AHI remains ≥ 5/h
Treatment failure	Ongoing clinical symptoms, with or without <50% reduction in AHI

Note that some literature defines **success** more loosely as ≥50% AHI reduction, and reports a separate figure for reduction to AHI <10 ("near-complete response"). If asked, give the three-category scheme above and note that definitions vary between studies — which is precisely why efficacy figures differ so widely between papers.

The signature of a good outcome

Domain	Expected change with effective treatment
Sleep variables	REM sleep **increases**; NREM **decreases**. Total sleep time typically does *not* change — instead there is a significant redistribution towards more REM and less NREM (REM rebound, reflecting recovery of previously suppressed REM). Arousal index **decreases**
Respiratory variables	AHI reduces; minimum SaO_2 increases; snoring frequency and intensity reduce

TABLE 1. EFFECT OF MAS ON POLYSOMNOGRAPHIC VARIABLES*

Variable	Control	MAS	p Value
Sleep variables			
TST, min	372 ± 5	363 ± 5	NS
REM sleep, min	61 ± 4	75 ± 4	< 0.05
REM, %	16 ± 1	21 ± 1	< 0.005
NREM sleep, min	311 ± 4	287 ± 4	< 0.0001
NREM, %	84 ± 1	80 ± 1	< 0.005
TST spent supine, %	46 ± 4	55 ± 4	NS
Arousal Index/h	41 ± 2	27 ± 2	< 0.0001
Sleep efficiency, %	87 ± 1	85 ± 1	NS
Respiratory variables			
AHI/h	30 ± 2	14 ± 2	< 0.0001
$MinSa_{O2}$, %	87 ± 1	91 ± 1	< 0.0001
Snoring frequency/h	402 ± 29	242 ± 28	< 0.005
Mean snoring intensity, dB	52 ± 1	49 ± 1	< 0.0001
Maximum snoring intensity, dB	70 ± 1	68 ± 1	NS

Definition of abbreviations: MAS = mandibular advancement splint; NREM% = time in NREM sleep as a % of TST; REM% = time in REM sleep as a % of TST; TST = total sleep time.

* All values represent mean ± SEM.

Figure 2.2 — Effect of a mandibular advancement splint on polysomnographic variables (control versus MAS, with p values). Sleep variables: TST unchanged (372 → 363 min, NS) while REM rises (61 → 75 min, 16% → 21% of TST) and NREM falls; arousal index falls markedly (41 → 27/h). Respiratory variables: AHI 30 → 14/h, minimum SaO_2 87 → 91%, snoring frequency 402 → 242/h, mean snoring intensity 52 → 49 dB (maximum intensity unchanged).

*Why it is examinable: this single table is the empirical proof of the pattern above. The two details worth quoting are that **TST does not change significantly** — the benefit is redistribution, not more sleep — and that **maximum** snoring intensity is unchanged while **mean** intensity and frequency fall. It also gives you real numbers for the magnitude of effect.*

2.5 Attended (laboratory) versus unattended (home) polysomnography

The four levels of sleep study

Level	Setting and channels	Signals
Type 1	Attended, in-laboratory; 15–20 sensors, ≥7 recording channels	Full PSG: EEG, EOG, EMG, ECG, oximetry, airflow, respiratory effort, body position, limb movements. Measures sleep stage and quantifies respiratory events accurately
Type 2	Unattended, home (occasionally laboratory-applied then home); 10–15 sensors, ≥7 channels	Full PSG montage, but without a technician present
Type 3	Home; 4–7 sensors	Limited channel monitoring of breathing parameters **without sleep assessment**: airflow (≥2 channels of respiratory movement, or movement and flow), respiratory effort, oximetry, heart rate/ECG
Type 4	Home; 1–2 sensors	One or two bioparameters — typically oximetry ± airflow or heart rate

Type 2 studies show a high level of agreement with in-laboratory PSG for diagnosing OSA, although there is greater potential for signal loss and study failure (~7%, up to 6.6% in some series) because no technician is in attendance. Types 3 and 4 have good ability to **confirm** OSA in highly selected patients with a high pre-test probability of moderate-to-severe disease and without significant medical or sleep comorbidity. In Australia, MBS reimbursement is not currently available for type 3 and 4 testing, so their use there has remained limited.

Head-to-head comparison

	Laboratory-based (type 1)	Home (type 2)
Indications	Low probability of moderate-to-severe OSA; patient unsuitable for a type 2 study; failed type 2 study. Also the standard for diagnosing narcolepsy, central sleep apnoea, parasomnias and movement disorders	High probability of moderate-to-severe OSA, in a patient without significant medical comorbidity or other suspected sleep disorder
Contra-indications / factors mandating type 1	Severe intellectual disability	**Patient factors:** severe intellectual disability; neuromuscular disease; major communication difficulties; severe physical disability with inadequate carer attendance; unsuitable home environment (noise, family interactions, distance from the laboratory, staff safety); seeking a second opinion where the original diagnosis is uncertain; serious medicolegal consequences; results of former testing inconsistent with clinical impression; patient preference. **Sleep disorder factors:** consideration of central sleep apnoea, hypoventilation, heart failure, neurological disorders, sleep-related movement disorder, parasomnia, seizure disorder or unexplained hypersomnolence; need for infrared camera or extended EEG montage; transcutaneous CO_2 monitoring required; video confirmation of body positional or rotational aspects, or of associated movements, required
Advantages	Most accurate and reliable — the reference standard; low failure rate, including for SDB other than OSA; measures sleep stage and quantifies respiratory events accurately; expert supervision allows real-time troubleshooting of sensors and immediate intervention to improve comfort	Performed at home in a familiar environment, generally giving more representative sleep; cost-effective and widely available; improves access, including in remote areas; avoids hospital-acquired infection risk
Disadvantages		

	Laboratory-based (type 1)	Home (type 2)
	Expensive, technically demanding and labour-intensive; unfamiliar environment and a strange technician can be stressful — particularly for children — impairing sleep quality and cooperation; limited accessibility in resource-limited settings	Failure rate ~6.6%, especially where the patient is wired up unsupervised or wired in the laboratory and sent home; **underestimates AHI by around 10%** because of loss of respiratory effort signals; suitable for OSA diagnosis only in conjunction with a comprehensive sleep evaluation and in patients with well-defined OSA symptoms; limited scope, so insufficient for non-respiratory sleep disorders or complex breathing patterns such as CSA; poor signal quality or lost data may alter or miss important measurements

EXAM TRAP — THE DIRECTION OF THE ERROR

A home study **underestimates** the AHI, typically by about 10%, because the denominator is total *recording* time rather than total *sleep* time and because effort signals are more often lost. The clinical corollary is the one to state: **a positive home study can be believed, but a negative home study in a symptomatic patient cannot** — it must be followed by in-laboratory PSG. Candidates frequently get this backwards.

2.6 Vigilance testing: MSLT and MWT

	Multiple Sleep Latency Test (MSLT)	Maintenance of Wakefulness Test (MWT)
Measures	The **tendency to fall asleep** — physiological sleepiness, in the absence of alerting factors	The **ability to stay awake**
Indications	Diagnosis and quantification of physiological sleepiness in narcolepsy and idiopathic hypersomnia; evaluation of persistent sleepiness after treatment; objective confirmation of reported clinical improvement after intervention; assessment of the effects of hypnotic medication	Assessment of the ability to remain awake where sleepiness poses a public or personal safety risk (driving, operating machinery); evaluation of treatment effectiveness. An alternative to the MSLT, especially in very sleepy patients
Method	Follows an overnight PSG to confirm adequate prior sleep. Four to five nap opportunities at ~2-hour intervals through the day. In a dark, quiet room the patient lies down and is instructed to *try to fall asleep*. Each trial is terminated after 20 minutes if no sleep occurs. The clinical version allows each nap to continue to a prespecified termination point (15 minutes after sleep onset) to permit REM sleep to emerge; the research version terminates as soon as a sleep-onset criterion is met, limiting total sleep obtained	The patient **sits up in a chair** in a quiet, dimly lit room and is instructed to *stay awake*. Vocalisation and movement are not allowed. Four or five trials, beginning 1.5–3 hours after awakening and repeating every two hours. Each trial ends after 20 minutes (modern protocols use 40 minutes) if no sleep occurs, or immediately after sleep onset. Patients must remain awake and occupied between trials
Interpretation	**Mean sleep latency (MSL)** — the average latency across naps; shorter latency indicates more severe physiological sleepiness. Also a measure of "sleepability": high sleepability gives an MSL of 6–8 min even after two weeks with time in bed extended to 10 hours; low sleepability gives a long MSL, as in insomnia. **≥2 sleep-onset REM periods (SOREMPs)** with an MSL ≤8 min supports narcolepsy — but SOREMPs are also frequent in untreated OSA. There is a negative correlation between prior night total sleep time and next-day MSL	Mean sleep latency measures how long wakefulness can be maintained; a longer latency indicates greater ability to maintain wakefulness. Considered a more accurate reflection of the ability to function and maintain alertness in common situations of inactivity
Confounders	**Physiological:** age (longest latency in prepubescent children, shorter in young adults; longer at 50 and 80 than in younger adults); prior sleep quantity and quality (MSL falls after deprivation, disruption or fragmentation). **Drugs:** MSL falls after barbiturates and hypnotics, rises after stimulants. **Circadian:** for shift workers and late sleepers the timing of testing matters, as circadian phase affects latencies and the occurrence of SOREMPs. **Psychological:** anxiety and stress increase MSL; depression may shorten it; anticipation of going home lengthens latency; increased CNS arousal (psychophysiological insomnia) lengthens latency.	

	Multiple Sleep Latency Test (MSLT)	Maintenance of Wakefulness Test (MWT)
	Protocol: a floor effect on the MSLT and a ceiling effect on the MWT; a five-nap protocol gives a longer latency (last-nap effect).	
Limitations	MSL should **not** be the sole criterion for determining sleepiness or for certifying a diagnosis or treatment response. It discriminates poorly between patients with sleep disorders and the normal population, with a large standard deviation and floor and ceiling effects — although it does show appropriate change from initial to subsequent testing after treatment. Neither test perfectly predicts real-world alertness or driving performance; clinical judgement alongside the result is essential for safety decisions. Normative data are limited, especially for children and shift workers	

HIGH-YIELD — ONE SENTENCE THAT CAPTURES THE PAIR

"The MSLT asks the patient to *try to sleep* and measures how quickly they do; the MWT asks them to *try to stay awake* and measures how long they last. The first is used to diagnose hypersomnolence, the second to make safety and fitness-to-drive judgements."

2.7 Screening for sleep-disordered breathing

THE DENTIST'S ROLE, STATED PRECISELY

The dentist's role is to **screen for symptoms and signs of OSA and then refer to a sleep physician** for clinical review and diagnostic testing. Dentists are not authorised to diagnose sleep-related breathing disorders. Since 2017 the American Dental Association has held that **all dentists should screen** patients for sleep-related breathing disorders — screening is not an optional sub-speciality activity.

Understanding what a screening test does

- **Sensitivity** is the proportion of subjects *with* OSA who have a positive questionnaire. For a highly sensitive test, a **negative** result is good at *ruling out* disease.
- **Specificity** is the proportion of subjects *without* OSA who have a negative questionnaire. For a highly specific test, a **positive** result is good at *ruling in* disease.
- In general, OSA screening questionnaires have **good but not optimal sensitivity and poor specificity** for detecting AHI ≥15 events/hour — producing a large number of false positives. They are screening tools, not diagnostic ones, and predictive value improves with increasing OSA severity.

EXAM TRAP — NEGATIVE QUESTIONNAIRES DO NOT EXCLUDE OSA

A negative questionnaire — **particularly the ESS** — does not rule out clinically significant OSA. These patients still need specialist referral if clinical suspicion is present. The ESS correlates poorly with AHI, because not all patients with OSA experience detectable sleepiness and because excessive daytime sleepiness has many other causes (other sleep disorders, depression, insufficient sleep, medication). **The ESS should never be used alone** to identify patients at high risk of OSA. Used in conjunction with another questionnaire, however, it helps identify patients with high risk of *symptomatic* disease who will benefit from treatment.

The questionnaires

Tool	Structure	Threshold and performance
Epworth Sleepiness Scale	8 items, each 0–3; total 0–24	>10 indicates increased daytime sleepiness. Measures the propensity for dozing in eight everyday situations. Higher score = greater sleepiness, *not* greater OSA severity
STOP-BANG	8 items: **S**noring, **T**iredness, **O**bserved	≥3 indicates increased risk; higher scores correlate with greater risk. High sensitivity for moderate-to-severe OSA, making it a reliable initial screen.

Tool	Structure	Threshold and performance
	apnoea, **P**ressure (hypertension), **B**MI >35, **A**ge >50, **N**eck >40 cm, **G**ender male — four yes/no questions and four clinical attributes	Using a higher cut-off reduces sensitivity but increases specificity. Widely used and reliable
OSA50	4 items, /10: waist circumference (>102 cm male, >88 cm female) 3, snoring 3, witnessed apnoeas 2, age ≥50 y 2	≥5 points. In a validation sample: **94% sensitivity, 31% specificity** for moderate-to-severe OSA — the classic illustration of "sensitive but not specific"
Berlin Questionnaire	11 items in 3 categories: snoring severity; daytime sleepiness; hypertension or obesity	High risk = positive in ≥2 categories. Developed by respiratory and primary care physicians
NoSAS	Neck >40 cm 4; BMI ≥30 5; snoring 2; age >55 y 4; male 2	≥8 suggests a high probability of significant SDB
Paediatric Sleep Questionnaire	22 items in 4 subscales: breathing, behaviour, sleepiness, other	>0.33 (i.e. more than a third answered positively) suggests high likelihood of SDB. Not diagnostic, but the most sensitive paediatric questionnaire available (Topic 8)

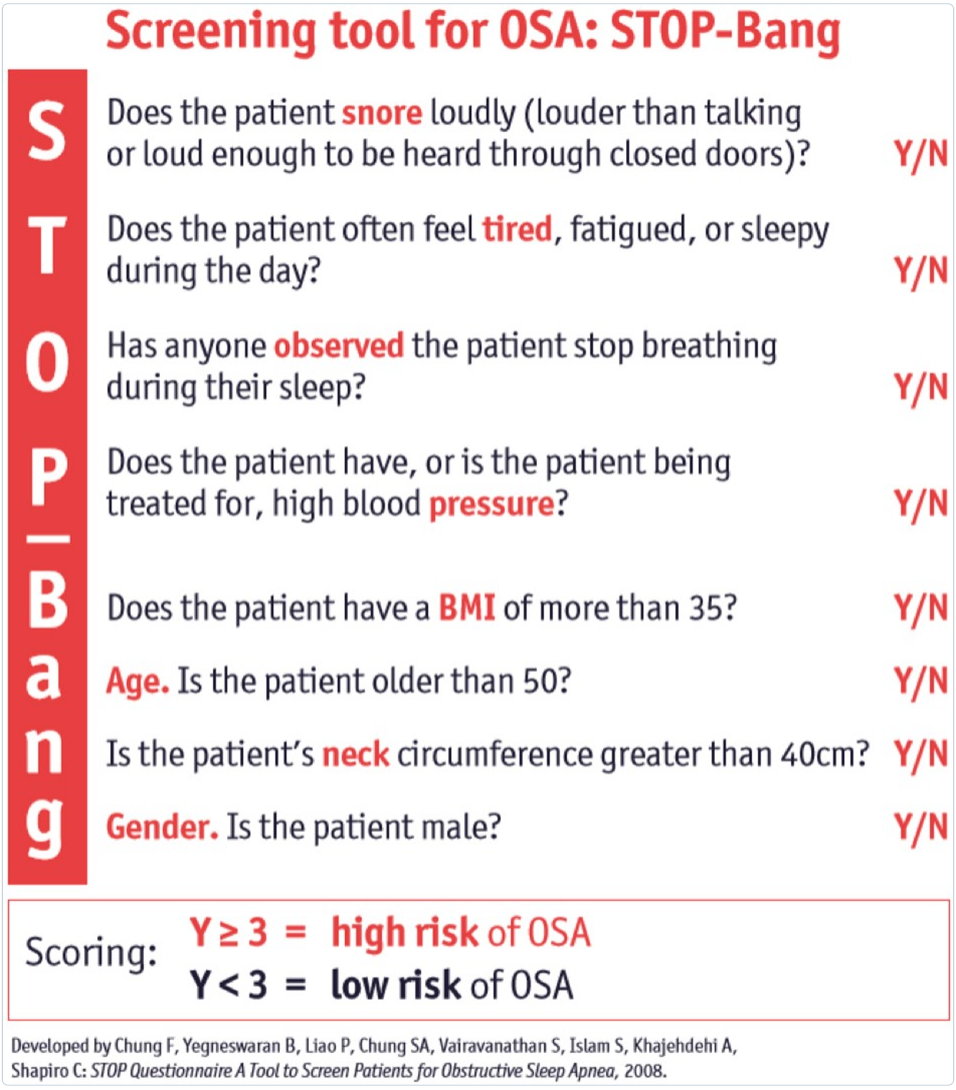


Figure 2.3 — The STOP-BANG questionnaire (Chung *et al.* 2008). Y ≥3 = high risk of OSA; Y <3 = low risk.

Why it is examinable: you should be able to recite the eight items from the acronym without prompting. It is the most likely single tool you will be asked to name, describe and score.

Epworth Sleepiness Scale

How likely are you to nod off or fall asleep in the following situations, in contrast to feeling just tired? This refers to your usual way of life in recent times.

Even if you haven't done some of these things recently, try to work out how they would have affected you. It is important that you answer each question as best you can.

Use the following scale to choose the most appropriate number for each situation.

	Would never nod off 0	Slight chance of nodding off 1	Moderate chance of nodding off 2	High chance of nodding off 3
Sitting and reading				
Watching TV				
Sitting, inactive, in a public place (e.g., in a meeting, theater, or dinner event)				
As a passenger in a car for an hour or more without stopping for a break				
Lying down to rest when circumstances permit				
Sitting and talking to someone				
Sitting quietly after a meal without alcohol				
In a car, while stopped for a few minutes in traffic or at a light				

Add up your points to get your total score. A score of 10 or greater raises concern: you may need to get more sleep, improve your sleep practices, or seek medical attention to determine why you are sleepy.

Figure 2.4 — The Epworth Sleepiness Scale. Eight situations rated 0 (would never doze) to 3 (high chance of dozing); a total of ≥10 raises concern.

Why it is examinable: note the framing on the form — "how likely are you to doze off or fall asleep, in contrast to feeling just tired". Candidates who explain that the ESS measures sleep propensity rather than fatigue, and that it therefore correlates poorly with AHI, demonstrate real understanding.

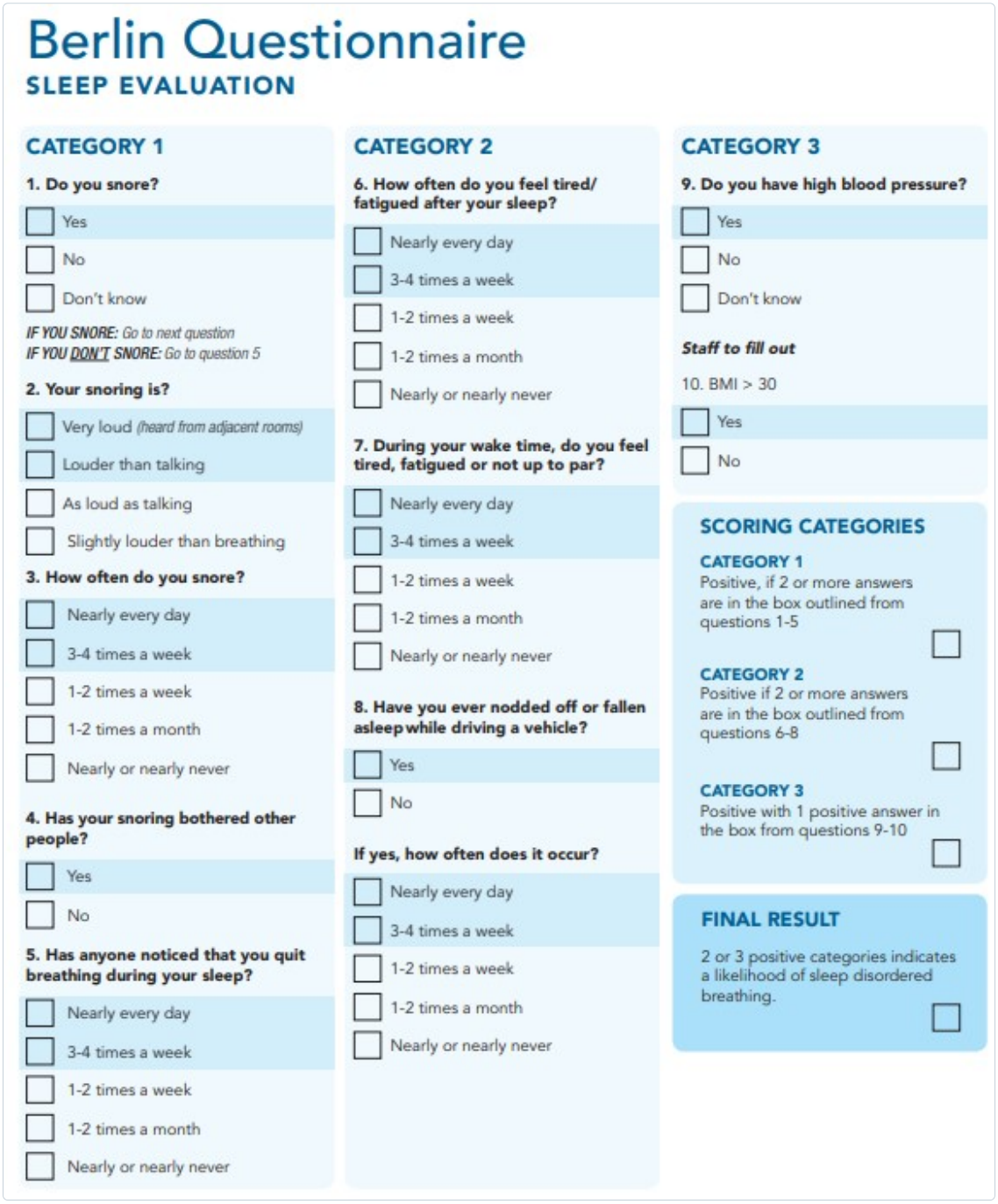
Table 1. OSA50 questionnaire

		If yes, score
Obesity	Waist circumference* (>102 cm for males or >88 cm for females)	3
Snoring	Has your snoring ever bothered other people?	3
Apnoeas	Has anyone noticed that you stop breathing during your sleep?	2
50	Are you aged 50 years or over?	2
	Total score	**..... / 10 points**

**Waist measurement to be measured at the level of the umbilicus (males 102 cm = size 40 [inches] and females 88 cm = size 16)*

Figure 2.5 — The OSA50 questionnaire. Obesity by waist circumference (>102 cm male, >88 cm female) 3 points; snoring that has bothered other people 3; witnessed apnoeas 2; age ≥50 years 2. Total /10.

Why it is examinable: the four highest-yield predictors of severe OSA distilled into one page, and the source of the 94%/31% sensitivity–specificity pair worth quoting.



Berlin Questionnaire

SLEEP EVALUATION

CATEGORY 1

1. Do you snore?

- ☐ Yes
- ☐ No
- ☐ Don't know

IF YOU SNORE: Go to next question
IF YOU DON'T SNORE: Go to question 5

2. Your snoring is?

- ☐ Very loud *(heard from adjacent rooms)*
- ☐ Louder than talking
- ☐ As loud as talking
- ☐ Slightly louder than breathing

3. How often do you snore?

- ☐ Nearly every day
- ☐ 3-4 times a week
- ☐ 1-2 times a week
- ☐ 1-2 times a month
- ☐ Nearly or nearly never

4. Has your snoring bothered other people?

- ☐ Yes
- ☐ No

5. Has anyone noticed that you quit breathing during your sleep?

- ☐ Nearly every day
- ☐ 3-4 times a week
- ☐ 1-2 times a week
- ☐ 1-2 times a month
- ☐ Nearly or nearly never

CATEGORY 2

6. How often do you feel tired/ fatigued after your sleep?

- ☐ Nearly every day
- ☐ 3-4 times a week
- ☐ 1-2 times a week
- ☐ 1-2 times a month
- ☐ Nearly or nearly never

7. During your wake time, do you feel tired, fatigued or not up to par?

- ☐ Nearly every day
- ☐ 3-4 times a week
- ☐ 1-2 times a week
- ☐ 1-2 times a month
- ☐ Nearly or nearly never

8. Have you ever nodded off or fallen asleep while driving a vehicle?

- ☐ Yes
- ☐ No

If yes, how often does it occur?

- ☐ Nearly every day
- ☐ 3-4 times a week
- ☐ 1-2 times a week
- ☐ 1-2 times a month
- ☐ Nearly or nearly never

CATEGORY 3

9. Do you have high blood pressure?

- ☐ Yes
- ☐ No
- ☐ Don't know

Staff to fill out

10. BMI > 30

- ☐ Yes
- ☐ No

SCORING CATEGORIES

CATEGORY 1
Positive, if 2 or more answers are in the box outlined from questions 1-5 ☐

CATEGORY 2
Positive if 2 or more answers are in the box outlined from questions 6-8 ☐

CATEGORY 3
Positive with 1 positive answer in the box from questions 9-10 ☐

FINAL RESULT

2 or 3 positive categories indicates a likelihood of sleep disordered breathing. ☐

Figure 2.6 — The Berlin Questionnaire, showing its three scoring categories (snoring behaviour; daytime sleepiness and drowsy driving; hypertension and BMI >30) and the final result rule: two or more positive categories indicates a likelihood of sleep-disordered breathing.

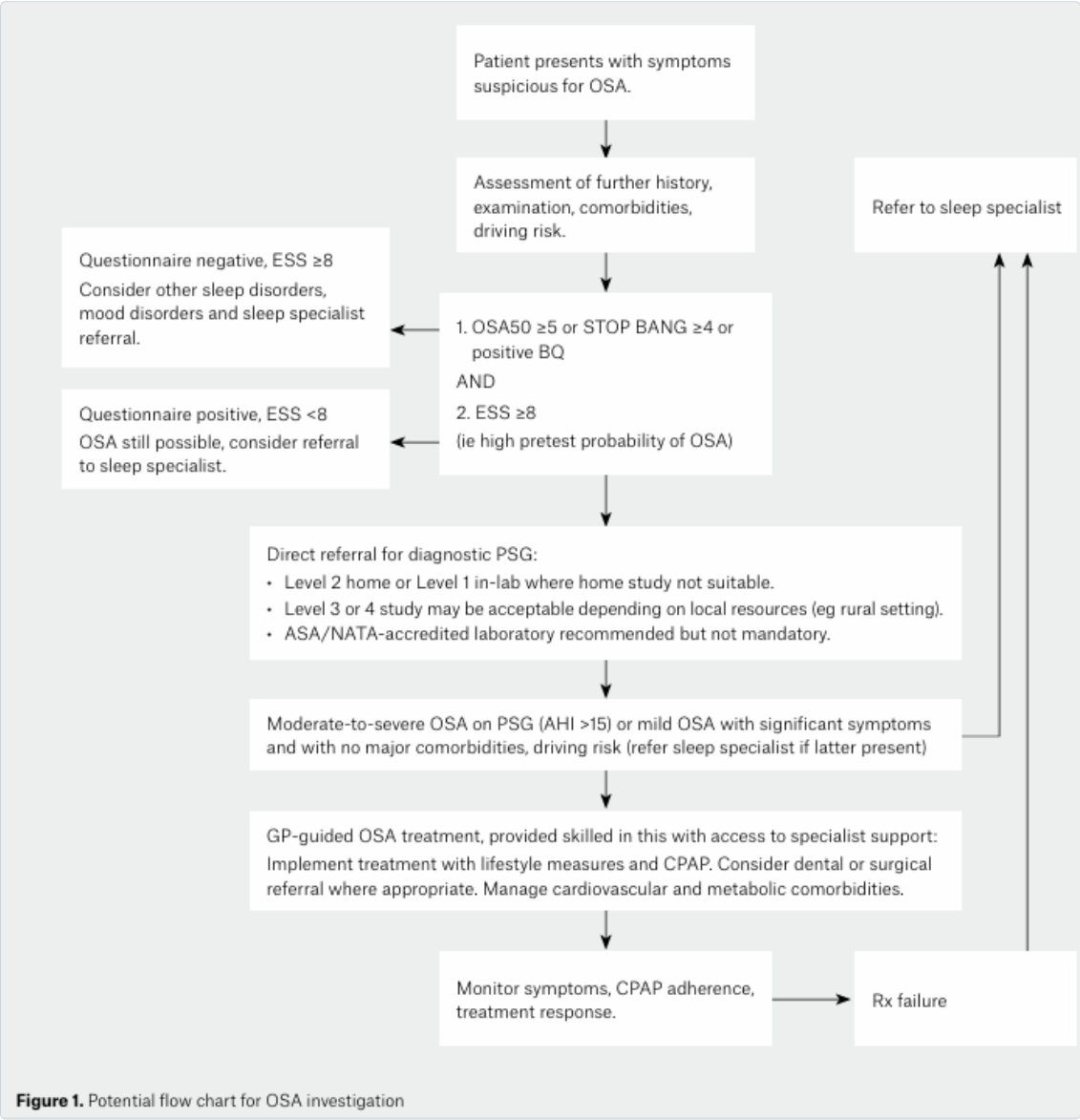


Figure 2.7 — A potential flow chart for OSA investigation, from presentation with suspicious symptoms through history, examination, comorbidities and driving risk, to questionnaire-based triage and the choice between direct diagnostic sleep study and specialist referral, and then to monitoring of symptoms, CPAP adherence and treatment response.

*Why it is examinable: this is the **Australian Medicare pathway**. Two routes exist: (i) the GP refers to a sleep or respiratory physician who then orders testing as appropriate; or (ii) the GP orders a sleep study directly, without prior specialist review, if criteria are met — a positive screening questionnaire (Berlin positive, OSA50 ≥5, or STOP-BANG ≥4) **in addition to** an ESS ≥8. The intent is to expedite diagnosis in a highly symptomatic group; the caveat you must state is that **many patients with significant OSA will not meet this questionnaire combination**. Remember also that levels 3 and 4 are screening tests, not diagnostic ones. These administrative thresholds are jurisdiction-specific.*

The dental physical examination as screening

Medical and dental history taking should record the patient's **height, weight and neck circumference** as part of OSA screening, and evaluate the history for known SDB risk factors — obesity, hypertension, type 2 diabetes, cardiovascular disease and family history of sleep apnoea. Lifestyle factors including smoking, alcohol use and medications affecting muscle tone should be reviewed. Key referral triggers are increased BMI, witnessed apnoeas, excessive daytime sleepiness and the presence of medical comorbidity.

Finding	Significance
Neck circumference	>43 cm (17 in) in men and >40 cm (16 in) in women is strongly associated with higher OSA risk. Document central obesity and elevated BMI — increased adipose tissue around the airway contributes to collapse

Finding	Significance
Retrognathia / micrognathia	Reduces airway space. A steep mandibular plane angle is linked to heightened OSA risk; lower pharyngeal changes include elongated pharyngeal structures and a narrowed posterior airway space, often with an inferiorly positioned hyoid
Macroglossia	Tongue **scalloping** (crenations) along the lateral borders suggests the tongue is too wide for the dental arch, contributing to oropharyngeal crowding. Macroglossia is often associated with an inferior-posterior hyoid position, placing tongue muscles at a mechanical disadvantage. Digital morphometric studies correlate increased tongue width and thickness with higher AHI
High palatal vault / narrow arch	Maxillary constriction with a high vault reduces oral cavity space; may present as a narrow, tapered V-shaped maxillary arch with posterior crossbites or compensatory mandibular constriction. Frequently associated with chronic mouth breathing
Large mandibular tori	Frequently observed in OSA; a large bony mass reduces oral volume, displacing the tongue and contributing to obstruction
Pharyngeal crowding	Enlarged uvula, elongated soft palate, excessive oropharyngeal soft tissue. A high Mallampati score indicates a longer uvula and soft palate. An enlarged uvula — typically longer than 15 mm and wider than 10 mm — has been directly linked to SDB
Mouth breathing	Often with dry mouth; a compensatory mechanism for nasal obstruction, associated with SDB
Bruxism	Commonly noted in patients with disrupted sleep and airway issues (see Topic 4 for the causation caveat)
Dental erosion	May indicate GORD, which frequently coexists with SDB

Modified Mallampati classification is used to describe the patency of the oral airway. To perform it correctly, ask the patient to sit upright, open the mouth widely, and protrude the tongue maximally **without phonation** (this last point distinguishes the *modified* Mallampati from the original anaesthetic Mallampati score).

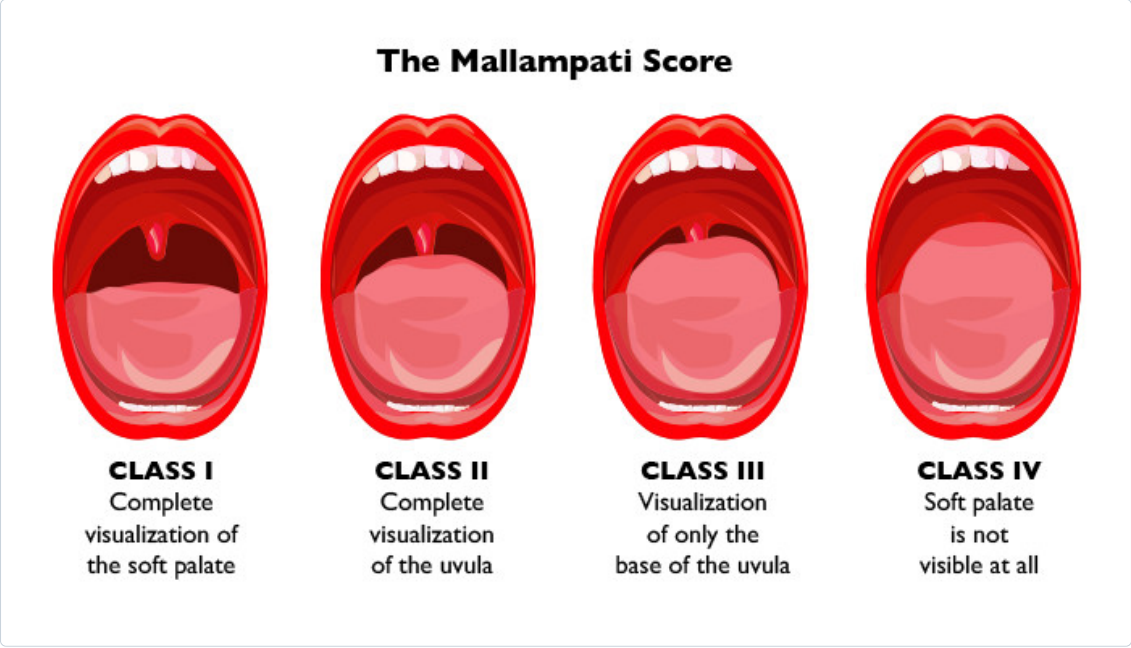


Figure 2.8 — The modified Mallampati classification. Class I: soft palate, uvula, fauces and pillars fully visible. Class II: soft palate and the majority of the uvula visible. Class III: only the soft palate and the base of the uvula visible. Class IV: only the hard palate visible.

*Why it is examinable: classes **III and IV** are the "at-risk anatomically" findings that belong in your targeted documentation. Be able to state the technique (upright, maximal tongue protrusion, no phonation) as well as the four classes.*

NUMBERS TO MEMORISE — TOPIC 2

- Apnoea ≥90% flow reduction ≥10 s · hypopnoea ≥30% reduction ≥10 s with ≥3% desaturation or arousal · SaO_2 ≥90% normal
- OSA diagnosis: RDI >15/h, or >5/h with symptoms. AHI ≥15 → material increase in metabolic and vascular risk
- Adult 5 / 15 / 30 · paediatric 1 / 5 / 10
- Sleep efficiency ≥85% normal; SOL and WASO <30 min; REM latency ~90 min
- Type 2 study failure rate ~6.6–7%; home study underestimates AHI by ~10%
- Type 1 = 15–20 sensors · type 2 = 10–15 · type 3 = 4–7 · type 4 = 1–2
- MSLT: MSL ≤8 min with ≥2 SOREMPs supports narcolepsy; high sleepability MSL 6–8 min; trials terminated at 20 min
- Screening thresholds: ESS >10 · STOP-BANG ≥3 · OSA50 ≥5 · Berlin ≥2 categories · NoSAS ≥8 · PSQ >0.33
- OSA50: 94% sensitivity, 31% specificity for moderate–severe OSA
- Australian direct-referral pathway: positive questionnaire (Berlin positive / OSA50 ≥5 / STOP-BANG ≥4) **plus** ESS ≥8
- Neck circumference >43 cm male, >40 cm female; uvula >15 mm long and >10 mm wide linked to SDB
- Effective treatment: REM ↑, NREM ↓, TST unchanged, arousal index ↓, AHI ↓, min SaO_2 ↑, snoring frequency and mean intensity ↓

SELF-TEST — TOPIC 2

1. Define apnoea, hypopnoea, AHI and RDI. Which is always larger and why?

> Apnoea: ≥90% airflow reduction for ≥10 s (obstructive if effort persists, central if absent). Hypopnoea: ≥30% airflow reduction for ≥10 s with ≥3% desaturation and/or an arousal. AHI: apnoeas plus hypopnoeas per hour of sleep. RDI: apnoeas plus hypopnoeas plus RERAs per hour of sleep. **RDI is always ≥ AHI** because it additionally counts respiratory effort-related arousals, which by definition do not meet apnoea or hypopnoea criteria.

2. A patient brings a home sleep study reporting AHI 4/h. They snore loudly, have witnessed apnoeas and an ESS of 15. What do you do?

> Do not accept the negative result. A home (type 2) study underestimates AHI by approximately 10%, has a failure rate of around 7%, and uses total recording time rather than total sleep time as its denominator. In a highly symptomatic patient a negative home study is not reassuring: refer back to the sleep physician for in-laboratory type 1 polysomnography. I would not commence oral appliance therapy on the basis of this study, because there is no confirmed diagnosis and no reliable baseline against which to judge efficacy.

3. You are shown a polysomnogram. Which five items change your dental treatment plan?

> (i) AHI/RDI and severity band — sets expectations and the efficacy baseline; (ii) minimum SaO_2 and degree of desaturation — significant desaturation with cardiovascular comorbidity argues for CPAP or combination therapy; (iii) positional data (supine versus non-supine AHI, percentage of the night supine) — supine dependence favours appliance success and positional therapy; (iv) sleep-stage dependence — REM-predominant disease responds relatively poorly to mandibular advancement; (v) architecture, arousal index and any other diagnoses — RBD, periodic limb movements or predominantly central events change the referral pathway rather than the appliance.

4. What does a successfully treated polysomnogram look like? Give the direction of change for six variables.

> REM sleep increases and NREM decreases, with total sleep time essentially unchanged — that is, redistribution rather than more sleep (REM rebound). Arousal index falls. AHI falls, minimum SaO_2 rises, and snoring frequency and mean intensity fall. In the published MAS data: AHI 30 → 14/h, minimum SaO_2 87 → 91%, arousal index 41 → 27/h, REM 16% → 21% of TST, snoring frequency 402 → 242/h, while TST (372 → 363 min) and maximum snoring intensity were unchanged.

5. Distinguish sensitivity from specificity in the context of OSA screening, and explain the practical consequence.

> Sensitivity is the proportion of people with OSA who screen positive; for a highly sensitive test a negative result is good at ruling disease out. Specificity is the proportion of people without OSA who screen negative; for a highly specific test a positive result is good at ruling disease in. OSA screening questionnaires have good but suboptimal sensitivity and **poor specificity** for AHI ≥15, so they generate many false positives — acceptable for a screening instrument whose purpose is to trigger referral, but the reason they can never be used to diagnose. The corollary is that a positive screen mandates referral, while a negative screen in a clinically suspicious patient does not permit reassurance.

6. Contrast the MSLT and MWT in one sentence each, and give the narcolepsy criterion.

> The MSLT measures the tendency to fall asleep — the patient is asked to try to sleep during four to five nap opportunities at two-hourly intervals after an overnight PSG, and mean sleep latency is calculated. The MWT measures the ability to stay awake — the patient sits in a chair in a quiet dim room and is instructed to remain awake over four to five trials. A mean sleep latency of ≤8 minutes with two or more sleep-onset REM periods supports narcolepsy, but SOREMPs also occur frequently in untreated OSA, so the test is not specific.

7. Describe how to perform the modified Mallampati assessment and state which classes concern you.

> Seat the patient upright, ask them to open the mouth widely and protrude the tongue maximally **without phonation**. Class I: soft palate, uvula, fauces and pillars fully visible; II: soft palate and most of the uvula; III: soft palate and base of uvula only; IV: hard palate only. Classes III and IV indicate a crowded oropharynx with a longer uvula and soft palate and belong in the "at risk anatomically" documentation, alongside tonsil grade 3–4, retrognathia, deep palatal vault, macroglossia with scalloping, thick neck and a low hyoid.

REFERENCES — TOPIC 2

1. Berry RB, Quan SF, Abreu AR, et al. *The AASM Manual for the Scoring of Sleep and Associated Events*. Version 3. American Academy of Sleep Medicine; 2023.
2. American Academy of Sleep Medicine. Summary of updates in version 3 of the AASM scoring manual. AASM; February 2023.
3. Kapur VK, Auckley DH, Chowdhuri S, et al. Clinical practice guideline for diagnostic testing for adult obstructive sleep apnea: an American Academy of Sleep Medicine clinical practice guideline. *J Clin Sleep Med*. 2017;13(3):479–504.
4. Collop NA, Anderson WM, Boehlecke B, et al. Clinical guidelines for the use of unattended portable monitors in the diagnosis of obstructive sleep apnea in adult patients. *J Clin Sleep Med*. 2007;3(7):737–747.
5. American Academy of Sleep Medicine. *International Classification of Sleep Disorders*. 3rd ed, Text Revision. AASM; 2023.
6. Littner MR, Kushida C, Wise M, et al. Practice parameters for clinical use of the multiple sleep latency test and the maintenance of wakefulness test. *Sleep*. 2005;28(1):113–121.
7. Carskadon MA, Dement WC, Mitler MM, Roth T, Westbrook PR, Keenan S. Guidelines for the multiple sleep latency test (MSLT): a standard measure of sleepiness. *Sleep*. 1986;9(4):519–524.
8. Johns MW. A new method for measuring daytime sleepiness: the Epworth Sleepiness Scale. *Sleep*. 1991;14(6):540–545.
9. Chung F, Yegneswaran B, Liao P, et al. STOP questionnaire: a tool to screen patients for obstructive sleep apnea. *Anesthesiology*. 2008;108(5):812–821.
10. Chai-Coetzer CL, Antic NA, Rowland LS, et al. A simplified model of screening questionnaire and home monitoring for obstructive sleep apnoea in primary care. *Thorax*. 2011;66(3):213–219.
11. Netzer NC, Stoohs RA, Netzer CM, Clark K, Strohl KP. Using the Berlin Questionnaire to identify patients at risk for the sleep apnea syndrome. *Ann Intern Med*. 1999;131(7):485–491.
12. Marti-Soler H, Hirotsu C, Marques-Vidal P, et al. The NoSAS score for screening of sleep-disordered breathing: a derivation and validation study. *Lancet Respir Med*. 2016;4(9):742–748.
13. Nixon GM, Brouillette RT. Diagnostic techniques for obstructive sleep apnoea: is polysomnography necessary? *Paediatr Respir Rev*. 2002;3(1):18–24.
14. McNicholas WT. Diagnosis of obstructive sleep apnea in adults. *Proc Am Thorac Soc*. 2008;5(2):154–160.
15. Australian Government Department of Health. Medicare Benefits Schedule — items 12203, 12204 and 12250; explanatory notes on sleep studies.
16. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2).
17. American Dental Association. Policy statement: the role of dentistry in the treatment of sleep-related breathing disorders. ADA; 2017.
18. Friedman M, Ibrahim H, Bass L. Clinical staging for sleep-disordered breathing. *Otolaryngol Head Neck Surg*. 2002;127(1):13–21.

TOPIC 3

Oral appliance therapy and other evidence-based treatments

BLUEPRINT

1. History of OAT · 2. Mechanism of action · 3. Indications, contraindications, advantages, disadvantages · 4. PAP therapies (CPAP, APAP, ASV, BPAP) · 5. Surgical options · 6. Positional therapy, sleep hygiene, weight loss, CBT · 7. Emerging therapies · 8. Combination therapies · 9. OAT efficacy vs other options · 10. Mean disease alleviation, efficacy, compliance · 11. Practice guidelines and evidence-based practice

The organising question is not "which treatment is best?" but "which is best for this patient, and how do I know?" The answer runs through 3.10.

3.1 History

- **George Catlin (19th c.)** — first suggested breathing affects sleep quality: nasal rather than mouth breathing gives more restful, better quality sleep, better daytime function and better general health.
- **Pierre Robin, 1903** — the **monobloc**, for glossoptosis. ~30 years later used to reposition the mandible.
- **~50 years of little development** thereafter.
- **Cartwright & Samelson, 1982** — the **tongue retaining device**, holding the tongue forward by suction; historically the second most common type.
- **Meier-Ewert *et al.*, 1985** — an intraoral protraction device; the 1980s work was the catalyst for the modern field.
- **Modern era** — focus on **mandibular advancement appliances**; case reports gave way to RCTs confirming efficacy in mild–moderate and more recently severe OSA. Soft palate lifters obsolete; TRDs used only rarely.

DEFINITIONS — AADSM TERMINOLOGY

- **Oral appliance** — a device intended to protrude and stabilise the mandible to maintain a patent airway during sleep.
- **Custom OA** — made from digital or physical impressions/models; not prefabricated; biocompatible material; engages both arches. **Non-custom ("boil-and-bite")** — prefabricated, only partially modified.
- **Titratable** — allows varying protrusion (the analogue of CPAP titration). **Non-titratable** — single position, no change possible.
- **Effective oral appliance (AADSM 2019)** — advancement in increments of **1 mm or less**; protrusive adjustment range **≥5 mm**; reversal possible; setting **verifiable**; suitable for placement and removal by patient or caregiver; maintains a stable retentive relationship to teeth, implants or edentulous ridge and retains the prescribed setting during use; FDA-cleared or equivalent.

3.2 Mechanism of action

- **TRD** — pliable silicone bulb over the protruded tongue held by suction, supported by teeth and soft tissues; forward tongue posture prevents collapse into the oropharyngeal airway (the commonest site). Indicated where dental factors preclude a MAD; **evidence insufficient**; used very seldom.
- **MAD** — dentally retained, one-piece or dual-component with a coupling mechanism, repositioning and supporting the mandible forward through the night.

MECHANISM, IN SEQUENCE

1. **Protrusion** advances the jaw and attached musculature, enlarging the airway.
2. **Velopharyngeal advancement** — increased palatoglossus and palatopharyngeus tension draws the soft palate anteriorly and inferiorly, displacing the velopharynx forward. This is why a mandibular appliance improves a *palatal-level* obstruction.
3. **Tongue base advancement** — increased tension in and stabilisation of the suprahyoid and genioglossus muscles prevents the tongue falling back; the hyoid is stabilised by forward mandibular movement.
4. **Structural change** — the retroglossal airway enlarges, and the airway enlarges **principally in its lateral dimension at the velopharyngeal level**, through traction on soft tissue connections between the pharynx and the mandibular ramus.
5. **Net effect** — improved *passive pharyngeal anatomy* and reduced collapsibility, **dose-dependently**, *without* systematic change to genioglossus function, loop gain or arousal threshold.

3.3 Indications, contraindications, advantages, disadvantages

- **Indications. Primary snoring** without apnoea — first-line for a confirmed diagnosis in the confirmed absence of OSA; snoring cannot be distinguished from OSA clinically, so **patients who snore must not be treated without a diagnosis**. **Mild–moderate**

OSA, possibly first-line where the patient prefers it to CPAP. **Severe OSA** where CPAP is refused, not tolerated, ineffective, or the patient is an inappropriate candidate — current evidence including an individual participant data meta-analysis supports comparable health outcomes, and AADSM 2025 states appliances are indicated for **mild, moderate and severe** OSA. Also failure of weight loss or positional therapy; failed or unsuitable for surgery.

- **Contraindications.** Insufficient dentition or inadequate retention; severe periodontal disease or tooth mobility; active severe TMJ disorder or pain (treat first — TMD is *relative*, and disc displacement alone is not a contraindication); limited maximum protrusion <6 mm; severe untreated caries or poor hygiene; severe bruxism (consider a more robust material rather than automatic exclusion); severe gag reflex; planned major dental or orthodontic work; anterior open bite (no posterior clearance); severe nasal obstruction, grade 3–4 tonsils, elongated palate or large uvula (reduce efficacy — address the nose first).
- **Advantages, including over CPAP.** Reduces snoring and the harmful effects of OSA; less obtrusive; more portable, especially travelling; silent; no power source; more acceptable to patients and families with higher acceptance and adherence; superior quality-of-life outcomes in some comparisons; reduces blood pressure comparably to CPAP in several trials.
- **Disadvantages and side effects.** Excessive salivation; dry mouth; gingival irritation; tooth tenderness or pain; abnormal occlusion on waking and permanent bite change through orthodontic tooth movement; masticatory muscle tenderness; TMJ discomfort or stiffness (significant persistent dysfunction is **rare**); cost and risk of failure; less efficacious than CPAP for AHI, arousal index and desaturation.
- Side effects are usually well tolerated once symptomatic relief is achieved. **Division of responsibility:** diagnosis, prescription and monitoring of effectiveness rest with the medical practitioner; the dentist assesses dental suitability, treats oral disease first, chooses the device, supervises implementation, and follows up to prevent, recognise and manage orofacial complications.

3.4 Positive airway pressure

THE FOUR MODALITIES

- **CPAP** — single continuous pressure. Gold standard and first-line for OSA, particularly severe.
- **APAP** — auto-adjusting within a set range. For varying requirements (position, stage, alcohol, weight change); equivalent to CPAP for ongoing treatment.
- **BPAP** — separate inspiratory and expiratory pressures. For hypercapnic SDB; requirement **>20 cmH_2O**; intolerance of high CPAP; chronic respiratory failure; sleep-related hypoventilation.
- **ASV** — breath-by-breath adaptation. For central and mixed apnoea and Cheyne–Stokes respiration. Most complex and expensive. **Contraindicated in symptomatic heart failure with reduced ejection fraction.**

- **Machine** = mask/interface with straps + tube + motor.
- **Advantages:** more efficacious than a MAD for AHI, arousal index, ODI and oxygen saturation; prevents collapse and relieves daytime sleepiness; favourable cardiovascular outcomes, with significant reduction in nocturnal, daytime and 24-h blood pressure (largest effect nocturnally); decreases pain sensitivity and increases pain tolerance; almost 100% successful *if* tolerated.
- **Disadvantages:** lower compliance and tolerance, especially in mild OSA — **40–60% at 12 months**, failure rates 29–83%; similar health outcomes to a MAD because greater efficacy is offset by inferior compliance; not curative (must be used nightly); cost, noise, equipment maintenance and follow-up burden; loss of intimacy; difficult in retrognathia because of mask leakage.
- **Side effects:** mostly benign — nasal congestion, oral and nasal dryness, dry mouth, sore throat, aerophagia, epistaxis, sneezing, sinusitis. Rare serious — pneumocephalus, pulmonary barotrauma, raised intraocular pressure, subcutaneous emphysema. Middle ear pressure rise may cause **otic barotrauma**: pain, tinnitus, vertigo; middle ear inflammation with conductive loss; pressure transmitted via the Eustachian tube may disrupt the oval or round window, damaging cochlear membranes or producing a perilymph fistula with sensorineural loss.
- **Titration:** establish the minimum pressure at which snoring ceases and apnoeic events are absent. Start **4 cmH_2O**, titrate to the lowest effective pressure, maximum ~**20 cmH_2O**.
- **Masks:** nasal masks are effective in mild, moderate and severe disease. **Oronasal (full face) masks require significantly higher pressures in moderate–severe disease** — at higher pressure the mouth is forced open by the jaw being pushed down, compromising efficacy; leakage exceeds nasal pillows; simultaneous nasal and oral pressure prevents the transmural pressure rise needed to open the airway. **Therefore repeat titration when changing from nasal to oronasal.** Nasal or intranasal interfaces are suggested for routine initiation.
- **Heated humidification** suggested — reduces dry mouth and throat, nasal discharge and congestion, dry or bleeding nose, sinus pain, headache, sore throat, hoarse voice, reduced smell. Trade-offs: "rain-out", cost, maintenance.
- **Modified pressure profile** (lower expiratory pressure) may increase comfort in poorly adherent patients or those struggling in laboratory titration, but no clinically significant difference supports routine use.

AASM 2019 PAP guideline — recommendations worth quoting

- **Good practice:** treatment must be based on a diagnosis established by **objective sleep apnoea testing**; a new diagnosis requires home or laboratory testing before treatment — but a patient already on PAP with good symptom control, diagnosed without prior

testing, should continue. Adequate follow-up with troubleshooting and monitoring of objective efficacy and usage data should occur after initiation and during treatment.

- **Whom to treat:** *recommend* PAP over no therapy in excessive sleepiness; *suggest* PAP in impaired sleep-related quality of life (assessed by SAQLI and FOSQ); *suggest* PAP in comorbid hypertension to reduce blood pressure. **Insufficient evidence** to suggest PAP in non-sleepy patients to reduce cardiovascular events or mortality — so conservative management with monitoring is acceptable where such a patient declines treatment.
- **Initiation:** *recommend* either APAP at home or in-laboratory titration in patients without significant comorbidity. Home — lower cost, less time away, faster initiation, better access; but risks inadequate education and inability to fix mask fit or leak on the night. Laboratory — technologist education, real-time confirmation of efficacy, immediate comfort intervention; but overnight stay, cost, patient time, possible delay. No difference in adherence, sleepiness or quality of life; patients generally prefer home. Choose on access, cost-effectiveness, patient preference and clinician judgement.
- **Modality:** *recommend* either CPAP or APAP for ongoing treatment (no difference in adherence, sleepiness or quality of life; APAP's advantage is automatic adjustment for alcohol, position and weight change). *Suggest* CPAP or APAP **over BPAP** routinely — similar benefit but BPAP risks a sub-optimally low expiratory pressure failing to prevent obstructive events, plus higher cost. Reserve BPAP for the specific indications above.
- **Education and behaviour:** *recommend* educational interventions **before** initiation (what OSA is, its consequences, what PAP is, potential benefits); *suggest* behavioural and/or troubleshooting interventions during the initial period — CBT or motivational enhancement, and close communication to identify and solve problems. Both improve adherence with minimal harm.
- **Monitoring:** *suggest* telemonitoring-guided intervention during the initial period — remote monitoring of usage, residual severity, unintentional leak and settings, triggering early intervention. Improves adherence and lowers side-effect severity; trade-offs are privacy and cost.
- **Follow-up:** review in the initial weeks to months — this promotes adherence and assesses response, and **adherence in the first days to weeks predicts long-term adherence**. Then yearly, more often for persistent complaints or difficulty. OSA rarely resolves except with substantial weight loss or successful corrective surgery. Routine repeat testing is unnecessary where symptom control is good and status unchanged.

"WHAT NEXT WHEN CPAP FAILS" — ANSWER IN FOUR STAGES, NOT ONE

1. **Was treatment necessary?** Check the study: severe enough to mandate therapy? Level 1 or 2? Apnoeas or hypopnoeas? Desaturations? Significant fatigue or sleepiness? Compelling comorbidities?
2. **Were the merits explained?** Why treat; why CPAP is usually first (gold standard, almost always successful if tolerated, though not cheap); the pathophysiology and organ systems; the likely symptomatic benefit; and that alternatives are neither complication-free nor guaranteed.
3. **Was the trial properly conducted?** Start slow and build up; daily remote therapy review; regular mask changes; gradual pressure increases; treat at a tolerable pressure before aspiring to therapeutic targets.
4. **Only then, alternatives — multidisciplinary.** Weight loss; nasal EPAP (Provent ~8, Theravent 2–6 cmH_2O — expensive, not very comfortable); MAD; ENT assessment and surgery; positional therapy; combination therapy; specialist-unit trial therapies (mandibular advancement surgery, hypoglossal nerve stimulation, tracheostomy). Some patients are genuine treatment failures.

3.5 Surgery

*Role is either **adjunctive** (alongside CPAP or OAT — e.g. nasal surgery to enable CPAP tolerance) or **salvage** (replacing them). Non-invasive options first. Organising triad: **expand, advance, stiffen**.*

- **Nasal** (usually adjunctive/pre-phase) — septoplasty; inferior turbinectomy or turbinoplasty; nasal valve surgery for lateral wall collapse; endoscopic sinus surgery; polypectomy. Improves airflow and CPAP/OAT tolerance; rarely curative alone. Intranasal corticosteroid for allergic nasopharyngeal oedema — treat allergic rhinitis before appliance therapy (refer to ENT) without waiting for complete resolution.
- **Soft palate / oropharynx — UPPP:** excision of tonsils and posterior soft palate/uvula with closure of the tonsillar pillars; **life-threatening adverse events and deaths reported**. Friedman staging predicts success (stage I highest, III mostly unsuccessful, IV where comorbidity or deformity present); success = >50% AHI reduction with AHI <20. **Extended uvulopalatal flap** — modified UPPP with additional mucosa and submucosal adipose tissue removed superior and lateral to the tonsillar fossa and from the posterior soft palate; better response with lower Mallampati scores. **LAUP** and **radiofrequency ablation**. **Soft palate implants** — Dacron rods under local anaesthesia; lower morbidity but high extrusion rate and limited durability.
- **Tongue base** — genioglossus advancement; midline glossectomy; lingual tonsillectomy; transoral robotic surgery; tongue coblation for retroglossal prolapse or hypertrophic lingual tonsil.
- **Skeletal** (generally better than soft tissue surgery) — **maxillomandibular advancement:** multilevel skeletal surgery enlarging the velo-oro-hypopharyngeal airway **without direct manipulation of pharyngeal tissue**, advancing the anterior pharyngeal tissues (soft palate, tongue base, suprahyoid muscles) attached to maxilla, mandible and hyoid, via Le Fort I and bilateral sagittal split ramus osteotomy with plates, screws and bone graft. Criteria: hypopharyngeal and/or velo-oro-hypopharyngeal narrowing, usually

with skeletal hypoplasia manifesting as retrognathia. Substantial AHI reduction, uncommon adverse events. Also distraction osteogenesis, SARPE/DOME, genioplasty.

- **Neurostimulation** — hypoglossal nerve stimulation (see 3.7); more recurrence over time. **Bariatric** — laparoscopic adjustable gastric banding, Roux-en-Y bypass; more effective than non-surgical weight loss for BMI and AHI but **may not cure OSA**; indicated after failed non-surgical loss, BMI ≥35 (often higher), with cardiometabolic comorbidity and surgical candidacy. **Multilevel** — simultaneous or phased, because collapse occurs at several sites; better results but less consistent outcomes and more adverse events. **Tracheostomy** — last resort; highly effective, substantial morbidity.
- **Decision-making** rests on severity, anatomy and level of obstruction, CPAP compliance, patient expectation and comorbidity. **DISE** is the principal investigation, identifying site by the **VOTE** classification (**V**elum, **O**ropharynx, **T**ongue base, **E**piglottis) and the pattern and degree of collapse. **Anteroposterior collapse is easiest to treat; lateral or circumferential generally requires MMA.** DISE can also show directly whether mandibular advancement opens the airway. **~4% adverse event rate** (cardiovascular, postoperative bleeding); remind patients surgery may not completely cure the problem.

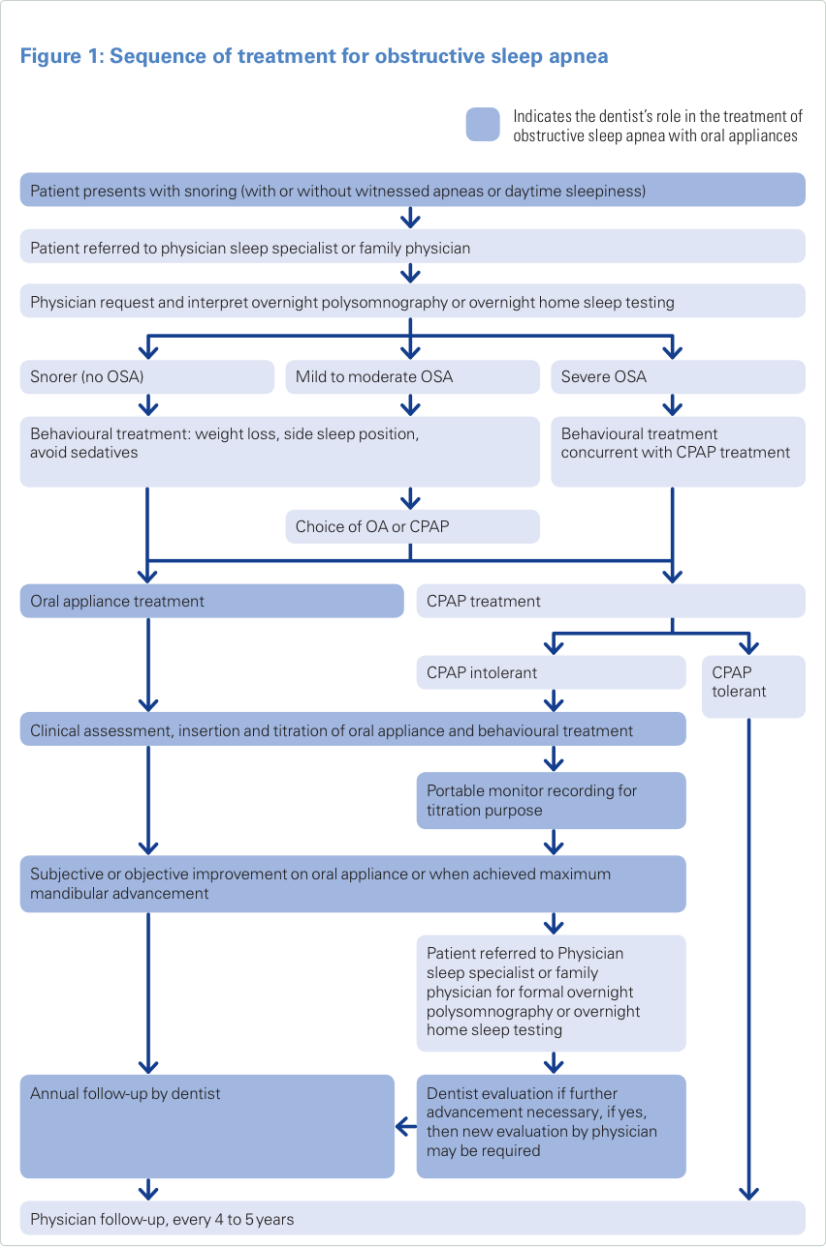


Fig 3.1 — Sequence of treatment for OSA, shaded boxes marking the dentist's role. Presentation with snoring ± witnessed apnoeas or sleepiness → referral to physician → physician requests and interprets PSG → severity stratification (snore/no OSA · mild–moderate · severe) → behavioural treatment (weight loss, side sleeping, avoiding sedatives) alone or concurrent with CPAP → choice of appliance or CPAP → clinical assessment, insertion and titration with behavioural treatment → subjective or objective reassessment at maximal comfortable advancement, with a portable monitor for titration purposes → **referral back to the physician** for review with PSG or home testing → annual dental follow-up → physician follow-up ~every 5 years with reassessment if the appliance is inadequate. **Examinable because:** answers the commonest structured question in the whole examination — "describe the patient journey and your role in it". Learn the sequence and where the two referral arrows point *back* to the physician.

3.6 Other therapies

- **Weight reduction.** 10% gain → ~32% AHI rise; 10% loss → ~26% fall. Mechanisms: reduced nasopharyngeal collapsibility; reduced mechanical loading particularly from parapharyngeal fat pads; reduced central adiposity lowering adipokines acting on the CNS, enhancing neuromuscular control of pharyngeal calibre; increased lung volume increasing caudal tracheal traction. Target BMI <30, though the effect is greatest with a high starting BMI (35–45). Method: dietitian, exercise physiologist, nutritionist; very low calorie diet + exercise + supervised counselling produces weight and waist reduction and AHI reduction; arguably first-line for mild OSA. Pharmacological: GLP-1 agonists (liraglutide, tirzepatide), phentermine; **sibutramine withdrawn** (MI and stroke risk, inferior to CPAP); orlistat inhibits gastrointestinal lipase. Feasible, low-cost, potentially curative, improves other cardiovascular risk factors — but hard to achieve and, above all, to sustain.
- **Positional therapy.** Promotes side sleeping; snoring severity and AHI are greater supine (in children AHI is lowest **prone**). Indication established by **PSG positional data**; strictly, **positional OSA = non-supine AHI <5**, and the goal is to reduce supine sleep time to zero; also requires a motivated patient and financial resources. Methods: tennis or golf ball, bra worn backwards, body pillows, shirts and vests — poor long-term adherence from discomfort and ineffectiveness; newer smart devices (e.g. NightShift) detect position and deliver a vibratory stimulus, but generally provide **no objective compliance data**. Reduces snoring in most non-apnoeic snorers but not those with OSA. Versus OAT: similar AHI reduction in positional OSA, but **OAT gives greater improvement in sleepiness and non-supine AHI**.
- **Sleep hygiene and behavioural therapy.** Sleep behavioural therapy = sleep hygiene, sleeping position, weight control, avoidance of sedatives including alcohol. Provide written information on all of these plus the device considered and all alternatives. Evidence quantifying the specific effect of sleep hygiene on OSA remains limited.
- **CBT.** Changes unhelpful thought patterns and behaviours; develops problem-solving; promotes relaxation and stress management. Recommended treatment for insomnia and particularly valuable in **COMISA**, where both conditions must be treated — reducing AHI alone will not resolve the complaint. CBT and motivational enhancement also improve PAP adherence.

3.7 Emerging therapies

- **Noradrenergic–antimuscarinic combinations** (atomoxetine + oxybutynin) — target poor muscle responsiveness; significant AHI reduction and improved sleep quality, particularly in combination; long-term safety and selection under investigation.
- **GLP-1 receptor agonists** (e.g. tirzepatide) — reduce both AHI and body weight in OSA with obesity, addressing metabolic and respiratory components together; the most significant recent pharmacological development. **Orexin agonists/antagonists** also under investigation.
- **Acetazolamide** — carbonic anhydrase inhibitor stabilising ventilatory control in the **high loop gain** endotype. **Eszopiclone ~3 mg** — raises the arousal threshold in the **low arousal threshold** endotype. **Supplemental oxygen** — also targets high loop gain.
- **Modafinil / armodafinil** — recommended for **residual** excessive sleepiness in patients who remain sleepy despite effective PAP and lack any other identifiable cause. Not a treatment for the OSA.
- **Nasal EPAP valves** (Provent ~8, Theravent 2–6 cmH_2O) — disposable one-way resistor valves: inhalation easy, exhalation resisted, creating positive end-expiratory pharyngeal pressure that stabilises the airway. Small, light, quiet, no power source; reduces AHI, ODI and sleepiness. Variable tolerance, no long-term data, contraindicated in nasal obstruction, ongoing expense, **no objective compliance data**, and treatment-emergent central sleep apnoea can occur as with CPAP. A reasonable "CPAP holiday"; consider PSG with the device in situ.
- **Hypoglossal nerve stimulation.** Theory — increase pharyngeal muscle tone to prevent collapse; stimulate genioglossus protrusion to maintain patency; use neuromodulation to dilate or reinforce the airway while avoiding traditional surgical side effects (severe dysphagia, throat pain). Hardware — a neurostimulator with one stimulation lead to **cranial nerve XII** and one sensing lead in the **fourth intercostal space** to detect inspiration. **Selection:** BMI <35, AHI 20–50, **non-concentric** retropalatal collapse. **Exclusions:** complete concentric collapse on DISE, grade 3–4 tonsils. **Outcomes:** reduced AHI, ODI and ESS; improved SAQLI and FOSQ; adherence ~86% — better than CPAP; potential to modify disease course; safe and effective for CPAP-refractory OSA. Availability is jurisdiction-dependent.
- **Orofacial myofunctional therapy** — targeted exercises strengthening upper airway muscles; reduces snoring and AHI in mild–moderate OSA or as an adjunct. **Other devices** — high-flow nasal cannula, oral negative pressure devices, under study.

3.8 Combination therapy

- **OAT + nasal or oral EPAP** — significantly reduces AHI in incomplete responders to OAT alone; larger AHI reduction than MAS alone, though sleep efficiency is lower with the combination.
- **OAT + CPAP** (connected or not) — for severe OSA, reduces AHI, ODI and required CPAP pressure versus either alone, and can improve PAP adherence by permitting a lower, more comfortable pressure. Custom masks and interfaces can be fabricated by the dentist to facilitate this.
- **OAT and CPAP used interchangeably** — improves overall effectiveness and adherence by increasing time on treatment while reducing the burden of CPAP use and of appliance side effects. Patients are willing to trade symptom-free days for fewer adverse events, and most are willing to pay for that.

- **OAT + positional therapy. CPAP + weight loss** — incremental reduction in insulin resistance, triglycerides and blood pressure versus either alone. **CPAP + modafinil** for residual sleepiness. **MAS + surgery**; multilevel surgical combinations. **Pharmacological** — atomoxetine + oxybutynin; acetazolamide + eszopiclone; oxygen with a hypnotic.

3.9 Comparative efficacy

- **AHI reduction:** CPAP highest (~75%) · OAT moderate (~42%) · UPPP variable (~30%) · exercise and weight loss similar to MAD.
- **Adherence:** CPAP lower · OAT higher · positional therapy similar to OAT.
- **ESS reduction** is similar across CPAP, MAD, exercise and weight loss.
- **Versus positional therapy:** OAT better for sleepiness; other measures similar in supine-dependent OSA.
- **Device design:** MADs generally outperform TRDs, especially in mild OSA; adjustable outperform fixed; newer designs report up to 95% efficacy for specific devices.

THE CONCLUSION TO STATE

"CPAP remains the gold standard because it is **more efficacious** at reducing AHI, arousal index and ODI and improving oxygen saturation. But oral appliance therapy offers advantages in patient preference and adherence, giving **comparable real-world effectiveness** for many outcomes — blood pressure, daytime sleepiness, neurocognitive function, quality of life and, on current evidence, cardiovascular mortality. OAT is therefore appropriate for mild–moderate OSA, and for severe OSA where CPAP is refused, not tolerated or ineffective."

3.10 Mean disease alleviation

DEFINITIONS — THE MOST CONCEPTUALLY IMPORTANT BOX IN THIS NOTEBOOK

- **Efficacy** — the maximum potential benefit of a treatment, typically measured in controlled trials as the reduction in a disease marker such as AHI. Sets the *upper limit* of benefit.
- **Compliance** — the extent to which patients correctly follow medical advice, often quantified as the percentage of prescribed therapy actually used. Determines *how much* of the efficacy is realised.
- **Mean disease alleviation** = **efficacy × compliance** — the overall percentage reduction in disease severity achieved in real-world conditions. Reflects *true* effectiveness.

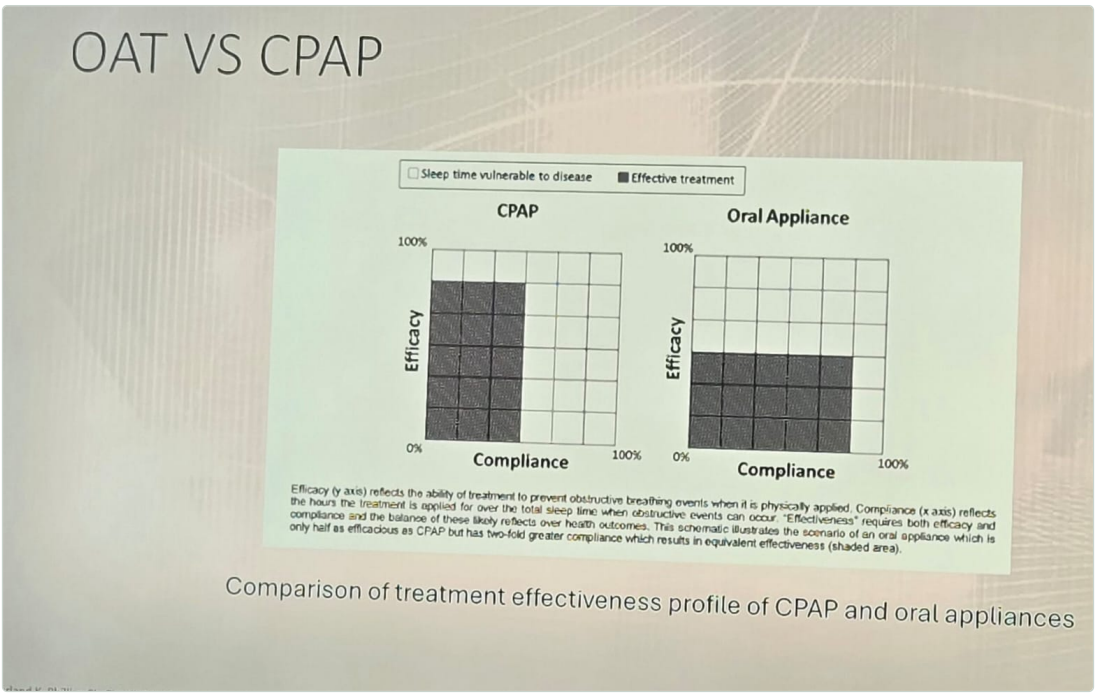


Fig 3.2 — Treatment effectiveness profile, CPAP vs oral appliance. Efficacy (y) = ability to prevent obstructive events while physically applied. Compliance (x) = hours applied over total sleep time when events can occur. "Effectiveness" requires **both**, and the shaded area — their product — reflects likely health outcomes. The schematic shows an appliance only half as efficacious as CPAP but with two-fold greater compliance, giving **equivalent effectiveness**. **Examinable because:** this figure *is* the answer to the paradox running through the whole examination — how a less efficacious treatment produces equal health outcomes. Reported MDA ~51% for appliances vs ~50% for CPAP. Draw it in a viva and you will be understood immediately.

- MDA allows direct comparison by accounting for both biological effect and patient behaviour. **High efficacy does not guarantee high effectiveness if compliance is poor**; moderate efficacy with high compliance may yield better real-world outcomes. **Legitimate criticism:** inconsistent definitions of compliance limit comparability across studies. Note also the emerging refinement of **effective AHI**, accounting for the AHI during untreated hours as well.

3.11 Guidelines, evidence-based practice, best practice

DEFINITIONS

Dentistry — the evaluation, diagnosis, prevention and/or treatment (non-surgical, surgical, or reversible and irreversible procedures) of diseases, disorders, irregularities or conditions of the oral cavity, maxillofacial area and/or adjacent and associated structures, and their impact on the human body.

Evidence-based dentistry — an approach to oral health care requiring the judicious integration of systematic assessments of clinically relevant scientific evidence, relating to the patient's oral and medical condition and history, with the dentist's clinical expertise and the patient's treatment needs, values and circumstances.

- **Purpose of EBD:** provide appropriate care, reduce inappropriate care, increase effectiveness of care, by promoting best available evidence in decision-making.
- **Five steps: Question** (define a focused, clinically relevant question) · **Search** (systematically find the best evidence, grade its strength, identify gaps) · **Appraisal** (critically appraise validity using critical evaluation tools) · **Application** (apply results considering clinical skills and patient preferences, values and conditions) · **Evaluation** (assess health and patient-reported outcomes to inform future care). Medical equivalent: ask, acquire, appraise, aggregate, apply, assess. **Three core components:** best available evidence, clinical expertise, patient values.
- **Best practice in dental sleep medicine:** written informed consent before treatment; satisfy regulator and association policies and codes of conduct; appropriate recognised training and experience; awareness of driving guidelines and fitness-to-drive obligations; keep records (casts, radiographs, photographs); ongoing education; **avoid conflicts of interest** — appliance selection must never be determined solely by insurer requirements or commercial relationships.
- **Guidelines** standardise care, reduce unwarranted variation and ensure effective and safe interventions are used. Developed by aggregating and critically evaluating empirical evidence (RCTs, cohort and case-control studies) with non-research evidence (case reports, established scientific principles, expert opinion). **Best practice** = the highest achievable standard for a procedure, technique or intervention, informed by evidence and leading to optimal outcomes; EBP is the mechanism by which it is identified and maintained. Benefits: improved outcomes, enhanced quality, greater efficiency, reduced cost, minimised error and risk.

NUMBERS — TOPIC 3

- Effective OA: increments ≤1 mm · range ≥5 mm · reversible · verifiable
- CPAP start 4, max ~20 cmH_2O; >20 → BPAP; 4–6 for snoring; compliance 40–60% at 12 mo; ≥4 h/night; failure 29–83%
- AHI reduction: CPAP ~75% · OA ~42% · UPPP ~30%. MDA: OA ~51% vs CPAP ~50%. MAD 6.5 vs CPAP 5.2 h/night
- Weight: 10% gain → +32% AHI; 10% loss → −26%; BMI target <30; bariatric BMI ≥35
- HNS: BMI <35, AHI 20–50, non-concentric retropalatal collapse; adherence ~86%. Provent ~8, Theravent 2–6 cmH_2O
- UPPP success = >50% AHI reduction and AHI <20; Friedman I best, III mostly unsuccessful; ~4% surgical adverse event rate
- Positional OSA = non-supine AHI <5. Minimum protrusive distance for a MAD = 6 mm

SELF-TEST — TOPIC 3

1. Give the AADSM definition of an effective oral appliance.

Advancement in increments of 1 mm or less, with a protrusive adjustment range of at least 5 mm; reversal must be possible; the setting must be verifiable; suitable for placement and removal by patient or caregiver; and it must maintain a stable retentive relationship to teeth, implants or edentulous ridge and retain the prescribed setting during use. FDA-cleared or local equivalent. A boil-and-bite device fails several criteria — it cannot be reliably titrated, its setting is not verifiable, and retention is poor.

2. How does a mandibular appliance improve a soft-palate-level obstruction?

Protrusion increases tension in palatoglossus and palatopharyngeus, drawing the soft palate anteriorly and inferiorly and displacing the velopharynx forward. In addition, traction on the soft tissue connections between the pharynx and the mandibular ramus widens the airway **laterally at the velopharyngeal level** — the predominant mechanism. So the appliance is not simply pulling the tongue forward; it remodels the whole velo-oropharyngeal segment.

3. Define efficacy, compliance and MDA, and explain why CPAP and appliances give similar outcomes.

Efficacy is maximum benefit under ideal conditions — AHI reduction while physically applied. Compliance is the proportion of prescribed therapy actually used. MDA is their product, representing real-world effectiveness. CPAP has roughly twice the efficacy but substantially lower adherence; the appliance's greater nightly use (~6.5 vs ~5.2 h) offsets its lower efficacy, giving MDA of ~51% and ~50% respectively. High efficacy therefore does not guarantee high effectiveness. Fair criticism: inconsistent definitions of compliance limit comparability across studies.

4. Which patients suit hypoglossal nerve stimulation, and who is excluded?

Suitable: BMI <35, AHI 20–50, non-concentric retropalatal collapse, typically refractory to or intolerant of CPAP. Excluded: complete concentric collapse on DISE, and grade 3–4 tonsils. The device has a stimulation lead to cranial nerve XII and a sensing lead in the fourth intercostal space to detect inspiration. Outcomes: reduced AHI, ODI and ESS, improved SAQLI and FOSQ, adherence ~86%. Availability is jurisdiction-dependent.

5. A patient with primary snoring and no daytime symptoms asks for an appliance. What must happen first?

A sleep physician must exclude OSA. Snoring cannot be distinguished from OSA clinically — one third of people with OSA do not snore and one third of snorers have no OSA — and patients who snore must not be treated in the absence of a diagnosis. Once absence of OSA is confirmed, an appliance is appropriate first-line therapy for snoring in a dentate patient and no further referral is needed for the appliance itself. Address conservative measures too — weight loss, reduced alcohol, positional therapy if supine-dominant. Follow-up sleep testing is **not** indicated for primary snoring unless symptoms worsen or fail to resolve.

TOPIC 3

Oral appliance therapy and other evidence-based treatments

EXAMINER'S BLUEPRINT

1. History of oral appliance therapy
2. Mechanism of action of oral appliance therapy
3. Indications, contraindications, advantages and disadvantages of oral appliance therapy
4. Positive airway pressure therapies (CPAP, APAP, ASV, BPAP) — indications, contraindications, advantages, disadvantages
5. Surgical options for sleep-disordered breathing
6. Other therapies: positional therapy, sleep hygiene, weight loss, cognitive behavioural therapy
7. Emerging therapies: pharmacology, EPAP valves, hypoglossal nerve stimulation, exercises
8. Combination therapies
9. Oral appliance efficacy compared with other treatment options
10. Mean disease alleviation (effectiveness), efficacy and compliance
11. Current practice guidelines, evidence-based practice and best practice

Topic 3 is the widest of the ten and the one most often used to separate the top of the cohort. The organising question is not "which treatment is best?" but "**which treatment is best for this patient, and how do I know?**" — and the answer runs through section 3.10, mean disease alleviation. Get that concept right and much of the rest of this topic becomes deducible.

3.1 History of oral appliance therapy

Figure / era	Date	Contribution
George Catlin	19th c.	The first to suggest that breathing affects sleep quality — that breathing through the nose rather than the mouth produces more restful, better quality sleep, translating into better daytime function and better general health
Pierre Robin	1903	Described the **monobloc** for the treatment of glossoptosis (a posteriorly displaced tongue obstructing the airway). Some 30 years later the appliance was used to reposition the mandible
The quiet period	1903–1950s	Approximately 50 years of very little development in oral appliances for SDB
Cartwright & Samelson	1982	Described the **tongue retaining device**, using suction to hold the tongue forward — historically the second most common appliance type
Meier-Ewert *et al.*	1985	Described an intraoral protraction device for sleep apnoea — a key step towards using mandibular advancement to treat OSA. The work of the 1980s was the catalyst for the modern field
Modern era	1990s–	Focus consolidated on **mandibular advancement appliances**. Early case reports gave way to randomised controlled trials confirming efficacy for mild-to-moderate OSA, and more recently for severe OSA. Soft palate lifters are no longer used; tongue retaining devices are used only rarely

DEFINITIONS — THE AADSM TERMINOLOGY SET

Oral appliance (OA) — a device intended to protrude and stabilise the mandible in order to maintain a patent airway during sleep.

Custom OA — fabricated using digital or physical impressions or models of the patient's oral structures; not prefabricated; made of biocompatible material; engaging both the upper and lower arches.

Non-custom OA ("boil-and-bite") — prefabricated and only partially modified to the patient's oral structures.

Titratable OA — allows a varying amount of mandibular protrusion (the analogue of CPAP titration).

Non-titratable OA — holds the mandible in a single protrusive position with no change possible.

Definition of an effective oral appliance (AADSM 2019). The appliance has a mechanism allowing the mandible to be advanced in increments of **1 mm or less**, with a protrusive adjustment range of **at least 5 mm**; reversal of the advancement must be possible; the protrusive setting must be **verifiable**; the appliance is suitable for placement and removal by the patient or caregiver; and it maintains a stable retentive relationship to the teeth, implants or edentulous ridge, retaining the prescribed setting during use. It should also be FDA-cleared (or equivalent).

3.2 Mechanism of action

Tongue retaining device (TRD)

A pliable silicone bulb fitted over a protruded tongue and held by suction, supported additionally by the teeth and soft tissues. Holding the tongue in a forward posture prevents it collapsing back into the oropharyngeal airway — the commonest site of collapse. Indicated where dental factors preclude a mandibular advancement appliance; evidence for efficacy is insufficient, and it is used very seldom today.

Mandibular advancement device (MAD)

The most commonly used and best researched device. Dentally retained, either one-piece or a dual-component design with upper and lower retainers connected by a coupling mechanism, designed to reposition the mandible forward and support that protrusive posture through the night.

Mode of action: an increase in the patency of the upper airway during sleep; provision of a stable and consistent anterior position of the mandible; advancement of the tongue; and possibly an increase in upper airway volume and/or a change in airway shape, with altered genioglossus muscle activity.

HIGH-YIELD — THE MECHANISM, IN SEQUENCE

1. **Mandibular protrusion** advances the jaw and attached musculature, enlarging the airway.
2. **Velopharyngeal advancement:** increased tension in the palatoglossal and palatopharyngeal muscles draws the soft palate anteriorly and inferiorly, displacing the velopharynx forward. This is why an appliance that acts on the mandible improves a *palatal-level* obstruction.
3. **Tongue base advancement:** increased tension in and stabilisation of the suprahyoid and genioglossus muscles, preventing the tongue falling back; the hyoid is stabilised by forward mandibular movement.
4. **Structural change to the pharynx:** the retroglossal airway enlarges, and — the key point — the airway enlarges **principally in its lateral dimension at the velopharyngeal level**, through traction on the soft tissue connections between the pharynx and the mandibular ramus.
5. **Net physiological effect:** improved *passive pharyngeal anatomy* and reduced pharyngeal collapsibility, in a **dose-dependent** manner, *without* systematic change to genioglossus muscle function, loop gain or arousal threshold.



Figure 3.1 — The two appliance classes: mandibular advancement devices (left), which reposition the mandible forward, and the tongue retaining device (right), which holds the tongue anteriorly by suction.

Why it is examinable: the TRD survives in examinations chiefly as the answer to "what would you offer a patient with insufficient dentition to retain a MAD?" — while noting that evidence for its efficacy is insufficient.

3.3 Indications, contraindications, advantages and disadvantages of OAT

Indications	Comment
Primary snoring without apnoea	OAT is first-line therapy for a confirmed diagnosis of snoring in the confirmed absence of OSA. Snoring cannot be distinguished from OSA clinically, so patients who snore **must not be treated in the absence of a diagnosis**
Mild to moderate OSA	May be first-line, particularly where the patient prefers an appliance to CPAP
Severe OSA where CPAP is refused, not tolerated, ineffective, or the patient is an inappropriate candidate	Historically restricted to mild–moderate disease, but current evidence — including an individual participant data meta-analysis in severe OSA — supports comparable health outcomes, and the AADSM 2025 standards state that OAs are indicated for **mild, moderate and severe** OSA
Failure of weight loss or positional therapy; failed or unsuitable for surgery	

Contraindication	Reason
Insufficient dentition / inadequate retention	No anchorage; risk of damage to remaining teeth
Severe periodontal disease or tooth mobility	Appliance unstable; may worsen the periodontal condition
Active severe TMJ disorder or pain	An appliance loads the joint; treat the TMD first (see Topic 4 — TMD is a *relative*, not absolute, contraindication and disc displacement alone is not a contraindication)
Limited maximum protrusion (<6 mm)	Insufficient range to achieve a therapeutic position
Severe untreated caries, poor oral hygiene	Restore and stabilise first
Severe bruxism	Appliance damage; consider a more robust material rather than automatic exclusion
Severe gag reflex	Tolerance
Planned major dental or orthodontic work	Complete the dental work first, or the appliance will not fit
Severe nasal obstruction; grade 3–4 tonsils, elongated palate, large uvula	Reduce efficacy — address the nose first (this improves outcome even though it does not itself treat the OSA)
Anterior open bite	No posterior clearance available for the appliance

Advantages (and advantages over CPAP)	Disadvantages and side effects
• Reduces snoring; reduces the harmful effects of OSA • Less obtrusive; more portable, especially with travel • Silent; not reliant on a power source • More acceptable to patients and their families — higher rates of acceptance and adherence • Superior quality-of-life outcomes in some comparisons • Reduces blood pressure to a degree comparable with CPAP in several trials	• Excessive salivation; dry mouth • Gingival irritation; tooth tenderness or pain • Abnormal occlusion on waking; permanent bite change through orthodontic tooth movement • Masticatory muscle tenderness; TMJ discomfort or stiffness (significant persistent TMJ dysfunction is **rare**) • Cost and the risk of treatment failure • Less efficacious than CPAP at reducing AHI, arousal index and desaturation

Side effects are usually well tolerated and accepted by patients, especially once symptomatic relief has been achieved. Note the division of responsibility: initial diagnosis of OSA, prescription of treatment and monitoring of therapy effectiveness rest with the medical practitioner; the dentist assesses suitability from a dental viewpoint, treats oral conditions before fitting, chooses an appropriate device, supervises its implementation, and follows up to prevent, recognise and manage orofacial complications.

3.4 Positive airway pressure therapies

THE FOUR MODALITIES

Mode	What it delivers	Primary indication
CPAP	A single continuous pressure	The gold standard and first-line therapy for OSA, particularly severe OSA
APAP	Auto-adjusting pressure within a set range	Varying pressure requirements (position, sleep stage, alcohol, weight change); equivalent to CPAP for ongoing treatment
BPAP	Separate inspiratory and expiratory pressures	SDB with hypercapnia; therapeutic pressure requirement **>20 cmH_2O**; intolerance of high CPAP settings; chronic respiratory failure; sleep-related hypoventilation
ASV	Adaptive servo-ventilation — breath-by-breath adaptation to the patient's own pattern	Central sleep apnoea, mixed apnoea, Cheyne–Stokes respiration. Most complex and expensive. **Contraindicated in symptomatic heart failure with reduced ejection fraction**

A CPAP machine has three parts: a mask or other interface fitting over the nose, or nose and mouth, with straps; a tube connecting the mask to the motor; and a motor blowing air into the tube.

Advantages, disadvantages and side effects

Advantages	Disadvantages
• More efficacious than a MAD at reducing AHI, arousal index and oxygen desaturation index, and at increasing oxygen saturation • Prevents upper airway collapse and relieves symptoms including daytime sleepiness • Favourably affects cardiovascular outcomes; significant reduction in nocturnal, daytime and 24-hour blood pressure — largest effect on nocturnal measurements • Decreases pain sensitivity and increases pain tolerance • Almost 100% successful *if* the patient can tolerate it	• Lower patient compliance and tolerance, especially in mild OSA. Long-term compliance only **40–60% at 12 months**; failure rates of 29–83% reported • Similar health outcomes to a MAD — greater efficacy is offset by inferior compliance • Not curative; must be used every night on a regular basis to control symptoms • Cost, noise, inconvenience of equipment maintenance and follow-up visits • Loss of intimacy • Difficult in the retrognathic patient because of mask leakage

Side effects. Mostly benign and mild: nasal congestion, dryness of the oral and nasal cavity, dry mouth, sore throat, aerophagia (excessive swallowing of air), epistaxis, sneezing, sinusitis. Rare but serious adverse effects have been reported: pneumocephalus, pulmonary barotrauma, raised intraocular pressure and subcutaneous emphysema. Increased middle ear pressure may cause otic barotrauma — a pressure differential between the middle ear and the environment causing pain, tinnitus and vertigo, with middle ear inflammation and conductive hearing loss; pressure transmitted to the inner ear can cause sensorineural loss, since a sudden middle ear pressure rise transmitted via the Eustachian tube may disrupt the oval or round window, damaging cochlear membranes or producing a perilymph fistula.

Titration and mask selection

- **Titration** establishes the minimum pressure at which snoring ceases and apnoeic events are absent. Start at **4 cmH_2O** in an adult and titrate to the lowest effective pressure, to a maximum of about **20 cmH_2O**.
- **Nasal masks** are effective in mild, moderate and severe apnoea. **Oronasal (full face) masks** require significantly higher pressures in moderate-to-severe disease: at higher pressures the mouth is forced open by the jaw being pushed down, compromising efficacy; leakage is greater than with nasal pillows; and simultaneous nasal and oral pressure application prevents the necessary increase in transmural pressure needed to open the airway. **Therefore, when changing from a nasal to an oronasal mask, repeat titration is required**, especially in moderate-to-severe disease. Nasal or intranasal interfaces are suggested over oronasal or oral interfaces for routine initiation.
- **Heated humidification** is suggested to reduce side effects — dry mouth and throat, nasal discharge and congestion, dry or bleeding nose, sinus pain, headache, sore throat, hoarse voice and reduced smell. The trade-offs are "rain-out" (condensation in the circuit and on the face), increased ongoing costs and maintenance.
- **Modified pressure profile PAP** (lower pressure during expiration) may increase comfort in poorly adherent patients or those struggling during in-laboratory titration, but no clinically significant difference has been demonstrated to support its routine use over standard PAP.

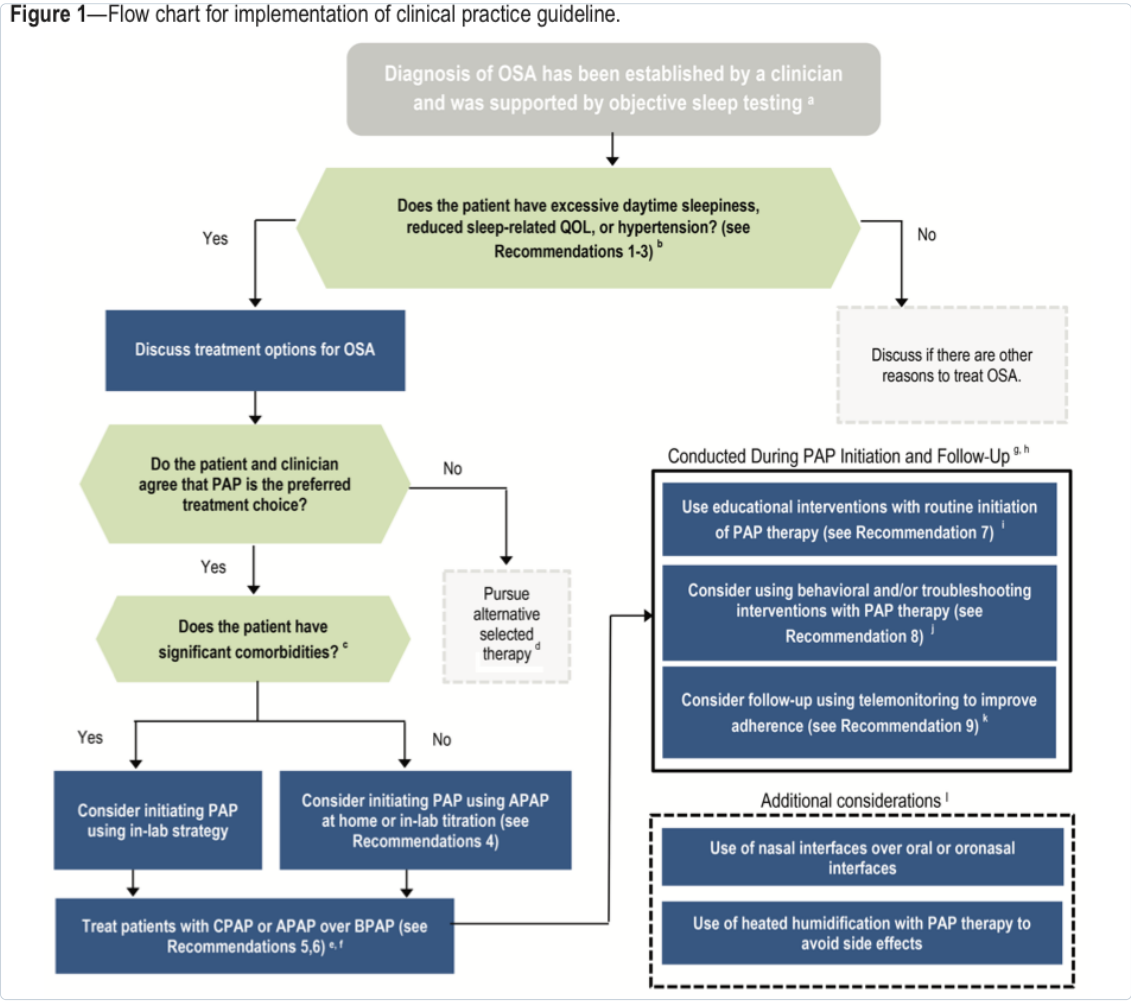


Figure 3.2 — Implementation flow chart for the AASM PAP clinical practice guideline. From established diagnosis via objective testing, through the question of excessive sleepiness, reduced sleep-related quality of life or hypertension, to shared decision-making on whether PAP is the preferred treatment; then initiation (home APAP versus in-laboratory titration), educational, behavioural and troubleshooting interventions, telemonitoring, and additional considerations of interface selection and heated humidification.

*Why it is examinable: the whole 2019 guideline in one page. Note the very first box — treatment must be based on a diagnosis established by **objective sleep apnoea testing**, which is the PAP counterpart of the dental rule that you never treat snoring without a diagnosis.*

The AASM 2019 PAP guideline — recommendations worth quoting

Area	Recommendation and rationale
Good practice statements	Treatment of OSA with PAP should be based on a diagnosis established using objective sleep apnoea testing. A new diagnosis should be established by home or in-laboratory testing before treatment — but a patient already on PAP with good symptom control, diagnosed without prior testing, should continue. Adequate follow-up, including troubleshooting and monitoring of objective efficacy and usage data, should occur after initiation and during treatment
Whom to treat	*Recommend* PAP over no therapy in adults with excessive sleepiness. *Suggest* PAP in adults with impaired sleep-related quality of life (assessed by the Calgary Sleep Apnea Quality of Life Index and the Functional Outcomes of Sleep Questionnaire). *Suggest* PAP in adults with comorbid hypertension, to reduce blood pressure. There is **insufficient evidence** to suggest PAP in non-sleepy patients to reduce cardiovascular events or mortality — so where a non-sleepy patient declines treatment because of that uncertainty, conservative management with monitoring for symptom development is acceptable
Initiation	*Recommend* initiation using either APAP at home or in-laboratory titration, in adults without significant comorbidity. Home: lower cost, less time away from home, faster initiation, better access — but risks inadequate education and inability to rectify mask fit or leak problems on the night. Laboratory: education by a trained technologist, real-time visual confirmation of efficacy, immediate comfort interventions — but requires an overnight stay with associated cost, patient time and possible delay. No difference in adherence, sleepiness or quality of life; patients generally

Area	Recommendation and rationale
	prefer the faster, more convenient home option. Choice should rest on access, cost-effectiveness, patient preference and clinician judgement
Modality	*Recommend* either CPAP or APAP for ongoing treatment — no difference in adherence, sleepiness or quality of life, with similar benefits and harms; APAP's potential advantage is automatic adjustment for acute and chronic change (alcohol, position, weight). *Suggest* CPAP or APAP **over BPAP** routinely: benefits are similar while BPAP carries the potential harm of a sub-optimally low expiratory pressure that fails to prevent obstructive events, plus higher cost. Reserve BPAP for the specific indications listed above
Education and behaviour	*Recommend* educational interventions **before** initiation (what OSA is, its consequences, what PAP is, its potential benefits). *Suggest* behavioural and/or troubleshooting interventions during the initial period — CBT or motivational enhancement, and close communication to identify and solve PAP-related problems. Both improve adherence with minimal harm and negligible burden
Monitoring	*Suggest* telemonitoring-guided interventions during the initial period: remote monitoring of usage, residual severity, unintentional mask leak and settings, triggering early intervention. Improves adherence and lowers side-effect severity scores; the trade-offs are potential loss of privacy and cost
Follow-up schedule	Review in the initial weeks to months after initiation — this promotes adherence and assesses response, and adherence over the first few days to weeks predicts long-term adherence. Then yearly evaluation, more frequently for persistent or recurrent complaints or difficulty with use. OSA rarely resolves except with substantial weight loss or successful corrective surgery, so periodic follow-up is required. Routine repeat sleep testing is not needed where symptom control is good and clinical status unchanged

What to do when CPAP fails — a structured answer

This is a classic viva. Work through it in four stages rather than jumping to alternatives.

1. **Was treatment of the OSA necessary in the first place?** Check the study: is the OSA severe enough to mandate therapy? What level of study was it — level 1 or 2? Were the events apnoeas or hypopnoeas? Were there desaturations? Is there significant fatigue or daytime sleepiness? Are there compelling comorbidities?
2. **Have the merits of treatment been explained?** Why treat, and why is CPAP the usual first choice — it is the gold standard and almost universally successful if tolerated, though not cheap. Explain the pathophysiology and the organ systems involved. Set the expectation that the patient may feel substantially better. Ensure the patient appreciates that alternatives are neither free of complications nor guaranteed to succeed.
3. **Was the CPAP trial properly conducted, to establish true intolerability?** Start slowly and build up. Daily therapy review through remote monitoring platforms. Were there regular mask changes and gradual pressure increases? Was the patient treated at a tolerable pressure before aspiring to therapeutic targets?
4. **Only then move to alternatives, multidisciplinary.** Weight loss; nasal EPAP valves (Provent ~8 cmH_2O, Theravent 2–6 cmH_2O — expensive and not very comfortable); mandibular advancement splint; ENT assessment and surgery; positional therapy; combination therapy; and referral to a high-powered specialist unit for trial therapies such as mandibular advancement surgery, hypoglossal nerve stimulation or tracheostomy. Some patients are genuine treatment failures in whom nothing works well.

3.5 Surgical options

Surgery is comparatively uncommon in the OSA pathway. Its role is either **adjunctive** (alongside CPAP or OAT — for example nasal surgery to enable CPAP tolerance) or **salvage** (replacing CPAP or OAT). Non-invasive options should be offered first. The organising triad is **expand, advance, stiffen**.

Level	Procedure	Detail
	Septoplasty; inferior turbinectomy or turbinoplasty; nasal valve surgery (lateral wall collapse); endo-	

Level	Procedure	Detail
Nasal (usually adjunctive / pre-phase)	scopic sinus surgery; polypectomy	Improves nasal airflow, reduces resistance and improves CPAP and OAT tolerance. Rarely curative for OSA in its own right
	Intranasal corticosteroid for allergic nasopharyngeal oedema	Medical pre-phase; treat allergic rhinitis before appliance therapy (referral to ENT), without waiting for complete resolution
Soft palate / oropharynx (stiffen or reduce)	**Uvulopalatopharyngoplasty (UPPP)**	Excision of the tonsils and posterior soft palate/uvula with closure of the tonsillar pillars. Life-threatening adverse events and deaths have been reported. Friedman staging predicts success (stage I highest success; stage III mostly unsuccessful; stage IV where comorbidity or deformity is present). Success defined as >50% AHI reduction with AHI <20
	Extended uvulopalatal flap (EUPF)	A modified UPPP with additional mucosa and submucosal adipose tissue removed superior and lateral to the tonsillar fossa and from the posterior soft palate. Better response rates in patients with lower Mallampati scores
	LAUP and **radiofrequency ablation**	Laser-assisted uvulopalatoplasty: serial laser incisions and vaporisations to shorten the uvula and tighten palatal tissue. Upper airway RFA without probe-tip temperature control also described
	Soft palate implants	Dacron rods inserted under local anaesthesia — lower morbidity, but high extrusion rate and limited durability
Tongue base	Genioglossus advancement; midline glossectomy; lingual tonsillectomy; transoral robotic surgery (TORS) for tongue base reduction; tongue coblation for retroglossal prolapse or hypertrophic lingual tonsil	Targets retroglossal obstruction
Skeletal (advance — generally better than soft tissue surgery)	**Maxillomandibular advancement (MMA)**	Multilevel skeletal surgery enlarging the velo-oro-hypopharyngeal airway **without direct manipulation of pharyngeal tissue**. It advances the anterior pharyngeal tissues — soft palate, tongue base, suprahyoid muscles — attached to the maxilla, mandible and hyoid, via a Le Fort I and bilateral sagittal split ramus osteotomy with plates, screws and bone graft. Anatomical criteria: hypopharyngeal and/or velo-oro-hypopharyngeal narrowing, usually with coexistent skeletal hypoplasia, clinically manifesting as retrognathia. Substantial AHI reduction with uncommon adverse events. Also distraction osteogenesis, SARPE/DOME, genioplasty
Neurostimulation	Hypoglossal nerve stimulation (upper airway stimulation) — see 3.7	More recurrence of OSA reported over time
Bariatric	Laparoscopic adjustable gastric banding; Roux-en-Y gastric bypass	More effective than non-surgical weight loss at reducing BMI and AHI, but **may not cure OSA**. Indicated after failed non-surgical weight loss, BMI ≥35 (often higher), with cardiometabolic comorbidity and surgical candidacy
Multilevel	Simultaneous or phased multilevel surgery (e.g. MMA plus radiofrequency)	Because collapse occurs at several sites; better results, but less consistent outcomes and more common adverse events
Last resort	Tracheostomy	Bypasses the obstruction entirely — highly effective but with substantial morbidity; reserved for rare life-threatening cases where all else has failed

Surgical decision-making depends on apnoea severity, anatomy and level of obstruction, CPAP compliance, patient expectation and comorbidity. **Drug-induced sleep endoscopy (DISE)** is the principal investigation for surgical

candidates, identifying the site (the **VOTE** classification: **V**elum, **O**ropharynx, **T**ongue base, **E**piglottis) and the pattern and degree of collapse. Anteroposterior collapse is the easiest to treat; lateral or circumferential collapse generally requires MMA. DISE can also be used to observe directly whether mandibular advancement opens the airway. Fitness for surgery matters — a roughly 4% adverse event rate (cardiovascular events, postoperative bleeding) — and patients must be reminded that surgery may not completely cure the problem.

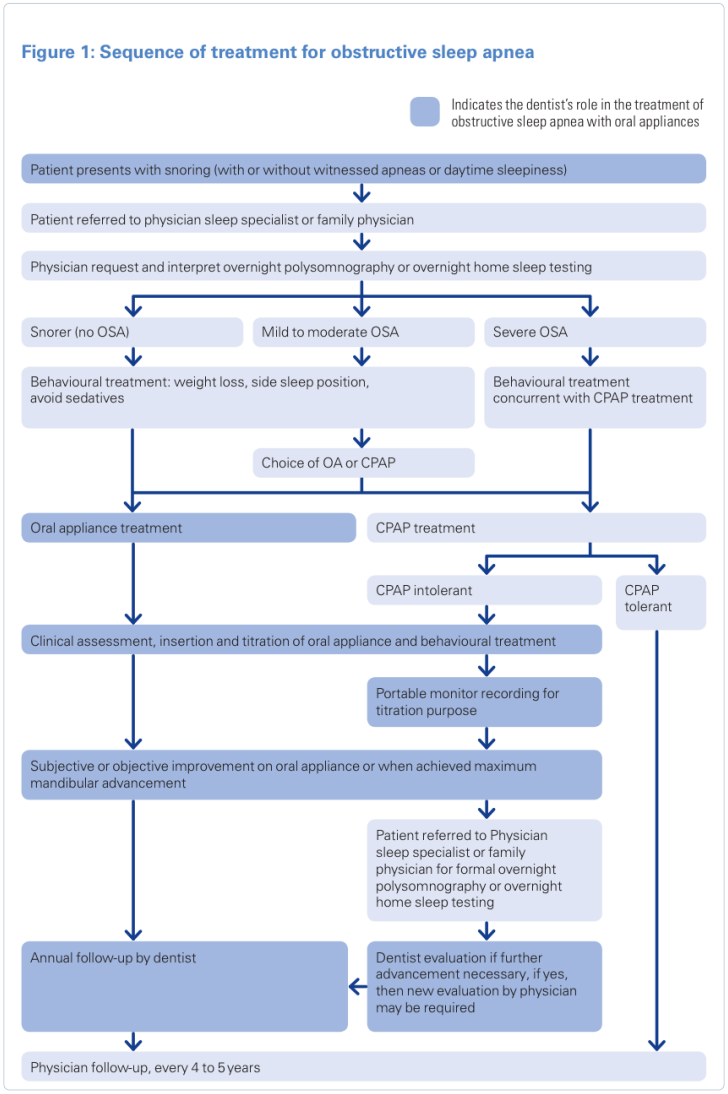


Figure 3.3 — Sequence of treatment for obstructive sleep apnoea, with the shaded boxes marking the dentist's role. The pathway runs from presentation with snoring (with or without witnessed apnoeas or daytime sleepiness) → referral to physician → physician request and interpretation of overnight polysomnography or sleep testing → severity stratification (snore/no OSA · mild-to-moderate · severe) → behavioural treatment with weight loss, side sleep position and avoidance of sedatives, either alone or concurrent with CPAP → choice of oral appliance or CPAP → clinical assessment, insertion and titration of the appliance with behavioural treatment → subjective or objective reassessment at maximal comfortable mandibular advancement, with the option of a portable monitor for titration purposes → referral back to the physician for a sleep specialist or family physician review, with overnight polysomnography or home sleep testing → annual follow-up by the dentist → physician follow-up approximately every five years, with reassessment if the appliance is inadequate.

Why it is examinable: this single figure answers the commonest structured question in the whole examination — "describe the patient journey and your role within it". Learn the sequence, and in particular where the two referral arrows point back to the physician.

3.6 Other therapies

Weight reduction

Aspect	Detail
Rationale	10% weight gain predicts a ~32% AHI increase; 10% weight loss predicts a ~26% AHI decrease. Mechanisms: reduced nasopharyngeal collapsibility; reduced mechanical loading, particularly from parapharyngeal fat pads; reduced central adiposity, lowering adipokines acting on the CNS and so enhancing neuromuscular control of pharyngeal calibre; and increased lung volume, increasing caudal tracheal traction on the pharynx
Target	BMI <30, though the effect is greatest in patients with a high starting BMI (35–45)
Method	Lifestyle and behavioural treatment with dietitian, exercise physiologist and nutritionist. A very low calorie diet plus exercise plus supervised lifestyle counselling produces weight loss, reduced waist circumference and AHI reduction. Arguably first-line for mild OSA in adults
Pharmacological	GLP-1 receptor agonists (liraglutide/Saxenda; tirzepatide) and phentermine. Sibutramine is no longer used because of myocardial infarction and stroke risk and inferior effect compared with CPAP; orlistat (inhibiting gastrointestinal lipase to reduce fat absorption) is an alternative
Advantages / disadvantages	Feasible, low cost, potentially curative, and improves other obesity-related cardiovascular risk factors — but difficult to achieve and, more importantly, to sustain

Positional therapy

- **Principle:** promotes side sleeping instead of supine sleeping. Snoring severity and AHI are greater supine; in children the lowest AHI is often prone.
- **Indication:** positional data from the PSG establish the indication. Strictly defined, **positional OSA means a non-supine AHI <5**; the therapeutic goal is to reduce total sleep time spent supine to zero. Also requires a motivated patient and financial resources.
- **Methods:** older techniques — tennis or golf ball in a sock or a bra worn backwards, body pillows, special shirts and vests — with poor long-term adherence due to discomfort and ineffectiveness. Newer smart devices (e.g. NightShift) detect body position and deliver a vibratory stimulus; note that these generally provide **no objective compliance data**.
- **Efficacy:** reduces snoring frequency and rate in most non-apnoeic snorers (AHI <5), but not in those with OSA. A reasonable alternative in mild-to-moderate OSA and where CPAP adherence is poor. Compared with OAT, positional therapy achieves similar AHI reduction in positional OSA, but OAT gives a greater improvement in sleepiness and in non-supine AHI. Less sustainable long term; requires follow-up.

Sleep hygiene and behavioural therapy

Sleep behavioural therapy encompasses sleep hygiene, sleeping position, weight control and avoidance of sedatives including alcohol. Provide patients with explanations and written information on all of these, alongside the device being considered and all alternatives. Lifestyle modification is recommended, though evidence specifically quantifying the effect of sleep hygiene on OSA remains limited.

Cognitive behavioural therapy

CBT changes unhelpful thought patterns and behaviours, developing problem-solving skills and promoting relaxation and stress management. It is the recommended treatment for insomnia and is particularly valuable in **comorbid insomnia and sleep apnoea (COMISA)**, where both conditions must be treated — reducing the AHI alone will not resolve the patient's complaint. CBT and motivational enhancement also improve PAP adherence.

3.7 Emerging therapies

Therapy	Mechanism, evidence and selection
Noradrenergic–antimuscarinic combinations (atomoxetine + oxybutynin)	Target poor pharyngeal muscle responsiveness. Significant AHI reductions and improved sleep quality in clinical trials, particularly in combination; long-term safety and optimal patient selection remain under investigation
Orexin agonists / antagonists	Under investigation for AHI and sleep quality
GLP-1 receptor agonists (e.g. tirzepatide)	Reduce both AHI and body weight in OSA with obesity — addressing the metabolic and respiratory components together. The most significant recent development in OSA pharmacotherapy
Acetazolamide	Carbonic anhydrase inhibitor; stabilises ventilatory control in the **high loop gain** endotype
Hypnotics (eszopiclone ~3 mg)	Raise the arousal threshold in the **low arousal threshold** endotype
Supplemental oxygen	Also targets high loop gain
Modafinil / armodafinil	Recommended for **residual** excessive daytime sleepiness in OSA patients who remain sleepy despite effective PAP therapy and who lack any other identifiable cause for the sleepiness. Not a treatment for the OSA itself
Nasal EPAP valves (Provent ~8 cmH_2O; Theravent 2–6 cmH_2O)	Disposable one-way resistor valves over the nostrils: inhalation is easy, exhalation is resisted, creating positive end-expiratory pharyngeal pressure that stabilises the airway. Advantages: small, light, quiet, no power source; reduces AHI, ODI and daytime sleepiness. Disadvantages: variable tolerance, no long-term data, contraindicated in nasal obstruction, ongoing expense, no objective compliance data, and treatment-emergent central sleep apnoea can occur as with CPAP. A reasonable "CPAP holiday" option; consider PSG with the device in situ
Hypoglossal nerve stimulation	**Theory:** increase pharyngeal muscle tone to prevent collapse; stimulate genioglossus protrusion to maintain patency; use neuromodulation to dilate or reinforce the airway while avoiding the side effects of traditional surgery (severe dysphagia, throat pain). **Hardware:** a neurostimulator with one stimulation lead to cranial nerve XII (innervating tongue musculature including genioglossus) and one sensing lead in the fourth intercostal space to detect inspiration. **Selection for success:** BMI <35, AHI 20–50, and **non-concentric** collapse of the retropalatal airway. **Exclusions:** complete concentric collapse on DISE, and grade 3–4 tonsils. **Outcomes:** reduced AHI, ODI and ESS; improved SAQLI and FOSQ; better compliance than CPAP (~86%); potential to modify the disease course; considered safe and effective for CPAP-refractory OSA. Availability is jurisdiction-dependent — FDA-approved in the USA, clinical trial access only in some other countries
Orofacial myofunctional therapy	Targeted exercises to strengthen upper airway muscles; reduces snoring and lowers AHI, especially in mild-to-moderate OSA or as an adjunct
Other devices	High-flow nasal cannula and oral negative pressure devices, under study with benefit in selected populations

3.8 Combination therapies

Combination therapy is gaining attention as a way of addressing OSA's multifactorial pathophysiology and of helping patients who do not fully respond to a single treatment. It targets multiple pathophysiological traits, or combines device-based with pharmacological intervention, to improve both efficacy and adherence.

- **OAT + nasal or oral EPAP** — significantly reduces AHI in incomplete responders to OAT alone; larger AHI reduction than MAS alone, though sleep efficiency is lower with the combination
- **OAT + positional therapy**
- **OAT + CPAP**, connected or unconnected — for severe OSA, reduces AHI, ODI and the required CPAP pressure compared with either alone, and can improve PAP adherence by permitting a lower, more comfortable pressure. Custom masks and interfaces can be fabricated by the dentist to facilitate this
- **OAT and CPAP used interchangeably** — improves overall treatment effectiveness and adherence by increasing the percentage of time on treatment, while

reducing the burden of CPAP use and of appliance side effects

- **CPAP + weight loss** — incremental reduction in insulin resistance, serum triglycerides and blood pressure compared with either alone
- **CPAP + modafinil** — for residual sleepiness
- **MAS + surgery**; multilevel surgical combinations
- **Pharmacological combinations** — atomoxetine plus oxybutynin; acetazolamide plus eszopiclone; supplemental oxygen with a hypnotic — targeting specific physiological profiles

3.9 Oral appliance efficacy compared with other treatments

Treatment	AHI reduction	Preference / adherence	Key outcomes
CPAP	Highest (~75%)	Lower	Best for AHI, oxygenation, arousal index
OAT (MAD/TRD)	Moderate (~42%)	Higher	Good for mild–moderate OSA and sleepiness; comparable health outcomes
Positional therapy	Moderate	Similar to OAT	Effective in positional OSA; OAT better for sleepiness and non-supine AHI
Surgery (UPPP)	Variable (~30%)	Not applicable	Alternative for selected patients
Exercise / weight loss	Similar to MAD	Variable	ESS reduction similar across CPAP, MAD, exercise and weight loss

HIGH-YIELD — THE CONCLUSION TO STATE

"CPAP remains the gold standard because it is **more efficacious** at reducing the AHI, the arousal index and the oxygen desaturation index, and at improving oxygen saturation. However, oral appliance therapy offers advantages in patient preference and adherence, which leads to **comparable real-world effectiveness** for many outcomes — including blood pressure, daytime sleepiness, neurocognitive function, quality of life and, on current evidence, cardiovascular mortality. Oral appliance therapy is therefore appropriate for mild-to-moderate OSA, and for severe OSA where CPAP is refused, not tolerated or ineffective."

3.10 Mean disease alleviation: efficacy, compliance and effectiveness

DEFINITIONS — THE MOST CONCEPTUALLY IMPORTANT BOX IN THE NOTEBOOK

Concept	Definition	Role
Efficacy	The maximum potential benefit of a treatment, typically measured in controlled trials as the reduction in a disease marker such as the AHI	Sets the upper limit of potential benefit
Compliance / adherence	The extent to which patients correctly follow medical advice, often quantified as the percentage of prescribed therapy actually used	Determines how much of the efficacy is realised
Mean disease alleviation (MDA)	**Efficacy × compliance** — the overall percentage reduction in disease severity achieved in real-world conditions	Reflects true treatment effectiveness

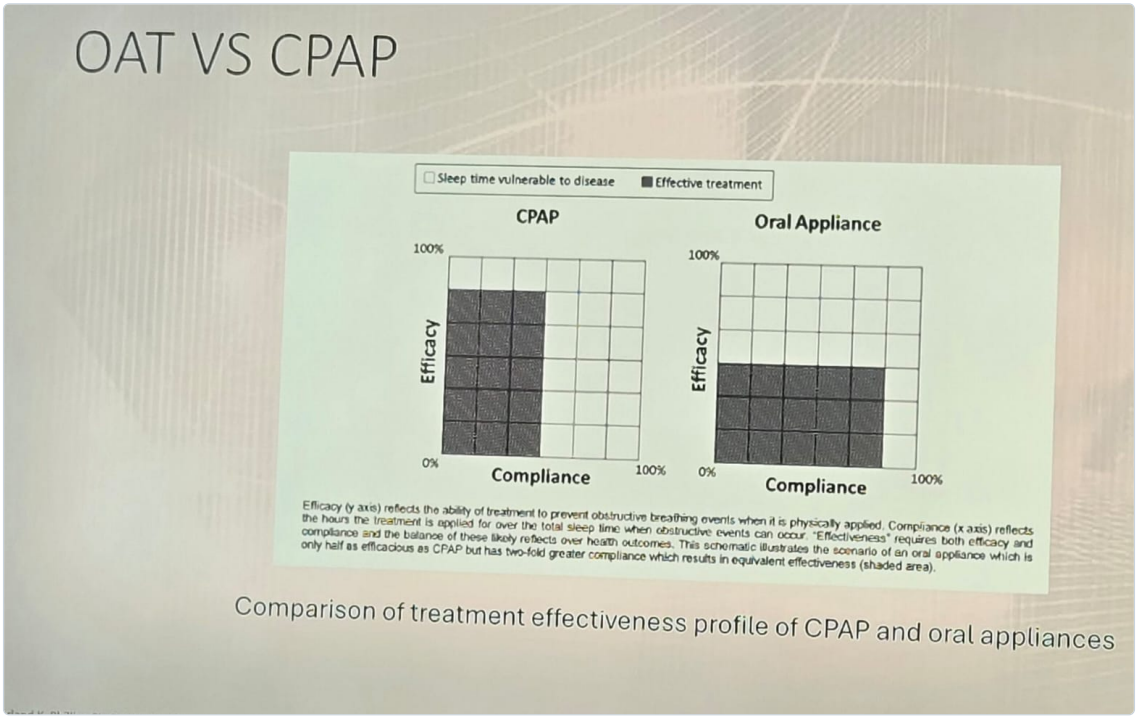


Figure 3.4 — Treatment effectiveness profile of CPAP versus an oral appliance. Efficacy (y axis) reflects the ability of the treatment to prevent obstructive breathing events while it is physically applied. Compliance (x axis) reflects the hours the treatment is applied, over the total sleep time when obstructive events can occur. "Effectiveness" requires **both**, and the shaded area — the product of the two — reflects likely health outcomes. The schematic illustrates an oral appliance that is only half as efficacious as CPAP but has two-fold greater compliance, resulting in **equivalent effectiveness**.

Why it is examinable: this figure is the answer to the apparent paradox that runs through the whole examination — how a less efficacious treatment can produce equal health outcomes. Reported MDA is approximately 51% for oral appliances versus 50% for CPAP. Draw this diagram in a viva and you will be understood immediately.

Practical implications. MDA allows direct comparison of therapies by accounting for both biological effect and patient behaviour. High efficacy does not guarantee high effectiveness if compliance is poor; a treatment with moderate efficacy but high compliance may yield better real-world outcomes. The definition and measurement of compliance are critical, because inconsistent definitions limit comparability of MDA across studies — a legitimate criticism to raise if asked to critique the concept. Note also the emerging refinement of **effective AHI**, which accounts not only for the AHI on treatment but for the AHI during the untreated hours.

3.11 Current practice guidelines, evidence-based practice and best practice

Why evidence-based dentistry

To provide appropriate care, reduce inappropriate care and increase the effectiveness of care, by promoting the use of the best available evidence to inform clinical decision-making.

DEFINITIONS

Dentistry — the evaluation, diagnosis, prevention and/or treatment (non-surgical, surgical, or reversible and irreversible procedures) of diseases, disorders, irregularities or conditions of the oral cavity, maxillofacial area and/or adjacent and associated structures, and their impact on the human body.

Evidence-based dentistry (EBD) — an approach to oral health care that requires the judicious integration of systematic assessments of clinically relevant scientific evidence, relating to the patient's oral and medical condition and history, with the dentist's clinical expertise and the patient's treatment needs, values and circumstances.

The five steps of EBD

1. **Question.** Define a clinically relevant and focused question where evidence may promote the patient's oral health.
2. **Search for information.** Systematically search for the best evidence across all studies, grade the strength of evidence using predefined criteria, and identify gaps.
3. **Appraisal.** Critically appraise the evidence for validity, using critical evaluation tools and an understanding of research methods.
4. **Application.** Apply the results to patient care, considering clinical skills and the patient's preferences, values and conditions.
5. **Evaluation.** Assess health care outcomes and patient-reported outcomes to inform future patient care.

The parallel medical formulation is the six-A cycle: *ask*, *acquire*, *appraise*, *aggregate*, *apply*, *assess*. Either is acceptable; the three core components of evidence-based practice are **best available evidence, clinical expertise and patient values**.

Best practice in dental sleep medicine

- Obtain written informed consent before treatment
- Satisfy the relevant regulator and professional association policies and codes of conduct (e.g. Dental Board, ADA, NZDA)
- Hold appropriate and recognised training and experience
- Be aware of driving guidelines and fitness-to-drive obligations
- Keep records: study models, radiographs, photographs
- Maintain ongoing education
- Avoid conflicts of interest — appliance selection should never be determined solely by insurer requirements or commercial relationships

Guidelines exist to standardise care, reduce unwarranted variation and ensure that the most effective and safe interventions are used. They are developed by aggregating and critically evaluating the best available evidence — empirical evidence from randomised controlled trials, cohort and case-control studies, together with non-research evidence from case reports, established scientific principles and expert opinion. **Best practice** is the highest achievable standard for a procedure, technique or intervention, informed by scientific evidence and leading to optimal outcomes; evidence-based practice is the mechanism by which best practice is identified and maintained, and its benefits are improved outcomes, enhanced quality, greater efficiency, reduced cost and minimised error and risk.

NUMBERS TO MEMORISE — TOPIC 3

- Effective OA definition: increments ≤1 mm · adjustment range ≥5 mm · reversible · verifiable setting
- CPAP: start 4 cmH_2O, max ~20 cmH_2O, >20 → BPAP; 4–6 cmH_2O for snoring; compliance 40–60% at 12 months; ≥4 h/night threshold; failure 29–83%
- AHI reduction: CPAP ~75% · OA ~42% · UPPP ~30%
- MDA: OA ~51% vs CPAP ~50%. Compliance: MAD ~6.5 h/night vs CPAP ~5.2 h/night
- Weight: 10% gain → +32% AHI; 10% loss → −26% AHI; BMI target <30; bariatric threshold BMI ≥35
- HNS selection: BMI <35, AHI 20–50, non-concentric retropalatal collapse; adherence ~86%
- Provent ~8 cmH_2O; Theravent 2–6 cmH_2O
- UPPP success: >50% AHI reduction and AHI <20; Friedman stage I best, III mostly unsuccessful; ~4% surgical adverse event rate
- Positional OSA: non-supine AHI <5
- Minimum protrusive distance for MAD: 6 mm (limited protrusion <6 mm is a contraindication)

SELF-TEST — TOPIC 3

1. Give the AADSM definition of an effective oral appliance.

A mechanism allowing the mandible to be advanced in increments of 1 mm or less, with a protrusive adjustment range of at least 5 mm; reversal of advancement must be possible; the protrusive setting must be verifiable; the appliance must be suitable for placement and removal by the patient or caregiver; and it must maintain a stable retentive relationship to the teeth, implants or edentulous ridge and retain the prescribed setting during use. It should be FDA-cleared or the local equivalent. Note that a boil-and-bite device fails several of these criteria — it cannot be reliably titrated, its setting is not verifiable, and its retention is poor.

2. Explain how a mandibular appliance improves an obstruction at the level of the soft palate.

Mandibular protrusion increases tension in the palatoglossus and palatopharyngeus, which draws the soft palate anteriorly and inferiorly, displacing the velopharynx forward. In addition, traction on the soft tissue connections between the pharynx and the mandibular ramus widens the airway **laterally** at the velopharyngeal level — this lateral dimensional change is the predominant mechanism. So the appliance is not simply pulling the tongue forward; it remodels the whole velo-oropharyngeal segment.

3. Define efficacy, compliance and mean disease alleviation, and use them to explain why CPAP and oral appliances produce similar health outcomes.

Efficacy is the maximum potential benefit under ideal conditions — the AHI reduction achieved while the treatment is physically applied. Compliance is the proportion of prescribed therapy actually used. Mean disease alleviation is their product, and represents true real-world effectiveness. CPAP has roughly twice the efficacy of an oral appliance but substantially lower adherence; the appliance's greater nightly use (~6.5 versus ~5.2 hours) offsets its lower efficacy, giving similar MDA figures of about 51% and 50% respectively. High efficacy therefore does not guarantee high effectiveness. A fair criticism of MDA is that inconsistent definitions of compliance limit comparability across studies.

4. Your patient has failed CPAP. Structure your approach.

Four stages. **First**, question whether treatment was necessary: review the study level, event type, desaturation, severity, symptoms and comorbidities. **Second**, confirm the merits of treatment were explained — why treat, why CPAP is usually first, the pathophysiology, the likely symptomatic benefit, and that alternatives are neither complication-free nor guaranteed. **Third**, verify the CPAP trial was properly conducted before accepting true intolerance: slow build-up, remote therapy review, mask changes, gradual pressure increases, and treating at a tolerable rather than immediately therapeutic pressure. **Only fourth**, move to alternatives in a multidisciplinary setting: weight loss, EPAP valves, mandibular advancement splint, ENT assessment and surgery, positional therapy, combination therapy, and specialist-unit trial therapies including mandibular advancement surgery, hypoglossal nerve stimulation and tracheostomy. Acknowledge that some patients are genuine treatment failures.

5. What is the role of surgery, and what investigation guides it?

Surgery is either adjunctive — alongside CPAP or OAT, for example nasal surgery to enable CPAP tolerance — or salvage, replacing them. Non-invasive options are offered first. The organising principle is expand, advance, stiffen: nasal surgery, soft tissue procedures (UPPP and modifications, tongue base reduction), skeletal surgery (maxillomandibular advancement, distraction, SARPE/DOME, genioplasty — generally better outcomes than soft tissue surgery), neurostimulation, and bariatric surgery, with tracheostomy as a last resort. Drug-induced sleep endoscopy guides selection by identifying the site (VOTE: velum, oropharynx, tongue base, epiglottis) and the pattern and degree of collapse: anteroposterior collapse is easiest to treat, while lateral or circumferential collapse generally requires MMA. DISE can also confirm directly whether mandibular advancement opens the airway.

6. Which patients are suitable for hypoglossal nerve stimulation, and which are excluded?

Suitable: BMI <35, AHI 20–50, and non-concentric collapse of the retropalatal airway, typically in a patient refractory to or intolerant of CPAP. Excluded: complete concentric collapse on drug-induced sleep endoscopy, and grade 3–4 tonsils. The device comprises a stimulation lead to cranial nerve XII and a sensing lead in the fourth intercostal space to detect inspiration. Outcomes include reduced AHI, ODI and ESS, improved SAQLI and FOSQ, and adherence of around 86% — better than CPAP. Availability is jurisdiction-dependent.

7. A patient with primary snoring and no daytime symptoms asks you to make an appliance. What must happen first?

> A sleep physician must exclude OSA. Snoring cannot be distinguished from OSA clinically, one third of people with OSA do not snore and one third of snorers have no OSA, and patients who snore must not be treated in the absence of a diagnosis. Once the absence of OSA is confirmed, an oral appliance is appropriate first-line therapy for snoring in a dentate patient, and no further referral is needed for the appliance itself. Conservative measures should also be addressed — weight loss, reduced alcohol, positional therapy where snoring is supine-dominant. Follow-up sleep testing is **not** indicated for primary snoring unless symptoms worsen or fail to resolve.

8. Name the five steps of evidence-based dentistry.

> Question — define a focused, clinically relevant question. Search — systematically identify the best evidence and grade its strength. Appraisal — critically appraise validity using critical evaluation tools. Application — apply results considering clinical skills and patient preferences, values and conditions. Evaluation — assess health and patient-reported outcomes to inform future care. The three core components integrated throughout are best available evidence, clinical expertise and patient values.

REFERENCES — TOPIC 3

1. Ramar K, Dort LC, Katz SG, et al. Clinical practice guideline for the treatment of obstructive sleep apnea and snoring with oral appliance therapy: an update for 2015. An AASM and AADSM clinical practice guideline. *J Clin Sleep Med*. 2015;11(7):773–828.
2. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2).
3. Mogell K, Blumenstock N, Mason E, Rohatgi R, Shah S, Schwartz D. Definition of an effective oral appliance for the treatment of obstructive sleep apnea and snoring: an update for 2019. *J Dent Sleep Med*. 2019;6(3).
4. Robin P. La glossoptose. Son traitement par un appareil. *Rev Stomatol*. 1903.
5. Cartwright RD, Samelson CF. The effects of a nonsurgical treatment for obstructive sleep apnea: the tongue-retaining device. *JAMA*. 1982;248(6):705–709.
6. Meier-Ewert K, Schäfer H, Kloss W. Treatment of sleep apnea by a mandibular protracting device. In: *Proceedings of the 7th European Congress on Sleep Research*; 1985.
7. Chan ASL, Sutherland K, Schwab RJ, et al. The effect of mandibular advancement on upper airway structure in obstructive sleep apnoea. *Thorax*. 2010;65(8):726–732.
8. Bamagoos AA, Cistulli PA, Sutherland K, et al. Polysomnographic endotyping to select patients with obstructive sleep apnea for oral appliances. *Ann Am Thorac Soc*. 2019;16(11):1422–1431.
9. Patil SP, Ayappa IA, Caples SM, Kimoff RJ, Patel SR, Harrod CG. Treatment of adult obstructive sleep apnea with positive airway pressure: an AASM clinical practice guideline. *J Clin Sleep Med*. 2019;15(2):335–343.
10. Phillips CL, Grunstein RR, Darendeliler MA, et al. Health outcomes of continuous positive airway pressure versus oral appliance treatment for obstructive sleep apnea: a randomised controlled trial. *Am J Respir Crit Care Med*. 2013;187(8):879–887.
11. Trzepizur W, Cistulli PA, Glos M, et al. Health outcomes of CPAP versus mandibular advancement device for the treatment of severe obstructive sleep apnea: an individual participant data meta-analysis. *Sleep*. 2021;44(7):zsab015.
12. Vanderveken OM, Dieltjens M, Wouters K, De Backer WA, Van de Heyning PH, Braem MJ. Objective measurement of compliance during oral appliance therapy for sleep-disordered breathing. *Thorax*. 2013;68(1):91–96.
13. Sutherland K, Phillips CL, Cistulli PA. Efficacy versus effectiveness in the treatment of obstructive sleep apnea: CPAP and oral appliances. *J Dent Sleep Med*. 2015;2(4):175–181.
14. Grote L, Hedner J, Grunstein R, Kraiczi H. Therapy with nCPAP: incomplete elimination of sleep related breathing disorder. *Eur Respir J*. 2000;16(5):921–927.
15. Aurora RN, Casey KR, Kristo D, et al. Practice parameters for the surgical modifications of the upper airway for obstructive sleep apnea in adults. *Sleep*. 2010;33(10):1408–1413.
16. Strollo PJ, Soose RJ, Maurer JT, et al. Upper-airway stimulation for obstructive sleep apnea (STAR trial). *N Engl J Med*. 2014;370(2):139–149.
17. Vanderveken OM, Maurer JT, Hohenhorst W, et al. Evaluation of drug-induced sleep endoscopy as a patient selection tool for implanted upper airway stimulation. *J Clin Sleep Med*. 2013;9(5):433–438.
18. Kezirian EJ, Hohenhorst W, de Vries N. Drug-induced sleep endoscopy: the VOTE classification. *Eur Arch Otorhinolaryngol*. 2011;268(8):1233–1236.
19. Malhotra A, Grunstein RR, Fietze I, et al. Tirzepatide for the treatment of obstructive sleep apnea and obesity. *N Engl J Med*. 2024;391(13):1193–1205.
20. Taranto-Montemurro L, Messineo L, Sands SA, et al. The combination of atomoxetine and oxybutynin greatly reduces obstructive sleep apnea severity. *Am J Respir Crit Care Med*. 2019;199(10):1267–1276.

21. Dieltjens M, Braem MJ, Van de Heyning PH, Wouters K, Vanderveken OM. Prevalence and clinical significance of supine-dependent obstructive sleep apnea in patients using oral appliance therapy. *J Clin Sleep Med*. 2014;10(9):959–964.
22. American Dental Association. Evidence-based dentistry: definitions and the EBD process. ADA Center for Evidence-Based Dentistry.
23. Sackett DL, Rosenberg WM, Gray JA, Haynes RB, Richardson WS. Evidence based medicine: what it is and what it isn't. *BMJ*. 1996;312(7023):71–72.

TOPIC 4

History, examination and imaging for patient selection and planning

BLUEPRINT

1. DSM history taking, goal setting, managing expectations · 2. Medical history: age, obesity, drugs, alcohol, smoking, other sleep disorders, comorbidities · 3. SDB and other sleep problems (narcolepsy, RLS, PLMD, insomnia, insufficient sleep, shift work) · 4. Medical comorbidities · 5. Ethics and informed consent · 6. Multidisciplinary team and communication · 7. Sleep bruxism and SDB · 8. Pain and sleep

4.1 History, goals, expectations

- **Reason for attendance:** snoring, suspected or diagnosed OSA, sleep bruxism, or orofacial pain related to sleep disorders. Ask about sleep symptoms, previous diagnoses, comorbidities, and impact on daily life and well-being.
- **Goal setting:** discuss the primary concern and align it with evidence-based options, explaining benefits and limitations in the light of anatomical, behavioural and medical factors. In older or comorbid patients the goal may reasonably be improved quality of life and minimised side effects rather than a normalised AHI. Critically, **agree with the medical provider and patient what treatment success means and the metrics for judging it**, including how efficacy will be verified.
- **Expectations:** results vary; some side effects (dental change, discomfort) may occur; ongoing follow-up and adjustment are usually needed; emphasise interdisciplinary care and referral where required; explain what additional therapies might follow an inadequate response. Discuss **sleep hygiene and duration** — these may not affect appliance efficacy directly but collectively fragment sleep and exacerbate sleepiness, **reducing the patient's perception of benefit**. A patient sleeping five hours will not feel well however good the appliance.

The examination — components

- **Medical–dental history:** age, weight, height; neck and belt circumference as indicated; presence of tonsils; tongue size; palate width and depth; malocclusion classification with **overjet and overbite measurements**; tooth erosion or wear; medications; other disease; family history of sleep disorders; chief complaint and current sleep problems including snoring, witnessed apnoeas, nocturia, morning headaches, hypersomnolence.
- **Sleepiness assessment** with validated instruments (ESS, STOP-BANG); record baseline BMI, blood pressure and neck circumference.
- **Soft tissue/intraoral:** modified Mallampati, tonsil grade, tongue size and scalloping, palatal vault, posterior pharyngeal crowding, signs of mouth breathing, signs of GORD such as enamel erosion, nasal patency.
- **Periodontal evaluation** — attachment loss, mobility, prognosis of the teeth that will bear the appliance.
- **TMJ** — history and clinical examination: symptoms, sounds, range of movement, path of opening and any deflection or deviation, palpation of the muscles of mastication.
- **Bruxism, orofacial pain, headaches. Occlusion** — examine and record the existing condition; this is your medicolegal baseline.
- **Teeth and restorations** — number of functional teeth, mobility, spacing, curve of Spee, heavily restored teeth, anticipated restorative needs.
- **Imaging** — initial dental radiographs as indicated; **panoramic and lateral cephalometric as a pre-treatment record**. **CBCT and other CT are not indicated at this stage.**
- **Diagnostic models** (casts or digital scans) and **intraoral and extraoral photographs** retained as records.
- **Treatment planning** — provide explanation and written information on sleep behavioural therapy (sleep hygiene, position, weight control, alcohol); explain the device considered **and all alternatives** (CPAP, surgery, positional therapy); propose various appliances according to oral health status and craniofacial morphology; give positional therapy information where OSA is positional.

TRAP — IMAGING

Default is **panoramic and lateral cephalometric** as a pre-treatment record. **CBCT is not routinely indicated.** Advocating routine CBCT airway analysis loses marks on two grounds: unjustified radiation dose, and the poor validity of static airway measurement (Topic 8 — the airway is dynamic, inter-examiner reliability is poor, normative values have wide standard deviations, and the commonest sites of collapse are the least well visualised). Cephalometric evaluation is *not always required*, but where deemed necessary should be performed by appropriately trained professionals.

4.2 Medical history — modifiers of airway and morbidity

Answer format: name the factor → state the airway mechanism → state the morbidity consequence.

- **Age** — children and infants are at higher risk of airway complications from anatomical and physiological differences; preoperative assessment and history are crucial for predicting difficult airway management and preventing perioperative respiratory events. Advanced age raises airway-related morbidity, particularly with comorbidity and polypharmacy.
- **Obesity** — increases collapse and respiratory muscle dysfunction, contributing to OSA and postoperative respiratory failure; requires tailored perioperative management.
- **Drugs and medications** — anaesthetics, sedatives, opioids and neuromuscular blockers impair dilator function, increasing obstruction and respiratory failure risk, particularly perioperatively. Also drug classes altering sleep architecture and inducing bruxism or RLS (1.12).
- **Smoking** — chronic inflammation, upper airway pathology, susceptibility to infection and cancer, higher morbidity and mortality. Passive smoking and family smoking history increase perioperative respiratory adverse events in children.
- **Alcohol** — impairs airway muscle tone and increases obstruction risk, especially with sedatives; even moderate intake may cause transient desaturation despite appliance use.
- **Other sleep disorders and comorbidities** — chronic rhinosinusitis, allergic rhinitis, asthma or COPD worsen airway function and complicate management; recent upper respiratory infection significantly raises perioperative airway risk, especially in children.

4.3 SDB and other sleep problems

- **Insomnia — COMISA.** Highly prevalent in SDB; the overlap produces more severe fragmentation, increased sleepiness and poorer quality of life. **Both must be treated** — reducing AHI alone will not help. CBT-I plus SDB treatment.
- **RLS and PLMD.** Frequently co-occur; SDB exacerbates limb movements and vice versa, causing further disruption and arousals; both contribute to sleepiness and impaired alertness. Some antidepressants induce or worsen RLS and periodic limb movements.
- **Narcolepsy.** May coexist, compounding sleepiness and cognitive impairment; symptom overlap makes clinical differentiation unreliable, so careful sleep studies are needed. SOREMPs on the MSLT — a narcolepsy criterion — are also frequent in untreated OSA.
- **Insufficient sleep and shift work disorder.** SDB worsens both by further disrupting architecture, increasing cognitive and cardiovascular risk; shift workers with SDB are at higher risk of sleepiness and accidents. A patient simply not spending enough time in bed will not respond as expected to an appliance.
- **Mood disorders.** SDB is associated with mood disorders, depression and anxiety, which may present with insomnia or hypersomnia, creating a cycle of worsening sleep and mental health.

OSA and depression — the nuance

- Symptoms **overlap** (tiredness, fatigue, sleepiness, poor motivation and concentration), complicating the diagnosis and treatment of each. **OSA may be responsible for antidepressant treatment failure**, and depression is a risk factor for incomplete CPAP response, so may need independent management.
- Psychiatric diagnoses in OSA include depression, anxiety, PTSD, psychotic and bipolar disorders. Depression prevalence in OSA ~**21.8%**. Screen initially by tool, confirm with established criteria if positive.
- The relationship is **mutual rather than cause-and-effect** and not fully understood — possibly coincidental, or due to fragmentation or repeated hypoxia; confounders (obesity, hypertension, diabetes, cardiovascular disease) relate to both, and insulin resistance contributes to depression as it does in OSA.
- **Proposed mechanisms:** (i) fragmentation and hypoxia as a cause of depression; (ii) **serotonin** — abnormalities in central and peripheral serotonergic neurotransmission are a potential factor in depression, and serotonin also affects upper airway dilator motor neurons via the hypoglossal nucleus, an effect further reduced in sleep; (iii) increased pro-inflammatory cytokines (IL-6, TNF) responsible for daytime sleepiness.
- **CPAP's effect on depression is not conclusive** — evidence is limited; some studies report benefit for both sleepiness and depression, others no effect.

4.4 Medical comorbidities

- **Hypertension and cardiovascular disease.** SDB is an **independent** risk factor for hypertension, stroke, heart failure and arrhythmia, driven by repeated intermittent hypoxia and sympathetic activation. There is a **dose–response relationship**: greater severity → greater risk of incident hypertension and adverse events, independent of obesity. Linked to atrial fibrillation and ventricular tachycardia; worsens outcomes in existing heart disease. CPAP may improve blood pressure control, particularly in severe OSA or treatment-resistant hypertension.
- **Metabolic syndrome and diabetes.** Hypertension, diabetes and metabolic syndrome are independently associated with moderate–severe SDB, and coexistence with obesity further increases risk. Intermittent hypoxia and fragmentation contribute to insulin resistance, dyslipidaemia and systemic inflammation.
- **GORD.** Frequently coexist, each worsening the other; mechanisms include increased intrathoracic pressure swings and arousals promoting reflux. Dentally: enamel erosion is a screening sign, and advise no food or drink but water for ≥2 h before bed.

- **Depression, anxiety, neurocognitive decline.** Higher rates particularly in women and post-menopausal populations. Neurocognitive decline — executive function, attention, memory — in adults *and* children, **even at mild disease levels**.
- **Chronic kidney disease** — high prevalence in CKD; severe SDB increases CKD risk. **Periodontitis** — emerging bidirectional relationship with inflammatory markers implicated in both.

> **WHY THE RELATIONSHIP MATTERS**
>
> Increased morbidity and mortality; reduced quality of life; substantial healthcare burden; and — the point examiners want — **therapeutic opportunity**: treating SDB can positively affect the management of associated chronic disease, improving outcomes beyond sleep. That is the argument for treating a patient whose only complaint is snoring but whose blood pressure is poorly controlled.

4.5 Ethics and consent

Developed in Topics 9.4 and 10.1. In summary: **autonomy, non-maleficence, beneficence, justice** plus **veracity**; specific considerations are informed consent, scope of competence, collaboration and referral, the screening-versus-diagnosis boundary, patient-centred care, confidentiality and quality assurance. Consent must be obtained **in writing, before fitting**, recording potential and probable side effects and the appliance's expected longevity.

4.6 Multidisciplinary team

> **THE FOUR GENERAL PRINCIPLES — MEMORISE**
>
> 1. The **diagnosis of OSA should always be made by a physician**.
> 2. Appliances should be **fitted by a qualified dentist** trained and experienced in dental sleep medicine.
> 3. **Follow-up assessment by the referring physician**, with PSG or sleep studies, is required to verify treatment efficacy.
> 4. The dentist's role is to **screen and provide the appliance**, not to diagnose SDB.

- **Shared:** written professional communication between GP, sleep physician and dentist; a written report to the referring practitioner detailing dental treatment and symptomatic feedback; a systematic follow-up programme agreed between practitioners; onward referral to periodontics, orthodontics or prosthodontics for complications and coincidental conditions.
- **Physician:** medical assessment, often with overnight PSG, interpreted by a sleep physician. Unattended portable monitors only in conjunction with a comprehensive sleep evaluation, supervised by a physician trained in sleep medicine; a physician prescription is required to order one. **OSA diagnosis should always be determined by a physician who has also seen the patient clinically.** After deciding on appliance therapy, sends a written referral or prescription **and the diagnostic report**. Remains involved in ongoing evaluation.
- **Dentist:** screen using appropriate tools by recognising symptoms; refer patients with primary snoring or snoring with potential OSA to their physician; **provide appliance and behavioural therapy only after a written request or prescription — never start treatment without a physician's assessment**; assess oral health and the sleep-disruption consequences of concomitant bruxism, GORD, orofacial pain or temporal headache; manage SDB, sleep bruxism, dental consequences of GORD and orofacial pain, including sleep hygiene, weight control, exercise and CBT with a psychologist, and referral to orthodontics, OMFS or otolaryngology; propose various appliances and neurostimulation therapy; jointly monitor changes in SDB and mental and physical health; monitor efficacy and safety; manage side effects; apply, manage and follow up the appliance.
- **Borderline cases.** Primary snoring *without* sleepiness → still refer to exclude OSA before providing an appliance. Sleepiness with suspected OSA → refer. **Sleep bruxism with orofacial pain or TMD → may be diagnosed by the dentist** — *but* if accompanied by sleepiness, snoring or sleep apnoea, a sleep physician is required and the patient should consult a professional trained in those fields.
- **Portable monitor levels.** Level 3 = ≥4 channels including airflow (≥2 channels of respiratory movement, or movement and airflow), heart rate/ECG, oxygen saturation. Level 4 = 1–2 bioparameters, oxygen saturation or airflow. **Both should be interpreted by a physician.** The dentist assesses the result and observes change in respiratory sleep parameters with treatment; the dentist does **not** use these data for diagnosis or as a formal final follow-up assessment.
- **Staged treatment sequence.** (1) Complete board certification or an equivalent diploma; maintain a file of history and examination results (possibly including depression, mood and sleepiness assessment) plus pre-treatment panoramic and cephalometric radiographs and casts. (2) Obtain the sleep record or PSG analysis with the OSA diagnosis, the physician's recommendation and assessment of concomitant conditions. (3) Perform clinical assessment — general and oral health, prognosis of the soft and hard tissues to be affected — and assess recent radiographs. (4) Obtain **written consent including risks before fitting**. (5) Maintain regular written communication about plan, progress and follow-up. (6) After starting, collect information on symptom resolution to determine optimal titration; any objective portable monitor data are **for titration only**. (7) Monitor progress and obtain a follow-up efficacy report from the sleep physician; if unsatisfactory, extend titration or review the plan; primary snoring needs no objective follow-up. (8) Establish a follow-up protocol — annual assessment of efficacy, compliance, side effects and integrity of the oral structures; the appliance may need replacement depending on usage and hygiene. (9) Ensure attendance at control visits and keep physician–dentist correspondence on file; consult the medical team if additional titration is needed.

TRAP — THE COMMONEST WAY TO LOSE MARKS IN THIS EXAMINATION

Any answer implying the dentist **diagnoses** OSA, or that treatment may begin on the strength of the dentist's own screening, is wrong — however clinically sensible the rest. So is any answer having the dentist **certify treatment success** from their own portable monitor data. Correct formulation: "I screen, refer, select, fit, calibrate, monitor and manage side effects; the physician diagnoses and verifies efficacy; and where I use oximetry or a home test it is to identify the therapeutic position, for titration purposes only."

- **Severity and outcome in the shared-care setting.** Adult mild 5–14 · moderate 15–30 · severe >30. Children mild 1–5 · moderate 5–10 · severe >10. **Successful outcome** = reduction of AHI, respiratory disturbance index and respiratory event index to **<5** *and* a **>50%** reduction in these indices.

4.7 Sleep bruxism and SDB

DEFINITIONS — VERBATIM

- **Sleep bruxism** — a masticatory muscle activity during sleep, characterised as rhythmic (phasic — grinding) or non-rhythmic (tonic — clenching), and **not a movement disorder or a sleep disorder in otherwise healthy individuals**.
- **Corollary:** SB is **not a disorder** but a *behaviour* that may be a risk factor or a protective factor for clinical consequences. Classified by consequence as: not a risk or protective factor (harmless); a risk factor (≥1 negative outcome); a protective factor (≥1 positive outcome). Also primary or secondary.
- **TMD** — a musculoskeletal and neuromuscular condition involving clinical signs and symptoms of the muscles of mastication, the temporomandibular joint and related structures.
- **Assessment grading: possible** (self-report), **probable** (self-report + clinical examination), **definite** (self-report + examination + instrumental assessment, notably PSG with EMG). Standard cut-offs should not be applied in otherwise healthy individuals; assess along a continuum of the behaviour.

- **Prevalence:** reported range 6–91%; self-reported adults ~8%; children and adolescents 14–46%; 30–50% of adults with OSA also exhibit SB. TMD in OSA 2–52%; OSA in TMD ~28%.
- **Pathophysiology:** episodes occur more commonly in N2–N3 in a cyclical pattern, typically with micro-arousals; catecholaminergic mechanisms (described in children); genetics; swallowing (SB may increase salivary flow to neutralise reflux acid); stress (more clearly linked to *awake* bruxism); pain (paradoxically TMD may *reduce* bruxism); airway; autonomic cardiac activation.

Claim	Evidence Strength	Reasoning
SB and SDB frequently co-occur	Strong	Multiple studies show high comorbidity
SDB directly causes SB	Weak	Evidence is inconsistent and often lacking
Treating SDB can reduce SB (if temporally linked)	Moderate	Some case and intervention studies support this

Fig 4.1 — Evidence strength for three claims. "SB and SDB frequently co-occur" — **strong** (multiple studies show high comorbidity). "SDB directly causes SB" — **weak** (inconsistent and often lacking). "Treating SDB can reduce SB, if temporally linked" — **moderate** (some case and intervention studies). **Examinable because:** memorise these three gradings. They give a defensible, precisely calibrated answer where most candidates either overclaim causation or dismiss the association entirely.

- **Shared clinical features:** both commoner supine; both involve oropharyngeal and masticatory muscle activation; both relate to arousals (micro-arousals in SB, major arousals in OSA); both associated with GORD, TMD and headache; both may respond to CPAP (case reports of CPAP treating SB), UPPP (treating SB in children with enlarged tonsils) and mandibular advancement (decreasing SB short-term). Shared risk factors: age, obesity, smoking, alcohol.
- **The "reactive–protective" hypothesis.** Sequence: muscle tone falls → obstruction → more negative intraluminal pressure → bruxism triggered, increasing tone and restoring patency → less negative pressure. Supporting: muscle tone is inversely related to

obstruction; PSG shows a peak of respiratory effort with chin muscle activity, then masseter activity and EEG arousal; some OSA treatments improve SB and perhaps TMD; **occlusal splints (Michigan-type) aggravate SDB** with increased AHI and snoring.

- **Against it:** unknown whether this is a true reactive–protective response or a non-specific arousal response (the same arousal produces leg movement too). Meta-analyses show **no significant or consistent correlation** across populations and severities; gold-standard PSG finds no consistent temporal association; and **more SB events precede apnoea–hypopnoea events than follow them**, which argues against a protective role. Likely multifactorial via shared arousal mechanisms rather than causality, possibly stronger in body-position-related OSA. Yet a real subgroup effect exists: a subpopulation of young SB patients shows concomitant respiratory events; an OSA subtype had SB with significantly more respiratory-related arousals and desaturation; SB prevalence is higher in OSA; AHI and desaturation are more frequent in SB patients; and grinding frequency correlates with apnoea, micro-arousal and desaturation.

THE SENTENCE TO SAY

"Sleep bruxism and OSA frequently co-occur and share plausible mechanisms — chiefly sleep arousal and autonomic activation — but a direct causal relationship is **not established**. The dentist should **refrain from implying a causal relationship** when discussing these conditions with patients. Treating the OSA can reduce sleep bruxism in some patients, particularly where the bruxism is temporally linked to apnoeic events."

- **SB alone** — does **not** need a sleep study; does **not** need mandibular advancement; an **occlusal splint is first line**. Medication can reduce SB but risks addiction with limited choice; botulinum toxin helps in the acute stage and also treats TMD; otherwise avoid pharmacology. Also hypnotherapy, biofeedback, CBT. **Always ask about OSA signs and symptoms before fitting a splint** — if present, consider a MAD instead, because a stabilisation splint increases AHI and snoring.
- **SB + TMD** — assess risk factors including behavioural; treat the TMD (often self-limiting, non-surgical) and **avoid a MAD until the TMD is treated**. If high likelihood of OSA, refer to a sleep physician and consider a MAD thereafter; if no obvious OSA signs, do not refer for a sleep study and avoid a MAD.
- **TMD with a MAD. Not** a contraindication. Transient jaw discomfort does not necessarily indicate TMD; only ~2% cannot tolerate an appliance. No significant TMJ morphological alteration; condylar translation is *less* than in maximum opening; long-term use does not impair the joint — TMD symptom intensity, joint noise and jaw fatigue may reduce. MADs produce more pain-related TMD than CPAP but this is transient and non-serious. **Disc displacement is not a contraindication.** Onset of arthrogenous and myogenous pain is unrelated to pre-existing TMD, and pre-existing TMD does not generally worsen — **but moderate–severe TMD should be treated first to reduce drop-out**. Compression devices cause greater early masticatory myalgia and residual tongue pain than traction devices.
- **Managing TMD on OAT** — manage as a musculoskeletal disorder informed by microtrauma principles and formal diagnostic criteria. **Mandibular exercises allow patients with TMD to wear a MAD**, improving sleep and quality of life, reducing pain intensity and increasing compliance.

TRAP — THE OCCLUSAL SPLINT

A conventional flat-plane stabilisation (Michigan-type) splint can **worsen** SDB — AHI and snoring both increase — because increased vertical dimension and posterior mandibular positioning reduce airway space. Therefore **screen for OSA before fitting any splint**. If OSA is likely, a MAD may serve both purposes. One of the most practically important facts in the syllabus and a very likely question.

4.8 Pain and sleep

- **Bidirectional and cyclical.** Pain prevents onset and causes frequent awakenings; acute experimental painful stimuli induce arousal and wakefulness in healthy people; pain is a primary cause of insomnia, and a high proportion of chronic pain patients have insomnia; nociceptive stimuli produce transient EEG change, shifts to lighter stages and brief arousals.
- Conversely: **hyperalgesia** — short or disturbed sleep increases sensitivity to painful stimulation; spontaneous muscle pain and headache develop or worsen; pain tolerance falls with increased fatigue and more severe subjective pain; the antinociceptive potency of morphine falls.
- **Mechanisms:** dysregulated serotonin, noradrenaline and dopamine (regulating both pain and sleep); pro-inflammatory TNF-α and IL-6 in both pathways; HPA-axis dysfunction.
- **Implications:** chronic pain management must address sleep simultaneously, usually multidisciplinary; psychological input for the mood and anxiety components; improving sleep improves pain tolerance and quality of life.
- **Fibromyalgia** — chronic generalised musculoskeletal pain with multiple tender areas, fatigue and unrefreshing sleep, cognitive difficulties and psychological distress. Prevalence 1–2%, ~80% female; second most common rheumatic condition after osteoarthritis. **ACR 1990 criteria:** widespread musculoskeletal pain ≥3 months *and* tenderness at **≥11 of 18 sites** on 4 kg palpation, bilaterally at occiput (suboccipital muscle insertions), low cervical (anterior intertransverse spaces C5–C7), trapezius (midpoint upper border), supraspinatus (origins above the scapular spine near the medial border), second rib (second costochondral junctions), lateral epicondyle (2 cm distal), gluteal (upper outer quadrants), greater trochanter (posterior to the prominence) and knee (medial fat pad). May follow a viral illness, emotional distress, work-related soft tissue injury or whiplash. Other pain problems common — migraine, irritable bowel syndrome, atypical chest pain, irritable bladder and **TMJ pain**. Aggravated by unaccustomed exertion,

emotional distress, prolonged stationary posture, menses or menopause, and cold damp weather; improved in warm dry weather and by heat, massage, acupuncture, stretching and mild exercise.

- **Chronic fatigue syndrome** — chronic fatigue without defined medical or psychiatric disease, with joint and muscle pain, headache, poor concentration, impaired short-term memory, disturbed sleep, subjective feverishness and sore throat. Prevalence 0.4–1.5%. Both FM and CFS are complex chronic illnesses of unknown aetiology characterised by pain, fatigue and sleep disturbance with considerable overlap, both featuring the perception of poor-quality unrefreshing sleep. Diagnosis is by history and symptom report. There is **no rational established approach** to treating pain-related sleep disturbance, and it is not clear that pain is actually the *cause* of disrupted sleep in chronic pain patients.

TRAP — FIBROMYALGIA CRITERIA

The 11-of-18 tender point count is the **1990 ACR classification criteria**. The ACR published revised criteria in 2010 (2011 modification, 2016 revision) which **abandoned the tender point examination** for the Widespread Pain Index and Symptom Severity Scale. Quote the tender point count — it remains the classic examinable fact — but signal you know the criteria have since been revised.

NUMBERS — TOPIC 4

- SB: reported 6–91% · self-reported adults ~8% · children/adolescents 14–46% · 30–50% of OSA patients
- TMD in OSA 2–52% · OSA in TMD ~28% · ~2% cannot tolerate a MAD · depression in OSA ~21.8%
- Successful outcome: AHI/RDI/REI <5 **and** >50% reduction
- Fibromyalgia (ACR 1990): pain ≥3 months + ≥11 of 18 tender sites at 4 kg; prevalence 1–2%, 80% female. CFS 0.4–1.5%
- No food or drink but water for ≥2 h before bed
- SB/SDB evidence: co-occurrence **strong** · SDB causes SB **weak** · treating SDB reduces SB **moderate**

SELF-TEST — TOPIC 4

1. What imaging would you obtain, and what would you decline?

Panoramic and lateral cephalometric radiographs as a pre-treatment record, plus initial dental radiographs as indicated, intraoral and extraoral photographs, and casts or digital scans. **Decline CBCT and other CT** — unjustified radiation without clear diagnostic yield, and poor validity of static airway measurement: the airway is dynamic, inter-examiner reliability is poor, normative values have wide standard deviations, and the sites most often responsible for collapse are the least well visualised. Cephalometric evaluation is not always required but should be done by appropriately trained clinicians where needed.

2. Describe the shared care model, allocating responsibilities.

The physician diagnoses OSA — always — having seen the patient clinically, interprets the study, prescribes appliance therapy in writing with the diagnostic report, verifies efficacy with follow-up testing, and remains involved in ongoing evaluation. The dentist screens, refers, assesses dental suitability, treats oral disease first, selects and fits a custom titratable appliance, calibrates it, gives behavioural advice, monitors adherence and the oral structures, manages side effects, and refers on to periodontics, orthodontics or prosthodontics as needed. Both maintain regular written communication about plan, progress and follow-up under an agreed systematic programme. Any objective data the dentist obtains are **for titration purposes only**.

3. A patient requests a night guard for tooth wear. What must you establish first?

Screen for OSA before fitting any splint. A flat-plane Michigan-type stabilisation splint can **worsen** SDB, increasing AHI and snoring, because of increased vertical dimension and posterior mandibular positioning. If there are OSA signs or symptoms, refer for diagnosis and consider a mandibular advancement device, which can address both protection and airway. Note that sleep bruxism alone needs neither a sleep study nor mandibular advancement, and an occlusal splint is appropriate first-line therapy in that situation.

4. Is TMD a contraindication to appliance therapy?

Not absolutely — it is **relative**. Active severe TMJ disease or pain should be treated first, principally to reduce drop-out, but disc displacement alone is not a contraindication and pre-existing TMD does not generally worsen. Transient jaw discomfort is common and does not itself indicate TMD; only ~2% cannot tolerate an appliance. Long-term use does not impair the joint — condylar translation is less than in maximum opening, there is no significant morphological alteration, and symptom intensity, joint noise and jaw fatigue may reduce. Mandibular exercises allow patients with TMD to wear an appliance, improving sleep and quality of life, reducing pain and increasing compliance. Traction devices impose lower stress than compression devices.

TOPIC 4

History, examination and imaging for patient selection and treatment planning

EXAMINER'S BLUEPRINT

1. Dental sleep medicine history taking: reason for attendance, goal setting and managing expectations
2. Medical history: effects of age, obesity, drugs and medications, alcohol, smoking, other sleep disorders and comorbidities on the upper airway and overall morbidity
3. Relationship of SDB to other sleep-related problems (narcolepsy, RLS, PLMD/PLMS, insomnia, insufficient sleep, shift worker syndrome)
4. Relevance of medical comorbidities: hypertension, cardiovascular disease, metabolic syndrome, GORD, depression, anxiety, neurocognitive decline
5. Ethics and informed consent
6. Working as part of a multidisciplinary team; communication with the referring physician and interested parties
7. Sleep bruxism and its relationship with sleep-disordered breathing
8. Pain and sleep

This topic covers the consultation itself. Two sections carry disproportionate marks: **4.6**, because the multidisciplinary model defines your legal and professional boundaries, and **4.7**, because sleep bruxism is where dentists are most tempted to overclaim. The rest is systematic history taking, which examiners test by asking you to justify *why* you ask each question.

4.1 Dental sleep medicine history taking, goal setting and expectations

Reason for attendance

Patients present with snoring, suspected or diagnosed OSA, sleep bruxism, or orofacial pain related to sleep disorders. Ask about sleep-related symptoms, previous diagnoses, comorbidities and the impact of sleep problems on daily life and well-being. Screening for OSA and related conditions is now recommended as part of routine dental care given the high prevalence and health risks of undiagnosed disease.

Goal setting

Collaborative goal setting means discussing the patient's primary concern — reducing snoring, improving sleep quality, alleviating pain — and aligning it with evidence-based options, explaining the benefits and limitations of appliance therapy and other interventions in the light of individual anatomical, behavioural and medical factors. In older adults or those with comorbidity, the goal may reasonably be improved quality of life and minimised side effects rather than normalisation of the AHI. Critically, before treatment the dentist should work with the medical provider and patient to **determine the goal of treatment success and agree the metrics by which it will be judged**, including how efficacy will be verified.

Managing expectations

- Results vary, and some side effects — dental changes, discomfort — may occur.
- Ongoing follow-up and adjustment are usually necessary.
- Emphasise interdisciplinary care and, where required, referral to a sleep physician for comprehensive management.
- Explain what additional therapies might be considered if the response to an appliance is inadequate.

- Discuss sleep hygiene and duration — ambient lighting and temperature, electronics in bed, regular schedules. These may not directly affect appliance efficacy, but collectively they fragment sleep and exacerbate daytime sleepiness, **reducing the patient's perception of benefit**. A patient who sleeps five hours will not feel well however good the appliance.

The dental sleep medicine examination — the components list

Component	Detail
Medical–dental history	Age, weight and height; neck and belt circumference as indicated; presence of tonsils; tongue size; palate width and depth; malocclusion classification including overjet and overbite **measurements**; presence of tooth erosion or tooth wear; medications; other disease. Family history of sleep disorders. Chief complaint and history of current sleep problems including snoring, witnessed apnoeas, nocturia, morning headaches and hypersomnolence
Sleepiness assessment	Validated instruments, e.g. Epworth Sleepiness Scale, STOP-BANG. Record baseline physical values (BMI, blood pressure, neck circumference)
Soft tissue / intraoral	Modified Mallampati; tonsil grade; tongue size and scalloping; palatal vault; posterior pharyngeal crowding; signs of mouth breathing; signs of GORD such as enamel erosion; nasal patency
Periodontal evaluation	Attachment loss, mobility, prognosis of the teeth that will bear the appliance
Temporomandibular joint	History and clinical examination: symptoms, sounds, range of movement, path of opening and any deflection or deviation, palpation of the muscles of mastication
Bruxism, orofacial pain, headaches	Assess concomitant sleep bruxism (grinding and/or clenching), orofacial pain and headache
Occlusion	Examination and recording of the existing condition — this is your medicolegal baseline (Topic 10)
Teeth and restorations	Number of functional teeth, mobility, spacing, curve of Spee, heavily restored teeth, anticipated restorative needs
Imaging	Initial dental radiographs as indicated. **Cephalometric and panoramic radiographs as a pre-treatment record.** Cone-beam CT and other CT are **not indicated** at this stage
Diagnostic models	Study casts (or digital scans) retained as part of the record
Photographs	Intraoral and extraoral photographs as a record of the pre-treatment dental condition

EXAM TRAP — IMAGING

The default imaging for a dental sleep medicine work-up is **panoramic and lateral cephalometric** radiography as a pre-treatment record. **CBCT is not routinely indicated.** Candidates who advocate routine CBCT airway analysis lose marks on two grounds: unjustified radiation dose, and the poor validity of static airway measurement (see Topic 8 — the airway is dynamic, inter-examiner reliability is poor, normative values have wide standard deviations, and the commonest sites of collapse are precisely those least well visualised). Cephalometric evaluation is *not always required* for a patient who will use an oral appliance, but where deemed necessary it should be performed by appropriately trained professionals.

Treatment planning for OSA — what to cover

- Provide explanation and written information on sleep behavioural therapy: sleep hygiene, sleeping position, weight control, alcohol intake.
- Explain the type of device being considered **and all alternatives** — CPAP, surgery, positional therapy.
- Propose various appliances according to the patient's oral health status and craniofacial morphology (mandibular advancement splint, tongue retaining device or equivalent).
- Provide positional therapy information for patients with positional OSA.

4.2 Medical history: modifiers of the upper airway and overall morbidity

Factors Affecting Upper Airway and Morbidity

Factor	Effect on Upper Airway	Impact on Morbidity
Age (children/elderly)	Higher risk of airway complications	Increased perioperative risk
Obesity	Airway collapse, muscle dysfunction	Higher respiratory failure
Drugs/Medications	Impaired airway muscle function	Increased respiratory events
Smoking	Chronic inflammation, cancer risk	Higher infection/mortality
Alcohol	Reduced muscle tone	Increased airway obstruction
Sleep Disorders/Comorbidities	Worsened airway function	Complicated management

Figure 4.1 — Factors affecting the upper airway and morbidity, tabulating each factor against its effect on the upper airway and its impact on morbidity: age in children and the elderly (higher risk of airway complications → increased perioperative risk), obesity (airway collapse and muscle dysfunction → higher respiratory failure risk), drugs and medications (impaired airway muscle function → increased respiratory events), smoking (chronic inflammation, cancer risk → higher infection and mortality), alcohol (reduced muscle tone → increased airway obstruction), and sleep disorders and comorbidities (worsened airway function → complicated management).

Why it is examinable: this three-column structure is exactly how to answer 4.2 in a written paper — name the factor, state the airway mechanism, then state the morbidity consequence. It converts a list into an argument.

Factor	Effect on the airway and on morbidity
Age	Children and infants are at higher risk of airway complications because of anatomical and physiological differences; preoperative assessment and history are crucial for predicting difficult airway management and preventing perioperative respiratory events. Advanced age in adults is associated with increased airway-related morbidity, particularly in combination with comorbidity and polypharmacy
Obesity	Increases upper airway collapse and respiratory muscle dysfunction, contributing to OSA and postoperative respiratory failure; requires tailored perioperative management
Drugs and medications	Anaesthetics, sedatives, opioids and neuromuscular blocking agents impair upper airway dilator function, increasing the risk of obstruction and respiratory failure — particularly perioperatively. Note also drug classes that affect sleep architecture and may induce bruxism or RLS (Topic 1.12)
Smoking	A major risk factor for upper airway pathology, chronic inflammation and increased susceptibility to infection and cancer, with higher morbidity and mortality. Passive smoking and family smoking history increase the risk of perioperative respiratory adverse events in children
Alcohol	

Factor	Effect on the airway and on morbidity
	Impairs airway muscle tone and increases the risk of obstruction, especially combined with sedatives. Even moderate intake may cause transient desaturation despite appliance use — advise avoidance in the hours before bed
Other sleep disorders and comorbidities	Coexisting upper airway disease (chronic rhinosinusitis, allergic rhinitis) and comorbidities such as asthma or COPD worsen airway function and complicate management. Recent upper respiratory tract infection significantly raises perioperative airway risk, especially in children

4.3 SDB and other sleep-related problems

Coexisting condition	Interaction with SDB
Insomnia — COMISA	Insomnia is highly prevalent in SDB, giving rise to the term **comorbid insomnia and sleep apnoea**. The overlap produces more severe sleep fragmentation, increased daytime sleepiness and poorer quality of life. **Both conditions must be treated** — reducing the AHI alone will not help the patient. CBT-I plus SDB treatment is the correct answer
Restless legs syndrome and PLMD	Frequently co-occur with SDB. SDB can exacerbate limb movements and vice versa, causing further sleep disruption and increased arousals; both contribute to excessive daytime sleepiness and impaired alertness. Note that some antidepressants induce or worsen RLS and periodic limb movements
Narcolepsy	SDB may coexist with narcolepsy, compounding sleepiness and cognitive impairment. Symptom overlap makes clinical differentiation unreliable, so careful sleep studies are needed. Remember that SOREMPs on the MSLT — a narcolepsy criterion — are also frequent in untreated OSA
Insufficient sleep and shift work disorder	SDB worsens the effects of both by further disrupting sleep architecture, increasing cognitive and cardiovascular risk. Shift workers with SDB are at higher risk of excessive sleepiness and accidents. A patient who is simply not spending enough time in bed will not respond to appliance therapy as expected
Mood and psychiatric disorders	SDB is associated with mood disorders, depression and anxiety, which may themselves present with insomnia or hypersomnia, creating a cycle of worsening sleep and mental health

OSA and depression — a nuanced relationship worth knowing in detail

- Symptoms of OSA and depression **overlap** — tiredness, fatigue, sleepiness, poor motivation and poor concentration — complicating the diagnosis and treatment of each. OSA may be responsible for antidepressant treatment failure, and depression is a risk factor for an incomplete response to CPAP, so it may need to be independently addressed.
- Psychiatric diagnoses in OSA patients include depression, anxiety, post-traumatic stress disorder, psychotic disorders and bipolar disorder. Prevalence of depression is higher in OSA (~21.8%). Clinicians should evaluate OSA patients for depression initially by screening tool, and use established criteria to confirm the diagnosis if positive.
- The relationship appears **mutual rather than cause-and-effect**. The pathophysiological relationship is not fully understood: it may be coincidental, or due to sleep fragmentation or repeated hypoxia, and confounders such as obesity, hypertension, diabetes and cardiovascular disease relate to both — insulin resistance, for example, contributes to depression as it does in OSA.
- Proposed mechanisms: (i) sleep fragmentation and hypoxia of OSA as a cause of depression; (ii) the role of **serotonin** in both the neurobiology of depression and upper airway control — abnormalities in central and peripheral serotonergic neurotransmission are a potential factor in depression, and serotonin also affects upper airway dilator motor neurons via the hypoglossal nucleus, an effect further reduced in the sleep state; (iii) increased pro-inflammatory substances and cytokines such as IL-6 and TNF, which are responsible for daytime sleepiness.
- The effect of CPAP on depression is **not conclusive** — current evidence that treating OSA improves depression is limited; some studies report a positive impact on both sleepiness and depression, others no effect.

4.4 Relevance of medical comorbidities

Comorbidity	Relevance
Hypertension and cardiovascular disease	SDB is an independent risk factor for hypertension, stroke, heart failure, arrhythmia and other cardiovascular disorders. Repeated intermittent hypoxia and sympathetic activation drive development and progression. There is a **dose–response relationship**: greater severity confers greater risk of incident hypertension and adverse cardiovascular events, independent of confounders including obesity. SDB is linked to atrial fibrillation and ventricular tachycardia and worsens outcomes in pre-existing heart disease. Practically: OSA is associated with hypertension and CPAP may improve blood pressure control, particularly where the OSA is severe or the hypertension treatment-resistant
Metabolic syndrome and diabetes	SDB is highly prevalent in obesity and metabolic comorbidity; hypertension, diabetes and metabolic syndrome are independently associated with moderate-to-severe SDB, and coexistence with obesity further increases risk. Intermittent hypoxia and sleep fragmentation contribute to insulin resistance, dyslipidaemia and systemic inflammation
GORD	SDB and GORD frequently coexist, each potentially worsening the other. Mechanisms include increased intrathoracic pressure swings and arousals during apnoeic events promoting reflux. Dentally relevant: enamel erosion is a screening sign, and appliance-related advice includes avoiding food and drink other than water for at least two hours before bed
Depression, anxiety, neurocognitive decline	Higher rates of depression and anxiety, particularly in women and post-menopausal populations. Neurocognitive decline — deficits in executive function, attention and memory — is observed in both adults and children with SDB, **even at mild levels of disease**. These comorbidities further impair quality of life and may complicate management
Chronic kidney disease	High prevalence of SDB in CKD; severe SDB increases the risk of CKD
Periodontitis	Emerging evidence of a bidirectional relationship, with inflammatory markers implicated in both — relevant to the dentist's own domain

HIGH-YIELD — WHY THE RELATIONSHIP MATTERS

Increased morbidity and mortality; reduced quality of life; substantial healthcare burden; and — the point examiners want — **therapeutic opportunity**: treating SDB can positively affect the management of associated chronic disease, improving outcomes beyond sleep itself. That is the argument for treating a patient whose only complaint is snoring but whose blood pressure is poorly controlled.

4.5 Ethics and informed consent

The substance of this section is developed in Topics 9.4 and 10.1, which should be read together with it. In summary, the four classical principles apply — **autonomy, non-maleficence, beneficence and justice** — with **veracity** as a fifth, and the specific dental sleep medicine considerations are informed consent, scope of competence, collaboration and referral, the limits of the dentist's role in screening versus diagnosis, patient-centred care, confidentiality and quality assurance. Informed consent must be obtained **in writing, before fitting the appliance**, and must record the potential and probable side effects and the appliance's expected longevity.

4.6 Working in a multidisciplinary team

THE FOUR GENERAL PRINCIPLES — COMMIT THESE TO MEMORY

1. The **diagnosis of OSA should always be made by a physician**.
2. Oral appliances should be **fitted by a qualified dentist** who is trained and experienced in dental sleep medicine.
3. **Follow-up assessment by the referring physician**, with polysomnography or sleep studies, is required to verify treatment efficacy.
4. The dentist's role is to **screen and provide the appliance**, not to diagnose SDB.

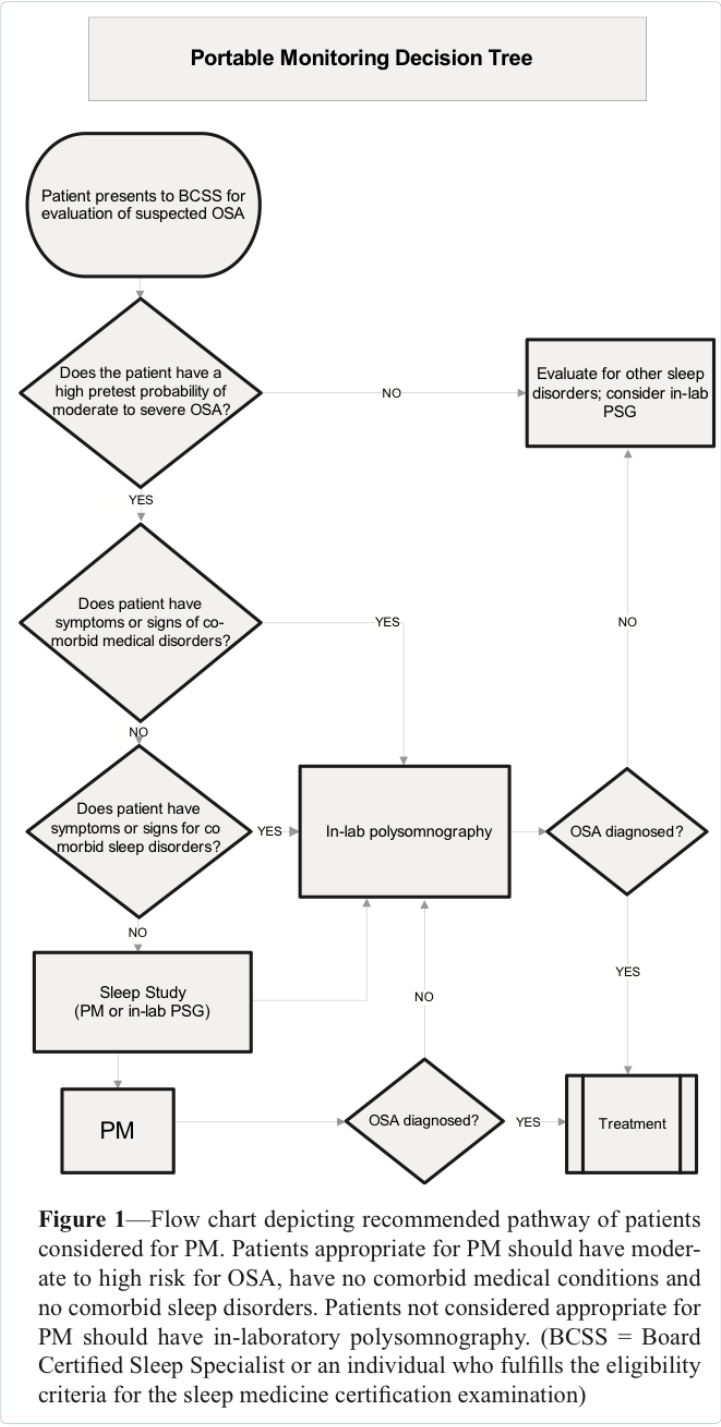


Figure 1—Flow chart depicting recommended pathway of patients considered for PM. Patients appropriate for PM should have moderate to high risk for OSA, have no comorbid medical conditions and no comorbid sleep disorders. Patients not considered appropriate for PM should have in-laboratory polysomnography. (BCSS = Board Certified Sleep Specialist or an individual who fulfills the eligibility criteria for the sleep medicine certification examination)

Figure 4.2 — Portable monitoring decision tree. Flow chart depicting the recommended pathway for patients considered for portable monitoring: patients appropriate for PM should be at moderate-to-high risk for OSA, have no comorbid medical conditions and no comorbid sleep disorders. Patients not considered appropriate for PM should have in-laboratory polysomnography, and PM should be conducted under the supervision of a board-certified sleep specialist or an individual who fulfils the eligibility criteria for the sleep medicine certification examination.

*Why it is examinable: this figure defines the boundary of the dentist's involvement with objective testing. Where a dentist may lawfully order or administer a home sleep apnoea test (jurisdiction-dependent), the resulting data are used **for titration purposes only** — never for diagnosis or formal follow-up assessment. See Topic 9.1 for the full portable monitoring guideline.*

The shared care model

Party	Responsibilities
Shared	Written professional communication about each patient between the general practitioner, sleep physician and dentist; a written report to the referring practitioner detailing dental treatment and the patient's symptomatic feedback; a systematic follow-up programme agreed between practitioners; appropriate

Party	Responsibilities
	onward referral to periodontics, orthodontics or prosthodontics for complications and coincidental conditions
Physician	Medical assessment, often with overnight polysomnography, interpreted by a sleep physician. Unattended portable monitor devices should be used only in conjunction with a comprehensive sleep evaluation and supervised by a physician trained in sleep medicine; a physician prescription is required to order a portable monitor. **OSA diagnosis should always be determined by a physician who has also seen the patient clinically.** After deciding that appliance therapy is the appropriate option, the physician sends a written referral or prescription **and the diagnostic report** to a dentist trained in the field. The referring physician should remain involved in the ongoing evaluation of treatment
Dentist	Screen for potential OSA using appropriate tools by recognising symptoms (snoring with other symptoms such as sleepiness, choking, witnessed apnoeas). Refer patients with either primary snoring or snoring with potential OSA to their physician for review of the overall medical history and to exclude OSA. **Provide appliance and behavioural therapy after receiving a written request or prescription from a physician; never start treatment for snoring or OSA without a physician's assessment** — because OSA carries increased mortality risk, appliances should be fitted by a qualified dentist. Assess oral health and the sleep-disruption consequences of concomitant bruxism, GORD, orofacial pain or temporal headache. Manage SDB, sleep bruxism, dental consequences of GORD and orofacial pain, including sleep hygiene, weight control, exercise programmes and CBT with a psychologist, and referral to orthodontics, oral and maxillofacial surgery or otolaryngology. Propose various appliances and neurostimulation therapy. Jointly monitor changes in SDB and in mental and physical health with the sleep physician. Monitor treatment efficacy and safety. Manage the side effects of appliance therapy. Apply, manage and follow up the use of oral appliances

Where diagnosis lies — the borderline cases

- Patient presents with **primary snoring without sleepiness** → still send to a sleep physician to exclude OSA before providing appliance therapy.
- Patient has **daytime sleepiness and suspected OSA** → refer.
- Patient has **sleep bruxism and orofacial pain or TMD** → these may be diagnosed by the dentist. **But** if accompanied by sleepiness, snoring or sleep apnoea, a sleep physician is required for diagnosis, and the patient should be advised to consult a professional trained in those fields.
- Level 3 and 4 portable monitor results should be **interpreted by a physician**. The dentist mainly assesses the result and observes change in respiratory sleep parameters with treatment; the dentist does not use these data for diagnosis or as a formal final follow-up assessment. (Level 3: four or more channels including airflow — at least two channels of respiratory movement, or respiratory movement and airflow — heart rate or ECG and oxygen saturation. Level 4: one or two bioparameters, oxygen saturation or airflow.)

Code of practice and the staged treatment sequence

1. The dentist should complete board certification requirements for dental sleep medicine or obtain an equivalent diploma, and maintain a file containing patient history and examination results (which may include assessment of depression, mood and sleepiness), plus pre-treatment panoramic and cephalometric radiographs and dental casts.
2. Obtain a copy of the sleep record or polysomnography analysis, together with the OSA diagnosis, the physician's recommendation and an assessment of concomitant conditions from the sleep physician.
3. Perform clinical assessment — general and oral health, and the prognosis of the soft and hard tissues that will be affected by appliance therapy — and assess recent radiographs.
4. Obtain **written consent, including potential risks, before fitting** the appliance.
5. Maintain regular written communication with the physician about the treatment plan, progress and follow-up.
6. After starting therapy, collect information on the resolution of the patient's symptoms to determine optimal titration. If objective portable monitor data can be obtained, use them **for titration purposes only**.
7. Monitor symptom progress and obtain a follow-up report on treatment efficacy from the sleep physician. If unsatisfactory, extend titration or review the treatment plan with the patient. Primary snoring requires no objective follow-up assessment.

8. Establish a follow-up protocol after physician assessment: annual follow-up assessing efficacy, compliance, side effects and the integrity of the oral structures; the appliance may need replacement depending on usage and hygiene.
9. Ensure the patient attends control visits, and keep the communication between physician and dentist on file. If an additional titration period is needed, consult the medical team.

EXAM TRAP — THE SINGLE COMMONEST WAY TO LOSE MARKS IN THIS EXAMINATION

Any answer that implies the dentist **diagnoses** OSA, or that treatment may begin on the strength of the dentist's own screening, is wrong — regardless of how clinically sensible the rest of the answer is. Equally, any answer that has the dentist **certifying treatment success** from their own portable monitor data is wrong. The correct formulation is: "I screen, refer, select, fit, calibrate, monitor and manage side effects; the physician diagnoses and verifies efficacy; and where I use oximetry or a home test it is to identify the therapeutic position, for titration purposes only."

The dentist's screening role, in the words of the guidance

Dentists screen for potential patients with SDB or OSA using questionnaires (STOP-BANG, Epworth Sleepiness Scale, Friedman Tongue Classification, Kushida Index, Berlin Questionnaire), evaluation of neck size and BMI, and the modified Mallampati classification to describe the patency of the oral airway; then refer patients suspected of SDB or OSA to a sleep physician or otolaryngologist; and then apply, manage and follow up the appliance.

Severity and outcome definitions used in the shared care setting

	Adult	Children (lower thresholds)
Mild	AHI 5–14	AHI 1–5
Moderate	AHI 15–30	AHI 5–10
Severe	AHI >30	AHI >10
Successful treatment outcome: reduction in the AHI, respiratory disturbance index and respiratory event index to **less than 5** *and* a **greater than 50%** reduction in these indices		

4.7 Sleep bruxism and its relationship with sleep-disordered breathing

DEFINITIONS TO KNOW VERBATIM

Sleep bruxism — a masticatory muscle activity during sleep, characterised as rhythmic (phasic — grinding) or non-rhythmic (tonic — clenching), and **not a movement disorder or a sleep disorder in otherwise healthy individuals**.

The critical corollary: sleep bruxism is **not a disorder** but a *behaviour* that may be a risk factor or a protective factor for certain clinical consequences. In terms of clinical consequences it may be classified as: not a risk or protective factor (a harmless behaviour); a risk factor (related to one or more negative health outcomes); or a protective factor (related to one or more positive health outcomes). It is also classified as primary or secondary.

Temporomandibular disorder (TMD) — a musculoskeletal and neuromuscular condition involving clinical signs and symptoms of the muscles of mastication, the temporomandibular joint and related structures.

Assessment grading (international consensus): possible (self-report), probable (self-report plus clinical examination), and definite (self-report, clinical examination and instrumental assessment, notably polysomnography with EMG). Standard cut-off points should not be applied in otherwise healthy individuals; bruxism-related masticatory muscle activity should be assessed along a continuum of the behaviour.

Prevalence

- Wide reported range: 6–91%, depending on assessment method.

- Self-reported sleep bruxism: ~8% of the adult population.
- Children and adolescents: 14–46%.
- Up to 30–50% of adults with OSA also exhibit sleep bruxism.
- Prevalence of TMD in OSA populations: 2–52%. Prevalence of OSA in TMD populations: ~28%.

Pathophysiology of sleep bruxism

- **Macro-sleep structure:** episodes occur more commonly in stages N2 to N3, in a cyclical pattern, and are typically associated with micro-arousals
- **Neurochemistry:** catecholaminergic mechanisms, particularly described in children
- **Genetics**
- **Swallowing:** bruxism may increase salivary flow to neutralise acid from gastro-oesophageal reflux
- **Stress:** more clearly linked to *awake* bruxism and anxiety
- **Pain:** paradoxically, TMD may *reduce* bruxism
- **Airway**
- **Autonomic cardiac activation**

The relationship with SDB — how to phrase it

Claim	Evidence Strength	Reasoning
SB and SDB frequently co-occur	Strong	Multiple studies show high comorbidity
SDB directly causes SB	Weak	Evidence is inconsistent and often lacking
Treating SDB can reduce SB (if temporally linked)	Moderate	Some case and intervention studies support this

Figure 4.3 — Evidence strength for three claims about sleep bruxism and SDB. "SB and SDB frequently co-occur" — **strong**, because multiple studies show high comorbidity. "SDB directly causes SB" — **weak**, because the evidence is inconsistent and often lacking. "Treating SDB can reduce SB (if temporally linked)" — **moderate**, supported by some case and intervention studies.

Why it is examinable: memorise these three gradings. They give you a defensible, precisely calibrated answer to a question where most candidates either overclaim causation or dismiss the association altogether.

Common clinical features shared by SB and SDB: both are commoner in supine sleep; both involve oropharyngeal and masticatory muscle activation; both relate to sleep arousals (micro-arousals in SB, major arousals in OSA); both are associated with GORD, TMD and headache; both may respond to CPAP (case reports of CPAP treating SB), to UPPP (which can treat SB in children with enlarged tonsils), and to mandibular advancement (which decreases SB in the short term). Shared risk factors include age, obesity, smoking and alcohol use.

The "reactive–protective" hypothesis, and its limits

The proposed sequence is: muscle tone falls during sleep, leading to obstruction → more negative intraluminal pressure → bruxism is triggered, increasing muscle tone and restoring airway patency → less negative pressure. Supporting observations: muscle tone is inversely related to upper airway obstruction; on polysomnography a sudden peak of respiratory effort can be seen together with chin muscle activity, followed by masseter activity and EEG evidence of

arousal; some OSA treatments have a positive effect on SB and perhaps on TMD; and occlusal splints (e.g. Michigan-type) *aggravate* SDB, with increased AHI and snoring.

Against it: it is not known whether this is a true reactive–protective response or simply a non-specific arousal response — the same arousal produces other motor activity such as leg movement. Meta-analyses and systematic reviews indicate that, overall, **no significant or consistent correlation** exists between SB and OSA across all populations and severities. Gold-standard polysomnographic studies have found no consistent temporal association, and more SB events have been shown to *precede* apnoea–hypopnoea events than follow them — which, if SB occurred after the event, would argue against a protective role. The relationship is likely multifactorial, involving shared arousal mechanisms rather than causality, and may be stronger in certain phenotypes such as body-position-related OSA. Other findings do suggest a real subgroup effect: a subpopulation of young SB patients shows concomitant respiratory events; a phenotypic subtype of OSA patients had SB with significantly more respiratory-related arousals and desaturation; SB has a higher prevalence in OSA patients; AHI and desaturation events are more frequent in SB patients; and grinding frequency correlates with apnoea, micro-arousal and desaturation.

> **HIGH-YIELD — THE SENTENCE TO SAY**
>
> "Sleep bruxism and OSA frequently co-occur and share plausible mechanisms — chiefly sleep arousal and autonomic activation — but a direct causal relationship is **not established**. The dentist should **refrain from implying a causal relationship** when discussing these conditions with patients. Treating the OSA can reduce sleep bruxism in some patients, particularly where the bruxism is temporally linked to apnoeic events."

Clinical decision-making

Presentation	Management
Sleep bruxism alone	Does **not** need a sleep study. Does **not** need mandibular advancement. An occlusal splint should be first-line therapy. Medication can reduce SB but carries addiction risk with limited choice; botulinum toxin can help in the acute stage and also to treat TMD; otherwise stay away from pharmacology. Other options: hypnotherapy, biofeedback, CBT. **Always ask about signs and symptoms of OSA before fitting a splint** — if present, consider a mandibular advancement device instead of a standard stabilisation splint, because a stabilisation splint increases AHI and snoring
SB + TMD	Assess for possible risk factors, including behavioural ones. Treat the TMD — it is often self-limiting; use non-surgical treatment; avoid a mandibular advancement device and treat the TMD first. If there is a high likelihood of OSA, refer to a sleep physician and consider a MAD thereafter. If there are no obvious signs or symptoms of OSA, do not refer for a sleep study and avoid a MAD
TMD and appliance therapy	TMD is **not** a contraindication to a MAD. Transient jaw discomfort does not necessarily indicate TMD; only around 2% cannot tolerate the appliance. Mandibular posture with a MAD produces no significant TMJ morphological alteration, and condylar translation with a MAD is less than that in the maximum open position. Long-term MAD use does not impair the joint — TMD symptom intensity may reduce, with reduction of joint noise and jaw fatigue, and without jaw stiffness or pain. MADs do produce more pain-related TMD than CPAP, but this is transient and non-serious. **TMJ disc displacement is not a contraindication.** The onset of arthrogenous and myogenous pain is not related to pre-existing TMD, and patients with pre-existing TMD do not generally experience exacerbation — **but moderate-to-severe TMD should be treated before starting appliance therapy to reduce the drop-out rate**. Compression-type devices produce greater early masticatory myalgia and residual tongue pain than traction devices, which impose lower stress
Managing TMD in a patient on OAT	Manage as a musculoskeletal disorder, informed by microtrauma principles and formal diagnostic criteria. **Mandibular exercises allow patients with TMD to wear a MAD** — improving sleep and quality of life, reducing pain intensity and increasing compliance

EXAM TRAP — THE OCCLUSAL SPLINT

A conventional flat-plane stabilisation (Michigan-type) splint can **worsen** sleep-disordered breathing: AHI and snoring both increase, because the increase in vertical dimension and posterior mandibular positioning reduce airway space. Therefore, before fitting any splint for bruxism, screen for OSA. If OSA is likely, a mandibular advancement device may serve both purposes. This is one of the most practically important pieces of knowledge in the whole syllabus and a very likely examination question.

4.8 Pain and sleep

Pain and sleep have a **bidirectional, cyclical** relationship. Pain disrupts sleep; poor sleep worsens pain; and a bad night increases pain sensitivity, leading to more pain and further sleep disturbance.

How pain affects sleep	How sleep affects pain
• Prevents sleep onset and causes frequent awakenings • Acute experimental painful stimuli induce arousal and wakefulness in healthy individuals • Pain is a primary cause of insomnia; a high proportion of chronic pain patients have insomnia or other sleep difficulties • Nociceptive stimuli produce transient EEG changes, shifts to lighter sleep stages and brief arousals	• **Hyperalgesia** — short or disturbed sleep increases sensitivity to painful stimulation • Spontaneous pain — sleep deficiency can cause or worsen muscle pain and headache • Reduced pain tolerance, with increased fatigue and more severe subjective pain • Reduced antinociceptive potency of morphine

Underlying mechanisms: dysregulated neurotransmitters — serotonin, noradrenaline and dopamine — involved in both pain and sleep regulation; pro-inflammatory factors such as TNF-α and IL-6 involved in both pathways; and dysfunction of the hypothalamic–pituitary–adrenal axis, the central stress system, in individuals with chronic pain and sleep disorders.

Clinical implications: treatment of chronic pain should simultaneously address sleep, usually requiring a multidisciplinary approach; psychological input is valuable for the mood and anxiety components; and improving sleep quality improves pain tolerance and overall quality of life.

Fibromyalgia and chronic fatigue syndrome

	Fibromyalgia (FM)	Chronic fatigue syndrome (CFS)
Definition	Chronic generalised musculoskeletal pain with multiple areas of tenderness in specific anatomical regions, fatigue and unrefreshing sleep, cognitive difficulties and psychological distress	Chronic illness with fatigue in the absence of a defined medical or psychiatric disease; disabling fatigue accompanied by joint and muscle pain, headache, poor concentration, impaired short-term memory, disturbed sleep, recurrent subjective feverishness and sore throat
Prevalence	1–2%, of whom ~80% are female; the second most common rheumatic condition after osteoarthritis	0.4–1.5% in the US and Japan
Classification criteria (ACR 1990)	(i) A history of widespread musculoskeletal pain for at least three months; and (ii) tenderness in at least **11 of 18** anatomical sites on 4 kg palpation pressure, at the following bilateral regions: **occiput** (suboccipital muscle insertions), **low cervical** (anterior aspects of the intertransverse spaces at C5–C7), **trapezius** (midpoint of the upper border), **supraspinatus** (origins above the scapular spine near the medial border), **second rib** (second costochondral junctions, just lateral to the junctions on the upper surfaces), **lateral epicondyle** (2 cm distal to the epicondyles), **gluteal** (upper outer quadrants of the buttocks in the anterior fold of muscle), **greater trochanter** (posterior to the trochanteric prominence) and **knee** (medial fat pad proximal to the joint line)	

	Fibromyalgia (FM)	Chronic fatigue syndrome (CFS)
Aetiology	May appear for no apparent reason, or follow a viral illness, an emotionally distressing event, a work-related soft tissue injury, or a whiplash neck injury following a road traffic accident	
Other features	Patients often describe other pain problems — migraine or chronic headache, irritable bowel syndrome, atypical chest pain, irritable bladder syndrome and **temporomandibular joint pain**. Pain is usually generalised without specific boundaries, and varies in distribution, quality and severity. Aggravated by unaccustomed physical exertion or exercise, emotional distress and prolonged stationary posture; worse with menses or emerging at the menopause; worse in cold, damp, humid weather and in autumn and winter. Improved in warm, dry weather and in spring and summer. Temporary relief from heat, massage, acupuncture, stretching and mild exercise. May occur in the context of other potentially disabling conditions — rheumatoid arthritis, systemic lupus erythematosus, HIV or Lyme disease	
Shared features and diagnosis	Both are complex chronic illnesses of unknown aetiology characterised primarily by pain, fatigue and sleep disturbance, with considerable clinical overlap; both feature the perception of poor-quality, unrefreshing sleep. Diagnosis is on the basis of history and symptom report. Patients with severe fatigue, sleep disturbance and pain should have a thorough history including medical and psychosocial circumstances and changes in sleep patterns around symptom onset, plus a careful history of prior illnesses, psychiatric disorders, alcohol or substance abuse, allergies, current prescription and over-the-counter medications and herbal supplements. There is no rational established approach to the treatment of pain-related sleep disturbance, and it is not clear that pain is actually the *cause* of disrupted sleep in chronic pain patients	

EXAM TRAP — FIBROMYALGIA CRITERIA

The 11-of-18 tender point count is the **1990 ACR classification criteria**. The ACR published revised criteria in 2010 (with a 2011 modification and a 2016 revision) which **abandoned the tender point examination** in favour of the Widespread Pain Index and Symptom Severity Scale. Your source notes use the 1990 criteria. Quote the tender point count if asked — it remains the classic examinable fact and is what most syllabi expect — but signal that you know the criteria have since been revised. That single sentence turns a correct answer into an impressive one.

NUMBERS TO MEMORISE — TOPIC 4

- SB prevalence: reported range 6–91%; self-reported adult ~8%; children and adolescents 14–46%; 30–50% of OSA patients
- TMD in OSA populations 2–52%; OSA in TMD populations ~28%; ~2% of patients cannot tolerate a MAD because of jaw discomfort
- Depression prevalence in OSA ~21.8%
- Successful treatment outcome: AHI/RDI/REI <5 **and** >50% reduction
- Fibromyalgia (ACR 1990): widespread pain ≥3 months + tenderness at ≥11 of 18 sites at 4 kg pressure; prevalence 1–2%, 80% female
- CFS prevalence 0.4–1.5%
- Advise no food or drink other than water for ≥2 hours before bed (reflux risk)
- SB/SDB evidence gradings: co-occurrence **strong** · SDB causes SB **weak** · treating SDB reduces SB **moderate**

SELF-TEST — TOPIC 4

1. List the imaging you would obtain before commencing oral appliance therapy, and justify what you would *not* obtain.

> Panoramic and lateral cephalometric radiographs as a pre-treatment record, plus initial dental radiographs as indicated, together with intraoral and extraoral photographs and study casts or digital scans. **CBCT and other CT are not indicated at this stage.** The justification for declining CBCT is twofold: unjustified radiation exposure without a clear diagnostic yield, and the poor validity of static airway measurement — the airway is dynamic, inter-examiner reliability is poor, normative values have wide standard deviations, and the sites most often responsible for collapse are the least well visualised. Cephalometric evaluation is not always required, but where it is deemed necessary it should be performed by appropriately trained clinicians.

2. A patient is referred by their GP with loud snoring and an ESS of 14, requesting an appliance. There is no sleep study. What do you do?

> I do not commence treatment. Snoring cannot be distinguished from OSA clinically, and treatment must not begin in the absence of a diagnosis made by a physician. I would perform my screening assessment — history, ESS, STOP-BANG, BMI, blood pressure, neck circumference, and the dental and craniofacial examination — and refer to a sleep physician for clinical review and diagnostic testing, sending my findings with the referral. I would also assess and document dental suitability so that treatment can start promptly once a diagnosis and prescription are received. Where local law permits a dentist to order a home sleep apnoea test, I could do so, but the patient must still be referred to a medical provider for the diagnosis.

3. Describe the shared care model, allocating each responsibility.

> The physician diagnoses OSA — always — having seen the patient clinically, interprets the sleep study, prescribes appliance therapy in writing with the diagnostic report, verifies treatment efficacy with follow-up sleep testing, and remains involved in ongoing evaluation. The dentist screens for symptoms and signs, refers, assesses dental suitability, treats oral disease first, selects and fits an appropriate custom titratable appliance, calibrates it, provides behavioural advice, monitors adherence and the integrity of the oral structures, manages side effects, and refers on to periodontics, orthodontics or prosthodontics where needed. Both maintain regular written communication about the treatment plan, progress and follow-up, and a systematic follow-up programme is agreed between them. The dentist may obtain objective data by portable monitor for **titration purposes only**.

4. What is the relationship between sleep bruxism and OSA?

> They frequently co-occur — up to 30–50% of adults with OSA also exhibit sleep bruxism — and share plausible mechanisms, principally sleep arousal and autonomic activation, plus common risk factors of age, obesity, smoking and alcohol. However, meta-analyses show no consistent correlation across populations and severities, gold-standard polysomnographic studies find no consistent temporal association, and more bruxism events have been shown to precede apnoea–hypopnoea events than follow them, arguing against a straightforward protective role. A causal relationship is therefore **not established**, and the dentist should refrain from implying causation when discussing these conditions with patients. Treating the OSA can reduce bruxism in some patients where the two are temporally linked. Graded by evidence strength: co-occurrence strong, causation weak, treatment effect moderate.

5. A patient requests a night guard for tooth wear. What must you establish first, and why?

> Screen for OSA before fitting any splint. A conventional flat-plane stabilisation (Michigan-type) splint can **worsen** sleep-disordered breathing, increasing both AHI and snoring, because of the increased vertical dimension and posterior mandibular position. If there are signs or symptoms of OSA, refer for diagnosis and consider a mandibular advancement device, which can address both the bruxism-related protection and the airway. Note also that sleep bruxism alone requires neither a sleep study nor mandibular advancement, and an occlusal splint is appropriate first-line therapy in that situation.

6. Is temporomandibular disorder a contraindication to oral appliance therapy?

> Not absolutely. TMD is a **relative** contraindication: active severe TMJ disease or pain should be treated before starting therapy, principally to reduce the drop-out rate, but disc displacement alone is not a contraindication and pre-existing TMD does not generally worsen with appliance use. Transient jaw discomfort is common and does not itself indicate TMD; only around 2% of patients cannot tolerate an appliance. Long-term use does not impair the joint — condylar translation is less than in maximum opening, there is no significant morphological alteration, and TMD symptom intensity, joint noise and jaw fatigue may actually reduce. Mandibular exercises allow patients with TMD to wear an appliance, improving sleep and quality of life, reducing pain intensity and increasing compliance. Traction-type devices impose lower stress than compression-type devices, which produce greater early masticatory myalgia and residual tongue pain.

7. Explain the bidirectional relationship between pain and sleep, with mechanisms.

> Pain prevents sleep onset, causes frequent awakenings, produces transient EEG changes and shifts to lighter stages, and is a primary cause of insomnia. Conversely, short or disturbed sleep produces hyperalgesia — lowered pain thresholds and increased sensitivity — can cause or worsen spontaneous muscle pain and headache, reduces pain tolerance, and reduces the antinociceptive potency of morphine. The result is a self-reinforcing cycle. Mechanisms include dysregulation of serotonin, noradrenaline and dopamine, which regulate both pain and sleep; pro-inflammatory cytokines including TNF-α and IL-6 acting in both pathways; and HPA axis dysfunction. The clinical implication is that chronic pain management must address sleep simultaneously, usually through a multidisciplinary approach.

REFERENCES — TOPIC 4

1. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2).
2. Ramar K, Dort LC, Katz SG, et al. Clinical practice guideline for the treatment of obstructive sleep apnea and snoring with oral appliance therapy: an update for 2015. *J Clin Sleep Med*. 2015;11(7):773–828.
3. Lobbezoo F, Ahlberg J, Raphael KG, et al. International consensus on the assessment of bruxism: report of a work in progress. *J Oral Rehabil*. 2018;45(11):837–844.
4. Lobbezoo F, Ahlberg J, Glaros AG, et al. Bruxism defined and graded: an international consensus. *J Oral Rehabil*. 2013;40(1):2–4.
5. Manfredini D, Guarda-Nardini L, Marchese-Ragona R, Lobbezoo F. Theories on possible temporal relationships between sleep bruxism and obstructive sleep apnea events: an expert opinion. *Sleep Breath*. 2015;19(4):1459–1465.
6. Gagnon Y, Mayer P, Morisson F, Rompré PH, Lavigne GJ. Aggravation of respiratory disturbances by the use of an occlusal splint in apneic patients: a pilot study. *Int J Prosthodont*. 2004;17(4):447–453.
7. Sanders AE, Essick GK, Fillingim R, et al. Sleep apnea symptoms and risk of temporomandibular disorder: OPPERA cohort. *J Dent Res*. 2013;92(7 Suppl):70S–77S.
8. Schiffman E, Ohrbach R, Truelove E, et al. Diagnostic criteria for temporomandibular disorders (DC/TMD) for clinical and research applications. *J Oral Facial Pain Headache*. 2014;28(1):6–27.
9. Nikolopoulou M, Naeije M, Aarab G, Hamburger HL, Visscher CM, Lobbezoo F. The effect of raising the bite without mandibular protrusion on obstructive sleep apnoea. *J Oral Rehabil*. 2011;38(9):643–647.
10. Doff MHJ, Veldhuis SKB, Hoekema A, et al. Long-term oral appliance therapy in obstructive sleep apnea syndrome: a controlled study on temporomandibular side effects. *Clin Oral Investig*. 2012;16(3):689–697.
11. Cunali PA, Almeida FR, Santos CD, et al. Mandibular exercises improve mandibular advancement device therapy for obstructive sleep apnea. *Sleep Breath*. 2011;15(4):717–727.
12. Wolfe F, Smythe HA, Yunus MB, et al. The American College of Rheumatology 1990 criteria for the classification of fibromyalgia. *Arthritis Rheum*. 1990;33(2):160–172.
13. Wolfe F, Clauw DJ, Fitzcharles MA, et al. 2016 revisions to the 2010/2011 fibromyalgia diagnostic criteria. *Semin Arthritis Rheum*. 2016;46(3):319–329.
14. Moldofsky H. Sleep and pain. *Sleep Med Rev*. 2001;5(5):385–396.
15. Smith MT, Haythornthwaite JA. How do sleep disturbance and chronic pain inter-relate? Insights from the longitudinal and cognitive-behavioral clinical trials literature. *Sleep Med Rev*. 2004;8(2):119–132.
16. Ejaz SM, Khawaja IS, Bhatia S, Hurwitz TD. Obstructive sleep apnea and depression: a review. *Innov Clin Neurosci*. 2011;8(8):17–25.
17. Collop NA, Anderson WM, Boehlecke B, et al. Clinical guidelines for the use of unattended portable monitors in the diagnosis of obstructive sleep apnea in adult patients. *J Clin Sleep Med*. 2007;3(7):737–747.
18. Al-Jewair TS, Al-Jasser R, Almas K. Periodontitis and obstructive sleep apnea's bidirectional relationship: a systematic review and meta-analysis. *Sleep Breath*. 2015;19(4):1111–1120.
19. Sweetman A, Lack L, Bastien C. Co-morbid insomnia and sleep apnea (COMISA). *Brain Sci*. 2019;9(12):371.

TOPIC 5

Appliance selection, design, impressions, bite registration and fitting

BLUEPRINT

1. Clinical practice guideline for OAT · 2. Mechanism on the upper airway · 3. Indications and contraindications · 4. Designs, materials, fabrication · 5. Selecting the best design · 6. Bite registration rationale and techniques · 7. Impression materials and accuracy · 8. Laboratory request · 9. Delivery: fit, comfort, vertical dimension, protrusion · 10. Home care instructions

5.1 The clinical practice guideline

*Governing document: the joint **AASM/AADSM Clinical Practice Guideline (Ramar et al., "an update for 2015")**, supplemented by the AADSM Dental Sleep Medicine Standards, updated 2025.*

- **Diagnosis first.** Presence or absence of OSA **must be determined before initiating treatment** — to identify patients at risk from complications, and to provide a baseline against which effectiveness can be established. Severity must be established to make an appropriate decision, and **severity predicts appliance response**.
- **Who may fit. Qualified dental personnel** trained and experienced in the overall care of oral health, the TMJ, dental occlusion and associated oral structures. Management should be overseen by practitioners with serious training in sleep medicine and/or SDB, with emphasis on the proper protocol for diagnosis, treatment and follow-up; the guideline states the dentist should receive diplomate certification in dental sleep medicine or seek the "qualified dentist" designation. Cephalometric evaluation is not always required but where deemed necessary should be performed by appropriately trained professionals.
- **Treatment objectives.** *Primary snoring without OSA or UARS:* reduce snoring to a **subjectively acceptable level**. *OSA:* resolution of clinical signs and symptoms **and normalisation of the AHI and oxyhaemoglobin saturation**.

The six recommendations, with rationale

- **Prescribe an appliance rather than no treatment for adults requesting treatment of primary snoring (without OSA).** Appliances reduce snoring frequency and intensity and improve sleep quality and quality of life. But primary snoring must be diagnosed by a sleep physician to avoid misdiagnosis, and conservative measures (weight loss and other behavioural measures, positional therapy, avoiding alcohol) should be tried first. Note AADSM 2025: in the confirmed absence of OSA, no referral is required and treatment may commence at the qualified dentist's discretion.
- **Severe OSA should have an initial trial of nasal CPAP** — greater effectiveness demonstrated. Upper airway surgery (tonsillectomy and adenoidectomy, craniofacial operations, tracheostomy) may also supersede appliance use where predicted to be highly effective.
- **Where the patient is intolerant of CPAP or prefers alternative therapy, prescribe an appliance.** CPAP remains first-line because it is better for AHI, arousal index and ODI and for increasing oxygen saturation, and the appliance's blood pressure impact is more limited (though appliances may reduce blood pressure in selected patients). But appliances have **greater adherence and less discontinuation from side effects** and are unlikely to harm — so even though less efficacious, benefit outweighs risk. Indicated for mild–moderate OSA in patients who prefer them, do not respond to CPAP, are not appropriate CPAP candidates, or have failed CPAP or behavioural measures such as weight loss or sleep-position change.
- **Use a custom, titratable appliance over non-custom devices.** Custom titratable devices reduce AHI, arousal index and ODI and increase oxygen saturation to a greater extent. Neither significantly affects sleep architecture or efficiency — but custom appliances improve other physiological sleep parameters, thereby improving daily function and quality of life, and improve daytime sleepiness **not inferior to CPAP**.
- **Conduct follow-up sleep testing to improve or confirm efficacy** — PSG or attended cardiorespiratory study with the appliance in place, **after final adjustments or fitting**. Efficacy improves significantly when changes are made based on data obtained during or after sleep studies, because subjective feedback alone is insufficient to determine the optimal setting. Also consider testing on recurrent symptoms, substantial weight change, or a new comorbidity relevant to OSA. **Not indicated for primary snoring** unless symptoms worsen or fail to resolve.
- **Instruct patients to return for periodic office visits** with both physician and dentist. Adjustments made by the dentist based on physician-obtained PSG or home test data produce greater long-term improvement with fewer side effects. Once optimal fit and efficacy are established: **every 6 months for the first year, at least annually thereafter**.
- **Provide dentist oversight** to survey for dental side effects and occlusal changes and reduce their incidence — most side effects are dental. Intolerance and improper use are potential problems since appliances require patient effort; they may aggravate TMJ disease and cause dental misalignment and discomfort unique to each device; and they can be **rendered ineffective by patient alteration**. Purpose of follow-up: monitor adherence, evaluate device deterioration or maladjustment, evaluate oral structures and occlusal integrity, assess for worsening OSA. Objective re-evaluation of respiration is indicated if signs or symptoms worsen or recur. A specific aim is reducing the **drop-out rate, highest in the first year**, while use is likely lifelong. Consider allergy to metals or acrylic, strong bruxism, anatomical deviation.

Selection criteria at first consultation

- **BMI** — failure more likely with higher BMI, especially ≥35; better response in mild–moderate OSA with lower BMI.
- **Dentition** — generally ≥6 teeth per arch (stricter: 8–10 sound teeth per arch with adequate bone support); assess mobility, spacing, curve of Spee, restorative status.
- **Nasopharyngeal assessment** — nasolaryngoscopy to evaluate pharyngeal stenosis through the naso-, velo-, oro- and laryngopharynx, with jaw thrust and Müller manoeuvre.
- **Capacity** — protrusion ≥5–8 mm (normal 10–12 mm); maximum opening 25–40 mm (normal 40–60 mm).

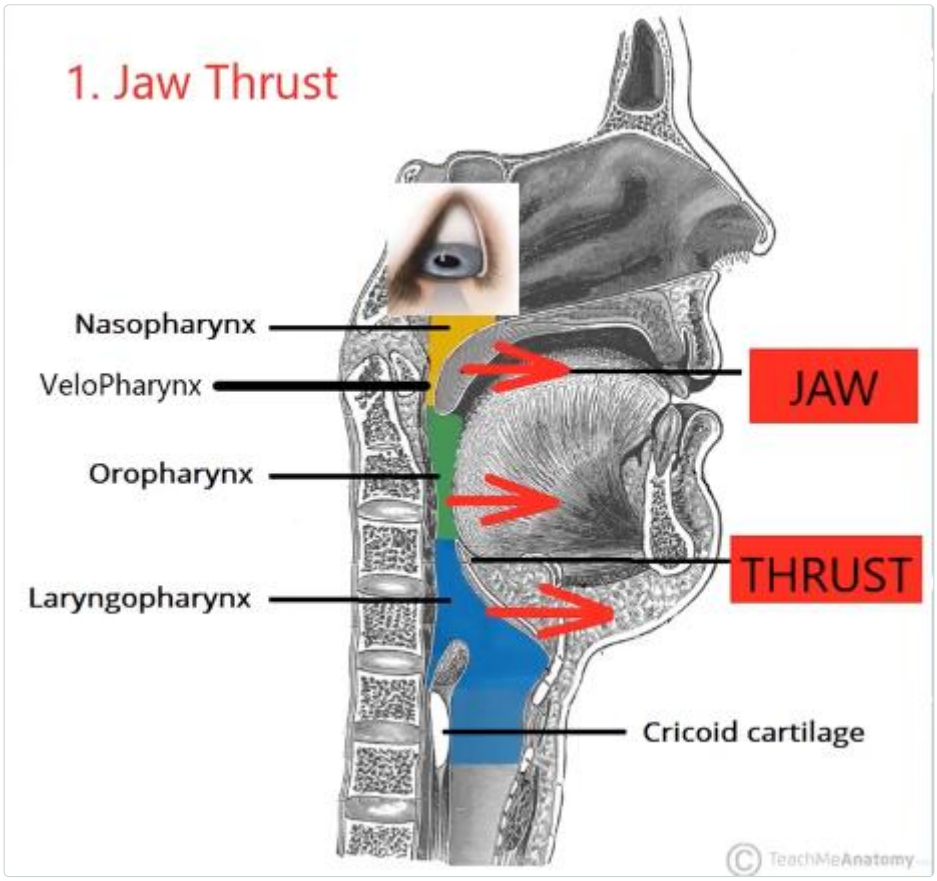


Fig 5.1 — The four pharyngeal levels and jaw thrust. Nasopharynx, velopharynx, oropharynx, laryngopharynx, down to the cricoid cartilage. **Examinable because:** nasolaryngoscopy is how you show that selection is anatomically reasoned rather than empirical. **Jaw thrust** sometimes increases space at only one or two of the four levels — exactly what you want to know before committing to advancement. The **Müller manoeuvre** (patient inhales while the clinician obstructs the nose, reducing intrapharyngeal pressure) shows which levels collapse and can be combined with jaw thrust. Caution: jaw thrust produces an oblique stretch on the pharyngeal constrictors, increasing wall tension and therefore compliance and collapsibility — so it is not a purely benign predictor.

5.2 Mechanism and prediction of success

- **Primary mode of action is improvement in passive pharyngeal anatomy.** Advancement reduces collapsibility **dose-dependently without systematically changing genioglossus function**, in a predominantly obese, severe-OSA population.
- The airway enlarges — particularly in its **lateral dimension at the velopharyngeal level** — and the tongue is displaced anteriorly; velopharyngeal, nasopharyngeal and anteroposterior dimensions all increase. **Loop gain, arousal threshold and dilator muscle activity are unchanged.**
- Dose dependence extends beyond anatomy: dose-dependent airway enlargement, augmentation of upper airway muscle activity and improvement in minimum SaO_2 with increasing protrusion; respiratory and sleep variables also improve dose-dependently.
- **Mild OSA suits the mechanism better** — less collapsible airways, and the pharynx increases more along its whole length than in severe disease. The mechanism is **less predictable than CPAP's**, even long term. Can be used alone or combined with CPAP or positional therapy, and can reduce CPAP side effects in patients troubled by high pressures.
- **Efficacy and effect:** success up to ~80% in mild, ~60% in more severe. Long-term efficacy good to ~10 years, but real-world effectiveness declines because the natural history is AHI deterioration. Reduces AHI and improves nightly oxygenation; increases slow-wave sleep percentage and REM duration. Snoring controlled better than by an intraoral placebo, though CPAP is better; **persistent snoring on treatment has been related to insufficient apnoea control and poor adherence** — a useful marker. Effect on daytime sleepiness unclear in snorers and mild–moderate OSA; in severe OSA there is reduction versus placebo but CPAP is more effective; the low-arousal-threshold subgroup may gain more subjective advantage. Health effects: reduces blood pressure (similar to CPAP), improves nightly blood pressure dips, improves endothelial and cardiac function, improves mean sleep latency on MSLT.

THE RESPONDER PROFILE

- **Patient traits:** younger; female; lower BMI and non-obese; smaller neck circumference; lower baseline AHI.
- **Polysomnographic:** supine-position-dependent (positional) OSA; NREM-predominant or non-stage-dependent rather than REM-predominant; a **lower required CPAP pressure** (higher pressure implies a more collapsible airway and poorer appliance response).
- **Endotype:** less collapsible airway; lower loop gain; higher arousal threshold; lower response to arousal.
- **Cephalometric:** shorter soft palate; longer maxilla; lower hyoid and decreased hyoid-to-mandibular-plane distance; greater angle between cranial base and mandibular plane; retrognathic mandible with narrow SNB and wider SNA.
- **Endoscopic:** verified pharyngeal widening on advancement; **tongue base collapse predicts success**. Conversely complete concentric palatal collapse and complete laterolateral oropharyngeal collapse indicate likely deterioration. Absence of a **pterygomandibular raphe** (present in only ~64% of people) predicts better response — greater maximal advancement, greater airway change, greater AHI reduction, less likely non-response. **Inspiratory tongue movement** with advancement may be beneficial (responder), minimal, or counterproductive (non-responder). A more posteriorly located tongue plus a more collapsible airway identifies responders.
- **Trial-based:** sufficient AHI reduction on an overnight test with a **remotely controlled mandibular positioner** — the best predictor available ("excellent"), against clinical/craniofacial prediction ("poor") and nasopharyngoscopy ("fair to good").
- **Adverse:** weight increase during treatment relates to failure. Differences between patients in the ability to advance the mandible modify the potential to eliminate events.
- **Essential caveat:** patient factors alone are **not** a good predictor, and demographic, anthropometric and polysomnographic data only weakly inform efficacy. ~⅔ achieve a clinically meaningful effect (OSA at least halved); ~⅓ are unlikely to benefit — hence the need for prediction methods and for objective verification rather than assumption.

5.3 Indications and contraindications

- **Indications:** mild–moderate OSA as a primary option; moderate–severe OSA where CPAP cannot be tolerated or adhered to; **snoring** — often the only reason the patient attended a sleep clinic, and non-apnoeic snorers without daytime symptoms tolerate CPAP poorly (most agree to try but the majority give up), so an appliance is a good alternative.
- **Contraindications:** insufficient dentition or inadequate retention (including implant-retained cases); severe periodontal disease or loose teeth; severe pre-existing or active TMJ disease or pain; inadequate protrusion <6 mm; severe bruxism; poor hygiene or significant caries; severe gag reflex; ongoing significant dental or orthodontic work; anterior open bite (no posterior clearance). Severe OSA is only a *relative* contraindication — trial CPAP first, but an appliance is appropriate if CPAP fails.

5.4 Designs, materials, fabrication

TRAP — BOIL-AND-BITE DEVICES

Do not use off-the-shelf, non-individualised appliances. Fixed position and not reliably titratable; poor fit and inadequate retention; poorly tolerated; potential for undesirable dental side effects. They fail the AADSM effective-appliance definition on at least three counts — reliable titration, verifiable setting, stable retention. Self-fitted appliances are at most temporary; they do not fit well and cannot be worn long term.

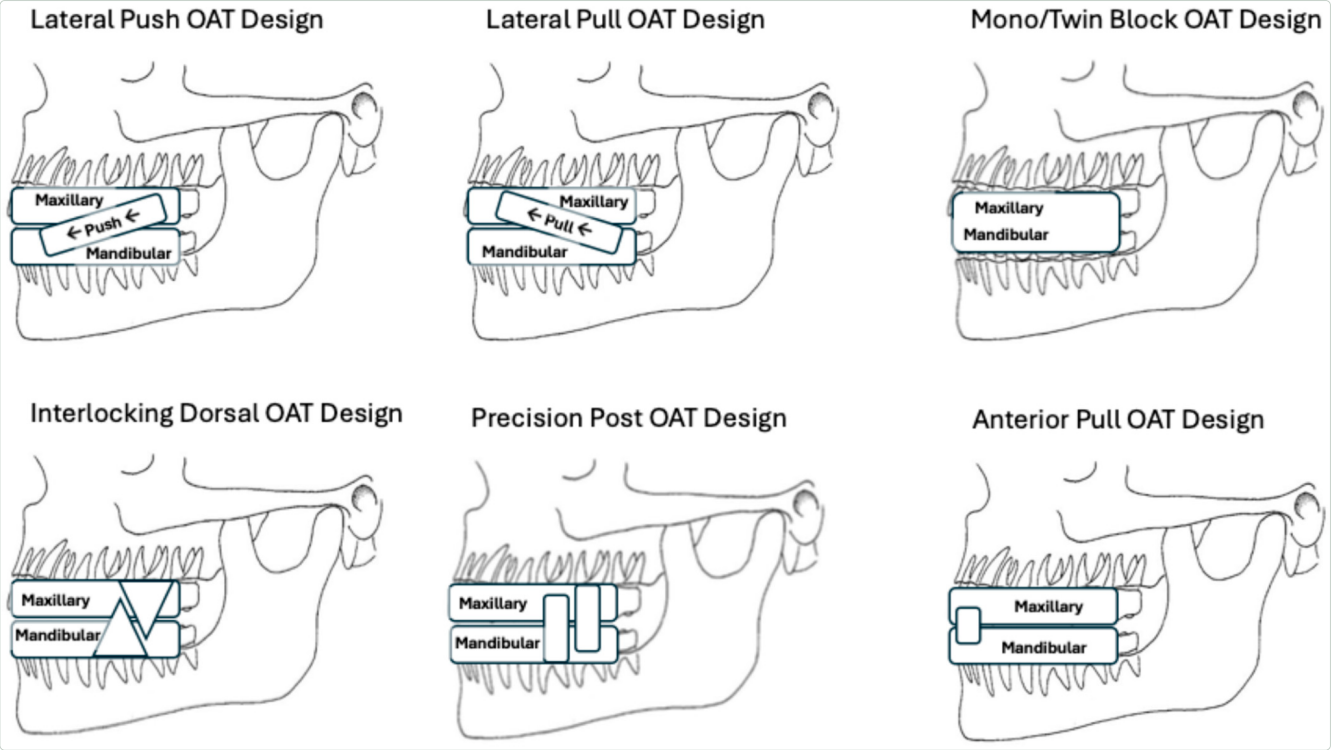


Fig 5.2 — The six design categories, each a maxillary/mandibular overlay pair with its coupling mechanism: Lateral Push, Lateral Pull, Interlocking Dorsal, Precision Post, Mono/Twin Block, Anterior Pull. **Examinable because:** lets you answer any design question by *mechanism* rather than brand — which is what examiners want.

- **Lateral Push** — bilateral metal piston-like **Herbst arms**; pushes the mandibular overlay forward using the maxillary overlay as anchor. Titrated by displacement screws in the rods. Robust, fine titration ~0.1 mm; metal may impinge on soft tissue; mandible may drop; needs good retention on both arches.
- **Lateral Pull** — bilateral elastomeric or nylon straps *pulling* the mandibular overlay forward. Titrated by swapping strap lengths. Small, well documented; but straps fracture or stretch, titration gross (0.5–1 mm), and poor retention lets the less retentive component be lifted by the strap.
- **Interlocking Dorsal** — angled dorsal posts from the mandibular overlay interlocking with maxillary blocks. Titrated by screws in the blocks (e.g. **10 turns = +1 mm**). An *unattached* mechanism.
- **Precision Post** — dual posts from both overlays interlocking perpendicularly to guide, reposition and stabilise the mandible. Titrated by swapping an overlay for one with a different setting (e.g. +1 mm).
- **Mono/Twin Block** — overlays fused at the occlusal surface (mono) or mated by interlocking blocks (twin). Titrated by swapping overlays. Mono-block can beat prefabricated devices and gives good advancement but restricts jaw movement and can cause TMJ discomfort.
- **Anterior Pull** — anterior screw or strap repositioning the mandible against the opposing overlay. **Good for poor dexterity** — adjustment accessible at the front, easy and fine titration 0.1–0.3 mm, robust — **but force concentrates on the anterior teeth**, with greater vertical dimension, tongue irritation and anterior loading; needs good retention.

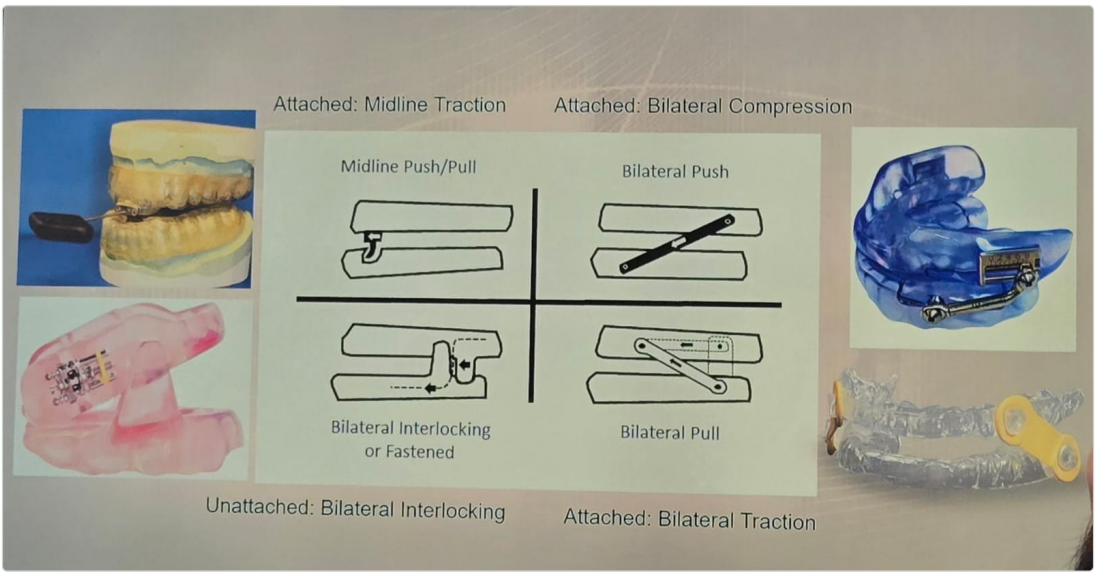


Fig 5.3 — Attachment and propulsion mechanisms in four quadrants: attached midline traction (midline push/pull); attached bilateral compression (bilateral push); unattached bilateral interlocking; attached bilateral traction (bilateral pull), with clinical examples. **Examinable because:** the same taxonomy expressed as *attached vs unattached* and *traction vs compression*. Two consequences: **traction devices impose lower stress** than compression devices (which cause greater early masticatory myalgia and residual tongue pain), and **unattached interlocking designs let the mandible drop** supine, potentially reducing efficacy.

Design principles affecting outcome

- The most common and most effective are **custom-made, titratable appliances that do not allow mouth opening** — upper and lower parts with an intermaxillary adjustment mechanism; either fixed to one another (e.g. by screw) or allowing lateral and/or vertical mandibular movement.
- **Appliances allowing mouth opening are less effective**; adding elastics to prevent opening halved supine apnoea frequency, and elastics increase MAD success in positional OSA. A **fixed** device may beat a loose one because the mandible drops supine.
- **Larger advancement = larger efficacy; smaller vertical opening preferred** — increasing vertical opening increases pharyngeal collapsibility and is associated with reduced efficacy.
- **Coverage.** Fewer overjet and overbite changes with a soft elastomeric device covering parts of the alveolar processes in addition to all teeth, versus a hard acrylic appliance with full occlusal coverage fixed mainly to teeth. However **hard materials give better protection against tooth movement**, and all teeth need covering — leave molars uncovered and they will move or over-erupt. A flexible appliance without incisor coverage increased lower anterior irregularity versus a rigid appliance with incisor coverage. An orthodontic appliance incorporating forces counteracting the posteriorly directed load on the upper front teeth had a positive effect on overjet change.
- **Between-device differences are clinically small** in symptoms or AHI. The influence of design on efficacy and side effects is **almost unknown**, and side effects are a product of **protruding the mandible to achieve a therapeutic effect rather than of specific design**.
- **Not recommended:** designs marketed on theories of airway and bone remodelling, promoted by manufacturers, with no peer-reviewed publication, no demonstrated superiority, and non-adherence to AADSM guidance — typically bulky and not titratable.

Materials

- **Acrylic (PMMA)** — good mechanical properties, well established; retention by **frictional resistance**; can be relined and repaired; used in DLP and SLA printing. More breakage than nylon in bruxers; FDM-printed PMMA is weaker; needs ~5 mm clearance.
- **Nylon / polyamide (PA12), CAD-CAM** — lighter, smaller, robust, less breakage so **better for bruxers**; retention by **flexion into undercuts**; needs only ~3 mm clearance; titration scale visible on the device. But **stains easily so polishing is important**; gross titration 0.5–1 mm; **cannot be relined or repaired**, so a large new restoration means a remake.
- **EVA** — good impact absorption, non-toxic, easily shaped for mouthguards; additive manufacturing limited; effectiveness in sleep apnoea appliances debated. **Cobalt-chrome** — extremely durable, good for bruxers; expensive to cast, limited to framework devices or coating. **Titanium/CAD-CAM** — titratable and robust but greater vertical dimension, very bulky, tongue-irritating.
- **Selection factors:** number of teeth; arch relationship (class II vs III); bruxism; TMD; gagging.
- **Fabrication:** traditional (casts and bite registrations — well established, precise custom fit) or 3D printing (high customisation, potentially reduced time and cost, in-house printing for fast replacement; DLP and SLA offer accuracy, resolution and speed). **Digital scanning is more accurate** than conventional impressions — check the midline before and after scanning.

5.5 Selecting the device

"AT RISK ANATOMICALLY" — THE TARGETED DOCUMENTATION CHECKLIST

- Micrognathia / retrognathia
- Deep palatal vault
- Large tongue — **scalloping** is the practical sign, size being hard to define
- Modified Mallampati class 3 or 4
- Tonsil grade 3 or 4 — depress the tongue and ask the patient to say "ah"
- Thick neck; obesity
- Long airway (long neck)
- Low hyoid — hyoid to mandibular plane >15 mm

- **Dental factors** — sufficient healthy teeth and a suitable bite are crucial. Assess number of functional teeth, mobility (should not be mobile), spacing, curve of Spee, restorative status. A **deep curve of Spee** means insufficient vertical dimension at the second molars — consider not covering the sevens and tell the laboratory.
- **Arch relationship** — class II vs III alters mechanics and the direction of unwanted movement. An **anterior open bite precludes a MAD** — no posterior clearance at all.
- **Skeletal pattern** — severe maxillomandibular deformity may need specialised treatment; retrognathia favours an appliance and is a specific problem for CPAP because of mask leakage.
- **Mouth and tongue** — a TRD is an option for missing teeth or where jaw alignment precludes a MAD. **Bruxism** favours nylon or cobalt-chrome and a two-piece design. **TMD and gagging** favour traction rather than compression mechanisms, with minimal vertical opening and bulk.
- **Dexterity, cognition, visual acuity, range of motion** — an anterior adjustment mechanism suits poor dexterity, accepting anterior force concentration.
- **Nasal patency** — assess and treat first. It does not itself treat the OSA but **improves the treatment outcome**. Ask specifically about allergic rhinitis and treat it (referring to ENT) before appliance therapy; treatment may take time but you need not wait for complete resolution.
- **Sleep study findings** — severity and positional dependence; adjustable MADs have higher response rates than fixed devices.
- **Patient preference** — perceived comfort, ease of use, durability, cost. **The appliance must not be selected solely because of insurer requirements** — the choice belongs to the qualified dentist and the patient.

5.6 Bite registration

THE THREE COMPONENTS OF THE CONSTRUCTION BITE

- **Horizontal (protrusion)** — the principal therapeutic variable. Start at **50–70%** of maximum protrusion; AADSM 2025 states **25–75%** lies within both the comfort and therapeutic range. Do not exceed ~75% initially.
- **Vertical (bite opening)** — provides interocclusal space without discomfort or fatigue. **~2 mm interincisal** (moderate 1–3 mm increase typical). Aim for posterior clearance rather than a raised bite: nylon ~3 mm, acrylic ~5 mm, some designs up to 6 mm (very difficult).
- **Lateral (midline)** — accommodates asymmetry or deviation. The midline should be **reproducible** at both centric and protrusion; it need not be identical, but it must be reproducible.

- **Why not maximum protrusion.** The acceptable range for an individual is **not known**. Maximum protrusion gives the greatest airway enlargement but is inadequate as a setting because it causes TMJ and muscle discomfort. Classical practice: registration at **60% of maximum with 2 mm vertical separation** of the incisors; other conventions use ~75%, or a wax record in the most protrusive position the patient can comfortably maintain. Practical rule: **start at 50%, do not go beyond 75%**.
- The proposed consistent between-patient "threshold" of **70% of maximal comfortable advancement** has been challenged: airway collapsibility improves in **non-responders** during the initial range (0–50%), whereas in **responders** most improvement occurs close to maximal possible advancement — the dose–response curve differs by phenotype.
- **Excessive protrusion does not guarantee improved efficacy**, may affect compliance, and in some instances **overprotrusion may worsen the SDB**. The degree of subsequent bite change is influenced by design, initial bite type and treatment duration as well as protrusion.

George Gauge method

1. **Determine the range** — record most retruded and most protruded readings and add their absolute values (e.g. −3 and +7 = 10 mm; −4 and +5 = 9 mm).
2. **Calculate the target** — 50–70% of that range. 60% of 10 mm = 6 mm; from −3, register at **+3**.
3. **Select the fork** — the **2 mm grey** fork is usual; the **5 mm white** fork is usually too much, promoting mouth breathing and inhibiting lip seal.
4. **Seat and check** — the blue incisal guide must clamp the lower incisors firmly; check the midline is the same and reproducible at centric and protrusion.
5. **Register** — do not inject bite registration material into the lower incisor clamp area.

6. **Check posterior vertical clearance**, especially in a deep bite. If insufficient — usually at the sevens — ask the laboratory not to cover them.
7. **Consider a centric occlusion record** as well, for a morning aligner (Topic 10).

Techniques

- **Construction bite devices** — mechanical devices giving precise control and measurement in three dimensions, with guides and rulers documenting advancement and vertical opening numerically. The George Gauge is the archetype.
- **Neuromuscular (TENS)** — transcutaneous electrical nerve stimulation relaxes the muscles, allowing a relaxed physiological mandibular position before recording, reducing nociceptive input and potentially improving accuracy.
- **Phonetic** — uses speech (phonation of particular sounds) to find a position that keeps the airway open without excessive protrusion or opening.
- **Clinical measurement and simulation tools** — bite gauges and mandibular positioning simulators for reproducibility. **Digital/ virtual** — scanning and CAD, potentially outperforming traditional methods in accuracy and efficiency.
- Under AADSM 2025 standards, **impressions and the construction bite record must be completed in person** — telehealth may be used for screening and many follow-ups, but not for these.

5.7 Impression materials and accuracy

- **Addition silicone (PVS)** — excellent accuracy and dimensional stability; elastomer, so recovers after deformation and suits undercuts. **Polyether** — highly accurate and versatile, consistent, good tear strength, hydrophilic so useful in moist and challenging situations. **Polysulfide** — older elastic material, used where detail is critical. **Condensation silicone** — less dimensionally stable. **Alginate** — easy to handle, used for diagnostic models and study casts; poor dimensional stability, pour promptly. **Digital scan** — more accurate; avoids material handling and storage issues; verify the midline.
- **Requirements:** accuracy and detail reproduction (some materials to **20–25 µm**); dimensional stability from removal to pouring; elasticity and tear strength for undercuts; wettability on moist tissues; biocompatibility; ease of handling with suitable working time but short setting time; appropriate viscosity and thixotropy; a rigid tray; moisture control; and operator technique, which matters as much as the material.
- For appliance fabrication, record the **upper and lower dentition plus ~3 mm of gingiva**, with the protrusive construction bite.

5.8 Laboratory request

> **A WORKED EXAMPLE OF AN APPLIANCE SPECIFICATION**
>
> "Upper and lower removable clear acrylic plates with full occlusal coverage fitted to both dental arches, 1.5–2 mm thickness. Two acrylic flanges, one on either side of the palatal aspect of the upper molars, fitting into slots cut into the acrylic on the lingual aspect of the lower plate in the molar region, to prevent posterior movement of the mandible while permitting free opening of the mouth. Two 10 mm screws to allow incremental anterior advancement of the lower plate."

- **Elements:** date; prescribing clinician (name, licence/registration number, contact, practice address); laboratory details; patient details (name, record number, age); **type of appliance**; **material specification**; **vertical dimension/opening in mm** plus any required posterior clearance; **mandibular protrusion setting** (percentage of range or absolute advancement, plus the George Gauge reading used); **retention features**; **occlusal considerations** — for example **"do not cover the upper and lower second molars"** where clearance is inadequate; delivery instructions (polishing, adjustments, finishing); additional notes; attachments (models, bite registrations, scans, photographs); signature and date.
- **Protrusion, vertical dimension and retention** are the essential core — they determine whether the appliance works and whether it is comfortable. Delivery typically 2–3 weeks after the impression.

5.9 Delivery

1. **Pre-delivery.** Verify the prescription against the titration plan (type, material, vertical increase, protrusive setting). Inspect for bubbles, rough edges, distortion; confirm screws and connectors operate smoothly. Disinfect per manufacturer guidance and apply water-based lubricant for initial seating.
2. **Seating and retention.** Seat gently, engaging the posterior bulge or clasps before anterior structures. Ask the patient to swallow, speak ("pa–ta–ka") and open wide; apply gentle traction on each side — the device should resist displacement without discomfort. If retention is insufficient, evaluate undercut engagement and adjust acrylic **in stress-bearing areas only**.
3. **Vertical dimension.** Measure with a Willis gauge or from a fixed landmark (nose tip to chin) with and without the appliance. Confirm the interincisal increase of typically **2–4 mm over habitual closing** is tolerated; with the jaw relaxed there should be no strain, pulling or muscle fatigue.
4. **Protrusion.** Verify against the built-in millimetre scale or a protrusion gauge and confirm it matches the prescription. Ask the patient to close in the protruded position and check for **even bilateral posterior contacts without slide back to retrusion**; observe swallowing and speaking — the mandible should stay advanced.

5. **Comfort and soft tissue.** Ask about pressure points, gag triggers, tongue displacement. Inspect for mucosal blanching, sore spots on the ridges, adequate gingival clearance. Confirm no premature occlusal interferences.
6. **Final adjustments and education.** Remove high spots with a round bur, polish, reassess retention. Demonstrate insertion and removal over a soft surface. Review cleaning, storage and the adjustment protocol. Provide emergency contact details. **Book review at 2–4 weeks** (AADSM: reassess within 30 days).

Provide a **written copy of instructions, the delivery form and warranty**, signed and dated by patient and team member, one copy to the patient and one retained. Also consider measures to maximise comfort and minimise dental change — occlusal irregularities, interdental spacing — and to attenuate jaw discomfort and muscle fatigue: morning exercises, a morning repositioning device, associated palliative care (Topic 10).

5.10 Home care

- **Use.** Wear **every night**, for the whole duration of sleep. Contact the practice immediately if there is discomfort. Avoid eating or drinking anything but water for **≥2 h before bed** to reduce reflux risk.
- **Cleaning.** Rinse immediately after removal. Clean each morning with a **soft-bristled brush and mild soap** or denture cleaner, using **cool or warm — never hot — water** (hot water warps the device). **Avoid toothpaste** — too abrasive, causing scratches that harbour bacteria, dull the appliance and cause odour. Disinfect weekly in a denture-safe soak.
- **Handling and storage.** Remove over a soft surface or a basin of water. **Air dry** before placing in the ventilated case. Store dry in the case — never in tissue or loose in a pocket. Keep away from sunlight, heat, pets and children.
- **Oral health.** Brush and floss regularly. **Morning routine:** gentle jaw exercises — resting the chin on a fist, curling the tongue back to find the retruded position — and the morning occlusal guide for 15–20 minutes if provided.
- **Contact the practice** immediately for discomfort or pain, and for any change such as jaw pain, altered bite, or worsening snoring.

NUMBERS — TOPIC 5

- Protrusion 50–70% of max (AADSM 25–75%); start 50%, never exceed 75%; challenged "70% threshold"
- Vertical: 2 mm grey fork (5 mm white too much); delivery 2–4 mm over habitual; clearance nylon ~3, acrylic ~5, some 6 mm
- Protrusive capacity ≥5–8 mm (normal 10–12); max opening 25–40 mm (normal 40–60); <6 mm contraindication
- Dentition ≥6/arch (stricter 8–10 sound teeth with adequate bone); BMI ≥35 predicts failure
- Titration 0.1–0.4 mm/turn; 10 turns = +1 mm; effective appliance ≤1 mm increments, ≥5 mm range
- Impression: dentition + ~3 mm gingiva; detail to 20–25 µm; acrylic plate 1.5–2 mm thick
- Delivery 2–3 weeks after impression; review 2–4 weeks (≤30 days); follow-up 6-monthly year 1 then annually
- Success up to ~80% mild, ~60% severe; ~⅔ benefit, ~⅓ none; pterygomandibular raphe present in ~64%

SELF-TEST — TOPIC 5

1. State the six recommendations of the 2015 guideline.

(i) Prescribe an appliance rather than no treatment for adults requesting treatment of primary snoring without OSA. (ii) Where an adult with OSA is intolerant of CPAP or prefers alternative therapy, prescribe an appliance rather than no treatment. (iii) Qualified dentists should use a **custom, titratable** appliance over non-custom devices. (iv) Sleep physicians should conduct **follow-up sleep testing** to improve or confirm efficacy rather than follow up without testing. (v) Physicians and qualified dentists should instruct patients to return for **periodic office visits** with both. (vi) Qualified dentists should provide **oversight** to survey for dental side effects and occlusal change and reduce their incidence.

2. Walk through taking a construction bite with a George Gauge.

Establish the protrusive range by adding the absolute values of the most retruded and most protruded readings — for example −3 and +7 gives 10 mm. Target 50–70%: 60% of 10 mm is 6 mm, so from −3 register at +3. Use the 2 mm grey fork; the 5 mm white fork usually gives too much opening, promoting mouth breathing and inhibiting lip seal. Confirm the blue incisal guide clamps the lower incisors firmly and the midline is reproducible at both centric and protrusion. Register, avoiding material in the lower incisor clamp area. Check posterior clearance, particularly in a deep bite, and if inadequate at the sevens ask the laboratory not to cover them. Consider a centric occlusion record for a morning aligner.

3. Why not set the appliance at maximum protrusion?

It gives the greatest airway enlargement but causes TMJ and muscle discomfort, threatens compliance and increases dental side effects. Excessive protrusion has not been shown to guarantee improved efficacy and **overprotrusion may worsen the SDB**. The acceptable individual range is unknown, so start at 50–70% (25–75% is within the comfort and therapeutic range) and titrate incrementally to the minimum effective position, leaving leeway rather than going straight to 75%. The dose–response also differs by phenotype: responders gain most improvement near maximal advancement, whereas non-responders improve early over 0–50%.

4. Compare nylon and acrylic.

> Nylon (usually CAD-CAM polyamide) is lighter, smaller and more robust with less breakage, so preferable for bruxers; retains by flexion into undercuts; needs less clearance (~3 mm); and has a visible titration scale making adjustment easier. Drawbacks: stains readily so polishing matters, gross titration at 0.5–1 mm, and it cannot be relined or repaired — a large new restoration means a remake. Acrylic retains by frictional resistance, can be relined and repaired, and allows finer incremental screw adjustment, but breaks more readily in bruxers and needs ~5 mm clearance. Cobalt-chrome is extremely durable and good for bruxers but expensive to cast and limited to framework designs.

5. Your patient has a deep bite with a marked curve of Spee. Implications?

> Insufficient vertical clearance at the second molars. Options: increase the vertical dimension — undesirable, since greater opening increases pharyngeal collapsibility, reduces efficacy and promotes mouth breathing — or, preferably, instruct the laboratory **not to cover the second molars**. The trade-off is that uncovered molars may move or over-erupt, so document it, explain it and monitor at review. Material choice helps: nylon needs only ~3 mm clearance versus ~5 mm for acrylic. Separately, an anterior open bite precludes a MAD altogether — no posterior clearance at all.

TOPIC 5

Appliance selection, design, impressions, bite registration and fitting

EXAMINER'S BLUEPRINT

1. Clinical practice guideline for oral appliance therapy in the treatment of OSA
2. Mechanism of action of the oral appliance on the upper airway
3. Indications and contraindications for oral appliance therapy
4. Considerations for different appliance designs including advantages and disadvantages of each, manufacturing materials and fabrication techniques
5. Selecting the best device design based on history, examination, sleep study findings and patient preferences
6. Rationale for initial bite registration position — vertical, horizontal and lateral components — and discussion of bite registration techniques
7. Impression materials available and requirements for an accurate impression
8. Writing a laboratory request for device fabrication
9. Delivering a retentive device: assessing fit, comfort, vertical dimension and protrusion at delivery
10. Delivery of device home care instructions

This is the most practical topic and the one where your day-to-day clinical experience is directly examinable. It rewards precision: exact percentages of protrusion, exact vertical dimensions, the sequence at the delivery appointment, and the ability to name and describe appliance designs by mechanism rather than by brand.

5.1 The clinical practice guideline for oral appliance therapy

The governing document remains the joint **AASM and AADSM Clinical Practice Guideline (Ramar *et al.*, "an update for 2015")**, supplemented by the AADSM *Dental Sleep Medicine Standards*, most recently updated in 2025. The 2015 document was the first official joint guideline of the two academies and is built around the collaborative physician–dentist model.

Diagnosis first

- The presence or absence of OSA **must be determined before initiating treatment with an appliance**, for two reasons: to identify patients at risk from the complications of sleep apnoea, and to provide a baseline against which the effectiveness of subsequent treatment can be established.
- Detailed diagnostic criteria for OSA are available and include clinical signs, symptoms and the findings identified by polysomnography.
- The **severity** of sleep-related respiratory problems must be established in order to make an appropriate treatment decision — and severity is predictive of the response to an appliance.

Appliance fitting and who may do it

- Oral appliances should be fitted by **qualified dental personnel** who are trained and experienced in the overall care of oral health, the temporomandibular joint, dental occlusion and associated oral structures.
- Dental management of patients with appliances should be overseen by practitioners who have undertaken serious training in sleep medicine and/or sleep-related breathing disorders, with focused emphasis on the proper protocol for diagnosis, treatment and follow-up. The guideline states that the dentist should receive diplomate certification in dental sleep medicine or seek the designation of "qualified dentist".

- Although cephalometric evaluation is not always required for a patient who will use an oral appliance, appropriately trained professionals should perform these examinations when they are deemed necessary.

Treatment objectives

Condition	Treatment objective
Primary snoring without features of OSA or upper airway resistance syndrome	To reduce the snoring to a **subjectively acceptable level**
OSA	Resolution of the clinical signs and symptoms of OSA, and **normalisation of the AHI and oxyhaemoglobin saturation**

The recommendations, with their rationale

Recommendation	Rationale and detail
Sleep physicians should prescribe an oral appliance, rather than no treatment, for adults who request treatment of primary snoring (without OSA)	Appliances reduce the frequency and intensity of snoring and improve sleep quality and quality of life. But primary snoring must be diagnosed by a sleep physician to avoid misdiagnosis, and conservative measures — weight loss and other behavioural measures, positional therapy, avoiding alcohol — should be tried first; if these fail, provide an appliance. Note the AADSM 2025 position that in the confirmed absence of OSA, a referral is not required for an appliance to manage snoring and treatment may commence at the qualified dentist's discretion
Patients with severe OSA should have an initial trial of nasal CPAP	Greater effectiveness has been shown with CPAP than with oral appliances. Upper airway surgery — including tonsillectomy and adenoidectomy, craniofacial operations and tracheostomy — may also supersede appliance use in patients for whom these operations are predicted to be highly effective
Where an OSA patient is intolerant of CPAP or prefers alternative therapy, sleep physicians should prescribe an oral appliance	CPAP remains first-line for OSA because it is better at reducing AHI, arousal index and oxygen desaturation index and at increasing oxygen saturation, and the impact of an appliance on blood pressure is limited by comparison (though appliances may reduce blood pressure in selected patients). However, appliances have a **greater adherence rate and less discontinuation of therapy due to side effects**, and are unlikely to harm the patient — so even though less efficacious, the benefit outweighs the risk. Although not as efficacious as CPAP, appliances are indicated for patients with mild-to-moderate OSA who prefer them to CPAP, or who do not respond to CPAP, are not appropriate candidates for CPAP, or have failed treatment attempts with CPAP or with behavioural measures such as weight loss or sleep-position change
Qualified dentists should use a custom, titratable appliance over non-custom oral devices	Custom titratable appliances reduce AHI, arousal index and oxygen desaturation index, and increase oxygen saturation, to a greater extent than non-custom devices. Neither significantly affects sleep architecture or sleep efficiency — but custom appliances improve other physiological sleep parameters, thereby improving daily function and quality of life, and improve daytime sleepiness to a degree not inferior to CPAP
Sleep physicians should conduct follow-up sleep testing to improve or confirm treatment efficacy	Testing (PSG or attended cardiorespiratory sleep study with the appliance in place) should occur **after final adjustments or fitting have been performed**. Significant improvement in appliance efficacy occurs when changes are made based on data obtained during or after sleep studies, because subjective feedback alone is not sufficient to determine the optimal setting. Also consider follow-up testing in appliance-treated patients with recurrent symptoms, substantial weight change, or a new diagnosis of a comorbidity relevant to OSA. **Follow-up sleep testing is not indicated for patients with primary snoring** unless symptoms worsen or fail to resolve
Sleep physicians and qualified dentists should instruct patients to return for periodic office visits	Adjustments made by the dentist based on data from polysomnography or home sleep apnoea testing conducted by the sleep physician produce greater long-term improvement in the OSA with fewer side effects. Once optimal fit is obtained and efficacy shown, dental specialist follow-up **every 6 months for the first year and at least annually thereafter** is recommended
Qualified dentists should provide oversight of appliance therapy	To survey for dental-related side effects or occlusal changes and reduce their incidence — most appliance side effects are dental. Intolerance and improper use are potential problems, since appliances require patient effort to use properly; they may aggravate TMJ disease and may cause dental misalignment and discomfort unique to each device; and they can be

Recommendation	Rationale and detail
	rendered ineffective by patient alteration. The purpose of follow-up is to monitor adherence, evaluate device deterioration or maladjustment, evaluate the health of the oral structures and the integrity of the occlusion, and assess the patient for signs and symptoms of worsening OSA. Objective re-evaluation of respiration during sleep is indicated if signs or symptoms of OSA worsen or recur. A specific aim is to reduce the **drop-out rate, which is highest in the first year**, while appliance use is likely to be lifelong. Close communication with the dental specialist is most conducive to good patient care. Consider also possible allergy to metals or acrylic, strong bruxism, and anatomical deviation

Selection criteria at the first consultation

Criterion	Detail
BMI	Failure is more likely with a higher BMI, especially ≥35. Better response in patients with mild-to-moderate OSA and lower BMI
Dentition	Dental evaluation; generally at least 6 teeth per arch (some sources specify 8–10 sound teeth per arch with adequate bone support). Assess mobility, spacing, curve of Spee and restorative status
Nasopharyngeal assessment	Nasolaryngoscopy to evaluate the degree of pharyngeal stenosis through the naso-, velo-, oro- and laryngopharynx, with the **jaw thrust** manoeuvre and the **Müller manoeuvre** (see below)
Protrusive and opening capacity	Protrusion at least 5–8 mm (10–12 mm normal); maximum opening 25–40 mm (40–60 mm normal)

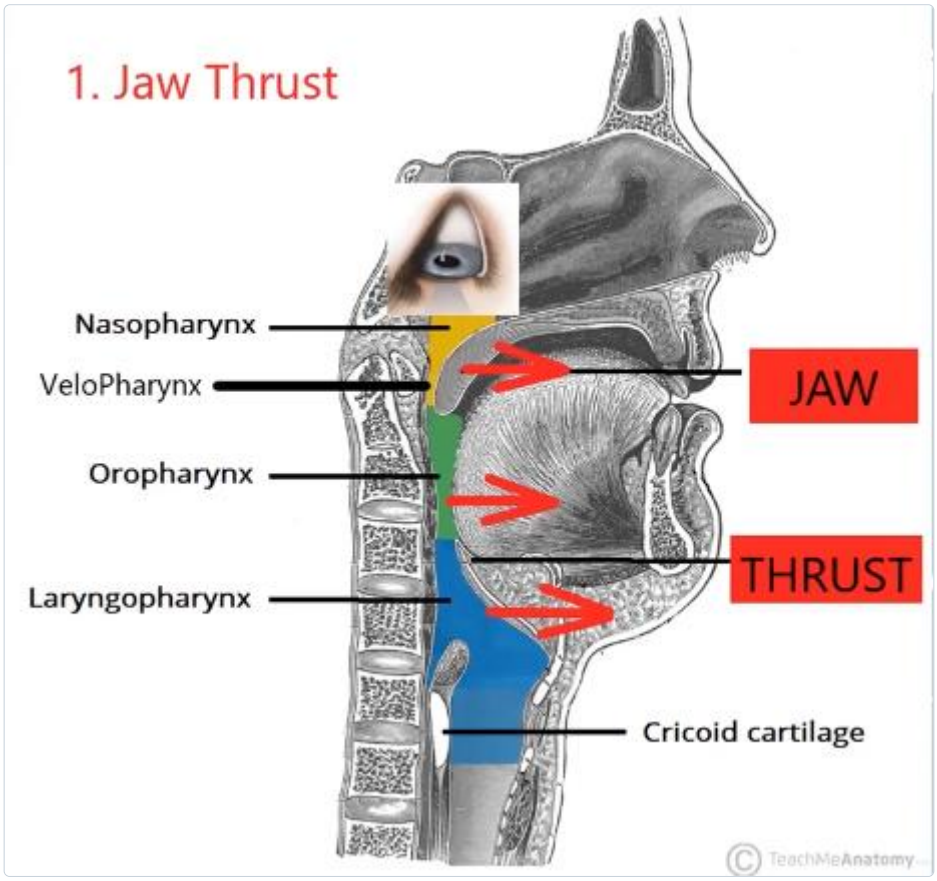


Figure 5.1 — The four pharyngeal levels and the effect of jaw thrust. Nasopharynx, velopharynx, oropharynx and laryngopharynx, down to the cricoid cartilage, with the direction of the jaw thrust manoeuvre indicated.

*Why it is examinable: nasolaryngoscopic assessment is how a candidate demonstrates that appliance selection is anatomically reasoned rather than empirical. **Jaw thrust** sometimes increases space at only one or two of the four levels — which is exactly the information you want before committing a patient to mandibular advancement. The **Müller manoeuvre** (the patient inhales while the clinician obstructs the nose, reducing intrapharyngeal pressure) demonstrates which levels collapse, and can be combined with jaw thrust. Note the caution: jaw thrust produces an oblique stretch on the pharyngeal constrictors, increasing wall tension and therefore compliance and collapsibility — so the manoeuvre is not a purely benign predictor.*

5.2 Mechanism of action on the upper airway, and prediction of success

Mechanism

- The **primary mode of action is improvement in passive pharyngeal anatomy**. Mandibular advancement reduces pharyngeal collapsibility in a **dose-dependent** manner without systematically changing genioglossus muscle function, in a predominantly obese, severe-OSA population.
- The appliance advances the mandible and changes the volume of the upper airway. The airway enlarges — particularly in its **lateral dimension at the velopharyngeal level** — and the tongue is displaced anteriorly. There is an increase in the velopharyngeal, nasopharyngeal and anteroposterior dimensions of the upper airway.
- Appliances improve pharyngeal collapsibility while **loop gain, arousal threshold and dilator muscle activity are unchanged**.
- Dose dependence extends beyond anatomy: there is dose-dependent enlargement of the upper airway, augmentation of upper airway muscle activity and improvement in minimum SaO_2 with increasing mandibular protrusion, and the respiratory and sleep variables also improve in a dose-dependent way.
- Patients with **mild OSA** are thought to be better suited to the mechanism than those with more severe disease: they generally have less collapsible airways, and the pharynx increases more, along its whole length, than in patients with more severe disease.
- The mechanism is **less predictable than that of CPAP**, even in the longer term. An appliance can be used alone or in combination with CPAP or positional therapy where it is not fully successful alone, and can reduce the side effects of CPAP in patients troubled by uncomfortably high pressures.

Efficacy and effect

- Success rate up to ~80% in mild OSA; ~60% in more severe cases
- Long-term efficacy is good to about 10 years, but real-world long-term effectiveness probably declines because the natural history of the disease is deterioration in AHI
- Reduces AHI and improves nightly oxygenation
- Increases the percentage of slow-wave sleep; increases REM sleep duration
- Snoring is controlled more effectively by an appliance than by an intraoral placebo, though CPAP is more effective than an appliance. **Persistent snoring during appliance treatment has been related to insufficient apnoea control and poor adherence** — a useful clinical marker
- Effect on daytime sleepiness is unclear in snorers and mild-to-moderate OSA; in more severe OSA there is a reduction compared with placebo, but CPAP is more effective. The low-arousal-threshold subgroup may experience more subjective advantage from an appliance
- Health effects: reduces blood pressure (similar effect to CPAP), improves the nightly dips in blood pressure, and improves endothelial and cardiac function. Improves mean sleep latency on the MSLT

Predicting treatment success

HIGH-YIELD — THE RESPONDER PROFILE

Domain	Features favouring success
Patient traits	Younger age; female sex; lower BMI and non-obesity; smaller neck circumference; lower baseline AHI (less severe disease)
Polysomnographic	Supine-position-dependent (positional) OSA; NREM-predominant or non-stage-dependent rather than REM-predominant disease; a lower required CPAP pressure (higher pressure implies a more collapsible airway and a poorer appliance response)
Physiological / endotype	Less collapsible upper airway; less sensitive ventilatory control system (lower loop gain); higher arousal threshold; lower response to arousal
Anatomical / cephalometric	Shorter soft palate length; longer maxilla; lower hyoid position and a decreased hyoid-to-mandibular-plane distance; greater angle between the cranial base and mandibular plane; retrognathic mandible with a narrow SNB and wider SNA angle
Endoscopic	Verified widening of the pharynx visualised by nasendoscopy on advancement; **tongue base collapse** predicts success. Conversely, complete concentric palatal collapse and complete laterolateral oropharyngeal collapse indicate likely treatment deterioration. Absence of a **pterygomandibular raphe** (present in only ~64% of people) predicts a better response — greater maximal advancement, greater airway change with advancement, greater AHI reduction and lower likelihood of non-response. **Inspiratory tongue movement** with mandibular advancement may be beneficial (responder), minimal, or counterproductive (non-responder). Combining structural and functional assessment — a more posteriorly located tongue plus a more collapsible airway — identifies responders
Trial-based	Sufficient AHI reduction during an overnight test with a **remotely controlled mandibular positioner** — the single best predictor available (rated "excellent"), against which clinical and craniofacial prediction is "poor" and nasopharyngoscopy "fair to good"
Adverse	Weight increase during treatment has been related to treatment failure. Differences between patients in the ability to advance the mandible modify the potential to reduce or eliminate obstructive events

The essential caveat: patient factors alone are **not** a good predictor of efficacy, and demographic, anthropometric and polysomnographic data only weakly inform appliance efficacy. Around two thirds of patients achieve a clinically meaningful effect (OSA at least halved) and one third are unlikely to benefit clinically — hence the need for prediction methods, and hence the requirement for objective verification rather than assumption.

5.3 Indications and contraindications

Covered in detail at 3.3; the essentials are repeated here because this is a separate blueprint line.

Indications	Contraindications
• Mild to moderate OSA — a primary treatment option • Moderate to severe OSA where CPAP cannot be tolerated or adhered to • Snoring — often the only reason the patient attended a sleep clinic in the first place. Non-apnoeic snorers without daytime symptoms tolerate CPAP poorly: most agree to try it but the majority give up, so an appliance is a good alternative	• Insufficient dentition or inadequate retention (including implant-retained cases) • Severe periodontal disease; loose teeth • Severe pre-existing or active TMJ disease or pain • Inadequate mandibular protrusion (<6 mm) • Severe bruxism • Poor oral hygiene, significant caries • Severe gag reflex • Undergoing significant dental or orthodontic work • Anterior open bite (no posterior clearance) • Severe OSA is a relative contraindication only — CPAP should be trialled first, but an appliance is appropriate if CPAP fails

5.4 Appliance designs, materials and fabrication

EXAM TRAP — THE BOIL-AND-BITE DEVICE

Do not use off-the-shelf, boil-and-bite, non-individualised appliances. They sit in a fixed position and cannot be titrated reliably, fit poorly with inadequate retention, are poorly tolerated, and carry the potential for undesirable dental side effects. A boil-and-bite device also fails the AADSM definition of an effective appliance on at least three counts — reliable titration, verifiable setting and stable retention. Self-fitted appliances are, at most, temporary devices; they do not fit well and cannot be worn long term.

The design taxonomy — six mechanism classes

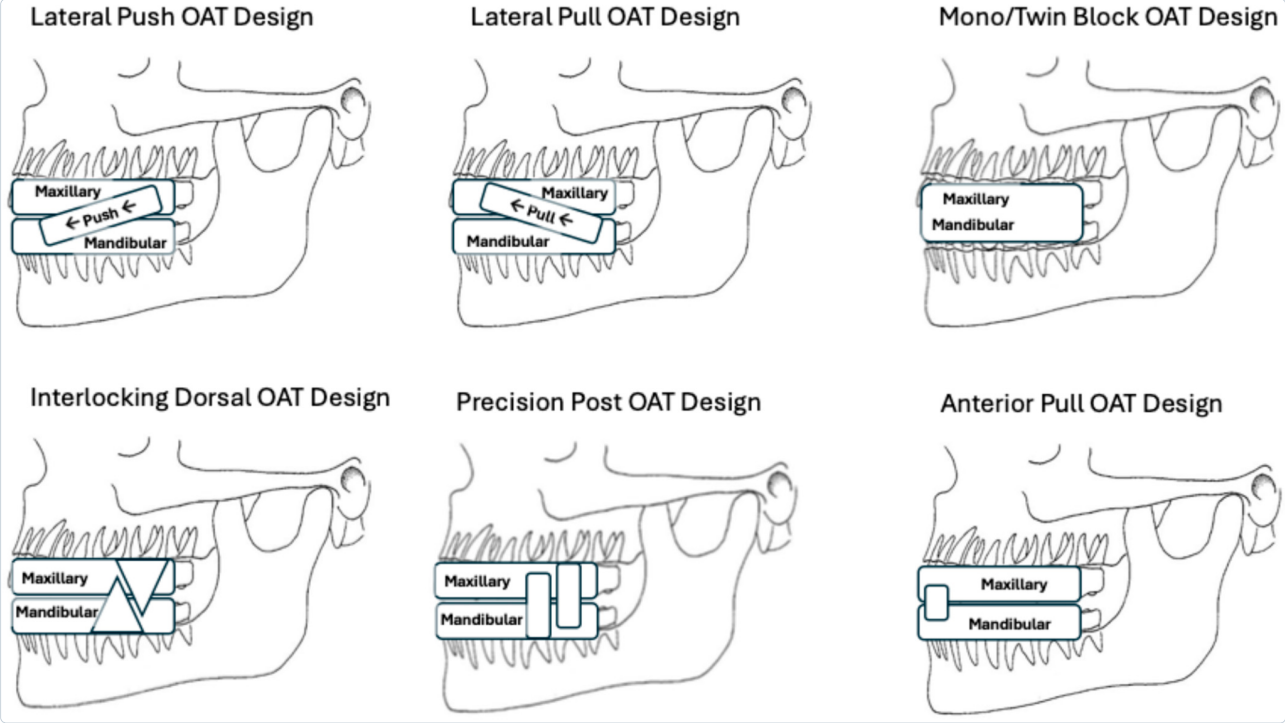


Figure 5.2 — The six oral appliance design categories, each shown as a maxillary and mandibular overlay pair with its coupling mechanism: Lateral Push, Lateral Pull, Interlocking Dorsal, Precision Post, Mono/Twin Block and Anterior Pull.

Why it is examinable: this classification lets you answer any design question by mechanism rather than by brand name, which is what examiners want. Learn the six names, how each repositions the mandible, and how each is titrated.

Design	Mechanism	Titration method and comment
Lateral Push	Metal, piston-like rod mechanisms — **Herbst arms** — on each side of the maxillary and mandibular overlays. Repositions the mandible by *pushing* the mandibular overlay forward, using the maxillary overlay as an anchor	Turning displacement screws embedded in the metal Herbst rods. Robust and strong with fine titration (~0.1 mm); metal connectors may impinge on soft tissue; the mandible may drop; requires good retention on both arches
Lateral Pull	Elastomeric or nylon straps on each side of the overlays. Repositions the mandible by *pulling* the mandibular overlay forward, using the maxillary overlay as anchor	Swapping straps of different lengths. Small and well documented, but the strap fractures or stretches easily and is imprecise; titration is gross (0.5–1 mm); requires good retention, otherwise the less retentive component may be lifted by the strap

Design	Mechanism	Titration method and comment
Interlocking Dorsal	Angled dorsal posts extending occlusally from each side of the mandibular overlay, interlocking with blocks affixed to the maxillary overlay to reposition and stabilise the mandible	Turning displacement screws embedded in the block mechanisms — for example 10 screw turns yields +1 mm of repositioning. An *unattached* mechanism
Precision Post	Dual posts extending occlusally from each side of both overlays, interlocking perpendicularly to guide, reposition and stabilise the mandible at the prescribed location	Iteratively swapping a maxillary or mandibular overlay for a new one containing a different repositioning setting (e.g. +1 mm)
Mono / Twin Block	*Mono-block*: the two overlays are fused at their occlusal surfaces. *Twin block*: mated by interlocking block components affixed to the occlusal surfaces	Swapping an overlay for a new one at a different setting. Mono-block can be more effective than prefabricated devices and gives good advancement, but restricts jaw movement and can cause TMJ discomfort
Anterior Pull	Anterior fixation mechanism — typically a screw or nylon strap — repositioning the mandible using the opposing overlay as anchor	Turning the anterior displacement screw or swapping the strap. **Good for patients with poor dexterity** because the adjustment is at the front and easy to reach, with easy and fine titration (0.1–0.3 mm) and robust construction — **but the force is concentrated on the anterior teeth**, and it produces a greater vertical dimension, tongue irritation and loading on anterior teeth, and requires good retention

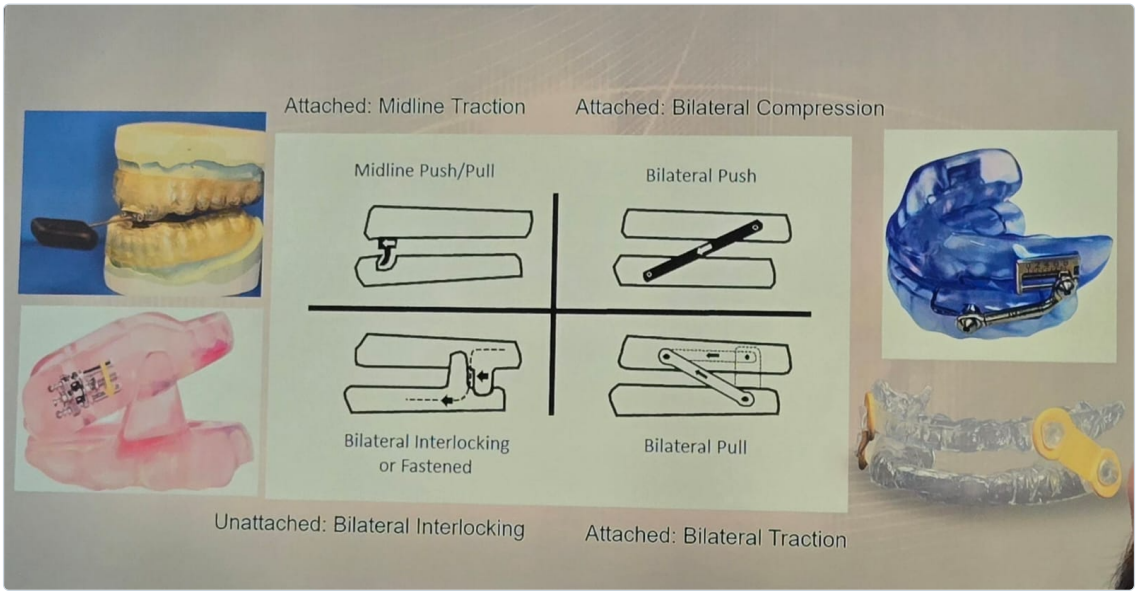


Figure 5.3 — Attachment and propulsion mechanisms, organised into four quadrants: attached midline traction (midline push/pull), attached bilateral compression (bilateral push), unattached bilateral interlocking (bilateral interlocking or fastened), and attached bilateral traction (bilateral pull), with clinical examples of each.

*Why it is examinable: this is the same taxonomy expressed as attached versus unattached and traction versus compression. The two clinically important consequences are that **traction devices impose lower stress** than compression devices — which produce greater early masticatory myalgia and residual tongue pain — and that **unattached (interlocking) designs permit the mandible to drop** in the supine position, which may reduce efficacy.*

Design principles that affect outcome

- The most common and most effective devices are **custom-made, titratable appliances that do not allow mouth opening**. They comprise an upper and a lower part with an intermaxillary adjustment mechanism allowing the mandible to be advanced successively to find an optimal position. The two parts may be fixed to one another by, for example, a screw mechanism, or the mandible may be allowed to move laterally and/or vertically.
- **Appliances that allow mouth opening are less effective** at reducing sleep apnoea; adding elastics to prevent opening halved apnoea frequency in the supine position. Elastics increase MAD success in positional OSA. A **fixed** device may therefore be better than a loose one, because the mandible drops supine.
- **Larger advancement gives larger efficacy; smaller vertical opening is preferred** — increasing the vertical opening increases pharyngeal collapsibility and has been associated with reduced appliance efficacy.
- **Coverage matters for side effects.** Fewer changes in overjet and overbite were seen with a soft elastomeric device covering parts of the alveolar processes in addition to all the teeth, compared with a hard acrylic appliance with full occlusal coverage fixed mainly to the teeth. However, **hard materials provide better protection against tooth movement**, and all teeth need to be covered — do not leave molars uncovered, or they will move or over-erupt. A flexible appliance without incisor coverage increased lower anterior irregularity compared with a rigid appliance with incisor coverage. A specific orthodontic appliance incorporating forces to counteract the posteriorly directed load on the upper front teeth had a positive effect on overjet change.
- **Between-device differences are clinically small.** There are clinically insignificant differences between appliances in symptoms or AHI. The overall conclusion is that the influence of device design on efficacy and side effects is **almost unknown**, and that side effects are a product of protruding the mandible to achieve a therapeutic effect rather than of specific appliance design.
- **Designs not recommended:** those marketed on theories of airway and bone remodelling, promoted by manufacturers, with no peer-reviewed scientific publication, no demonstrated superiority over current appliances, and non-adherence to AADSM guidance — typically bulky and not titratable.

Materials

Material	Advantages	Disadvantages
Acrylic (PMMA)	Good mechanical properties; well established; retention by **frictional resistance**; can be relined and repaired; used in 3D printing with DLP and SLA technologies	More breakage than nylon in bruxers; FDM-printed PMMA has shown lower strength; requires more vertical clearance (~5 mm)
Nylon / polyamide (PA12) — CAD-CAM	Minimal — lighter and smaller; robust and less likely to break, so **better for bruxers**; retention by **flexion into undercuts**; requires less vertical clearance (~3 mm); titration scale visible on the appliance, making adjustment easier; published scientific data	**Stains easily — polishing is important**; gross titration (0.5–1 mm); cannot be relined or repaired, so a large new restoration means a remake; requires good retention
Ethylene-vinyl acetate (EVA)	Good impact absorption; non-toxic; easily shaped — suitable for mouthguards	Additive manufacturing with EVA is limited; effectiveness in sleep apnoea appliances is debated
Cobalt-chrome alloy	Extremely durable — a good choice for bruxers	Casting is expensive; use limited to framework devices or as a coating
Titanium / CAD-CAM (e.g. O_2 vent devices)	Titratable; robust	Greater vertical dimension; very bulky; irritating to the tongue; requires good retention

Material selection factors: number of teeth; the relationship between the arches (class II versus class III); bruxism; TMD; gagging.

Fabrication techniques

- **Traditional:** dental casts and bite registrations used to create custom appliances. Well established, allowing precise custom fitting to the patient's anatomy.
- **3D printing / additive manufacturing:** digital models built layer by layer. High customisation, potentially reduced production time and cost, and in-house printing offering fast replacement. DLP and SLA offer advantages in accuracy, resolution and speed; the choice depends on the application and the desired workflow.
- **Digital scanning is more accurate** than conventional impressions. Check the scan before and after for a correct midline.

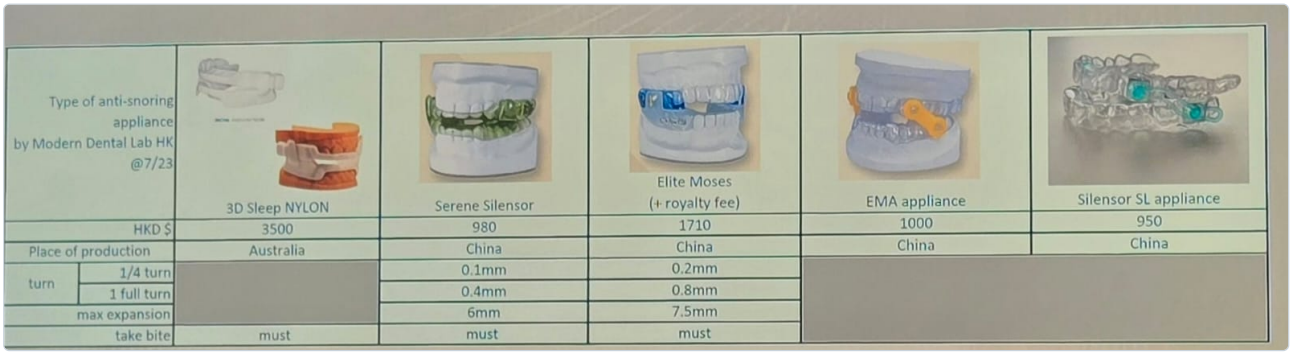


Figure 5.4 — Comparative specification of commercially available appliances from one laboratory, showing appliance type, place of production, and — the clinically useful columns — **the advancement per quarter turn and per full turn, and the maximum expansion**: 0.1 mm per quarter turn and 0.4 mm per full turn with a 6 mm maximum for one design, and 0.2 mm/0.8 mm with 7.5 mm maximum for another; and whether a construction bite is mandatory.

Why it is examinable: these are the numbers you use at the chairside. Know that adjustment increments are typically 0.1–0.4 mm per turn (and as fine as 0.2 mm per turn on an anteriorly adjusted screw), that the effective-appliance definition requires increments ≤1 mm and a range ≥5 mm, and that maximum expansion ranges of about 6–7.5 mm are typical. Pricing and availability are regional and not examinable.

5.5 Selecting the best device for this patient

TARGETED DOCUMENTATION — THE "AT RISK ANATOMICALLY" CHECKLIST

Record these features explicitly at the initial examination. They justify your treatment plan, form the medicolegal baseline, and are what you will be asked to list in a viva:

- Micrognathia / retrognathia
- Deep palatal vault
- Large tongue — scalloping is the practical sign, since size is difficult to define
- Modified Mallampati class 3 or 4
- Tonsil grade 3 or 4 — check by depressing the tongue and asking the patient to say "ah"
- Thick neck
- Obesity
- Long airway (long neck)
- Low hyoid — hyoid to mandibular plane distance >15 mm

Consideration	How it drives selection
Dental factors	Sufficient healthy teeth and a suitable bite are crucial for a MAD. Assess the number of functional teeth (at least 10 per arch on stricter criteria; ≥6 on others), mobility (teeth should not be mobile), spacing, curve of Spee, and whether teeth are heavily restored. A **deep curve of Spee** means insufficient vertical dimension at the second molars — you may choose not to cover the sevens, and should ask the laboratory accordingly
Arch relationship	Class II versus class III alters both the mechanics and the direction of unwanted tooth movement. An anterior open bite precludes a MAD because there is no posterior clearance at all
Jaws and skeletal pattern	Severe maxillomandibular deformity may require specialised treatment. Retrognathia favours an appliance and is a specific problem for CPAP because of mask leakage

Consideration	How it drives selection
Mouth and tongue	A tongue retaining device is an option for patients with missing teeth or who cannot tolerate a MAD because of jaw alignment
Bruxism	Favours nylon or cobalt-chrome for durability, and a two-piece design
TMD and gagging	Traction rather than compression mechanisms; minimise vertical opening and bulk
Dexterity, cognition, visual acuity, range of motion	An anterior adjustment mechanism suits poor dexterity — the adjustment is accessible at the front — accepting that force is then concentrated on the anterior teeth
Nasal patency	Assess and treat first. It does not itself treat the OSA but it **improves the treatment outcome**. Ask specifically about allergic rhinitis and treat it (referring to ENT) before appliance therapy; treatment may take time but you need not wait for complete resolution
Sleep study findings	Severity and positional dependence. Adjustable MADs have higher response rates than fixed devices
Patient preference	Perceived comfort, ease of use, durability, financial considerations. **The appliance should not be selected solely because of the requirements of governmental or private insurers** — the choice belongs to the qualified dentist and the patient

5.6 Bite registration: rationale and technique

HIGH-YIELD — THE THREE COMPONENTS OF THE CONSTRUCTION BITE

Component	Purpose	Typical approach
Horizontal (protrusion)	Opens the airway by forward mandibular positioning; the principal therapeutic variable	Start at **50–70%** of maximum protrusion. AADSM 2025: a protrusive range of **25–75%** lies within both the comfort and the therapeutic range. Do not exceed ~75% initially
Vertical (bite opening)	Provides sufficient interocclusal space for the appliance without causing discomfort or jaw fatigue. A slight increase can improve airway patency, but it must be balanced against muscle strain and joint loading	**~2 mm interincisal** (a moderate 1–3 mm increase is typical). Aim for posterior clearance rather than a raised bite: nylon needs ~3 mm, acrylic ~5 mm, and some designs require up to 6 mm, which is very difficult to achieve
Lateral (midline)	Accommodates mandibular asymmetry or deviation, and optimises comfort and function	Adjust as needed. The midline should be **reproducible** at both centric and protrusion — it need not be identical, but it must be reproducible

The rationale, and why not maximum protrusion

- The acceptable range of mandibular advancement for an individual patient is **not known**. Maximum protrusion produces the greatest airway enlargement, but this is not an adequate basis for setting the appliance because it causes TMJ and muscle discomfort.
- Classical practice: occlusal registration at **60% of maximum mandibular protrusion with 2 mm of vertical separation** of the upper and lower incisors. Other conventions advance the mandible to about 75% of maximum protrusion, or take a wax interocclusal record with the mandible in the most protrusive position the patient can comfortably maintain. A practical rule is to **start at 50% and do not go beyond 75%**.
- A previously proposed concept of a consistent between-patient "threshold" of **70% of maximal comfortable advancement** for optimal outcome has been challenged by the observation that airway collapsibility improves in *non-responders* during the initial range of advancement (0–50%), whereas in *responders* the majority of improvement in collapsibility occurs close to the maximum possible advancement. In other words, the dose–response curve differs between phenotypes.

- Excessive protrusion does not guarantee improved efficacy, may affect compliance, and in some instances **overprotrusion may worsen the sleep-disordered breathing**.
- The degree of subsequent bite change is influenced by appliance design, the initial type of bite and the treatment duration — as well as by the amount of protrusion.

Using the George Gauge — a step-by-step method

1. **Determine the protrusive range.** Record the most retruded and the most protruded readings and add their absolute values. For example, −3 and +7 gives a range of 10 mm; −4 and +5 gives 9 mm.
2. **Calculate the target.** Take the bite at 50–70% of that range. For 60% of 10 mm = 6 mm: starting from −3, the bite is registered at +3.
3. **Select the bite fork for vertical opening.** The **2 mm grey** fork is the usual choice. The **5 mm white** fork is usually too much opening — it promotes mouth breathing and inhibits lip seal.
4. **Seat and check.** Ensure the blue incisal guide clamps the lower incisors firmly. Check the midline: it should be the same and reproducible at centric and at protrusion.
5. **Register.** Do not inject bite registration material into the area of the lower incisor clamp.
6. **Check posterior vertical clearance**, especially in a deep bite case. If clearance is insufficient — usually at the second molars — ask the laboratory not to cover the sevens.
7. **Consider a centric occlusion record as well**, for fabrication of a morning aligner (Topic 10).

Bite registration techniques

Technique	Description
Construction bite devices	Mechanical devices allowing precise control and measurement of mandibular position in three dimensions, using guides and rulers to document numerically the amount of advancement and vertical opening. The George Gauge is the archetype
Neuromuscular (TENS)	Transcutaneous electrical nerve stimulation to relax the muscles, allowing the mandible to be positioned in a relaxed physiological posture before recording, reducing nociceptive input and potentially improving accuracy by minimising muscle tension
Phonetic	Uses patient speech — phonation of particular sounds — to find an optimal mandibular position that keeps the airway open without excessive protrusion or opening
Clinical measurement and simulation tools	Bite gauges and mandibular positioning simulators to determine and replicate the ideal therapeutic position reproducibly
Digital and virtual	Digital scanning and computer-aided design to capture jaw relationships; may outperform traditional methods in accuracy and efficiency

Getting the bite registration right is crucial for successful therapy and compliance: it allows the design of a comfortable appliance that works from the first night, reducing the need for multiple adjustments. Note that under AADSM 2025 standards, **impressions and the construction bite record should be completed in person** — telehealth may be used for screening and many follow-up visits, but not for these.

5.7 Impression materials and the requirements for accuracy

Material	Properties and use
Addition silicone (polyvinyl siloxane, PVS)	Excellent accuracy and dimensional stability — a popular choice. An elastomer, so returns to shape after deformation and is well suited to undercuts
Polyether	Highly accurate and versatile, with consistent results and good tear strength; hydrophilic, which is useful in moist conditions and challenging situations
Polysulfide	

Material	Properties and use
	One of the older elastic materials, used for partial denture and appliance impressions where detail is critical
Condensation silicone	Generally less dimensionally stable than addition silicone
Alginate (irreversible hydrocolloid)	Easy to use and handle; commonly used for diagnostic models and study casts because it captures general oral structures adequately. Poor dimensional stability — must be poured promptly
Digital intraoral scan	More accurate than conventional impressions; avoids material handling and storage issues; supports in-house printing. Verify the midline before and after scanning

Requirements for an accurate impression

- **Accuracy and detail reproduction** — some materials reproduce detail as fine as 20–25 µm
- **Dimensional stability** — shape and size must remain consistent from removal until the cast is poured
- **Elasticity and tear strength** — must stretch and recover without tearing on removal, especially around undercuts and deep grooves
- **Wettability** — ability to spread over and contact moist oral tissues, crucial for detail capture
- **Biocompatibility** — non-toxic, non-irritant, safe intraorally
- **Ease of handling and setting** — easy to mix and manipulate, with a suitable working time but a short setting time to minimise discomfort
- **Appropriate viscosity and thixotropy** — held in the tray, yet fluid enough to adapt to tissues and capture detail
- **A rigid tray** for support
- **Moisture control** — proper isolation of the working area, since moisture interferes with wetting and detail capture
- **Clinical technique** — the operator's knowledge and application of the chosen technique matter as much as the material

For appliance fabrication the impression or scan must record the **upper and lower dentition plus approximately 3 mm of gingiva**, together with the protrusive construction bite record.

5.8 Writing the laboratory request

> **A WORKED EXAMPLE OF AN APPLIANCE SPECIFICATION**
>
> "Upper and lower removable clear acrylic plates with full occlusal coverage fitted to both dental arches, 1.5–2 mm thickness. Two acrylic flanges, one on either side of the palatal aspect of the upper molars, fitting into slots cut into the acrylic on the lingual aspect of the lower plate in the molar region, to prevent posterior movement of the mandible while permitting free opening of the mouth. Two 10 mm screws to allow incremental anterior advancement of the lower plate."

The elements of a complete laboratory prescription

Section	Content
Date	Date of prescription
Prescribing clinician	Full name; dental licence or registration number; contact telephone and email; practice address
Laboratory information	Name, address, telephone
Patient information	Full name; identification or record number; age
Type of appliance	e.g. mandibular advancement device, named design or mechanism class
Material specification	Acrylic type, nylon, metal components or other materials as applicable
Vertical dimension / opening	

Section	Content
	Specify the vertical increase in millimetres (e.g. 2 mm interincisal); note any required posterior clearance
Mandibular protrusion setting	Specify the desired protrusion — e.g. "60% of the protrusive range" or an absolute advancement in millimetres, and the George Gauge reading used
Retention features	Clasps, precision attachments, undercut engagement or other retention mechanisms
Occlusal considerations	Whether the appliance should cover the occlusal surfaces fully or selectively — for example, **"do not cover the upper and lower second molars"** where clearance is inadequate — and any bite adjustments
Delivery instructions	Polishing, adjustments, special finishing
Additional notes	Special patient considerations, bite registration details, whether impressions or scans are enclosed
Attachments	Impression models, bite registrations, digital scans, photographs
Signature and date	Clinician signature

Clear specification of protrusion, vertical dimension and retention is the essential core — these three determine whether the appliance works and whether it is comfortable. Delivery typically follows 2–3 weeks after the impression.

5.9 Delivering a retentive device

Stage	Actions
1. Pre-delivery preparation	**Verify the laboratory prescription** — confirm appliance type, material, vertical dimension increase and protrusive setting against the titration plan. **Inspect the appliance** — no bubbles, rough edges or distortion in the acrylic; movable components (screws, connectors) operate smoothly. **Sanitise and lubricate** — disinfect per manufacturer guidelines and apply a water-based lubricant to polished surfaces for initial seating
2. Seating and retention check	Seat the appliance gently, engaging the posterior bulge or clasps before the anterior structures. Assess retention: with the appliance in place, ask the patient to swallow, speak (a "pa–ta–ka" sequence) and open wide; then apply gentle traction on each side with gloved fingers — the device should resist displacement without discomfort. If retention is insufficient, evaluate undercut engagement and adjust acrylic **in stress-bearing areas only**
3. Vertical dimension verification	Measure the vertical opening — use a Willis gauge or measure the vertical dimension of occlusion intraorally from a fixed landmark (e.g. nose tip to chin) both with and without the appliance. Confirm the interincisal increase, typically **2–4 mm over habitual closing**, is tolerated. Ask the patient to relax the jaw; there should be no strain, pulling or muscle fatigue
4. Protrusion setting confirmation	Verify the advancement using the built-in millimetre scale or a protrusion gauge, and confirm it matches the prescription. Functional testing: ask the patient to close in the protruded position and check for **even bilateral posterior contacts without slide back to the retruded position**. Observe swallowing and speaking — the mandible should remain in the advanced position
5. Comfort and soft tissue evaluation	Ask about pressure points, gag reflex triggers and tongue displacement. Inspect soft tissues for blanching of mucosa, sore spots on the alveolar ridges and adequate clearance from the gingiva. Confirm no premature occlusal interferences — all teeth should contact evenly or as designed
6. Final adjustments and education	Remove high spots with a round bur, polish and reassess retention. Demonstrate insertion and removal technique, handling over a soft surface to prevent damage. Review cleaning, storage and the adjustment protocol. Provide emergency contact details for discomfort, loosening or breakage. **Schedule the first review at 2–4 weeks** (AADSM: reassess within 30 days of insertion)

It is recommended that a **written copy of the instructions, the delivery form and the warranty information** be signed and dated by the patient and a team member, with one copy given to the patient and the other retained in the record. At delivery, also consider how to maximise comfort and minimise the development of dental changes — including occlusal irregularities and interdental spacing — and take appropriate measures to attenuate jaw discomfort

and muscle fatigue, such as morning exercises and a morning repositioning device with associated palliative care (Topic 10).

5.10 Home care instructions

Domain	Instruction
Device use	Wear the appliance **every night** as directed, for the whole duration of sleep, to achieve the best result. Contact the practice immediately if there is discomfort — the appliance should fit properly. Avoid eating or drinking anything other than water for at least **two hours before bedtime** to reduce the risk of acid reflux
Cleaning	Rinse immediately after removal to wash away saliva and debris. Clean daily each morning with a **soft-bristled toothbrush and mild soap** or a denture cleaner, using **cool or warm — never hot — water**, as hot water warps the device. **Avoid toothpaste**: it is too abrasive and causes scratches that harbour bacteria, dull the appliance and lead to odour. Disinfect weekly in a denture-safe soak per manufacturer instructions
Handling and storage	Always remove over a soft surface or a basin of water to prevent drop damage. Allow the appliance to **air dry** before placing it in its ventilated case, to prevent moisture build-up. Store dry in the case when not in use — never wrapped in tissue or loose in a pocket. Keep away from direct sunlight, heat sources, pets and children
Oral health	Brush and floss regularly to prevent dental problems and prolong the life of the appliance
Morning routine	Perform gentle jaw exercises — for example resting the chin on a fist to let the jaw settle, or curling the tongue back to find the retruded position — to reduce morning stiffness and help re-establish the habitual bite. Use the morning occlusal guide if provided, for about 15–20 minutes after waking
When to contact the practice	Immediately for discomfort or pain. Also for any change such as jaw pain, a change in the bite, or worsening snoring

NUMBERS TO MEMORISE — TOPIC 5

- Initial protrusion 50–70% of maximum (AADSM 2025: 25–75% acceptable); start at 50%, do not exceed 75%; the challenged "70% threshold" concept
- Vertical: 2 mm grey George Gauge fork (5 mm white fork usually too much); interincisal increase 2–4 mm; posterior clearance nylon ~3 mm, acrylic ~5 mm, some designs 6 mm
- Protrusive capacity ≥5–8 mm (normal 10–12 mm); maximum opening 25–40 mm (normal 40–60 mm)
- Dentition: ≥6 teeth per arch (stricter: 8–10 sound teeth per arch with adequate bone support)
- BMI ≥35 predicts failure
- Titration increments 0.1–0.4 mm per turn; 10 turns = +1 mm on interlocking dorsal designs; effective appliance needs ≤1 mm increments and ≥5 mm range
- Impression: upper and lower dentition + ~3 mm gingiva; detail reproduction to 20–25 µm; acrylic plate 1.5–2 mm thick
- Delivery 2–3 weeks after impression; first review 2–4 weeks after delivery (within 30 days)
- Follow-up: every 6 months in year 1, then at least annually
- Success up to ~80% in mild OSA, ~60% in severe; ~⅔ meaningful benefit, ~⅓ no benefit
- Pterygomandibular raphe present in ~64% of people — absence predicts better response
- No food or drink but water for ≥2 h before bed

SELF-TEST — TOPIC 5

1. State the six recommendations of the 2015 AASM/AADSM oral appliance guideline.

(i) Sleep physicians should prescribe an oral appliance, rather than no treatment, for adults who request treatment for primary snoring without OSA. (ii) Where an adult with OSA is intolerant of CPAP or prefers alternative therapy, sleep physicians should prescribe an oral appliance rather than no treatment. (iii) Qualified dentists should use a **custom, titratable** appliance over non-custom oral devices. (iv) Sleep physicians should conduct **follow-up sleep testing** to improve or confirm treatment efficacy, rather than follow up without testing. (v) Sleep physicians and qualified dentists should instruct patients to return for **periodic office visits** with both. (vi) Qualified dentists should provide **oversight** of appliance therapy to survey for dental-related side effects and occlusal changes and reduce their incidence.

2. Walk me through taking a construction bite with a George Gauge.

First establish the protrusive range by recording the most retruded and most protruded readings and adding their absolute values — for example −3 and +7 gives 10 mm. Calculate the target as 50–70% of that range; 60% of 10 mm is 6 mm, so starting from −3 the bite is registered at +3. Select the vertical: the 2 mm grey fork is standard, since the 5 mm white fork usually gives too much opening and promotes mouth breathing while inhibiting lip seal. Check the blue incisal guide clamps the lower incisors firmly, and check the midline — it should be reproducible at both centric and protrusion, though not necessarily identical. Register the bite, avoiding injecting material into the lower incisor clamp area. Then check posterior vertical clearance, particularly in a deep bite, and if clearance is inadequate at the second molars ask the laboratory not to cover the sevens. Consider also taking a centric occlusion record for a morning aligner.

3. Why not simply set the appliance at maximum protrusion?

Maximum protrusion produces the greatest airway enlargement, but it is not an adequate basis for the setting because it causes TMJ and muscle discomfort, threatens compliance, and increases dental side effects. Excessive protrusion has not been shown to guarantee improved efficacy and in some instances **overprotrusion may worsen the sleep-disordered breathing**. The acceptable range for an individual patient is not known, so we start at 50–70% of maximum (25–75% is within the comfort and therapeutic range) and titrate incrementally to the minimum effective position — leaving some leeway rather than going to 75% immediately. The dose–response relationship also differs between phenotypes: in responders most of the improvement in collapsibility occurs near maximal advancement, whereas in non-responders collapsibility improves early, over the initial 0–50%.

4. Name and describe the six appliance design categories.

Lateral Push — bilateral metal Herbst rods pushing the mandibular overlay forward against the maxillary anchor, titrated by embedded displacement screws; robust with fine titration but metal may impinge on soft tissue. **Lateral Pull** — bilateral elastomeric or nylon straps pulling the mandibular overlay forward, titrated by swapping strap lengths; small but the strap fractures and titration is gross. **Interlocking Dorsal** — angled dorsal posts from the mandibular overlay interlocking with maxillary blocks, titrated by screws in the blocks (10 turns = +1 mm); unattached, so the mandible can drop. **Precision Post** — dual perpendicular interlocking posts on both overlays, titrated by swapping an overlay for one at a new setting. **Mono/Twin Block** — overlays fused (mono) or mated by interlocking blocks (twin), titrated by swapping overlays; restricts jaw movement and may cause TMJ discomfort. **Anterior Pull** — an anterior screw or strap repositioning the mandible against the opposing overlay; ideal for poor dexterity as adjustment is accessible at the front, with fine titration, but it concentrates force on the anterior teeth and increases vertical dimension. Expressed differently, the same taxonomy divides into attached versus unattached and traction versus compression, with traction devices imposing lower stress.

5. Compare nylon and acrylic appliances.

Nylon (polyamide, usually CAD-CAM) is lighter, smaller and more robust with less breakage, making it preferable for bruxers; it retains by flexion into undercuts, needs less posterior clearance (~3 mm), and has a visible titration scale making adjustment easier. Its drawbacks are that it stains readily so polishing is important, titration is gross at 0.5–1 mm, and it cannot be relined or repaired — a large new restoration means a remake. Acrylic retains by frictional resistance, can be relined and repaired, and permits finer screw adjustment bit by bit, but breaks more readily in bruxers and needs more clearance (~5 mm). Cobalt-chrome is extremely durable and good for bruxers but expensive to cast and limited to framework designs.

6. What do you check at the delivery appointment?

Six stages. Pre-delivery: verify the prescription against the plan, inspect for bubbles, rough edges, distortion and smooth screw operation, then disinfect and lubricate. Seating and retention: seat posteriorly first, ask the patient to swallow, say "pa–ta–ka" and open wide, then apply gentle bilateral traction — it should resist displacement without discomfort; adjust acrylic in stress-bearing areas only. Vertical dimension: measure with a Willis gauge or from a fixed landmark with and without the appliance, confirm the 2–4 mm interincisal increase is tolerated and that the relaxed jaw has no strain. Protrusion: verify against the millimetre scale, check even bilateral posterior contact with no slide back to retrusion, and observe swallowing and speaking. Comfort and soft tissue: check pressure points, gag triggers, mucosal blanching, ridge soreness, gingival clearance and premature contacts. Finally: adjust and polish, demonstrate insertion and removal over a soft surface, give written home care instructions and warranty signed by patient and team member, provide emergency contact details, and book review within 30 days.

7. Your patient has a deep bite with a marked curve of Spee. What are the implications?

A deep curve of Spee means there is insufficient vertical clearance for the appliance at the second molars. The options are to increase the vertical dimension — which is undesirable, since greater vertical opening increases pharyngeal collapsibility and reduces efficacy, and promotes mouth breathing — or, preferably, to instruct the laboratory **not to cover the second molars**. The trade-off is that uncovered molars may move or over-erupt, so this must be documented, explained to the patient and monitored at review. Material choice also helps: nylon requires only about 3 mm of clearance compared with about 5 mm for acrylic. Note separately that an anterior open bite precludes a mandibular advancement device altogether, because there is no posterior clearance at all.

REFERENCES — TOPIC 5

1. Ramar K, Dort LC, Katz SG, et al. Clinical practice guideline for the treatment of obstructive sleep apnea and snoring with oral appliance therapy: an update for 2015. *J Clin Sleep Med*. 2015;11(7):773–828.
2. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2).
3. Mogell K, Blumenstock N, Mason E, Rohatgi R, Shah S, Schwartz D. Definition of an effective oral appliance for the treatment of obstructive sleep apnea and snoring: an update for 2019. *J Dent Sleep Med*. 2019;6(3).
4. Sheats R, Essick G, Grosdidier J, et al. Identifying the appropriate therapeutic position of an oral appliance. *J Dent Sleep Med*. 2020;7(4).
5. Marklund M, Verbraecken J, Randerath W. Non-CPAP therapies in obstructive sleep apnoea: mandibular advancement device therapy. *Eur Respir J*. 2012;39(5):1241–1247.
6. Okuno K, Pliska BT, Hamoda M, Lowe AA, Almeida FR. Prediction of oral appliance treatment outcomes in obstructive sleep apnea: a systematic review. *Sleep Med Rev*. 2016;30:25–33.
7. Sutherland K, Takaya H, Qian J, Petocz P, Ng AT, Cistulli PA. Oral appliance treatment response and polysomnographic phenotypes of obstructive sleep apnea. *J Clin Sleep Med*. 2015;11(8):861–868.
8. Bamagoos AA, Cistulli PA, Sutherland K, et al. Polysomnographic endotyping to select patients with obstructive sleep apnea for oral appliances. *Ann Am Thorac Soc*. 2019;16(11):1422–1431.
9. Aarab G, Lobbezoo F, Hamburger HL, Naeije M. Effects of an oral appliance with different mandibular protrusion positions at a constant vertical dimension on obstructive sleep apnea. *Clin Oral Investig*. 2010;14(3):339–345.
10. Vroegop AV, Vanderveken OM, Van de Heyning PH, Braem MJ. Effects of vertical opening on pharyngeal dimensions in patients with obstructive sleep apnoea. *Sleep Med*. 2012;13(3):314–316.
11. Walker-Engström ML, Ringqvist I, Vestling O, Wilhelmsson B, Tegelberg A. A prospective randomized study comparing two different degrees of mandibular advancement with a dental appliance in treatment of severe obstructive sleep apnea. *Sleep Breath*. 2003;7(3):119–130.
12. Tegelberg Å, Walker-Engström ML, Vestling O, Wilhelmsson B. Two different degrees of mandibular advancement with a dental appliance in treatment of patients with mild to moderate obstructive sleep apnea. *Acta Odontol Scand*. 2003;61(6):356–362.
13. Remmers J, Charkhandeh S, Grosse J, et al. Remotely controlled mandibular protrusion during sleep predicts therapeutic success with oral appliances in patients with obstructive sleep apnea. *Sleep*. 2013;36(10):1517–1525.
14. Chan ASL, Sutherland K, Schwab RJ, et al. The effect of mandibular advancement on upper airway structure in obstructive sleep apnoea. *Thorax*. 2010;65(8):726–732.
15. Vanderveken OM, Devolder A, Marklund M, et al. Comparison of a custom-made and a thermoplastic oral appliance for the treatment of mild sleep apnea. *Am J Respir Crit Care Med*. 2008;178(2):197–202.
16. Vroegop AV, Vanderveken OM, Dieltjens M, et al. Sleep endoscopy with simulation bite for prediction of oral appliance treatment outcome. *J Sleep Res*. 2013;22(3):348–355.
17. Cistulli PA, Gotsopoulos H, Marklund M, Lowe AA. Treatment of snoring and obstructive sleep apnea with mandibular repositioning appliances. *Sleep Med Rev*. 2004;8(6):443–457.

18. Almeida FR, Lowe AA. Principles of oral appliance therapy for the management of snoring and sleep disordered breathing. *Oral Maxillofac Surg Clin North Am*. 2009;21(4):413–420.
19. Sheats RD, Schell TG, Blanton AO, et al. Management of side effects of oral appliance therapy for sleep-disordered breathing. *J Dent Sleep Med*. 2017;4(4):111–125.

TOPIC 6

Assessing effectiveness and titrating the oral appliance

BLUEPRINT

1. Returning to the chief complaint in titration · 2. Reviewing signs and symptoms at follow-up · 3. Referral for objective testing · 4. Treatment sleep study protocols · 5. Efficacy, long-term effectiveness, limitations · 6. Adherence and compliance monitoring

*The examinable judgement: knowing **when to stop advancing** — a negotiation between symptom resolution, objective indices, tolerance and the accumulating dental cost of protrusion.*

6.1 Returning to the chief complaint

- **Subjective.** Monitor and collect information on the **resolution of the patient's sleep disorder symptoms** to assess efficacy and determine optimal titration: reduced snoring, improved daytime symptoms, improvement in the original complaint. **Within 30 days of insertion** and at regular intervals thereafter, reassess **using the same questionnaires and tools used at baseline** — otherwise change cannot be quantified.

THE SCOPE-OF-PRACTICE RULE FOR OBJECTIVE DATA

Objective data obtained by the dentist — portable monitor or pulse oximeter — are used **for titration purposes only**: not for follow-up assessment and not for diagnosis. The dentist must be able to interpret the results accurately, monitoring mainly the *changes* in respiratory sleep parameters with treatment. It is the physician's role to use portable monitors for check-ups and reassessments. Decisions about using home tests or oximetry, and the resulting data, should be made in concert with patient, medical provider and qualified dentist.

- **Protocol.** AHI improvement is **dose-dependent** on advancement. A representative published protocol: set at **50% of maximum protrusion**; **2-week** acclimatisation; then **6 weeks** of patient-led adjustment; where symptoms persist (snoring, sleepiness, apnoeas, hypopnoeas) advance **one to two increments (0.2–0.4 mm) per night**; continue until symptoms are adequately improved or further protrusion causes discomfort; **hold the vertical dimension constant** throughout.
- **Practical points.** Titration takes **~5 months, longer than CPAP**, but compliance is better once tolerated; the TMJ needs time to accustom to a more advanced position. **Symptoms improve before the AHI does.** Always leave leeway rather than going straight to 75%. Nylon is easier to titrate because the scale is visible; acrylic needs incremental screw turns. Titration may be by patient or dentist; completion usually 2–3 months. Home oximetry supports titration, and a wearable home test can be arranged once titration seems adequate; sequential PSG studies facilitate titration. Sobering observation: in long-term series all appliances had been **continuously advanced to compensate for the shift in dental occlusion** (Topic 10).
- **Reverting.** At each follow-up, ask about the chief complaint and assess side effects (tooth pain, jaw stiffness, TMJ discomfort). If symptoms worsen or significant new side effects appear, either return to a previous well-tolerated less advanced position, or reduce the size of future steps rather than reversing fully. Continue at the lower setting and re-evaluate next visit to decide whether advancement can resume. **Document all adjustments, feedback and side effects meticulously.**
- **Why revert:** to resolve unacceptable side effects; to balance efficacy against tolerance; and to avoid prematurely abandoning titration over minor side effects, which would give a suboptimal outcome.

6.2 Reviewing signs and symptoms at follow-up

- **Jaw and TMJ** — discomfort, stiffness, pain, clicking or sounds; usually improve, but persistent or severe symptoms need management or adjustment.
- **Dental discomfort and bite change** — tooth pain, gingival irritation, soreness, the bite feeling "off" or teeth not fitting as before; long-term use decreases overbite and overjet — monitor and record.
- **Salivation** — excessive salivation or dry mouth, common initially and often transient (salivation typically settles after a few weeks) but document and manage.
- **Appliance condition** — cracks, warping, wear affecting fit and efficacy; confirm retention and stability.
- **Effectiveness** — persistence or return of snoring, apnoea symptoms, sleepiness or fatigue; follow-up testing may be warranted.
- **Tolerance and adherence** — comfort with nightly use, and whether side effects are reducing use.
- **Approach:** record and discuss symptoms at each visit; reassure about transient side effects and educate on managing them; adjust or refer as needed; monitor dental and skeletal change periodically, especially in long-term users, to prevent irreversible problems; coordinate with the sleep physician, including discussion of alternatives if the appliance is intolerable or ineffective.

6.3 When to refer for objective testing

WHY SUBJECTIVE FEEDBACK ALONE IS INSUFFICIENT

Two facts make this non-negotiable: **appliances may worsen OSA in some patients**, and there are **known placebo effects** with this therapy — a patient may feel better while the AHI is unchanged or worse. Referral for follow-up evaluation by the referring physician, with PSG or sleep studies, is therefore **required** to verify efficacy and benefit.

- **Reasons to refer:** confirming efficacy (for moderate and severe OSA especially, since subjective report is unreliable); guiding further adjustment where symptoms have not fully resolved or the setting is uncertain; persistent symptoms; monitoring disease progression after significant weight change or new comorbidity; complex or severe cases — a mandibular advancement splint is a viable alternative to CPAP even in more severe OSA, **but follow-up PSG is mandatory in these patients**.
- **Not indicated: primary snoring**, unless symptoms worsen or fail to resolve.
- Maintain regular written communication with the physician about plan, progress and follow-up; the referring physician should be involved in ongoing evaluation, and the dentist should obtain a follow-up efficacy report. If results are unsatisfactory, extend titration or review the plan with the patient. Objective testing may also compare a trial device with the final appliance to confirm optimal protrusion.

6.4 Treatment sleep study protocols

TWO ACCEPTED PATHWAYS TO THE THERAPEUTIC POSITION

- **Signs and symptoms only:** initial mandibular positioning → resolution or improvement of signs and symptoms → **verification of efficacy by the physician**.
- **Multimethod positioning:** initial positioning → resolution or improvement of signs and symptoms → pulse oximetry, high-resolution oximetry or HSAT administered by the qualified dentist, and/or HSAT or PSG ordered by the physician → **verification of efficacy by the physician**.
- Both end in the same place. The dentist's objective data **supplement** the titration decision; they never replace physician verification. Advance until target values are attained, can no longer be improved, or the maximum comfortable position is reached.

- **Process.** (1) The qualified dentist selects, fits and titrates, adjusting to find the optimal protrusion. (2) The patient has a follow-up study — home test or attended PSG — wearing the appliance at the dentist-determined setting; a home test is common for confirming efficacy after initial fitting and titration, while attended PSG is more comprehensive and serves both titration and final assessment. (3) The sleep physician analyses the results, using the AHI to assess severity. (4) If AHI reduction is inadequate, further titration or adjustment is required; the study also identifies new or persistent adverse effects. (5) Where the calibrated position is subtherapeutic, physician and dentist discuss further calibration, adjunctive or alternative treatment — and in consultation with the physician a patient may elect to continue despite subtherapeutic results.
- **Advanced methods. Remotely controlled mandibular positioner** — objective single-night titration during PSG to identify the optimal advancement; the best-performing predictive method ("excellent") against clinical/craniofacial prediction ("poor") and nasopharyngoscopy ("fair to good"). **DISE** with a remotely controlled positioner allows titration under direct visualisation. Honest limitation: demographic, anthropometric and polysomnographic data only **weakly** inform efficacy — which is why trial-based methods outperform prediction from patient characteristics.
- Further titration, modification, redesign or remake as required. **Monitor monthly for the first six months** and at least annually thereafter for as long as the appliance is worn.

6.5 Efficacy, long-term effectiveness, limitations

- **Efficacy:** AHI <5 (cure) ~**37%** · AHI <10 ~**52%** · ≥50% reduction (success) ~**64%** · clinically meaningful effect ~⅔ · negligible improvement ~⅓ · overall quoted success 60–70% · one series reported "success" 50% and "response" 14% · cures mild–moderate OSA in 40–50% with significant improvement in a further 10–20% · snoring reduced ~45% but not eliminated · ESS 11.2 → 7.8 · patient-reported improvement in sleep quality and snoring 90–95% · apnoea controlled 50–70% depending on initial severity · blood pressure mean reduction ~4.2 mmHg · indirect comparison OA ~42%, CPAP ~75%, UPPP ~30%.

TRAP — WHY EFFICACY FIGURES VARY SO MUCH

Published figures range from 37% to 95% "success". The variation is almost entirely the **definition used**: cure (AHI<5), near-complete response (AHI<10 with ≥50% reduction), partial response (≥50% reduction), or symptomatic improvement. Always state the definition with the number — "about 64% achieve ≥50% reduction, but only about 37% reach AHI below 5" — and the answer reads as informed rather than recited.

- **Long-term effectiveness.** Health outcomes are **comparable** to CPAP at least short-term, **even in severe OSA**, with greater adherence to the splint; appliances also benefit blood pressure, endothelial function and left ventricular function. Long-term

efficacy good to ~10 years, with sustained effectiveness up to five years in some studies. But real-world effectiveness **declines**: some patients discontinue for poor subjective effect or side effects, continuing patients may have poorer objective effect, and the natural history is AHI deterioration with age and weight gain. Effectiveness is also vulnerable to bite change and age, requiring continuous re-evaluation.

- **Limitations.** Not as effective as CPAP for apnoea, snoring and daytime function — but with a definite role. Efficacy and side effects depend on type, protrusion, vertical opening and settings. Long-term dental change: reduced overbite and overjet, fewer occlusal contacts. Common short-term effects: jaw pain, oral discomfort, salivation, dry mouth. Treatment may be discontinued for ineffectiveness, discomfort or side effects. Biological change requires continuous monitoring and adjustment. Not ideal above BMI 35. Mechanism less predictable than CPAP.

"WHEN IS NOT ENOUGH ACTUALLY NOT ENOUGH?"

- **Persistent snoring, no other symptoms** — may be acceptable to the patient; but persistent snoring on treatment relates to insufficient apnoea control and poor adherence, so verify objectively before accepting it. **With other symptoms** — treat as inadequate control: re-titrate, verify objectively, consider combination or alternative therapy.
- **Residual AHI, no other symptoms** — weigh residual risk against the burden of escalation; AHI 5–15 carries only a small risk to health, so the question becomes whether you are treating a number or treating a patient. Consider **effective AHI**, accounting for untreated hours too. **With other symptoms** — clearly inadequate: further titration, combination therapy, or referral back for alternatives.

6.6 Adherence and compliance monitoring

DEFINITION

AADSM compliance — the appliance worn for a minimum of **≥80% per night**, timed from placement in the mouth to removal, on **≥5 nights per week**. Patients should sleep a minimum of **7 hours** and wear the appliance for the duration of sleep every night. Contrast the CPAP research convention of **≥4 hours per night**.

- **Data:** ~80% adherent at 1 month averaging ~6 h/night → ~67% at 6 months, then stable through 1 year. Mean objective use 6.1–6.7 h/night on ~83% of days per week; ~83% use ≥4 h/night; ~63% average >6 h/night; ~92% would be "compliant" by CPAP standards. Discontinuation ~9.8% (20–30% overall; ~7.5% specifically from side effects). Still wearing at ~33 months 55–68%; regular use at 31±18 months 76%; continuing over several years if properly constructed 50–70%; rate of regular users 82% averaging 6.7 h/day. MAD 6.5 vs CPAP 5.2 h/night. **Objective and subjective estimates are broadly similar for appliances** — a notable contrast with CPAP, where patients substantially over-report.
- **Better adherence:** younger age; larger baseline overjet; married or partnered; perceived benefit; device comfort; patient knowledge. The four most important factors determining treatment choice are **effectiveness, transportability, embarrassment and cost**.
- **Worse adherence:** psychological comorbidity; side effects, particularly dental discomfort and bite change; poor comfort or fit; **loss of the bed partner, or no further complaints of snoring** — because most patients wear the appliance for snoring, the **benefit is perceived by the bed partner while the side effects are experienced by the wearer**.
- **Technology:** embedded micro-sensors measuring temperature and wear time; some also track head position, and the most current appliances can incorporate oximetry. Compliance chips allow continuous remote upload to a secure cloud for real-time monitoring. Objective measurement with an embedded micro-sensor thermometer is safe and feasible. **Challenges:** saliva ingress, power supply, battery toxicity, fitting without added discomfort, material compatibility, verifiability.
- **Why it matters:** it **distinguishes device inefficacy from poor compliance** — the single most important diagnostic function, since without it you cannot know whether to advance the appliance or address behaviour; it permits optimised titration and better outcomes; it supports legal and insurance documentation where proof is mandatory, notably in **safety-sensitive transport occupations**; and it allows categorisation as fully compliant, improving over time, or non-compliant. Because untreated or subtherapeutic OSA carries long-term risk, **non-compliant patients should be referred to their medical provider to consider alternative therapies**.

The five steps of provision, first year onward

1. **Insertion, adjustment, titration and evaluation during the first six months of regular use** — monitoring symptom resolution to assess efficacy and determine optimal titration; portable monitor data for titration only, interpreted accurately.
2. **Referral for follow-up evaluation** by the referring physician, with PSG or sleep studies required to verify efficacy — because appliances may worsen OSA in some patients and placebo effects are known. Not indicated for primary snoring unless symptoms worsen or fail to resolve.
3. **Maintain regular written communication** with the physician and others about plan, progress and follow-up; obtain a follow-up efficacy report; if unsatisfactory, extend titration or review the plan.
4. **Further titration, modification, redesign or remake** as required; monitor monthly for the first six months and at least annually thereafter.

5. **Follow-up and reassessment after six months** to assess adherence and efficacy. Advance further if symptoms recur. **If maximum advancement is reached and symptoms persist, refer to the referring physician**, who may refer to a sleep specialist. Assess appliance integrity; it may need replacement depending on usage and hygiene — appliances last approximately **2–3 years**. For long-term efficacy, refer to the physician **every five years** for continuing assessment, as OSA worsens with age and weight gain.

NUMBERS — TOPIC 6

- AHI <5 ~37% · <10 ~52% · ≥50% reduction ~64% · ⅔ benefit, ⅓ none · snoring −45% · ESS 11.2→7.8 · BP −4.2 mmHg · 90–95% report improvement
- Titration: start 50%; 2-week acclimatisation; 6-week self-adjustment; 0.2–0.4 mm/night; total ~2–5 months; vertical constant
- Reassess ≤30 days; monthly first 6 months; then annually; physician every ~5 years; appliance life 2–3 years
- Compliance: ≥80% of night, ≥5 nights/week, ≥7 h sleep (CPAP ≥4 h). Adherence ~80% at 1 mo → ~67% at 6 mo; 6.1–6.7 h/night; 55–68% at 33 months; ~9.8% discontinue; ~7.5% from side effects

SELF-TEST — TOPIC 6

1. Describe a titration protocol from delivery to therapeutic position.

Set initially at 50% of maximum protrusion. Two weeks' acclimatisation, then ~six weeks of patient-led adjustment, advancing one to two increments of 0.2–0.4 mm per night while symptoms persist. Hold the vertical dimension constant. Continue until symptoms are adequately improved or further protrusion causes discomfort — remembering symptoms improve before the AHI, and overprotrusion may worsen the breathing. Review within 30 days using the **same** baseline questionnaires so change is quantifiable, then monthly for six months. Total titration usually two to five months, longer than CPAP because the TMJ needs time. Once adequate, arrange objective assessment and refer to the physician for verification.

2. Your patient reports complete resolution of snoring and feels much better. Is treatment successful?

Not yet established. Subjective feedback alone is insufficient: appliances can worsen OSA in some patients and placebo effects are known. Complete response requires symptom resolution **and** treatment AHI below 5. Arrange objective assessment — oximetry or a home test for titration purposes if lawful locally — and refer back to the sleep physician for verification with PSG or an attended cardiorespiratory study, appliance in place at the final setting. Exception: primary snoring, where follow-up testing is not indicated unless symptoms worsen or fail to resolve.

3. Residual AHI of 12 at maximum comfortable protrusion. What next?

Confirm the data are used appropriately — my data inform titration, the physician verifies efficacy. Check whether the position is genuinely maximal and comfortable and whether adherence is adequate; a residual AHI in a poorly adherent patient is a compliance problem, not a titration problem, which is exactly why objective adherence data matter. If maximum advancement is reached with persistent symptoms, refer back to the referring physician, who may refer to a sleep specialist. Options: combination therapy (appliance plus positional therapy, plus nasal EPAP, or plus CPAP at lower pressure), weight management, treatment of nasal obstruction, or a change of modality. Document that a patient may, in consultation with the physician, elect to continue despite a subtherapeutic result — that decision belongs to patient and physician jointly, not to me alone.

TOPIC 6

Assessing effectiveness and titrating the oral appliance

EXAMINER'S BLUEPRINT

1. Reverting to the patient's chief complaint, history, signs and symptoms in the titration process
2. Reviewing signs and symptoms during follow-up appointments
3. Considering and assessing the need for referral for objective testing of appliance efficacy
4. Treatment sleep study protocols for confirming appliance efficacy and the need for further titration
5. Oral appliance efficacy, long-term effectiveness and limitations
6. Treatment adherence and the need for compliance monitoring

Titration is where dental sleep medicine becomes a longitudinal discipline rather than a single procedure. The examinable judgement is knowing **when to stop advancing** — and recognising that the answer is a negotiation between symptom resolution, objective indices, patient tolerance and the accumulating dental cost of protrusion.

6.1 Returning to the chief complaint in the titration process

Subjective measures

The dentist should monitor and collect information on the **resolution of the patient's sleep disorder symptoms** in order to assess efficacy and determine the optimal titration of the appliance: reduced snoring, improved daytime symptoms, and improvement in the specific complaint the patient originally brought. Within 30 days after insertion, and at regular intervals thereafter, the patient should be evaluated and reassessed for signs and symptoms **using the same questionnaires and tools used at the initial assessment** — otherwise change cannot be quantified.

Objective measures

HIGH-YIELD — THE SCOPE-OF-PRACTICE RULE FOR OBJECTIVE DATA

If the dentist can obtain objective data, for example with a portable monitor or pulse oximeter, this information should be used **for titration purposes only** — not for follow-up assessment and not for diagnosis. The dentist working with portable monitors for titration should be able to interpret the results accurately, which means monitoring mainly the *changes* in respiratory sleep parameters with treatment. The dentist should not use this equipment for diagnosis or as a formal assessment; it is the role of the physician to use portable monitors for check-ups or reassessments. Any decisions about the use of home sleep apnoea tests or pulse oximetry, and the resulting objective data, should be made in concert with the patient, the treating medical provider and the qualified dentist.

How to advance — and when to revert

Improvements in AHI are **dose-dependent** on the amount of mandibular advancement, so advancement is the primary lever. A representative published protocol:

1. At initiation, the mandible is set at **50% of the patient's maximum protrusion**.
2. The patient accustoms to this protrusive position over a **2-week period**.
3. Patients are then allowed to adjust the appliance over a **6-week period**.
4. Where symptoms persist — snoring, excessive daytime sleepiness, apnoeas or hypopnoeas — patients are instructed to advance the mandible each night by **one to two increments (0.2–0.4 mm)**.

5. Adjustment continues until symptoms are adequately improved (no signs of apnoea, hypopnoea or snoring) or until further protrusion causes discomfort.
6. The **vertical dimension is kept constant** throughout the follow-up period.

Practical points from clinical practice: the titration period for an appliance is around **5 months, longer than for CPAP**, but once tolerated compliance is better; sufficient time is needed for the TMJ to accustom to a more advanced position. Symptoms typically improve **before** the AHI does. Always leave some leeway rather than going straight to 75%. Nylon appliances are easier to titrate because the scale is visible on the device, whereas acrylic requires turning the screw incrementally. Titration may be performed by the patient or the dentist. Completion of titration usually takes 2–3 months. Home oximetry and other objective measures can be used to support titration, and a wearable home test can be arranged once you believe the titration is adequate. Sequential polysomnographic studies facilitate titration. Note that in reported long-term series, all appliances had been **continuously adjusted with more advancement in order to compensate for the shift in dental occlusion** — a sobering observation to which Topic 10 returns.

Reverting during titration

Step	Action
Assess	At each follow-up, ask about the chief complaint (excessive daytime sleepiness, recurrent snoring) and assess for side effects — tooth pain, jaw stiffness, TMJ discomfort
Identify	If the patient reports worsening symptoms or new significant side effects, decide whether to revert or adjust
Reverse	Either return the appliance to a previous, well-tolerated, less advanced position, or reduce the size of future advancement steps rather than reversing fully
Monitor	Continue at the less advanced setting and re-evaluate symptoms and side effects at the next visit to determine whether advancement can resume or whether the current setting is optimal
Document	Record all adjustments, patient feedback and side effects meticulously to guide the process

Why revert? To mitigate or resolve unacceptable side effects such as persistent tooth pain, jaw discomfort or TMJ symptoms; to balance efficacy against tolerance in finding the optimal advancement; and to avoid prematurely abandoning titration because of minor side effects, which would lead to a suboptimal outcome. The patient's subjective feedback and history of discomfort guide the process, gradual increments balance efficacy against tolerance, and objective data from sleep studies or home monitoring inform decisions about returning to previous settings.

6.2 Reviewing signs and symptoms at follow-up

What to review	Detail
Jaw and TMJ symptoms	Discomfort, stiffness, TMJ pain, clicking or sounds. These may appear during initial use and usually improve, but persistent or severe symptoms require management or device adjustment
Dental discomfort and bite change	Tooth pain, gingival irritation, soreness, and changes in occlusion — the bite feeling "off" or teeth not fitting together as before. Long-term use can decrease overbite and overjet; monitor and record
Salivation changes	Excessive salivation or dry mouth — common initially, often transient, but should be documented and managed. Excessive salivation typically decreases after a few weeks
Appliance condition	Cracks, warping or wear affecting fit and efficacy; confirm retention and stability remain adequate
Effectiveness	Persistence or return of snoring, apnoea symptoms, daytime sleepiness or fatigue. Follow-up sleep testing may be warranted to assess efficacy objectively
Tolerance and adherence	How comfortable the patient feels using the appliance nightly, and whether side effects are causing discontinuation or reduced use

Clinical approach: record and discuss symptoms at each visit; reassure about common transient side effects and educate on managing them; adjust the appliance or refer for specialised care as needed; monitor dental and skeletal change with periodic assessment, especially in long-term users, to prevent irreversible problems; and coordinate care with sleep physicians for optimal management and for discussion of alternative therapy if the appliance is intolerable or ineffective.

6.3 When to refer for objective testing

HIGH-YIELD — WHY SUBJECTIVE FEEDBACK ALONE IS INSUFFICIENT

Two facts make this non-negotiable. First, **oral appliances may worsen OSA in some patients**. Second, there are **known placebo effects** with this type of therapy — a patient may feel better while the AHI is unchanged or worse. Referral of the OSA patient for follow-up evaluation by the referring physician, with polysomnography or sleep studies, is therefore **required** to verify treatment efficacy and benefit.

Reason to refer for objective testing	Detail
Confirming efficacy	Subjective feedback alone is insufficient to determine whether the appliance is working optimally; objective testing confirms the device achieves the desired AHI reduction. For moderate and severe OSA, objective testing should be undertaken because subjective report is not reliable
Guiding further adjustment	Where symptoms have not fully resolved or the optimal setting is uncertain, objective data determine the therapeutic advancement
Persistent symptoms	Ongoing symptoms mean the appliance may not be in an adequate therapeutic position, or that adjustment or an alternative therapy is needed
Monitoring disease progression	Significant weight change or new comorbidity can affect appliance therapy; objective testing re-evaluates efficacy after such changes
Complex or severe cases	Particularly important in severe OSA, since even a reduced AHI remains associated with adverse health outcomes. Consider a mandibular advancement splint as a viable alternative to CPAP even in more severe OSA — **but follow-up polysomnography is mandatory in these patients**
Not indicated	**Primary snoring** — follow-up polysomnography is not indicated unless symptoms worsen or fail to resolve

Maintenance of regular written communication with the patient's physician and other healthcare professionals concerning the treatment plan, progress and follow-up is recommended. The referring physician should be involved in ongoing evaluation, and the dentist should obtain a follow-up report on treatment efficacy from the sleep physician. In the case of unsatisfactory results, the dentist may extend the titration or review the treatment plan with the patient. Objective testing may also be used to compare the efficacy of a trial device with the final appliance, ensuring the optimal protrusion is achieved.

6.4 Treatment sleep study protocols

TWO ACCEPTED PATHWAYS TO IDENTIFYING THE THERAPEUTIC POSITION

Signs and symptoms only	Multimethod positioning
1. Initial mandibular positioning 2. Resolution or improvement of signs and symptoms 3. **Verification of treatment efficacy by the physician**	1. Initial mandibular positioning 2. Resolution or improvement of signs and symptoms 3. Pulse oximetry, high-resolution pulse oximetry or HSAT administered by the qualified dentist, and/or HSAT or PSG ordered by the physician 4. **Verification of treatment efficacy by the physician**

Both pathways end at the same place. The dentist's objective data — where lawfully obtained — **supplement** the titration decision; they never replace physician verification. Advancement should continue until target parameter values are attained, can no longer be improved, or the maximum comfortable position is reached.

The process, step by step

1. **Initial titration by the dentist.** A qualified dentist selects, fits and titrates the appliance, adjusting the mandibular position to find the optimal level of protrusion.
2. **Follow-up sleep study.** The patient undergoes a follow-up study — home sleep test or attended polysomnography — while wearing the appliance at the dentist-determined setting. A home sleep test is a common method for confirming efficacy after initial fitting and titration; attended PSG is more comprehensive and can be used for both titration and final assessment.
3. **Efficacy evaluation.** The sleep physician analyses the results to determine effectiveness, using the AHI to assess severity, with a reduced AHI indicating successful treatment.
4. **Further titration if needed.** If the follow-up study shows an inadequately reduced AHI, further titration or adjustment is required. The follow-up study also identifies new or persistent adverse effects that may necessitate adjustment.
5. **Where the calibrated position is subtherapeutic,** the medical provider and dentist discuss further calibration, adjunctive treatment or alternative treatment. In consultation with the physician a patient may elect to continue despite subtherapeutic results.

Advanced titration methods

- **Remotely controlled mandibular positioner (RCMP)** — allows objective, single-night titration during polysomnography to identify the optimal therapeutic advancement. This is the best-performing predictive method available, rated "excellent" against clinical and craniofacial prediction ("poor") and nasopharyngoscopy ("fair to good").
- **Drug-induced sleep endoscopy** — a remotely controlled positioner can be used with DISE to titrate mandibular position under direct visualisation.
- Note the honest limitation: demographic, anthropometric and polysomnographic data only **weakly** inform mandibular advancement splint efficacy, which is precisely why trial-based methods outperform prediction from patient characteristics.

Further titration, modification, redesign or remake of the appliance should be undertaken as required. Patients should be monitored **monthly for the first six months** by the dentist and at least annually thereafter for as long as the appliance is worn.

6.5 Efficacy, long-term effectiveness and limitations

The efficacy figures

Outcome measure	Value
AHI <5/h (cure / complete response)	~37%
AHI <10/h (near-complete response)	~52%
AHI reduced by ≥50% (success / response)	~64%
Clinically meaningful effect (OSA at least halved)	~⅔ of patients
Negligible improvement	~⅓ of patients
Overall success rate quoted for AHI reduction	60–70%
Response rate as defined by AHI <10 in one series	"Success" 50%, "response" 14%
Cure of mild–moderate OSA	40–50% of patients, with significant improvement in a further 10–20%
Snoring reduction	~45% — reduced but not eliminated
AHI reduction, indirect comparison	OA ~42%, CPAP ~75%, UPPP ~30%
ESS improvement	11.2 → 7.8 (significant)
Patient-reported improvement in sleep quality and reduced snoring	90–95%
Apnoea controlled (depending on initial severity)	50–70%
Blood pressure	Mean reduction ~4.2 mmHg

EXAM TRAP — WHY EFFICACY FIGURES VARY SO MUCH

The published figures range from 37% to 95% "success", and candidates are sometimes caught out by quoting a single number. The variation arises almost entirely from the **definition used**: cure (AHI <5), near-complete response (AHI <10 with ≥50% reduction), partial response (≥50% reduction), or symptomatic improvement. Always state the definition alongside the number — "approximately 64% achieve a 50% or greater reduction in AHI, but only about 37% reach an AHI below 5" — and the answer immediately reads as informed rather than recited.

Long-term effectiveness

- Growing evidence suggests health outcomes, at least in the short term, are **comparable** between mandibular advancement splints and CPAP, **even in severe OSA**, with greater adherence to the splint. Oral appliances also show beneficial effects on blood pressure, endothelial function and left ventricular function.
- Long-term efficacy is good to about **10 years**, and some studies show sustained effectiveness for up to five years.
- However, real-world effectiveness **declines over time**, because some patients discontinue treatment on account of poor subjective effect or side effects, continuing patients may have a poorer objective effect, and the natural history of the disease is deterioration in AHI with age and weight gain. Effectiveness is also vulnerable to changes in the patient's bite and age, requiring continuous re-evaluation.

Limitations

- Not as effective as CPAP in reducing sleep apnoea, snoring and improving daytime function — but with a definite role in the treatment of snoring and sleep apnoea
- Efficacy and side effects depend on appliance type, degree of protrusion, vertical opening and other settings
- Long-term dental change: overbite and overjet reduction, and reduced occlusal contact points
- Common short-term side effects: jaw pain, oral discomfort, excessive salivation, dry mouth
- Treatment may be discontinued for ineffectiveness, discomfort or side effects
- Biological change in the mouth and bite means continuous monitoring and potential adjustment are required to maintain effectiveness
- Not ideal for patients with BMI >35
- Mechanism is less predictable than CPAP

HIGH-YIELD — "WHEN IS NOT ENOUGH ACTUALLY NOT ENOUGH?"

A genuinely sophisticated question, and worth rehearsing. Consider the two axes separately:

	No other symptoms	With other symptoms
Persistent snoring	May be acceptable to the patient; but persistent snoring on treatment has been related to insufficient apnoea control and poor adherence — so verify objectively before accepting it	Treat as inadequate control: re-titrate, verify objectively, consider combination or alternative therapy
Residual AHI	Weigh residual risk against the burden of escalation; an AHI of 5–15 carries only a small risk to health, so the question becomes whether you are treating a number or treating a patient. Consider **effective AHI** — which accounts for the AHI during untreated hours as well	Clearly inadequate: further titration, combination therapy, or referral back for alternatives

6.6 Adherence and compliance monitoring

DEFINITION

AADSM definition of compliance: the appliance being worn for a minimum of **≥80% per night** — starting when the appliance is placed in the mouth and ending when it is removed — for **≥5 nights per week**. Compliance discussions should stress that the patient should sleep a minimum of **7 hours per night** and wear the appliance for the duration of their sleep every night. Contrast the CPAP research convention of **≥4 hours per night**.

The adherence data

Measure	Value
Objective adherence at 1 month	~80%, averaging ~6 hours per night
Objective adherence at 6 months	~67%, then stable through 1 year
Mean objective use	6.1–6.7 h per night; ~83% of days per week
Using ≥4 h per night	~83%
Averaging >6 h per night	~63%
Considered compliant by CPAP standards	~92%
Discontinuation	~9.8%; ~20–30% discontinue overall; ~7.5% specifically due to side effects
Still wearing at ~33 months	55–68%
Regular use at 31 ± 18 months	76%
Continuing use over several years if properly constructed	50–70%
Rate of regular users	82%, average use 6.7 h/day
MAD versus CPAP nightly use	6.5 vs 5.2 h per night

Objective and subjective adherence estimates are broadly similar for oral appliances — a notable contrast with CPAP, where patients substantially over-report use.

Predictors of adherence, and reasons for discontinuation

Better adherence	Worse adherence
• Younger age • Larger baseline overjet • Being married or partnered • Perceived benefit; device comfort; patient knowledge • Device effectiveness, transportability, low embarrassment and acceptable cost — the four most important factors determining choice of treatment	• Psychological comorbidity • Side effects, particularly dental discomfort and bite change • Poor device comfort or fit • Loss of the bed partner, or no further complaints of snoring from the partner — because most patients wear the appliance for snoring, the **benefit is perceived by the bed partner while the side effects are experienced by the wearer**

Compliance monitoring technology

Objective monitoring uses embedded micro-sensors that measure temperature and wear time; some devices additionally track head position during sleep, and the most current appliances can also incorporate oximetry. Systems such as compliance-recording chips allow continuous and remote upload of data to a secure cloud, permitting real-time monitoring by clinicians. Objective measurement with an embedded micro-sensor thermometer is safe and feasible. Challenges for micro-sensors include saliva ingress, power supply, battery toxicity, fitting within the appliance without additional discomfort, compatibility with the appliance material, and verifiability.

Why it matters

- **It distinguishes device inefficacy from poor compliance** — the single most important diagnostic function of adherence data. Without it you cannot know whether to advance the appliance or address behaviour.
- It permits optimised titration and better treatment outcomes.
- It supports legal and insurance documentation where proof of compliance is mandatory — notably in **safety-sensitive occupations such as transport workers**.

- It allows patients to be categorised as fully compliant, improving over time, or non-compliant. Because patients with untreated or subtherapeutic OSA can eventually experience long-term health risks, **non-compliant patients should be referred to their medical provider to consider alternative therapies**.

Follow-up and reassessment in the first year

The steps involved in providing an appliance for a patient with OSA are conventionally set out as follows.

1. **Insertion, adjustments, titration and evaluation during the first six months of regular use.** The dentist monitors and collects information on the resolution of the patient's sleep disorder symptoms to assess efficacy and determine optimal titration. Where objective data are obtained with a portable monitor, they are used for titration only, and the dentist must be able to interpret the results accurately.
2. **Referral of the patient for follow-up evaluation** by the referring physician, with polysomnography or sleep studies required to verify treatment efficacy and benefit — because appliances may worsen OSA in some patients and placebo effects are known. Follow-up polysomnography is not indicated for patients with primary snoring unless symptoms worsen or do not resolve.
3. **Maintenance of regular written communication** with the physician and other professionals concerning the treatment plan, progress and follow-up. The referring physician should be involved in ongoing evaluation, and the dentist should obtain a follow-up report on treatment efficacy from the sleep physician. In the case of unsatisfactory results, the dentist may extend titration or review the treatment plan.
4. **Further titration, modification, redesign or remake** of the appliance as required. Patients should be monitored monthly for the first six months by the dentist and at least annually thereafter.
5. **Follow-up and reassessment after six months** to assess treatment adherence and efficacy. The dentist may advance or titrate further if symptoms recur. If maximum mandibular advancement is reached and symptoms are still present, the patient should be referred to the referring physician, who may then refer to a sleep specialist for further evaluation. The dentist should assess the integrity of the appliance, which may need replacement depending on usage and hygiene; appliances generally last approximately **2–3 years**. For long-term efficacy, patients should be referred to their referring physician **every five years** for continuing assessment, as OSA is known to worsen with age and weight gain.

NUMBERS TO MEMORISE — TOPIC 6

- Efficacy: AHI <5 ~37% · AHI <10 ~52% · ≥50% reduction ~64% · ⅔ meaningful benefit, ⅓ none
- Snoring reduced by ~45%; ESS 11.2 → 7.8; blood pressure ~4.2 mmHg reduction; sleep quality/snoring improvement reported by 90–95%
- Titration: start at 50% of maximum; 2-week acclimatisation, then 6-week self-adjustment; increments 0.2–0.4 mm per night; total titration ~2–5 months; vertical dimension held constant
- Reassess within 30 days of insertion; monthly for first 6 months; then at least annually; refer to physician every ~5 years
- AADSM compliance: ≥80% of the night, ≥5 nights per week, with ≥7 h sleep. CPAP convention ≥4 h per night
- Adherence: ~80% at 1 month → ~67% at 6 months, then stable; 6.1–6.7 h per night; 55–68% still wearing at 33 months; ~9.8% discontinue; ~7.5% discontinue from side effects
- Appliance lifespan 2–3 years

SELF-TEST — TOPIC 6

1. Describe a titration protocol from delivery to therapeutic position.

Set the appliance initially at 50% of maximum protrusion. Allow a two-week acclimatisation period at that setting. Then allow patient-led adjustment over approximately six weeks, advancing by one to two increments of 0.2–0.4 mm per night where symptoms persist — snoring, sleepiness, witnessed apnoeas. Hold the vertical dimension constant throughout. Continue until symptoms are adequately improved or further protrusion causes discomfort, remembering that symptoms improve before the AHI does and that overprotrusion may worsen the breathing. Review within 30 days of insertion, using the **same** questionnaires as at baseline so change is quantifiable, and monthly for the first six months. Total titration usually takes two to five months — longer than CPAP, because the TMJ needs time to accustom to the advanced position. Once you judge titration adequate, arrange objective assessment, then refer to the physician for verification of efficacy.

2. Your patient reports complete resolution of snoring and feels much better. Is the treatment successful?

Not yet established. Subjective feedback alone is insufficient: oral appliances can worsen OSA in some patients, and there are known placebo effects with this therapy. Complete response requires symptom resolution **and** a treatment AHI below 5. I would arrange objective assessment — pulse oximetry or a home sleep test for titration purposes if lawful in my jurisdiction — and refer the patient back to the sleep physician for verification of treatment efficacy with polysomnography or an attended cardiorespiratory study, with the appliance in place at the final setting. The exception is primary snoring, for which follow-up sleep testing is not indicated unless symptoms worsen or fail to resolve.

3. You obtain a home sleep test showing residual AHI of 12 at maximum comfortable protrusion. What next?

First, confirm the data are being used appropriately — my objective data inform titration, and the physician verifies efficacy. Second, check whether the position is genuinely maximal and comfortable, and whether adherence is adequate; a residual AHI in a poorly adherent patient is a compliance problem, not a titration problem, which is exactly why objective adherence data matter. Third, if maximum mandibular advancement has been reached and symptoms persist, refer back to the referring physician, who may refer on to a sleep specialist. Options then include combination therapy — appliance plus positional therapy, plus nasal EPAP, or appliance plus CPAP at a lower pressure — weight management, treatment of nasal obstruction, or a switch of modality. I would document that a patient may, in consultation with the physician, elect to continue despite a subtherapeutic result, and that an AHI of 12 with resolved symptoms carries a relatively small residual risk — but that decision belongs to the patient and physician jointly, not to me alone.

4. Give the AADSM definition of compliance and contrast it with the CPAP convention.

The AADSM defines compliance as the appliance being worn for at least 80% of the night — timed from placement in the mouth to removal — on at least five nights per week, with the patient advised to sleep a minimum of seven hours nightly and wear the appliance for the whole duration of sleep. The CPAP research convention is much weaker: at least four hours per night. This matters when comparing modalities, because by the CPAP standard around 92% of appliance users would be classed compliant. Objective appliance adherence is approximately 80% at one month falling to 67% at six months and then stabilising, with mean use of 6.1–6.7 hours per night — exceeding typical CPAP use of about 5.2 hours.

5. Why is objective compliance monitoring important, beyond simply confirming use?

Its most important function is to **distinguish device inefficacy from poor compliance** — without that distinction you cannot know whether to advance the appliance or to address behaviour, and you risk over-protruding a patient who simply is not wearing the device. It also permits optimised titration and better outcomes; supports legal and insurance documentation where proof of compliance is mandatory, notably in safety-critical transport occupations; and allows categorisation as fully compliant, improving, or non-compliant — with non-compliant patients referred back to the medical provider to consider alternatives, since untreated or subtherapeutic OSA carries long-term health risk. Technology uses embedded micro-sensors recording temperature and wear time, sometimes with head position and oximetry, uploading remotely; the practical challenges are saliva, power supply, battery toxicity, bulk, material compatibility and verifiability.

6. A patient's snoring has resolved but their partner has recently died. What is the risk?

Discontinuation. Patients frequently stop wearing an appliance when the bed partner is no longer present or no longer complains of snoring, because in most cases the appliance is worn for snoring: the **benefit is perceived by the bed partner while the side effects are experienced by the wearer**. Once the perceived benefit disappears, only the cost remains. The management is to re-establish the patient's own reasons for treatment — daytime symptoms, blood pressure, cardiovascular and cognitive risk — to review objective adherence data, and to be alert to bereavement affecting both sleep and motivation more generally.

REFERENCES — TOPIC 6

1. Ramar K, Dort LC, Katz SG, et al. Clinical practice guideline for the treatment of obstructive sleep apnea and snoring with oral appliance therapy: an update for 2015. *J Clin Sleep Med*. 2015;11(7):773–828.
2. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2).
3. Sheats R, Essick G, Grosdidier J, et al. Identifying the appropriate therapeutic position of an oral appliance. *J Dent Sleep Med*. 2020;7(4).
4. Radmand R, Chiang H, Di Giosia M, et al. Defining and measuring compliance with oral appliance therapy. *J Dent Sleep Med*. 2021;8(3).
5. Vanderveken OM, Dieltjens M, Wouters K, De Backer WA, Van de Heyning PH, Braem MJ. Objective measurement of compliance during oral appliance therapy for sleep-disordered breathing. *Thorax*. 2013;68(1):91–96.
6. Dieltjens M, Verbruggen AE, Braem MJ, et al. Determinants of objective compliance during oral appliance therapy in patients with sleep-disordered breathing. *JAMA Otolaryngol Head Neck Surg*. 2015;141(10):894–900.
7. Remmers J, Charkhandeh S, Grosse J, et al. Remotely controlled mandibular protrusion during sleep predicts therapeutic success with oral appliances in patients with obstructive sleep apnea. *Sleep*. 2013;36(10):1517–1525.
8. Sutherland K, Vanderveken OM, Tsuda H, et al. Oral appliance treatment for obstructive sleep apnea: an update. *J Clin Sleep Med*. 2014;10(2):215–227.
9. Ferguson KA, Cartwright R, Rogers R, Schmidt-Nowara W. Oral appliances for snoring and obstructive sleep apnea: a review. *Sleep*. 2006;29(2):244–262.
10. Marklund M. Update on oral appliance therapy for OSA. *Curr Sleep Med Rep*. 2017;3(3):143–151.
11. Aarab G, Lobbezoo F, Heymans MW, Hamburger HL, Naeije M. Long-term follow-up of a randomized controlled trial of oral appliance therapy in obstructive sleep apnea. *Respiration*. 2011;82(2):162–168.
12. Schwartz D, Adame M, Addy N, et al. American Academy of Dental Sleep Medicine position on the scope of practice for dentists ordering or administering home sleep apnea tests. *J Dent Sleep Med*. 2020;7(4).
13. Trzepizur W, Cistulli PA, Glos M, et al. Health outcomes of CPAP versus mandibular advancement device for the treatment of severe obstructive sleep apnea: an individual participant data meta-analysis. *Sleep*. 2021;44(7):zsab015.
14. Gjerde K, Lehmann S, Berge ME, Johansson AK, Johansson A. Oral appliance treatment in moderate and severe obstructive sleep apnoea patients non-adherent to CPAP. *J Oral Rehabil*. 2016;43(4):249–258.

TOPIC 7

Long-term follow-up of patients in oral appliance therapy

BLUEPRINT

1. Impact of age, weight change, alcohol, medication change, sleep hygiene/quantity · 2. Documenting the evolution of the initial complaint · 3. Treatment modification for progressive disease · 4. Appliance condition, stability and care at yearly review

The only topic organised entirely around one idea: ***OSA is a progressive disease and the appliance is a lifelong prosthesis****. A static setting in a changing patient is a slowly failing treatment.*

7.1 Modifiers of long-term therapy

- **Better long-term response:** younger; less obese with smaller neck circumference (men); lower baseline AHI. **Supine and REM sleep predict a lower response rate** — the splint is relatively poor at alleviating REM-predominant OSA compared with NREM-predominant or non-stage-dependent disease; conversely **positional OSA favours the appliance**. Gender and polysomnographic phenotype are, in some analyses, unrelated to outcome — predictors are probabilistic, not deterministic.
- **Age** — influences both adherence and effectiveness. Some studies show lower adherence in older adults because changes in dentition and mucosal tissue make adaptation harder; others show no age difference. Older adults can still benefit provided oral health supports retention — so the question is not the patient's age but whether the dentition can still anchor the device.
- **Weight change** — gain worsens OSA and may reduce effectiveness by altering airway anatomy and appliance fit; loss improves outcomes. Significant fluctuation requires reassessment of fit and efficacy, and is a specific trigger for repeat objective testing. Recall: 10% gain → ~32% AHI rise; 10% loss → ~26% fall.
- **Alcohol** — before sleep it exacerbates collapse and increases severity, undermining benefit; even moderate intake may cause transient desaturation despite appliance use. Advise avoidance in the hours before bed.
- **Medication change** — drugs affecting the CNS or muscle tone (sedatives, hypnotics, opioids, muscle relaxants) may reduce airway patency or alter architecture, potentially reducing efficacy. Emerging research explores combining pharmacotherapy with appliance therapy, but changes should be monitored closely. **Ask about new medications at every review.**
- **Sleep hygiene and quantity** — good hygiene and adequate duration support success by stabilising architecture and reducing factors that worsen apnoea. Poor habits blunt the daytime alertness gains and potentiate severity. A patient sleeping five hours will not feel the benefit of a well-titrated appliance, and may wrongly conclude it has failed.

7.2 Documenting the evolution of the complaint

- **Why:** it tracks effectiveness, symptom progression and adherence simultaneously, and allows timely adjustment — reinforcing adherence and addressing emerging problems such as discomfort or incomplete resolution.
- **Self-reported symptoms** — the patient's perception of severity and therapy effect: ESS, sleep diaries, mobile applications.
- **Sleep-observer reports** — the bed partner or family on snoring intensity, apnoeas and restless sleep, offering "objective-seeming" data that complement self-report: partner observations, STOP-BANG items.
- **Evolution tracking** — regular follow-ups, app-based tracking, and repeated administration of the **same** baseline questionnaires. **Objective data integration** — compliance-monitor data, CPAP usage logs, polysomnography, oximetry.
- Monitoring changes in self-reported sleepiness correlates with long-term adherence and better outcomes. Yearly follow-up should include changes in sleep, blood pressure, weight, behavioural risk factors, and re-evaluation using previously employed questionnaires.

WHY THE BED PARTNER IS PART OF THE RECORD

Often the only available reporter of the nocturnal phenotype, and frequently the person who perceives the benefit while the patient bears the side effects. Recording their observations detects recurrence of snoring or witnessed apnoeas before the patient notices; documents benefit the patient may under-report; and identifies the moment the perceived reason for treatment disappears — a strong predictor of discontinuation.

7.3 Modification for progressive disease

- **Why necessary.** *Disease progression* — OSA worsens over time and the advancement needed may increase. *Long-term dental change* — prolonged use produces dentoskeletal change (reduced overbite and overjet, altered incisor position), and critically a **forward shift of the lower teeth relative to the upper reduces the effective advancement**, putting efficacy at risk; this is why appliances in long-term series had to be advanced simply to compensate for occlusal drift. *Increasing side effects* — greater protrusion increases jaw stiffness, tooth discomfort and dry mouth, requiring careful balance.
- **How modified.** (1) Titrate in small increments, repeating until the most effective level is found that still manages symptoms while minimising side effects. (2) Regular check-ups with dentist and sleep specialist to monitor apnoea progress and appliance effects on

jaw and teeth. (3) Re-adjust protrusion on reassessment — with significant progression, a larger advancement or a switch of appliance type. (4) Assess short- and long-term side effects at each follow-up and modify to relieve discomfort.

- **Key considerations:** the optimal position is unique to each patient and determined by careful repeated titration; construction, fitting and adjustment should be by a dentist with expertise; the process is always a balance of therapeutic effect against side effects.
- **Other modalities.** *PAP* — pressure settings need regular fine-tuning; home and laboratory PSG diagnose, grade and evaluate effectiveness over time. *Lifestyle* — weight management (reducing upper airway collapsibility), sleep position training, avoidance of alcohol, opioids and other medications that worsen SDB. *Concomitant conditions* — optimise heart failure, asthma or chronic lung disease; treat nasal obstruction (nasal valve collapse, allergic rhinitis) with steroids or surgery. *Novel* — hypoglossal nerve stimulation as salvage for PAP-intolerant patients; maxillary or palatal expansion to increase volume and reduce collapsibility; myofunctional therapy. *Recognising progression* — an integrated, dynamic, personalised plan, often combining therapies. Note also that some processes are **reversible**: treating SDB can reduce inflammatory oedema and increase pharyngeal size, improving the trajectory.

7.4 The yearly review

KEY TAKEAWAY

Long-term success hinges on a thorough yearly evaluation of appliance condition and fit, combined with clear reinforcement of proper cleaning, handling and storage routines.

- **1. Appliance condition.** *Surface integrity* — cracks, crazing, rough spots that could irritate tissues. *Material wear* — discolouration, thinning, deformation of acrylic or thermoplastic. *Hardware* — clasps, hinges and screws free of corrosion, moving smoothly without binding. Also assess for bacterial and fungal growth; replace according to need. Reiterate: "Even small cracks can harbour bacteria and compromise effectiveness."
- **2. Fit and stability.** *Seating accuracy* — fully engages the intended undercuts or rests on occlusal surfaces without rocking. *Retention* — gently tug to verify it resists displacement in normal function. *Occlusal contacts* — no interference or premature contact. Verify efficacy and **occlusal stability**, and that previously resolved symptoms (snoring, sleepiness) remain resolved. Communicate: "If you ever feel looseness, schedule an interim visit — don't wait until next year."
- **3. Daily care.** *Cleaning* — rinse immediately after removal; brush gently with a soft brush and non-abrasive, fragrance-free cleanser (avoid toothpaste); disinfect weekly in a denture-safe soak. *Handling* — remove over a soft surface or basin of water; avoid hot water and direct sunlight. *Storage* — dry, in the ventilated case; never in tissue or loose in a pocket; away from pets, children and heat.
- **4. Scheduling and next steps.** Confirm understanding of care and fit; offer to demonstrate cleaning. Assess adherence and comfort, and screen for TMJ, intraoral tissue and occlusal side effects, damage to teeth or restorations, and appliance problems — documenting presence, management and resolution, and communicating findings to the medical provider and general dentist. Advise immediate contact for discomfort, looseness or damage. Schedule the next annual review while making clear earlier review is available on request. **Notify the medical provider if the patient terminates therapy.**
- **Replacement.** A comprehensive dental sleep medicine evaluation may be needed before fabricating a new appliance, and **is required** if the patient was treated previously by another dentist. Communicate with the medical provider early to determine what triggers a referral before replacement; notify them of the request, allow reassessment, and let them decide whether efficacy of the new appliance needs verification. If previously treated effectively, the new appliance can be set to the same protrusion — though it may still need calibration to restore the therapeutic position.

NUMBERS — TOPIC 7

- Follow-up 6-monthly year 1, then at least annually; physician reassessment ~every 5 years; appliance life ~2–3 years
- Weight: 10% gain → +32% AHI; 10% loss → −26% AHI
- Better response: younger, less obese, smaller neck (men), lower baseline AHI, positional OSA, NREM-predominant
- Poorer response: supine-predominant during the study, REM-predominant OSA, weight gain during treatment

SELF-TEST — TOPIC 7

1. A patient stable for four years returns with recurrent snoring and sleepiness. Work through your assessment.

Four categories. **Disease progression** — OSA worsens with age and weight gain, so the previously therapeutic advancement may now be insufficient; check weight, blood pressure and new comorbidities. **Occlusal drift** — a forward shift of the lower teeth relative to the upper reduces the effective advancement, so the same setting now delivers less protrusion; compare current overjet and overbite against baseline records and casts. **Appliance failure** — cracks, crazing, material fatigue, worn or corroded hardware, lost retention. **Behavioural and pharmacological modifiers** — new alcohol intake, new sedating medication, reduced sleep duration or worse hygiene, plus objective adherence data. Then re-titrate as indicated, arrange objective reassessment, and refer back to the sleep physician — recurrence of symptoms is a specific indication for repeat objective testing.

2. Why must the appliance be advanced over time even when the disease has not progressed?

> Because prolonged use produces progressive occlusal change — the lower teeth move anteriorly and the upper distally — so at a fixed setting the **effective** advancement steadily reduces. In long-term series appliances had to be continuously advanced simply to compensate for the occlusal shift. This is a genuinely uncomfortable feature of the therapy: the side effect erodes the treatment effect, and compensating requires more protrusion, which accelerates the side effect. It argues for minimising titration to the least effective advancement, using a morning occlusal guide, and comparing carefully against baseline records each year.

3. What do you assess at the annual review?

> Four domains. **Appliance condition** — surface integrity, material wear, hardware function, biofilm. **Fit and stability** — seating without rocking, retention on gentle traction, occlusal contacts, plus verification of continuing efficacy and occlusal stability. **Care practices** — reinforce cleaning, handling and storage. **Clinical status** — adherence and comfort; TMJ, soft tissue and occlusal side effects; damage to teeth or restorations; changes in sleep, blood pressure, weight and behavioural risk factors; and re-administration of the baseline questionnaires. Document side effects and their management, communicate to the medical provider and general dentist, and schedule the next review while making clear earlier review is available.

TOPIC 7

Long-term follow-up of patients in oral appliance therapy

EXAMINER'S BLUEPRINT

1. Impact of age, weight change, alcohol use, medication change and sleep hygiene/quantity on long-term oral appliance therapy
2. Relevance and documentation of the evolution of the patient's initial complaint, based on self-reported and sleep-observer measures
3. Treatment modification related to the progressive nature of sleep-disordered breathing
4. Reiterating appliance condition, stability and care at yearly review appointments

Topic 7 is short but distinctive: it is the only topic organised entirely around the idea that **OSA is a progressive disease and the appliance is a lifelong prosthesis**. Every answer here should carry that framing. The corollary examiners look for is that a static appliance setting in a changing patient is a slowly failing treatment.

7.1 Factors that modify long-term appliance therapy

Factor	Impact on Oral Appliance Therapy	Clinical Considerations
Age	Possible reduced adherence or fit issues in elderly	Assess retention, dentition, mucosal health
Weight Change	Weight gain worsens OSA, weight loss improves fit	Reassess appliance fit after weight changes
Alcohol Use	Increases airway collapse; reduces OAT efficacy	Advise avoidance before bedtime
Medication Change	May alter muscle tone, sleep structure, OAT effect	Monitor and adjust therapy as needed
Sleep Hygiene	Poor sleep reduces therapy benefits	Encourage good routines, sufficient sleep

Figure 7.1 — Factors modifying oral appliance therapy over time, tabulating each factor against its impact and the clinical consideration it generates: age (possible reduced adherence or fit issues in the elderly → assess retention, dentition and mucosal health), weight change (weight gain worsens OSA, weight loss improves fit and outcome → reassess appliance fit after weight change), alcohol use (increases airway collapse and reduces appliance efficacy → advise avoidance before bedtime), medication change (may alter muscle tone, sleep structure and appliance effect → monitor and adjust therapy as needed), and sleep hygiene (poor sleep reduces therapy benefit → encourage good routines and sufficient sleep).

Why it is examinable: this is the whole of 7.1 in one object, and its three-column structure — factor, impact, clinical action — is exactly the format to reproduce in a written answer.

Predictors of a better long-term response

Domain	Better response
Age	Younger
Anthropometry	Less obese, with smaller neck circumference (in men)
Polysomnography	Lower baseline AHI. **Supine and REM sleep predict a lower response rate** — the splint is relatively poor at alleviating REM-predominant OSA compared with NREM-predominant or non-stage-dependent disease. Conversely, **positional (non-positional-independent) OSA favours the appliance**
Not predictive	Gender and polysomnographic phenotype are, in some analyses, unrelated to treatment outcome — a reminder that predictors are probabilistic, not deterministic

The five modifiers in detail

Modifier	Effect and management
Age	Influences both adherence and effectiveness. Some studies show older adults have lower adherence, because changes in dentition and mucosal tissue make adaptation harder; others show no significant age-related difference. Overall, older adults can still benefit provided their oral health supports appliance retention — so the question to ask is not the patient's age but whether the dentition can still anchor the device
Weight change	Weight gain worsens OSA and may reduce appliance effectiveness by altering airway anatomy and appliance fit; weight loss improves outcomes. Significant fluctuation requires reassessment of both fit and efficacy — and is one of the specific triggers for repeat objective testing (Topic 6.3). Recall the quantitative relationship: 10% gain → ~32% AHI increase; 10% loss → ~26% AHI decrease
Alcohol	Consumption before sleep exacerbates airway collapse and increases apnoea severity, undermining the benefit of therapy. Even moderate intake may cause transient oxygen desaturation despite appliance use. Advise avoidance in the hours before bedtime
Medication change	Drugs affecting the central nervous system or muscle tone may alter airway patency or sleep architecture, potentially reducing efficacy — sedatives, hypnotics, opioids, muscle relaxants. Emerging research explores combining pharmacotherapy with appliance therapy, but medication changes should be monitored closely for their effect on apnoea control. Ask about new medications at every review
Sleep hygiene and quantity	Good hygiene and adequate duration support success by stabilising sleep architecture and reducing factors that worsen apnoea. Poor habits blunt the daytime alertness gains from therapy and potentiate apnoea severity, so consistent routines are recommended. Practically: a patient sleeping five hours will not feel the benefit of a well-titrated appliance, and may wrongly conclude that the appliance has failed

7.2 Documenting the evolution of the initial complaint

Documenting how the complaint evolves matters because it tracks treatment effectiveness, symptom progression and adherence simultaneously, and because it allows timely adjustment — reinforcing adherence and addressing emerging problems such as discomfort or incomplete symptom resolution. Two perspectives are needed.

Aspect	Importance	Examples of measures
Self-reported symptoms	The patient's perception of severity and of the effect of therapy — crucial insight into perceived health status and benefit	Epworth Sleepiness Scale, sleep diaries, mobile applications
Sleep-observer reports	External confirmation and additional symptom insight — the bed partner or family member reports on snoring intensity, apnoeas and restless sleep, offering "objective-seeming" data that complement self-report	Partner observations, STOP-BANG items
Evolution tracking	Guides treatment adjustment and encourages adherence	Regular follow-ups, app-based tracking, repeated administration of the **same** baseline questionnaires
Objective data integration	Complements subjective reporting to give a comprehensive assessment	Appliance compliance-monitor data, CPAP usage logs, polysomnography, oximetry

Monitoring changes in self-reported sleepiness correlates with long-term adherence and better outcomes, supporting continuous documentation. Typical yearly follow-up should include monitoring changes in sleep, blood pressure, weight loss or gain, behavioural risk factors, and re-evaluation of other metrics using previously employed questionnaires.

HIGH-YIELD — WHY THE BED PARTNER IS PART OF THE RECORD

The bed partner is often the only available reporter of the nocturnal phenotype, and — as Topic 6.6 notes — is frequently the person who perceives the benefit while the patient bears the side effects. Recording the partner's observations therefore serves three purposes: it detects recurrence of snoring or witnessed apnoeas before the patient notices; it documents benefit that the patient may under-report; and it identifies the moment when the perceived reason for treatment disappears, which is a strong predictor of discontinuation.

7.3 Treatment modification for progressive disease

Why modification is necessary

- **Progression of the disease.** Sleep apnoea worsens over time, and the optimal advancement needed to keep the airway open may increase.
- **Long-term dental change.** Prolonged appliance use produces dentoskeletal change — reduced overbite and overjet, altered incisor position. Critically, a **forward shift of the lower teeth relative to the upper reduces the effective degree of mandibular advancement**, putting the efficacy of treatment at risk. This is why appliances in long-term series had to be continuously advanced simply to compensate for occlusal drift.
- **Increasing side effects.** A greater degree of protrusion often increases the incidence of jaw stiffness, tooth discomfort and dry mouth, requiring careful adjustment to balance effectiveness against comfort.

How the appliance is modified

1. **Titration.** Adjust the advancement in small increments, repeating until the most effective level is found that still manages symptoms while minimising side effects.
2. **Regular follow-up.** Check-ups with both dentist and sleep specialist to monitor the progress of the apnoea and the effects of the appliance on the jaw and teeth.
3. **Appliance re-adjustment.** Based on reassessment, adjust the protrusion level; with significant progression this may mean a larger advancement or a switch to a different appliance type.
4. **Monitoring for side effects.** Assess short- and long-term side effects at each follow-up and modify settings to relieve discomfort.

Key considerations: the optimal mandibular position is unique to each patient and must be determined by careful repeated titration; construction, fitting and adjustment should be performed by a dentist with expertise in appliance therapy; and the process is always a balance between therapeutic effect and side effects.

Modification of other therapies for progressive SDB

The same principle — a dynamic, personalised, ongoing plan — applies across modalities, because SDB worsens with age and weight gain and can be exacerbated by other medical conditions.

Strategy	Detail
Adjusting PAP therapy	Pressure settings on CPAP or BPAP need regular fine-tuning to match the current degree of disease. Home and in-laboratory polysomnography diagnose SDB, determine severity and evaluate treatment effectiveness over time
Lifestyle and behavioural modification	Weight management, which significantly reduces severity by decreasing upper airway collapsibility; sleep position training to avoid the supine posture; and avoidance of alcohol, opioids and other medications that worsen SDB
Managing concomitant conditions	Optimising treatment of coexisting heart failure, asthma or chronic lung disease; treating nasal obstruction — nasal valve collapse, allergic rhinitis — with nasal steroids or surgery to improve airflow
Novel and advanced therapies	Hypoglossal nerve stimulation as a salvage option for PAP-intolerant patients; maxillary or palatal expansion to increase airway volume and reduce collapsibility; myofunctional therapy to improve upper airway muscle strength and endurance
Recognising progression	An integrated approach is required, often combining multiple therapies. Note also that some processes are **reversible**: treatment of SDB can reduce inflammatory oedema and increase pharyngeal size, improving the condition's trajectory

7.4 The yearly review appointment

> **KEY TAKEAWAY**
>
> Ensuring the long-term success of an oral appliance hinges on a thorough yearly evaluation of its condition and fit, combined with clear reinforcement of proper cleaning, handling and storage routines.

1. Appliance condition assessment

- **Surface integrity** — check for cracks, crazing or rough spots that could irritate the oral tissues.
- **Material wear** — note discolouration, thinning or deformation of the acrylic or thermoplastic.
- **Hardware functionality** — where metal components are present (clasps, hinges, screws), ensure they are free of corrosion and move smoothly without binding.
- Also assess for bacterial and fungal growth, and replace according to the patient's needs.

Reiterate to the patient: "It's important that your appliance remains intact. Even small cracks can harbour bacteria and compromise effectiveness."

2. Fit and stability evaluation

- **Seating accuracy** — confirm the appliance fully engages the intended undercuts, or rests on the occlusal surfaces, without rocking.
- **Retention** — gently tug to verify the device resists displacement during normal oral function.
- **Occlusal contacts** — ensure the appliance does not interfere with the bite or cause premature contacts.
- Verify appliance efficacy and **occlusal stability**, and that previously resolved symptoms such as snoring and daytime sleepiness remain resolved.

Communicate clearly: "A stable fit keeps you comfortable and ensures the appliance does its job. If you ever feel looseness, schedule an interim visit — don't wait until next year."

3. Reinforcing daily care

- **Cleaning:** rinse immediately after removal to wash away saliva and debris; brush gently with a soft toothbrush and a non-abrasive, fragrance-free cleanser, avoiding toothpaste; disinfect weekly in a denture-safe soak per manufacturer instructions.
- **Handling:** always remove over a soft surface or a basin filled with water to prevent accidental drop damage; avoid hot water and direct sunlight, which warp the appliance.
- **Storage:** store dry in the ventilated case when not in use — never wrapped in tissue or loose in a pocket; keep away from pets, children and heat sources.

Emphasise: "Consistent daily care helps prevent odour, staining and bacterial growth. A few minutes each day will save you discomfort and costly repairs later."

4. Scheduling and next steps

- Confirm the patient understands all aspects of appliance care and fit; offer to demonstrate in-office cleaning techniques if needed.
- Assess adherence and comfort, and screen for TMJ-related, intraoral tissue-related and occlusal side effects, damage to teeth or restorations, and appliance problems. Where side effects are noted, document their presence, management and manner of resolution, and communicate relevant subjective and objective findings to the medical provider and the general dentist.
- Recommend the patient contact the clinic immediately if discomfort, looseness or damage occurs.
- Schedule the next annual review, reminding the patient that an earlier check is available on request.
- Notify the medical provider if the patient terminates therapy.

Appliance replacement

Patients needing a replacement appliance may need a comprehensive dental sleep medicine evaluation before fabrication of a new device, and this is required if the patient was treated previously by another dentist. Communicate with the medical provider early to determine what circumstances would trigger a referral before replacement; the provider should be notified of the request, given the opportunity to reassess, and allowed to decide whether efficacy of the new appliance needs verification. If the patient was previously treated effectively, the new appliance can be set to the same protrusion setting — though it may still require calibration to restore the therapeutic position.

NUMBERS TO MEMORISE — TOPIC 7

- Follow-up: every 6 months in year 1, at least annually thereafter; physician reassessment approximately every 5 years
- Appliance lifespan ~2–3 years
- Weight: 10% gain → ~32% AHI increase; 10% loss → ~26% AHI decrease
- Better response: younger, less obese, smaller neck circumference (men), lower baseline AHI, positional OSA, NREM-predominant disease
- Poorer response: supine-predominant sleep during the study, REM-predominant OSA, weight gain during treatment

SELF-TEST — TOPIC 7

1. A patient stable on appliance therapy for four years returns with recurrent snoring and daytime sleepiness. Work through your assessment.

> I would consider four categories. **Disease progression** — OSA worsens with age and weight gain, so the previously therapeutic advancement may now be insufficient; check weight change, blood pressure and new comorbidities. **Occlusal drift** — a forward shift of the lower teeth relative to the upper reduces the effective advancement, so the appliance may be at the same setting but delivering less protrusion; compare current overjet and overbite against the baseline records and study casts. **Appliance failure** — cracks, crazing, material fatigue, worn or corroded hardware, or loss of retention. **Behavioural and pharmacological modifiers** — new alcohol intake, new sedating medication, reduced sleep duration or deteriorated sleep hygiene, and objective adherence data. I would then re-titrate as indicated, arrange objective reassessment, and refer back to the sleep physician for verification, remembering that recurrence of symptoms is a specific indication for repeat objective testing.

2. Why must the appliance be advanced over time even when the disease has not progressed?

> Because prolonged appliance use produces progressive occlusal change: the lower teeth move anteriorly and the upper teeth distally, so at a fixed appliance setting the **effective** mandibular advancement steadily reduces. In long-term series appliances had to be continuously advanced simply to compensate for the shift in dental occlusion. This is a genuinely uncomfortable feature of the therapy — the side effect erodes the treatment effect, and compensating for it requires more protrusion, which in turn accelerates the side effect. It is a strong argument for minimising titration to the least effective advancement, for a morning occlusal guide, and for careful annual comparison against baseline records.

3. What do you assess at the annual review?

> Four domains. **Appliance condition**: surface integrity (cracks, crazing, rough spots), material wear (discolouration, thinning, deformation), hardware function (corrosion, smooth movement), and bacterial or fungal growth. **Fit and stability**: seating accuracy without rocking, retention on gentle traction, occlusal contacts and absence of premature contact, plus verification of continuing efficacy and occlusal stability. **Care practices**: reinforce cleaning, handling and storage. **Clinical status**: adherence and comfort; TMJ, soft tissue and occlusal side effects; damage to teeth or restorations; changes in sleep, blood pressure, weight and behavioural risk factors; and re-administration of the baseline questionnaires. Document any side effects and their management, communicate findings to the medical provider and general dentist, and schedule the next review while making clear that earlier review is available on request.

4. How do alcohol and new medications affect appliance therapy, and what do you advise?

> Alcohol before sleep reduces pharyngeal dilator muscle tone and blunts arousal responses, exacerbating airway collapse and increasing apnoea severity; even moderate intake can cause transient oxygen desaturation despite appliance use. Advise avoidance in the hours before bed. New medications acting on the central nervous system or muscle tone — sedatives, hypnotics, opioids, muscle relaxants — may reduce airway patency or alter sleep architecture and so reduce appliance efficacy; some antidepressants additionally induce bruxism or restless legs. I therefore take a medication history at every review, and where a change coincides with symptom recurrence I consider it as a cause before advancing the appliance further.

REFERENCES — TOPIC 7

1. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2).
2. Ramar K, Dort LC, Katz SG, et al. Clinical practice guideline for the treatment of obstructive sleep apnea and snoring with oral appliance therapy: an update for 2015. *J Clin Sleep Med*. 2015;11(7):773–828.
3. Newman AB, Foster G, Givelber R, Nieto FJ, Redline S, Young T. Progression and regression of sleep-disordered breathing with changes in weight: the Sleep Heart Health Study. *Arch Intern Med*. 2005;165(20):2408–2413.
4. Peppard PE, Austin D, Brown RL. Association of alcohol consumption and sleep disordered breathing in men and women. *J Clin Sleep Med*. 2007;3(3):265–270.
5. Marklund M, Franklin KA. Long-term effects of mandibular repositioning appliances on symptoms of sleep apnoea. *J Sleep Res*. 2007;16(4):414–420.
6. Pliska BT, Nam H, Chen H, Lowe AA, Almeida FR. Obstructive sleep apnea and mandibular advancement splints: occlusal effects and progression of changes associated with a decade of treatment. *J Clin Sleep Med*. 2014;10(12):1285–1291.
7. Doff MHJ, Finnema KJ, Hoekema A, Wijkstra PJ, de Bont LGM, Stegenga B. Long-term oral appliance therapy in obstructive sleep apnea syndrome: a controlled study on dental side effects. *Clin Oral Investig*. 2013;17(2):475–482.

8. Sutherland K, Takaya H, Qian J, Petocz P, Ng AT, Cistulli PA. Oral appliance treatment response and polysomnographic phenotypes of obstructive sleep apnea. *J Clin Sleep Med*. 2015;11(8):861–868.
9. Ainge-Allen HW, Yee BJ, Ip MSM. Contemporary concise review 2020: sleep. *Respirology*. 2021;26(7):700–706.
10. Sheats RD, Schell TG, Blanton AO, et al. Management of side effects of oral appliance therapy for sleep-disordered breathing. *J Dent Sleep Med*. 2017;4(4):111–125.

TOPIC 8

Diagnosis and management of paediatric sleep-disordered breathing

BLUEPRINT

1. Prevalence · 2. Aetiology and pathophysiology · 3. Signs and symptoms · 4. Other causes of poor sleep in children and developmental impact · 5. Adolescents: cognition, physical performance, impulse control, decision making · 6. Screening · 7. Treatment: surgery, CPAP, orthodontics, other · 8. Differences from adults · 9. Underlying conditions and syndromes · 10. Referral pathways

Two ideas carry the topic: ***thresholds are lower and presentation is behavioural rather than sleepy****; and* ***adenotonsillectomy, not orthodontics, is first-line****.*

8.1–8.2 Prevalence, aetiology, pathophysiology

- **Prevalence:** OSA 1–5% (typically 2–3%); habitual snoring 3–12% (some cite 10–20%). Peaks in **younger children (2–8 y)** and again in **adolescents**; most commonly affects ages **3–6**, when tonsillar tissue is proportionally largest. Varies by population, gender and race.
- **Aetiology: adenotonsillar hypertrophy** — the most frequent cause, particularly in pre-schoolers; **obesity** — significant especially in teenagers; **craniofacial abnormality** — micrognathia, narrowed midface; **allergic rhinitis**; **neuromuscular disorders** — Down syndrome, cerebral palsy, affecting head and neck tone so muscles relax excessively; rarely **tumours or growths**. Other anatomical risk factors: micrognathia, retrognathia, macroglossia, nasal septal deviation, high-arched or narrow hard palate. **Caution: tonsil size does not correlate with OSA severity** (Brodsky classification).
- **Pathophysiology sequence:** (1) muscles keeping the airway open relax as the child falls asleep; (2) where the airway is already narrow (large tonsils, small jaw) or tone is reduced, tissues move closer or fold, producing partial or complete blockage; (3) the brain detects falling oxygen and rising CO_2 and triggers a brief arousal to restore tone and reopen the airway; (4) frequent arousals, often unremembered, fragment sleep and reduce restorative stages, producing daytime fatigue, hyperactivity and poor concentration; (5) over time this affects growth, neurocognitive development, academic performance and emotional well-being.
- **The four contributing mechanisms:** craniofacial morphology, tonsillar hypertrophy, altered neuromuscular tone, ventilatory control.
- **Adenoid facies (long face syndrome)** — **long face** with increased lower facial height, **convex profile**, mouth breathing with incompetent lip seal and open mouth posture, **high-arched palate**, **constricted maxilla** usually causing posterior crossbites and crowding, retruded mandible, sometimes anterior open bite. Consequences of a reduced upper airway are therefore not only sleep-breathing problems but also constricted maxilla with crossbite and crowding, impaired growth, cardiovascular problems, and learning and behavioural problems.

TRAP — THE MALOCCLUSION–OSA LINK IS WEAKER THAN DENTISTS ASSUME

Most children with OSA do **not** have obviously altered craniofacial morphology; measured differences are frequently not clinically relevant. **Dentists overestimate the correlation between malocclusion and OSA prevalence.** Greater prevalence of malocclusion has not been found to be associated with OSA, and abnormal respiration or OSA is not a frequent or common cause of malocclusion. Adenoid facies is a recognisable pattern, not a reliable screening test. Saying this demonstrates critical reading; asserting a strong causal link does the opposite.

8.3 Signs and symptoms

- **During sleep:** loud snoring or noisy breathing (gasping, snorting, choking); pauses lasting several seconds; **mouth breathing**; restless sleep with frequent position changes; night sweats; **unusual sleep positions** — seated or with the neck hyperextended; **bed-wetting in an older child** dry for a long period.
- **During the day:** morning headaches and fatigue; daytime sleepiness (less prominent than in adults); difficulty concentrating and paying attention; **behavioural problems** — hyperactivity, aggression, impulsivity; irritability; poor academic performance; excessive *or* poor weight gain; neuropsychological and cognitive deficits including reduced general intelligence level and IQ.

8.4 Other causes of poor sleep in children

- **Recommended duration:** <4 months not specified (variation too large) · 4–12 months 12–16 h · 1–2 y 11–14 h · 3–5 y 10–13 h · 6–12 y 9–12 h · 13–18 y 8–10 h.

- **Less than recommended:** attention, behaviour and learning problems; increased risk of accidents, injuries, hypertension, obesity, diabetes and depression; in teenagers, increased self-harm, suicidal thoughts and suicide attempts. **More than recommended:** hypertension, diabetes, obesity, mental health problems. **Sufficient:** improved attention, behaviour, learning, memory, emotional regulation, quality of life, and mental and physical health.
- **Causes.** *Environmental* — noisy or stressful home, inconsistent bedtime routines, family conflict. *Behavioural* — excessive screen time before bed (video games particularly) disrupting circadian rhythm and causing mental arousal; heavy meals or caffeine close to bedtime; inconsistent bedtimes, no wind-down routine, co-sleeping. *Medical and sleep disorders* — allergies, colds, ear infections, headaches; sleep apnoea, restless legs, periodic limb movements, night terrors, nightmares; reflux; insomnia. *Medications* — ADHD stimulants, SSRIs. *Mental and emotional health* — anxiety and depression in the child, and maternal anxiety and depression in pregnancy.
- **Developmental impact.** *Physical* — obesity, impaired glucose metabolism, high blood pressure. *Neurocognitive* — impaired memory retention and consolidation; greater ADHD-like symptoms; lower verbal, performance and full-scale IQ; poorer academic performance and learning ability. *Emotional and behavioural* — internalising (depression, anxiety) and externalising (aggression, rule-breaking) behaviours; ADHD-type inattention, impulsivity and hyperactivity; low energy, poor mood regulation, decreased social skills. *Neurodevelopmental risk* — insufficient sleep in the early years may raise the risk of poor neurobehavioural function later.

8.5 Adolescents

> **THE FRAMING SENTENCE**
>
> The reduction of sleep in adolescents is **not** driven by a lower need for sleep, but by biological, psychological and socio-cultural influences acting together.

- **Biological.** *Circadian phase delay* — early adolescent development and pubertal onset are associated with a more **evening-type phase preference**. *Sleep–wake homeostasis* — sleep pressure builds more slowly with time awake; with the phase delay this makes it a major challenge to fall asleep in the early evening and wake refreshed in the early morning. *Weekend catch-up* — later weekend bed and wake times reflect circadian biology and respond to insufficient weekday sleep, but **further delay the signal for the biological night**, perpetuating the cycle.
- **Electronic media** — four mechanisms: media use **directly displaces** sleep; devices allow greater interaction with friends at night; device light **suppresses melatonin** and disrupts circadian rhythm, producing inability to fall asleep at a reasonable time; and use increases sleep-disrupting mental, emotional and physiological arousal.
- **School start times** — early starts conflict with the delayed rhythm, producing chronic insufficient sleep. Later-starting middle schools report later rise times, more total sleep on school nights, less daytime sleepiness, less tardiness, fewer attention and concentration difficulties, better academic performance, and better safety with a **reduced crash rate**.
- **Caffeine** — shorter duration, increased sleep onset latency, increased WASO, increased daytime sleepiness. Consumers are **twice as likely** to have difficulty sleeping and report morning sleepiness — a vicious cycle. Reduces slow-wave sleep **dose-dependently** and alters the temporal organisation of REM and NREM, affecting learning and memory consolidation. Females are more likely to show tolerance and withdrawal. Associated with other substance use and abuse, other risk-taking behaviour, and inversely related to academic achievement.
- **Other causes:** chronic medical illness; anxiety and stress; prescribed psychotropics; chronic respiratory illness such as asthma; pain conditions such as migraine; stimulants (especially ADHD medication) used as a countermeasure to sleepiness or as performance enhancers; OTC or prescribed sleep medication causing residual sleepiness; alcohol; academic demands and extracurricular load.
- **Impacts.** *Cognition* — poor judgement, lack of motivation, inattention, affective dysregulation; higher-level **executive functions**, for which adolescence is a critical developmental period, are **selectively** affected. *Mood and self-harm* — shorter school-night sleep associates with daytime sleepiness and depressive symptoms; **<8 h is associated with an almost threefold increased risk of suicide attempts**. *Risk-taking* — irregular sleep patterns and self-reported sleep problems, rather than sleep loss per se, associate with increased risk-taking; heightened impulsivity and emotional dysregulation; impaired judgement increases risky choices by focusing on rewards rather than consequences; higher alcohol, tobacco and substance use. *Physical performance* — reduced hand–eye coordination; increased sports and occupational injury. *Road safety* — increased crash rate, especially young males, with more sleep-related crashes. *Obesity* — altered insulin, ghrelin, leptin and cortisol → insulin resistance, increased sympathetic activity, increased hunger, decreased satiety → more calories, higher fat percentage, less exercise; a **dose–response inverse relationship** between sleep and weight. *Academic* — poor performance.

8.6 Screening

- **History.** Medical, dental and family history — chronic allergies, asthma, recurrent ear or throat infections, family history of sleep apnoea; obesity, prematurity and syndromic conditions increase risk. **Cognitive, behavioural and developmental** — hyperactivity, irritability and difficulty concentrating closely resembling ADHD. **Growth** — assess for delayed growth or failure to thrive, which may signal disrupted sleep and insufficient oxygenation; dentists should assess **height, weight and neck size**.

- **PSQ** — the most widely used tool, in four sub-scores: **breathing, behaviour, sleepiness, other**. A total **>0.33** — more than 8 of 22 items positive — suggests high likelihood of SDB. Not diagnostic, but **one of the most sensitive questionnaires available**.
- **OSA-18** — quality of life and SDB symptoms; complementary to the PSQ. **ESS-CHAD** — modified paediatric Epworth for ages 6–16; ≥10 indicates sleepiness. **FAIREST-6** — dental red flags; **2 mild, 4 moderate, 6 severe**.

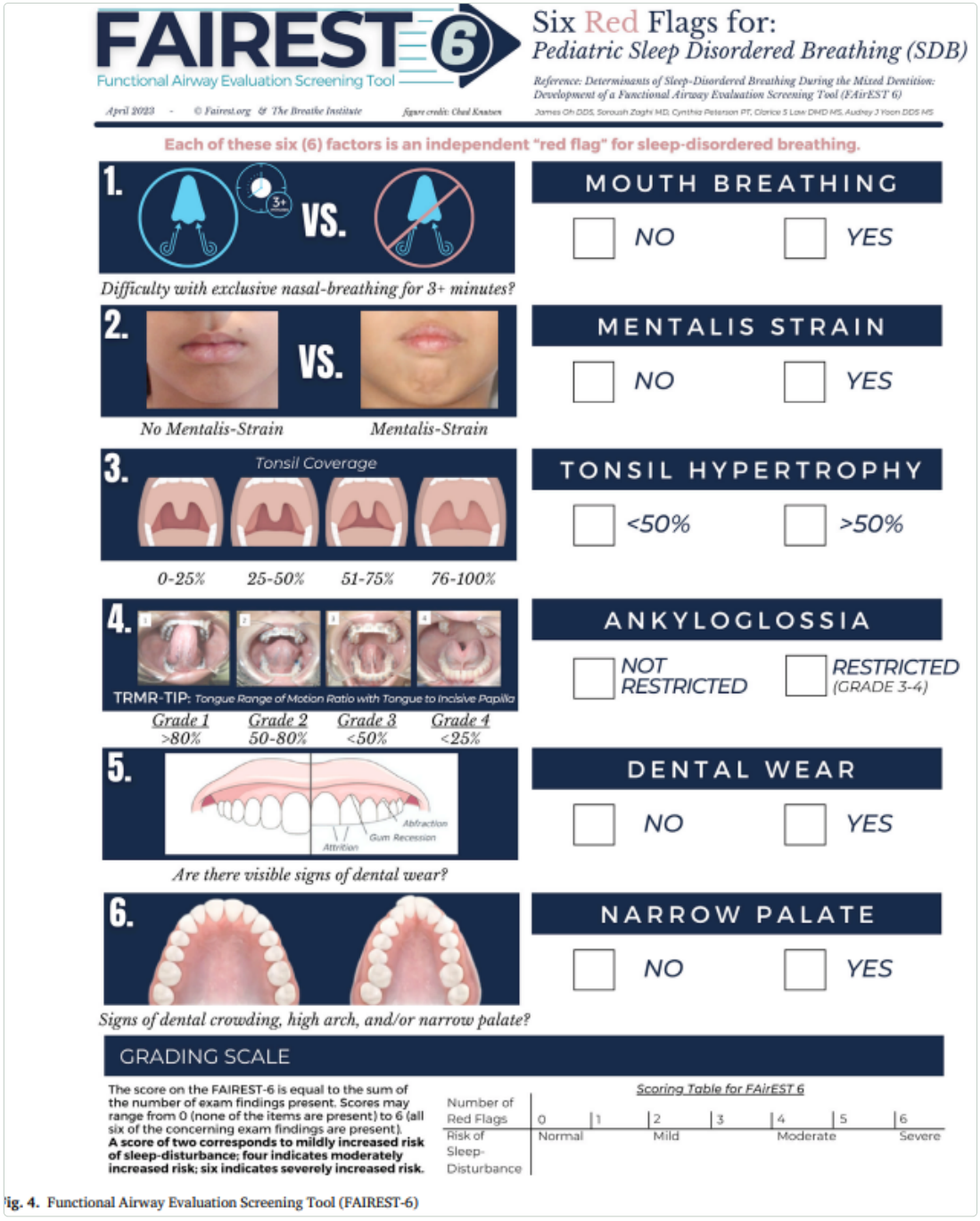


Fig. 4. Functional Airway Evaluation Screening Tool (FAIREST-6)

Fig 8.1 — FAIREST-6: six red flags for paediatric SDB. (1) **Mouth breathing** — difficulty with sustained nasal breathing for three minutes; (2) **mentalis strain**; (3) **tonsil hypertrophy** — <50% vs >50% obstruction; (4) **ankyloglossia** — restricted vs unrestricted tongue range of motion; (5) **dental wear**; (6) **narrow palate** — dental crowding, high arch and/or narrow palate. Scored independently, with a grading scale converting red-flag count to risk level. **Examinable because:** the most dentally-oriented paediatric screening tool in existence — every item is something a dentist observes routinely. Learn all six and the 2/4/6 grading.

- **Tonsils and adenoids** — use the modified Mallampati, Friedman tongue classification, or the **Brodsky scale**. **Hypertrophy obstructing >50% of the airway warrants further evaluation** — but tonsil size does not correlate with severity.
- **Palate and arches** — high-arched palate or narrow arches restricting the airway; narrow arches may indicate habitual mouth breathing.
- **Jaw size and position** — retrognathia or micrognathia; subtle radiographic findings such as inferior hyoid position and reduced mandibular or maxillary length. **Monitor jaw growth regularly in growing patients**, as intervention may be considered if structural concerns persist.
- **Tongue posture and function** — tongue tie, low posture, restricted movement; abnormal posture or tongue thrusting may affect patency and craniofacial development.

- **Breathing habits** — identify predominant mouth breathing, associated with nasal obstruction and contributing to abnormal jaw and facial development, manifesting as a long narrow face with elongated lower facial height, incompetent lip seal and open mouth posture. A common manifestation of paediatric OSA, yet detection and diagnosis remain inconsistent across studies. **Nasal resistance** can be measured by rhinometry.

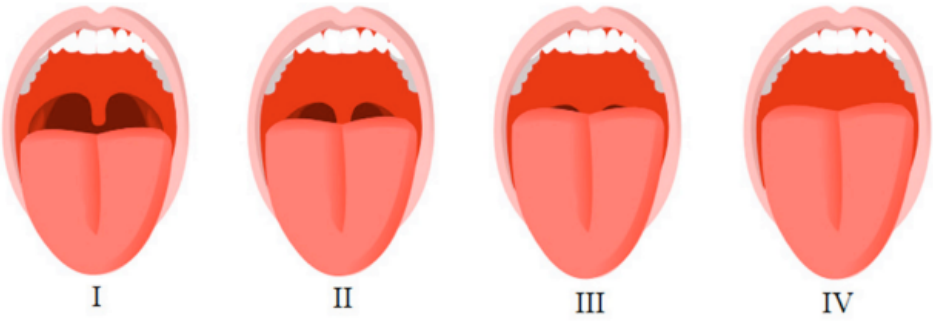


Fig. 5. Modified Mallampati grade
Class 1: Faucial/tonsillar pillars, uvula and soft palate are all visible.
Class 2: Partial visibility of the faucial/tonsillar pillars, uvula and soft palate.
Class 3: Base of the uvula, soft and hard palate visible.
Class 4: Only hard palate is visible.
(Mallampati Score Visual Contributed by Jmarchn. Creative Commons.)



Fig. 6. Brodsky Tonsil grading scale grades
Clinical grading of tonsil size: grade 0, tonsils limited to the tonsillar fossa; grade 1, tonsils occupying up to 25% of the space between the anterior pillars in the oropharynx; grade 2, tonsils occupying 25% to 50% of the space between the anterior pillars; grade 3, tonsils occupying 50% to 75% of the space between the anterior pillars; grade 4, tonsils occupying 75% to 100 % of the space between the anterior pillars.

Fig 8.2 — Modified Mallampati (I–IV) and Brodsky tonsil grades (0–4). *Mallampati:* I — soft palate, uvula, fauces and pillars all visible; II — soft palate and the majority of the uvula; III — soft palate and base of uvula only; IV — hard palate only. *Brodsky:* 0 — tonsils within the pillars; 1 — <25% of the space between the anterior pillars; 2 — 25–50%; 3 — 50–75%; 4 — >75%, or meeting in the midline. **Examinable because:** two classifications frequently confused. Mallampati grades **oral airway patency by tongue position**; Brodsky grades **tonsil size as a percentage of oropharyngeal width**. Technique for Brodsky: depress the tongue and ask the patient to say "ah". Grade 3–4 is an "at risk anatomically" finding (Topic 5) and an exclusion for hypoglossal nerve stimulation (Topic 3).

8.7 Treatment

- **Adenotonsillectomy — first line** and the most effective surgical therapy, improving polysomnographic parameters, behaviour and quality of life. **But post-operative persistence of OSA is common**, and it may be insufficient because it does not address oropharyngeal obstruction secondary to tonsillar hyperplasia, especially with Down syndrome, craniofacial anomalies or neuromuscular disease.
- **Risk factors for failure:** obesity; high pre-operative AHI; age **<3 or ≥7**; nasal abnormality or craniofacial disproportion; family history of OSA; asthma. **Post-operative PSG is suggested for these high-risk children even without persistent snoring** — though persistent OSA is less likely if there is no post-operative snoring.
- **Contraindication:** submucous cleft palate is relative — a partial adenotonsillectomy may be performed.
- **Post-operative management:** usually outpatient. Consider overnight admission where the **SpO_2 nadir is <80%, AHI ≥24/h, or peak PCO_2 ≥60 mmHg**, because of increased risk of respiratory compromise. **Risk factors for complications:** age <3; severe OSA on PSG; cardiac complications; failure to thrive; obesity; upper respiratory infection — if acute URI on the day, reschedule or monitor closely, but note that children with adenotonsillar hypertrophy have chronic rhinorrhoea and congestion even without viral infection; other comorbidity.
- **Partial tonsillectomy** — leaves tissue covering the tonsillar fossa musculature. *Advantages:* reduced morbidity, less immediate post-operative pain, earlier return to normal activity and diet. *Disadvantages:* regrowth risk **0.5–16%**, increased intra-operative blood loss, recurrent infection in the remnant, and the need for long-term monitoring for OSA related to regrowth.
- **Other surgery:** turbinate reduction, septoplasty; expansion sphincter pharyngoplasty for refractory cases; craniofacial procedures such as mandibular distraction osteogenesis; **tracheotomy as a last resort** — very effective but with much morbidity.
- **CPAP** — effective but **not first line where adenotonsillectomy is an option**, because of compliance problems (adherence ~50%, improved by respiratory therapist support). *Indications:* inadequate response to surgery, or surgery contraindicated — which can include patient and family preference, for example religious belief against surgery or transfusion. Often used with neuromuscular or

craniofacial disorders, or residual OSA after surgery. **Requires objective adherence assessment**; address adverse effects and use behaviour modification, and if still non-compliant use an alternative. **Pressure changes over time with growth and development, so reassess periodically**; PSG is used for titration.

- **Medication. Intranasal corticosteroid — only for mild OSA (AHI <5/h)**, not moderate or severe. Effects: small AHI reduction; improvement in snoring and observed apnoea; demonstrated shrinkage of adenoidal tissue. Follow-up needed as long-term effects are unknown. **Leukotriene receptor antagonists** may help in mild cases or where allergic rhinitis contributes.
- **Orthodontics. RME** — for maxillary constriction with dental malocclusion; increases the transverse diameter of the hard palate by reopening the mid-palatal suture using a fixed appliance with an expansion screw anchored on selected teeth; after **3–4 months** of expansion normally mineralised structure is rebuilt; reported mean reduction **~4.8 events/h (−70%)**. **MARPE** — with RME, increases nasal cavity volume, improves nasal airflow and reduces upper airway resistance, contributing to alleviation of mild–moderate SDB. **Mandibular advancement** — for class II correction; with maxillary expansion, one of the two orthodontic treatments that may change airway dimension; in children and adolescents with posterior crossbite or maxillary constriction and retrognathia, expansion and advancement produce airway dimensional change and AHI decrease. **Role:** where adenotonsillectomy is unsuitable, or combined with surgery, particularly with craniofacial abnormality or residual OSA. Long-term follow-up with sleep studies is important.

TRAP — THE HONEST EVIDENCE ON EXPANSION

Functional responses to orthopaedic change are **highly variable**. Expansion does not always help: some children do not improve and **some get worse**. In bi-maxillary expansion studies (maxilla by RME, mandible by dental expansion), about **one third became objectively worse**. There is also natural regression of symptoms in some children, which inflates uncontrolled before-and-after results. Expansion may therefore be considered **only for selected patients** — and crucially, any intervention intended to improve OSA needs the OSA to have been **diagnosed by a physician**. Presenting expansion as a reliable cure is the easiest way to fail this section.

- **Positional and other:** AHI is increased supine and lowest **prone** in children — but position is hard to maintain overnight. **Weight management** for obese children. **Watchful waiting** with close monitoring may suit mild snoring or mild OSA. **Myofunctional therapy** as an adjunct.
- **Summary by indication:** adenotonsillectomy for moderate–severe OSA with enlarged tonsils and adenoids (often first line, reduces AHI significantly, risk of residual OSA) · CPAP for residual OSA, non-surgical candidates, comorbidity (effective but adherence challenging, mask fitting critical, re-titrate with growth) · orthodontics for craniofacial abnormality or residual OSA (improves airway anatomically; adjunct or alternative; variable response) · medical therapy for mild OSA and allergic rhinitis (supportive only) · other surgery for severe or persistent disease. Effective treatment is **multidisciplinary** — paediatricians, ENT, sleep medicine, orthodontists.

8.8 Differences from adults

- **Causes.** *Children:* enlarged tonsils and adenoids lead; genetic predisposition; obesity in a minority but increasingly in adolescents; commonest at 3–6 y. *Adults:* obesity, age, male sex; smoking and alcohol; often diagnosed as OSA or CSA.
- **Presentation.** *Children:* snoring or noisy breathing; restless sleep or unusual positions; frequent bed-wetting; daytime fatigue and attention difficulties; **behavioural problems that may mimic ADHD**; poor academic performance. *Adults:* loud chronic snoring; stopped breathing episodes; morning headache or dry mouth; daytime drowsiness; decreased libido and mood swings; impaired memory or concentration.
- **Thresholds.** Children: AHI ≥1 abnormal; mild 1–5, moderate 5–10, severe ≥10. Adults: AHI ≥5 with symptoms or ≥15 regardless; mild 5–15, moderate 15–30, severe ≥30. **Age boundary 12 years.**
- **Diagnosis.** *Children:* thorough clinical evaluation of behaviour and sleep patterns; PSG; imaging (radiographs, endoscopy) may identify anatomical obstruction; **end-tidal CO_2 is a standard PSG measurement**. *Adults:* clinical or home sleep studies; evaluation of coexisting cardiovascular disease or metabolic syndrome.
- **Treatment.** *Children:* surgical removal of enlarged tonsils and adenoids first; weight management; PAP where surgery is ineffective; behaviour-focused therapy for sleep hygiene; orthodontics in selected cases. *Adults:* lifestyle change (weight loss, alcohol avoidance, position); CPAP; appliance therapy as a non-invasive alternative; surgery where necessary.
- **Outlook.** *Children:* early treatment often resolves the condition entirely, with improvements in behaviour, academic performance and quality of life; ongoing monitoring may be needed to prevent recurrence; where tonsils and adenoids shrink with age symptoms may improve, but many children benefit substantially from early diagnosis and treatment. *Adults:* chronic condition requiring continuous management, with increased risk of stroke, hypertension and cardiovascular disease; long-term adherence essential.

8.9 Underlying conditions and syndromes

- **Medical conditions:** obesity, asthma, allergic rhinitis, GORD and neuromuscular disorders all contribute to obstruction, inflammation or decreased airway muscle tone. Children with Duchenne muscular dystrophy and myasthenia gravis have weakened respiratory muscles, leading to hypoventilation and greater susceptibility.

- **Genetic syndromes:** Down, Prader–Willi, Crouzon and cleft lip and palate often involve craniofacial abnormalities — midfacial hypoplasia, macroglossia, retrognathia — narrowing the airway. In Prader–Willi, hypersomnolence and varying airway obstruction contribute.

DOWN SYNDROME

- Prevalence by AHI: >1 in **69%** · >1.5 in 76% · >2 in 75% · **>5 (moderate–severe) in 50%** · >10 in 34%. Compare 1–5% in the general paediatric population.
- Predisposing factors: **macroglossia, midface hypoplasia, overweight, poor muscle tone**. Moderate–severe disease is more prevalent at younger ages.
- **All children with Down syndrome should be referred for polysomnography by age 4** (AAPD).
- Clinical symptoms are **unreliable**: only a weak correlation between parental perception and PSG findings, and **53.8% of affected children had no symptoms** — the strongest argument for objective screening in a syndromic population.

- **In adults and generally.** *Cardiovascular disease* — SDB is often comorbid and may be causal, with evidence linking it to heart failure, hypertension and stroke; mechanisms include increased platelet aggregation, hypercoagulability, endothelial dysfunction, and contribution to left ventricular hypertrophy and heart failure. Guidelines now recommend considering SDB management in acute stroke and congestive heart failure. *Obesity and diabetes* — SDB is a risk factor for and exacerbates insulin resistance and type 2 diabetes, with higher prevalence in the diabetic population. *Chronic kidney disease* — high prevalence; severe SDB increases risk. *Stroke* — via atherosclerosis, hypertension and arrhythmia. *Pulmonary* — sleep-related hypoventilation can associate with restrictive pulmonary dysfunction, as in Poland syndrome.
- **Why it matters:** increased morbidity and mortality; reduced quality of life; healthcare burden and cost; and the **therapeutic opportunity** — treating SDB can positively affect associated chronic disease. In children, SDB with underlying conditions causes daytime behavioural problems, cognitive impairment, cardiovascular strain, growth delay and reduced quality of life. Early diagnosis matters especially because **some signs of neuromuscular disease may present initially as SDB**. Monitor at-risk children closely for snoring, mouth breathing, nocturnal gasping and behavioural problems.

8.10 Referral pathways and diagnosis

- **Sleep physician or ENT** — for children with multiple risk factors or symptoms, for comprehensive evaluation which may include PSG or further imaging. For enlarged adenoids and tonsils, surgical removal is first-line.
- **Orthodontic intervention** — where craniofacial development suggests a narrow airway or malocclusion, consider early intervention. **Any intervention intended to improve OSA needs to be diagnosed by a physician.** Maxillary expansion (RME, MARPE) increases nasal cavity volume, improves airflow and reduces resistance, contributing to alleviation of mild–moderate SDB.
- **Collaboration with paediatricians** — regular communication ensures coordinated treatment and monitoring, especially where SDB overlaps other conditions. **Parent and caregiver education** — awareness of symptoms, risks and health impacts, the importance of addressing them early, guidance on sleep hygiene and avoiding environmental triggers.
- **Diagnosis by a physician.** Clinical evaluation — detailed history of habitual snoring, observed apnoeas, laboured or paradoxical breathing, sleepiness, behavioural problems and learning difficulties; examination of upper airway anatomy including tonsil and adenoid size, craniofacial features, obesity and comorbidity. Validated questionnaires aid screening but are not definitive. **Overnight attended in-laboratory PSG is the gold standard**, recording EEG, EOG, EMG, ECG, airflow, effort, oxygen saturation, position and end-tidal CO_2, allowing correlation of events with stages and arousals; **obstructive AHI ≥1/h is abnormal**. **Indications for PSG:** clinical symptoms, especially **before adenotonsillectomy in children under 2 years**, in obesity, craniofacial or neuromuscular disorders, Down syndrome, or persistent symptoms after surgery; also for CPAP titration and complex cases. **Other tools:** respiratory polygraphy (no staging, less validated); nocturnal oximetry (simple, good predictive value in high-risk children); **home sleep apnoea testing is emerging and not recommended as a standalone paediatric diagnostic tool**; daytime nap studies less accurate; imaging and biomarkers promising but need further research.

TRAP — CBCT AIRWAY ANALYSIS IN CHILDREN

Reliability and validity are both questionable. **Reliability** — the oropharynx images clearly, but the most frequent sites of collapse are the nasopharynx and hypopharynx, and the adenoid site in children, so reliability is poor for the region that matters. **Validity** — upper airway dynamic function is complex, and a healthy subject with a narrow airway can overcome anatomical risk by actively dilating during inspiration. Cautions: poor inter-examiner reliability; normative values with large standard deviations; and most fundamentally, **airways are not static**. Do not offer CBCT airway analysis as a diagnostic tool for paediatric OSA.

NUMBERS — TOPIC 8

- OSA 1–5% (typically 2–3%) · snoring 3–12% · peaks 2–8 y and adolescence · commonest 3–6 y · age cut-off 12 y
- AHI: normal <1 · mild 1–5 · moderate 5–10 · severe ≥10; obstructive AHI ≥1 abnormal
- Sleep duration: 4–12 mo 12–16 h · 1–2 y 11–14 h · 3–5 y 10–13 h · 6–12 y 9–12 h · 13–18 y 8–10 h. <8 h in adolescence → ~3× suicide-attempt risk; caffeine users 2× more likely to have difficulty sleeping
- PSQ >0.33 (>8 of 22) · FAIREST-6 2/4/6 · paediatric ESS ≥10 · tonsillar obstruction >50% warrants evaluation
- Brodsky 0–4: within pillars · <25% · 25–50% · 50–75% · >75%
- Adenotonsillectomy failure risk: age <3 or ≥7, obesity, high pre-op AHI, nasal/craniofacial abnormality, family history, asthma. Overnight admission if SpO_2 nadir <80%, AHI ≥24/h or peak PCO_2 ≥60 mmHg. Partial tonsillectomy regrowth 0.5–16%
- Intranasal steroid only for AHI <5 · paediatric CPAP adherence ~50%
- RME: 3–4 months expansion; mean AHI reduction ~4.8/h (−70%); **but ~⅓ objectively worse after bi-maxillary expansion**
- Down syndrome: AHI >1 in 69% · >5 in 50% · >10 in 34%; PSG by age 4; 53.8% asymptomatic. PSG before adenotonsillectomy in children <2 y

SELF-TEST — TOPIC 8

1. A 5-year-old snores nightly, is hyperactive at school and wets the bed. Assessment and plan?

A classic paediatric presentation — behavioural rather than sleepy, with secondary enuresis. Take a medical, dental and family history covering allergies, asthma, recurrent ear and throat infections, prematurity, syndromic conditions and family history of sleep apnoea; record height, weight and neck size and plot growth looking for failure to thrive; administer the PSQ (positive if >0.33, i.e. more than 8 of 22 items) and consider FAIREST-6; and examine tonsil size by the Brodsky scale, modified Mallampati, palatal vault and arch width, jaw size and position, tongue posture and tie, and breathing habits. Then refer to a paediatric sleep physician or ENT for comprehensive evaluation and PSG, noting an obstructive AHI ≥1/h is abnormal and adenotonsillectomy is first-line if adenotonsillar hypertrophy is confirmed. I would not perform CBCT airway analysis, and would not begin orthodontic expansion to treat OSA before a physician diagnosis.

2. Give the paediatric AHI thresholds and explain why they differ.

Normal <1, mild 1–<5, moderate 5–<10, severe ≥10 — roughly one third of adult values (5, 15, 30); obstructive AHI ≥1/h is abnormal. Children normally have very stable breathing during sleep, so even a small number of obstructive events is pathological, and the consequences for growth, neurocognitive development, behaviour and academic performance occur at much lower event rates. The trap: a child with AHI 6 has *moderate* disease, not mild. Twelve years is the cut-off above which adult criteria apply.

3. First-line treatment and risk factors for failure?

Adenotonsillectomy — the most effective surgical therapy, improving PSG parameters, behaviour and quality of life. But post-operative persistence is common, and it may be insufficient because it does not address oropharyngeal obstruction secondary to tonsillar hyperplasia, especially in Down syndrome, craniofacial anomalies or neuromuscular disease. Risk factors for failure: obesity, high pre-operative AHI, age under 3 or 7 and over, nasal abnormality or craniofacial disproportion, family history of OSA, and asthma. Post-operative PSG is suggested for these children even without persistent snoring. Consider overnight admission if the SpO_2 nadir is <80%, AHI ≥24/h, or peak PCO_2 ≥60 mmHg.

4. A parent asks whether palatal expansion will cure their child's sleep apnoea.

Answer with the evidence. RME increases the transverse diameter of the hard palate by reopening the mid-palatal suture, and both RME and MARPE increase nasal cavity volume, improve airflow and reduce upper airway resistance; mean AHI reductions of about 4.8 events per hour (~70%) have been published in children with maxillary constriction. **But** responses are highly variable — expansion does not always help, some children do not improve and some get worse, and in bi-maxillary expansion studies about one third became objectively worse. Natural symptom regression also flatters uncontrolled results. So consider it only for selected patients, as an adjunct or where adenotonsillectomy is unsuitable, and any intervention intended to improve OSA requires physician diagnosis with long-term follow-up including sleep studies.

5. Why refer a child with Down syndrome for PSG even if parents report no problems?

Because symptoms are unreliable in this population. Prevalence is very high — ~50% at AHI >5 and 69% at AHI >1, versus 1–5% generally — driven by macroglossia, midface hypoplasia, overweight and poor muscle tone, with moderate–severe disease more prevalent at younger ages. Only a weak correlation exists between parental perception and PSG findings, and 53.8% of affected children had no symptoms. Guidance therefore recommends referral for polysomnography **by age 4 regardless of reported symptoms**.

TOPIC 8

Diagnosis and management of paediatric sleep-disordered breathing

EXAMINER'S BLUEPRINT

1. Prevalence of snoring and obstructive sleep apnoea in children
2. Aetiology and pathophysiology of snoring and OSA in children
3. Signs and symptoms of SDB in children and adolescents
4. Other causes of poor or insufficient sleep in children and the impact on development
5. Other causes of poor or insufficient sleep in adolescents and the impact on cognition, physical performance, impulse control and decision making
6. Screening of children and adolescents for SDB
7. Treatment of snoring and OSA in children and adolescents including surgical options, CPAP, orthodontic treatment and other therapies
8. Differences in SDB between children and adults
9. The relationship between SDB and underlying medical conditions and syndromes
10. Referral pathways for children and adolescents for diagnosis and treatment

Paediatric SDB is examined heavily because it is where dentists see patients earliest and where the potential to do harm through overconfident intervention is greatest. Two ideas carry the topic: the **thresholds are lower and the presentation is behavioural rather than sleepy**; and **adenotonsillectomy, not orthodontics, is first-line**. Section 8.7 contains an evidence-based caution about maxillary expansion that will distinguish a careful candidate from an enthusiastic one.

8.1 Prevalence

- **OSA:** 1–5%, with many sources suggesting around 2–3%.
- **Habitual snoring:** 3–12% (some sources cite 10–20%).
- Age and obesity influence prevalence, with peaks in **younger children (2–8 years)** and again in **adolescents**.
- Prevalence varies by population, gender and race.
- Most commonly affects children aged **3–6 years**, when tonsillar tissue is proportionally largest.

8.2 Aetiology and pathophysiology

Aetiology

Cause	Detail
Adenotonsillar hypertrophy	The **most frequent cause** of airway obstruction in children, particularly in pre-schoolers
Obesity	A significant risk factor, especially in teenagers, contributing to airway narrowing
Craniofacial abnormality	Micrognathia, a narrowed midface, and other structural differences that reduce airway dimension and increase collapsibility
Allergic rhinitis	Chronic nasal congestion contributing to obstruction
Neuromuscular disorders	Down syndrome, cerebral palsy — affecting head and neck muscle tone, so that muscles relax excessively during sleep leading to airway closure
Tumours or other growths	Rarely, a growth in the airway causes physical blockage

Additional anatomical risk factors described across the literature: micrognathia, retrognathia, macroglossia, adenotonsillar hypertrophy, nasal septal deviation, and a high-arched or narrow hard palate. Note the important caveat: **caution should be taken when using tonsil size evaluation, because tonsil size does not correlate with OSA severity** (Brodsky tonsil classification).

Pathophysiology — the sequence

1. **Muscle relaxation.** As the child falls asleep, the muscles that keep the upper airway open naturally relax.
2. **Airway narrowing or collapse.** Where the airway is already narrow — large tonsils, a small jaw — or where muscle tone is reduced, relaxation brings the airway tissues closer together or causes them to fold, producing partial or complete blockage.
3. **Respiratory effort and arousal.** During obstruction the brain detects falling oxygen and rising carbon dioxide, and triggers a brief awakening to restore muscle tone and reopen the airway.
4. **Sleep disruption.** These frequent arousals, often not remembered, fragment sleep and reduce restorative stages, producing daytime fatigue, hyperactivity and poor concentration.
5. **Consequences.** Over time this pattern affects growth, neurocognitive development, academic performance and emotional well-being.

The four contributing mechanisms in paediatric OSA are conventionally listed as **craniofacial morphology, tonsillar hypertrophy, altered neuromuscular tone and ventilatory control**.

Adenoid facies (long face syndrome)

An atypical orofacial growth pattern associated with chronic mouth breathing and nasal obstruction: a **long face** with increased lower facial height, a **convex profile**, mouth breathing with incompetent lip seal and open mouth posture, a **high-arched palate**, a **constricted maxilla** usually causing posterior crossbites and crowded teeth, a retruded mandible, and sometimes an anterior open bite. The consequences of a reduced upper airway are therefore not only sleep-breathing problems but also a constricted maxilla with posterior crossbite and crowding, together with impaired growth, cardiovascular problems, and learning and behavioural problems.

> **EXAM TRAP — THE MALOCCLUSION–OSA LINK IS WEAKER THAN DENTISTS ASSUME**
>
> Most children with OSA do **not** have obviously altered craniofacial morphology; the measured differences are frequently not clinically relevant. **Dentists overestimate the correlation between malocclusion and the prevalence of OSA.** A greater prevalence of malocclusion has not been found to be associated with OSA, and abnormal respiration or OSA is not a frequent or common cause of malocclusion. Adenoid facies is a recognisable pattern, not a reliable screening test. Stating this explicitly demonstrates critical reading; asserting a strong causal link between malocclusion and OSA does the opposite.

8.3 Signs and symptoms in children and adolescents

During sleep	During the day
• Loud snoring or noisy breathing — gasping, snorting, choking • Pauses in breathing lasting several seconds • **Mouth breathing** • Restless sleep, frequent changes of position • Night sweats • **Unusual sleep positions** — sleeping seated or with the neck hyperextended • **Bed-wetting in an older child** who has been dry for a long period	• Morning headaches; morning fatigue • Daytime sleepiness (less prominent than in adults) • Difficulty concentrating and paying attention • **Behavioural problems** — hyperactivity, aggression, impulsivity • Irritability • Poor academic performance • Excessive weight gain *or* poor weight gain • Neuropsychological and cognitive deficits, including reduced general intelligence level and IQ

8.4 Other causes of poor sleep in children, and the impact on development

Recommended sleep duration

Age	Recommended sleep per 24 hours
Infant <4 months	Not specified — variation too large
Infant 4–12 months	12–16 hours
Children 1–2 years	11–14 hours
Children 3–5 years	10–13 hours
Children 6–12 years	9–12 hours
Teenagers 13–18 years	8–10 hours

Less than recommended: attention, behaviour and learning problems; increased risk of accidents, injuries, hypertension, obesity, diabetes and depression; and in teenagers, increased risk of self-harm, suicidal thoughts and suicide attempts. **More than recommended:** hypertension, diabetes, obesity and mental health problems. **Sufficient sleep:** improved attention, behaviour, learning, memory, emotional regulation, quality of life, and mental and physical health.

Causes of poor or insufficient sleep

Category	Detail
Environmental	A noisy or stressful home environment, inconsistent bedtime routines, family conflict
Behavioural	Excessive screen time before bed — video games in particular — disrupting circadian rhythm and causing mental arousal; heavy meals or caffeine close to bedtime; inconsistent bedtimes, lack of a wind-down routine, co-sleeping; poor sleep hygiene generally
Medical and sleep disorders	Allergies, colds, ear infections and headaches; sleep apnoea, restless legs syndrome, periodic limb movements, night terrors and nightmares; acid reflux; insomnia
Medications	Stimulants used for ADHD; SSRIs for psychiatric conditions
Mental and emotional health	High levels of anxiety and depression — in the child, and maternal anxiety and depression during pregnancy — are linked to increased sleep problems

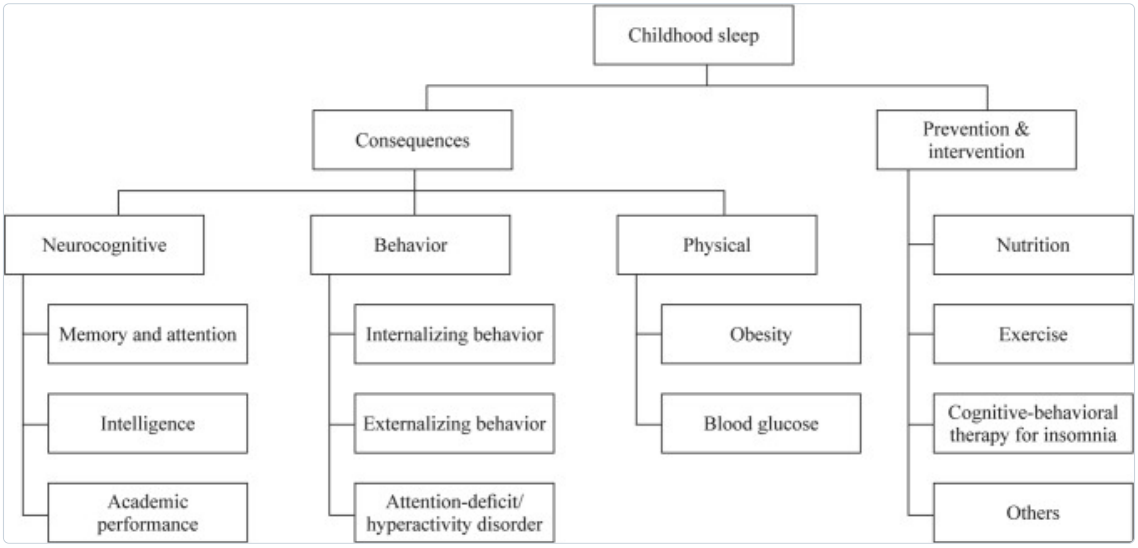


Figure 8.1 — Childhood sleep: consequences and prevention/intervention. The consequences branch divides into neurocognitive (memory and attention, intelligence, academic performance), behaviour (internalising behaviour, externalising behaviour, attention-deficit/ hyperactivity disorder), and physical (obesity, blood glucose). The prevention and intervention branch covers nutrition, exercise, cognitive-behavioural therapy for insomnia and others.

Why it is examinable: this figure supplies a ready-made three-domain structure — neurocognitive, behavioural, physical — for any question about the developmental consequences of paediatric sleep loss.

Impact on development

Domain	Effect
Physical health	Obesity, impaired glucose metabolism, high blood pressure
Neurocognitive	Impaired memory retention and consolidation; attention — greater levels of ADHD-like symptoms; intelligence — lower scores on verbal IQ, performance IQ and full-scale IQ; poorer academic performance; reduced learning ability
Emotional and behavioural	Internalising behaviours (depression, anxiety); externalising behaviours (aggression, rule-breaking); ADHD and other behavioural disorders with inattention, impulsivity and hyperactivity; low energy and poor mood regulation; decreased social skills
Neurodevelopmental risk	Insufficient sleep in the early years may lead to a higher risk of poor neurobehavioural function later

8.5 Adolescents: causes and consequences of insufficient sleep

> **THE FRAMING SENTENCE**
>
> The reduction of sleep in adolescents is **not** driven by a lower need for sleep, but by biological, psychological and socio-cultural influences acting together.

Biological processes

- **Circadian phase delay.** Early adolescent development and pubertal onset are associated with a more **evening-type circadian phase preference** — the preferred timing of sleep, wake and daytime activity shifts later.
- **Sleep–wake homeostasis.** Sleep pressure builds more slowly with time awake in adolescents. Together with the phase delay this makes it a major challenge for young people to fall asleep in the early evening and to wake refreshed in the early morning.
- **Weekend catch-up.** Going to bed later and waking later at weekends reflects circadian biology and is also a response to insufficient weekday sleep — but later weekend sleep timing and catch-up sleep **further delay the signal for the biological night**, perpetuating the cycle.

Electronic media

Television, music players, games consoles, computers and phones act through four mechanisms: media use **directly displaces** sleep; devices allow greater interaction with friends at night; light from devices **suppresses melatonin** and disrupts circadian rhythm, producing inability to fall asleep at a reasonable time; and use increases sleep-disrupting mental, emotional and physiological arousal.

School start times

Early school start times conflict with the delayed circadian rhythm and produce chronic insufficient sleep. Later-starting middle schools report later rise times, more total sleep on school nights, less daytime sleepiness, less tardiness, fewer attention and concentration difficulties, better academic performance, and better safety with a **reduced crash rate**.

Caffeine

- Shorter sleep duration, increased sleep onset latency, increased wake after sleep onset, and increased daytime sleepiness.
- Consumers are **twice as likely** to have difficulty sleeping and to report morning sleepiness — establishing a vicious cycle in which sleepiness prompts more caffeine.
- Caffeine reduces the percentage of time in slow-wave sleep in a **dose-related** manner and alters the temporal organisation of REM and NREM sleep, affecting learning and memory consolidation.
- Females are more likely to show tolerance and withdrawal symptoms.
- Associated with higher risk of other substance use and abuse and other risk-taking behaviour, and inversely related to academic achievement.

Other causes

- Chronic medical illness
- Mental health problems — anxiety, stress
- Prescribed psychotropic medication
- Chronic respiratory illness such as asthma
- Pain conditions such as migraine
- Use of stimulants — especially those prescribed for ADHD — as a countermeasure to sleepiness or as academic performance enhancers
- Over-the-counter or prescribed medication to induce sleep, leading to residual daytime sleepiness
- Alcohol consumption
- Academic demands: excessive homework and extracurricular load

Impacts

Domain	Effect
Cognition	Poor judgement, lack of motivation, inattention and affective dysregulation. Higher-level cognitive **executive functions**, for which adolescence is a critical developmental period, are **selectively** affected by sleep loss — problem-solving and decision-making in particular
Mood and self-harm	Shorter school-night total sleep time is associated with daytime sleepiness and depressive symptoms. Sleeping **less than 8 hours** is associated with an almost **threefold increased risk of suicide attempts**. Increased risk of mood disorders and suicidal ideation
Risk-taking	Irregular sleep patterns and self-reported sleep problems — rather than sleep loss *per se* — are associated with increased risk-taking behaviour. Heightened impulsivity and emotional dysregulation; impaired judgement increases risky choices by focusing on potential rewards rather than consequences. Higher rates of alcohol, tobacco and substance use
Physical performance	Reduced hand–eye coordination affecting sport and daily activity; increased sports-related and occupational injury risk
Road safety	Increased rate of motor vehicle crashes, especially in young males — more sleep-related crashes with drowsy driving

Domain	Effect
Obesity and metabolism	Increased risk of obesity through alterations in metabolic profile — insulin, ghrelin, leptin, cortisol — producing insulin resistance, increased sympathetic activity, increased hunger and decreased satiety, so that adolescents consume more calories with a higher percentage of fat and exercise less. There is a **dose–response inverse relationship** between sleep and weight: less sleep, higher BMI
Academic	Poor academic performance

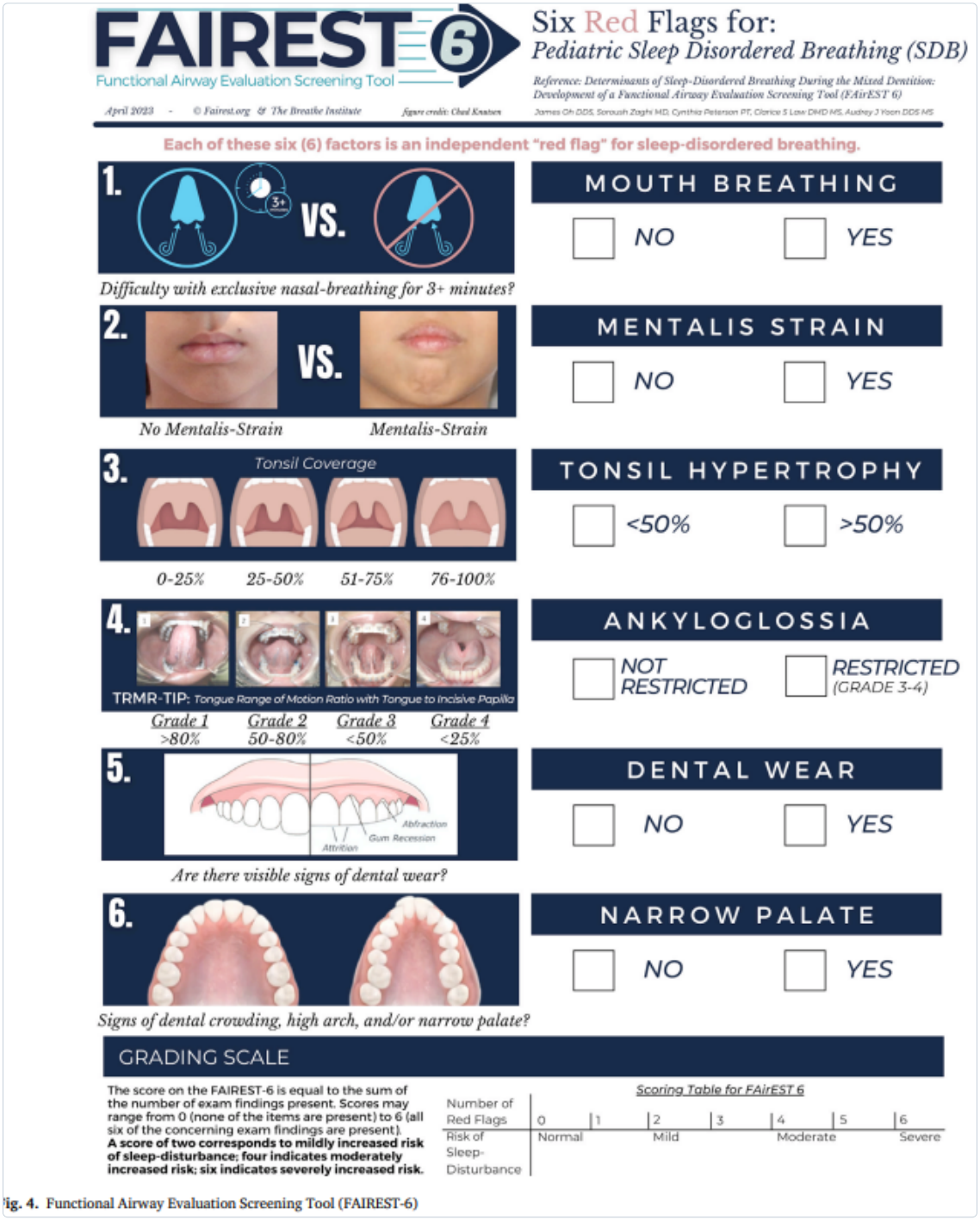
8.6 Screening children and adolescents

History and general assessment

- **Medical, dental and family history:** chronic allergies, asthma, recurrent ear or throat infections, and a family history of sleep apnoea or SDB. Obesity, prematurity and syndromic conditions increase risk.
- **Cognitive, behavioural and developmental assessment:** children with SDB often display hyperactivity, irritability and difficulty concentrating, closely resembling ADHD.
- **Growth and physical development:** assess for delayed growth or failure to thrive, which may signal disrupted sleep and insufficient oxygenation during growth. Dentists should assess **height, weight and neck size** to screen paediatric patients for OSA.

Screening tools

Tool	Structure, threshold and role
Paediatric Sleep Questionnaire (PSQ)	The most widely used paediatric instrument, structured into four sub-scores: **breathing, behaviour, sleepiness and other** domains. A total score exceeding **0.33** — meaning more than 8 of the 22 questions (33%) answered positively — suggests a high likelihood of SDB. Not diagnostic, but recognised as an effective initial screening tool and **one of the most sensitive questionnaires available** for detecting SDB
OSA-18	Assesses quality of life and SDB symptoms; a complementary tool to the PSQ
Modified paediatric Epworth (ESS-CHAD)	For children aged 6–16; evaluates daytime sleepiness and prompts further investigation if abnormal
FAIREST-6	Designed to identify red-flag indicators of paediatric SDB. Scoring: **2 = mild, 4 = moderate, 6 = severe** increase in risk



ig. 4. Functional Airway Evaluation Screening Tool (FAIREST-6)

Figure 8.2 — FAIREST-6: six red flags for paediatric sleep-disordered breathing. (1) **Mouth breathing** — difficulty with sustained nasal breathing for three minutes; (2) **mentalis strain**; (3) **tonsil hypertrophy** — <50% versus >50% obstruction, using the illustrated grading; (4) **ankyloglossia** — restricted versus not restricted tongue range of motion; (5) **dental wear** — visible signs of wear; (6) **narrow palate** — signs of dental crowding, high arch, and/or narrow palate. Each item is scored independently as a red flag, with the grading scale converting the number of red flags to a risk level.

Why it is examinable: this is the most dentally-oriented paediatric screening tool in existence — every one of the six items is something a dentist observes routinely. Learn all six, and the 2/4/6 grading.

Clinical oral and craniofacial examination

What to examine	What to look for
Tonsils and adenoids	Use the modified Mallampati classification, Friedman tongue classification, or the **Brodsky tonsil size scale**. **Tonsillar hypertrophy obstructing more than 50% of the airway warrants further evaluation.** But remember tonsil size does not correlate with OSA severity

What to examine	What to look for
Palatal structure and dental arches	High-arched palate or narrow dental arches, which may restrict the airway; narrow arches may also indicate habitual mouth breathing
Jaw size and position	Retrognathia or micrognathia. Subtle radiographic findings such as an inferiorly positioned hyoid and reduced mandibular or maxillary length may also indicate airway compromise. **Jaw growth should be monitored regularly in growing patients**, as interventions may be considered if structural concerns persist
Tongue posture and function	Tongue tie, low tongue posture, restricted tongue movement. Abnormal posture or habits such as tongue thrusting may affect airway patency and craniofacial development
Breathing habits	Identify predominant mouth breathing. Chronic mouth breathing is associated with nasal obstruction and can contribute to abnormal jaw and facial development, manifesting as a long, narrow face with elongated lower facial height, incompetent lip seal and open mouth posture. Mouth breathing is a common clinical manifestation of paediatric OSA, yet its detection and diagnosis remain inconsistent across studies
Nasal resistance	Can be measured by rhinometry

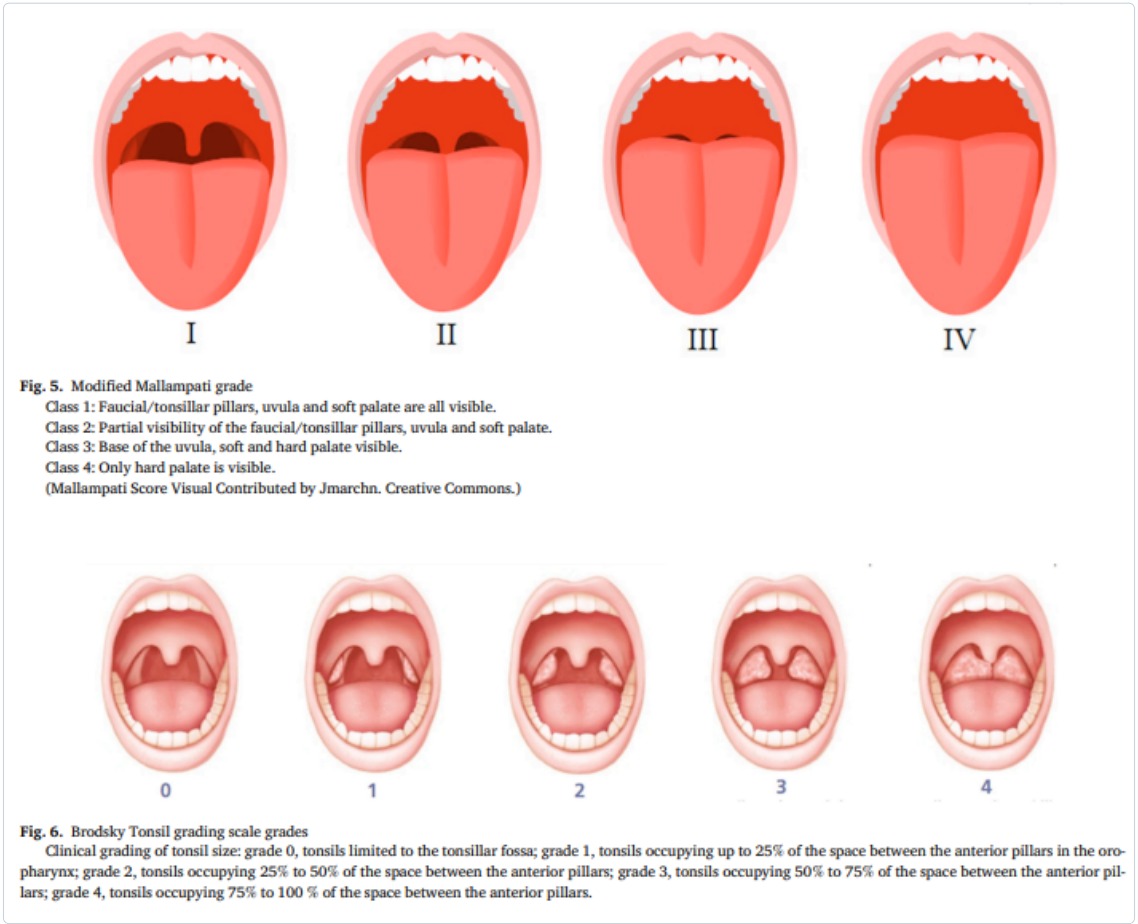


Fig. 5. Modified Mallampati grade
Class 1: Faucial/tonsillar pillars, uvula and soft palate are all visible.
Class 2: Partial visibility of the faucial/tonsillar pillars, uvula and soft palate.
Class 3: Base of the uvula, soft and hard palate visible.
Class 4: Only hard palate is visible.
(Mallampati Score Visual Contributed by Jmarchn. Creative Commons.)

Fig. 6. Brodsky Tonsil grading scale grades
Clinical grading of tonsil size: grade 0, tonsils limited to the tonsillar fossa; grade 1, tonsils occupying up to 25% of the space between the anterior pillars in the oropharynx; grade 2, tonsils occupying 25% to 50% of the space between the anterior pillars; grade 3, tonsils occupying 50% to 75% of the space between the anterior pillars; grade 4, tonsils occupying 75% to 100 % of the space between the anterior pillars.

Figure 8.3 — Modified Mallampati grades (I–IV) and Brodsky tonsil grades (0–4). *Mallampati:* class 1 — soft palate, uvula, fauces and tonsillar pillars all visible; class 2 — soft palate and the majority of the uvula visible; class 3 — soft palate and the base of the uvula only; class 4 — only the hard palate visible. *Brodsky:* grade 0 — tonsils within the pillars; grade 1 — tonsils occupying <25% of the space between the anterior pillars; grade 2 — occupying 25–50%; grade 3 — occupying 50–75%; grade 4 — occupying >75% of the space between the anterior pillars.

Why it is examinable: two classifications frequently confused with one another. Mallampati grades ***oral airway patency by tongue position****; Brodsky grades* ***tonsil size by percentage of oropharyngeal width****. Learn both scales precisely, and pair them with the caution that tonsil size does not correlate with OSA severity.*

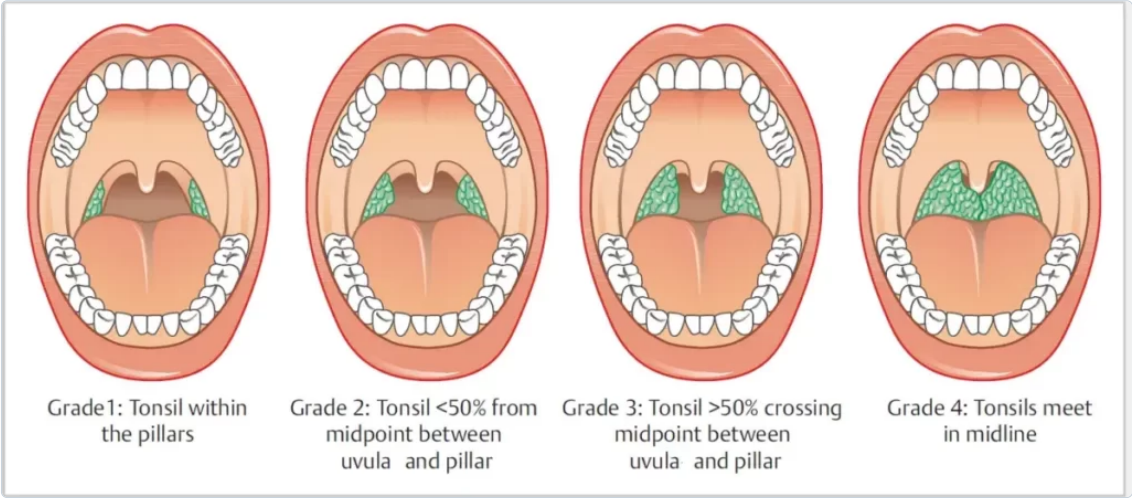


Figure 8.4 — Brodsky tonsil grading in detail: grade 1, tonsil within the pillars; grade 2, tonsil <50% from midpoint between uvula and pillar; grade 3, tonsil >50% crossing the midpoint between uvula and pillar; grade 4, tonsils meet in the midline.

Why it is examinable: the practical technique matters — depress the tongue and ask the patient to say "ah". Grade 3 or 4 is one of the "at risk anatomically" findings you should document (Topic 5), and is an exclusion criterion for hypoglossal nerve stimulation (Topic 3).

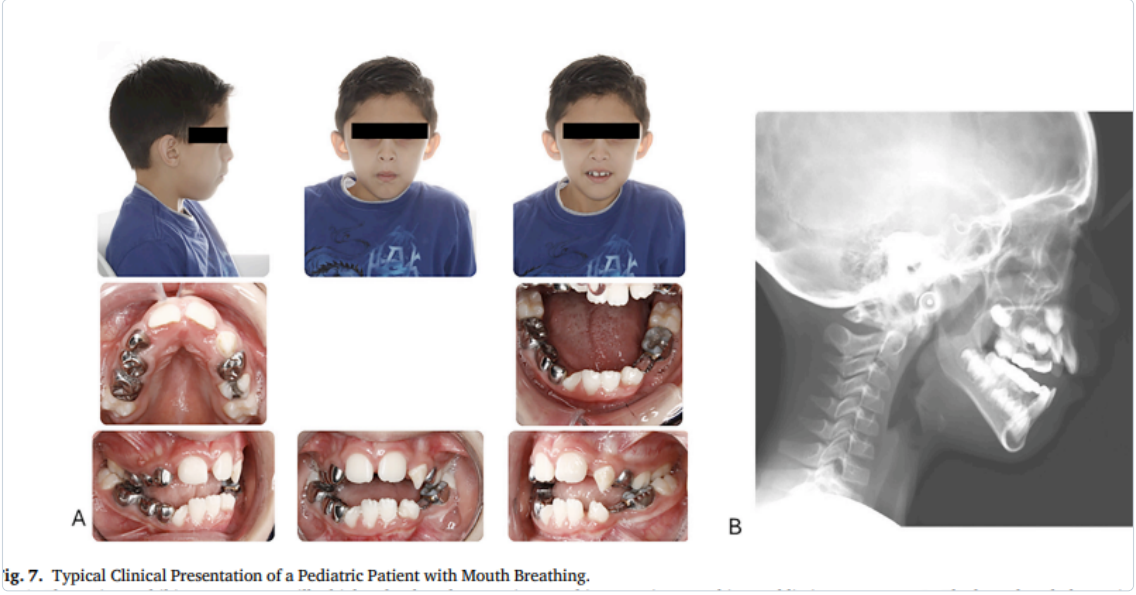


Figure 8.5 — Typical clinical presentation of a paediatric patient with mouth breathing: facial appearance with open mouth posture and incompetent lip seal, the intraoral appearance of the associated narrow, high-arched maxillary arch with crowding, and the corresponding radiographic appearance.

Why it is examinable: this is "adenoid facies" as it actually presents. Use it to anchor the constellation — long face, convex profile, mouth breathing, incompetent lip seal, high-arched constricted maxilla, posterior crossbite, crowding, retruded mandible — while remembering the caveat in 8.2 that most children with OSA do not show obvious morphological change.

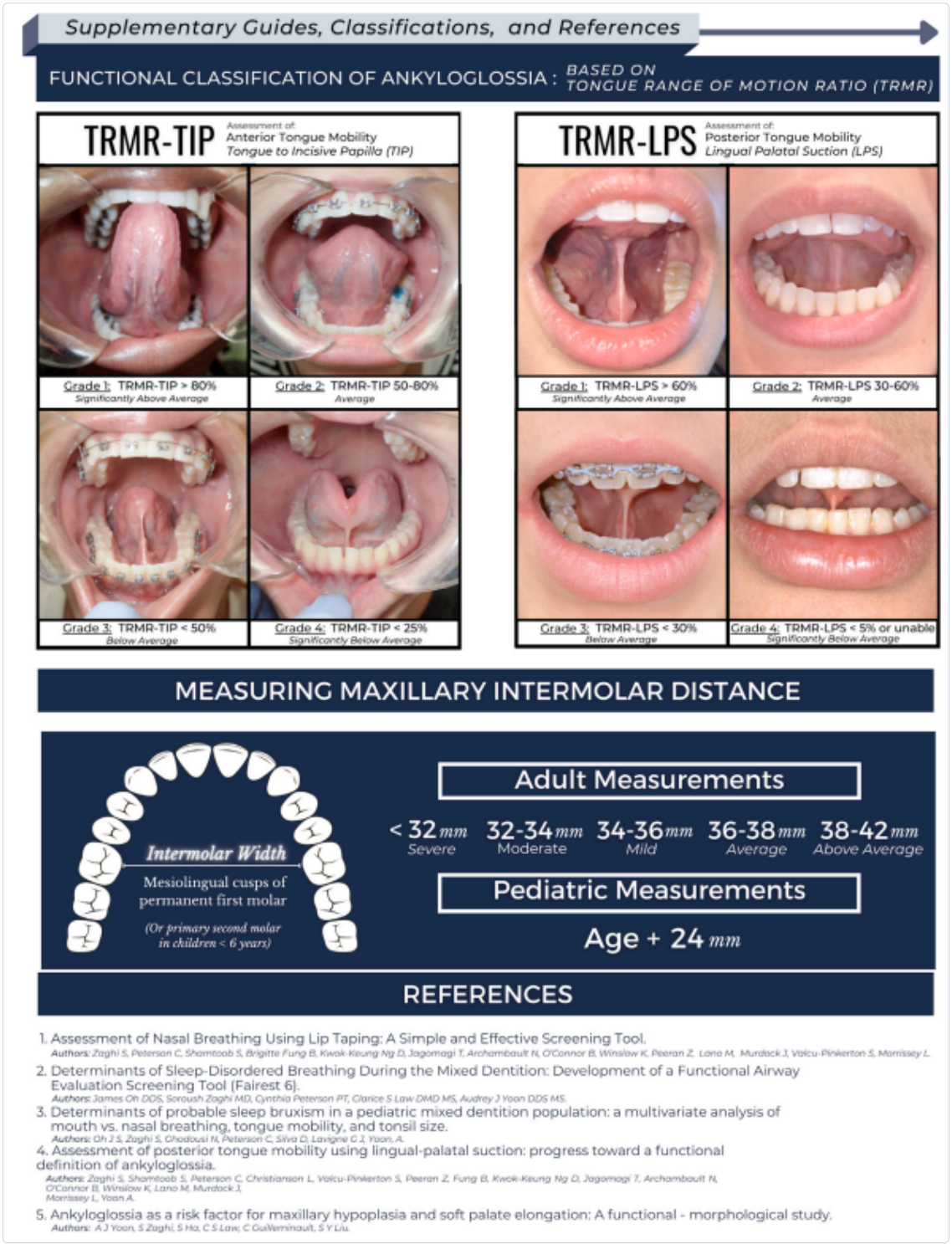
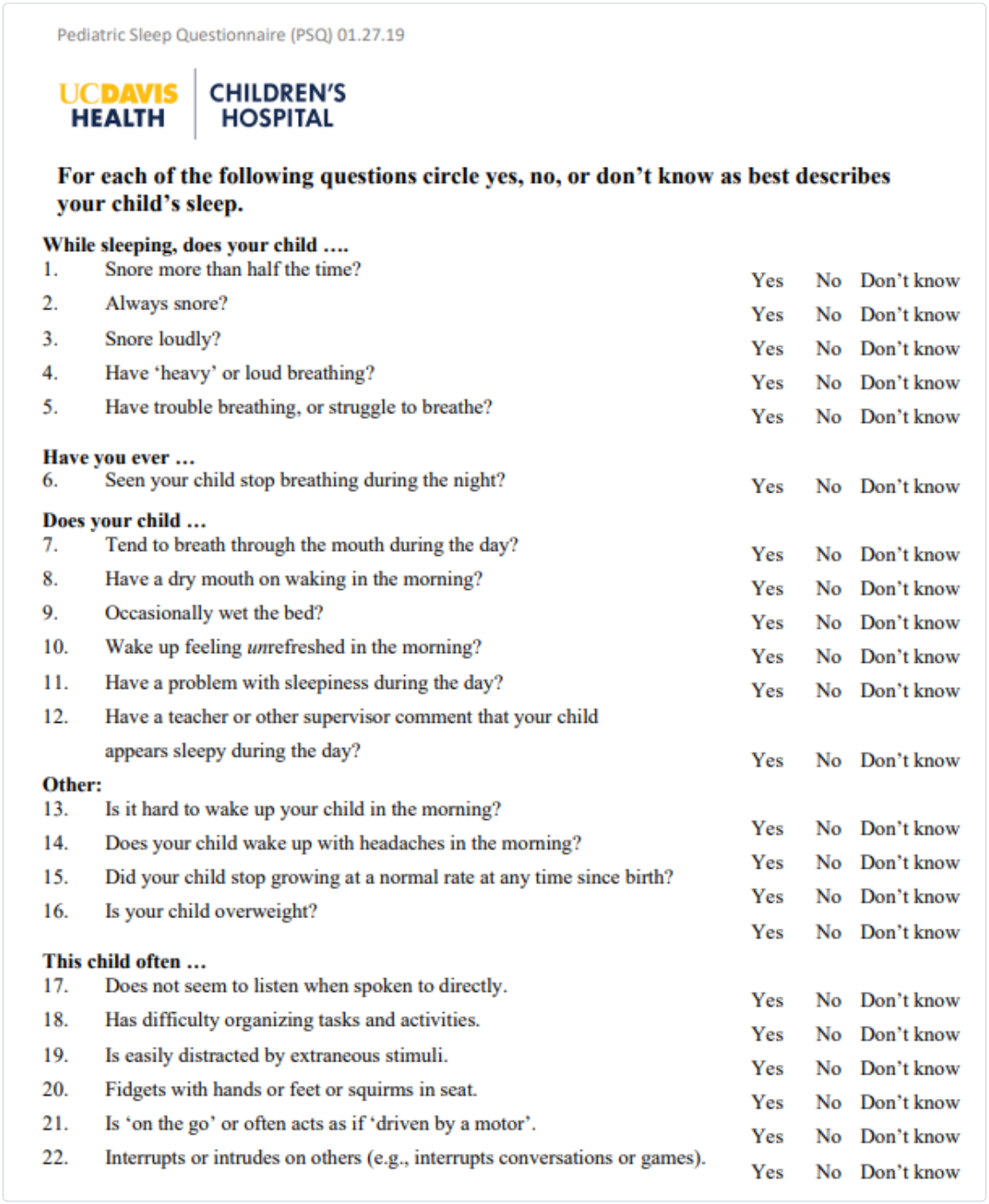


Figure 8.6 — Supplementary guides: the functional classification of ankyloglossia based on the tongue range of motion ratio (TRMR), shown for both the tongue-tip (TRMR-TIP) and the lingual-palatal suction (TRMR-LPS) positions with graded photographic examples; and reference values for **maxillary intermolar distance** — adult measurements graded <32, 32–34, 34–36, 36–38 and 38–42 mm, with paediatric measurements for age +24 months.

Why it is examinable: these give you objective, defensible numbers for two findings dentists otherwise describe only subjectively — tongue restriction and arch width. Quoting a measured intermolar distance rather than "the arch looks narrow" is exactly the shift in rigour examiners reward.



Pediatric Sleep Questionnaire (PSQ) 01.27.19

UCDAVIS HEALTH | CHILDREN'S HOSPITAL

For each of the following questions circle yes, no, or don't know as best describes your child's sleep.

While sleeping, does your child ….				
1.	Snore more than half the time?	Yes	No	Don't know
2.	Always snore?	Yes	No	Don't know
3.	Snore loudly?	Yes	No	Don't know
4.	Have 'heavy' or loud breathing?	Yes	No	Don't know
5.	Have trouble breathing, or struggle to breathe?	Yes	No	Don't know
Have you ever …				
6.	Seen your child stop breathing during the night?	Yes	No	Don't know
Does your child …				
7.	Tend to breath through the mouth during the day?	Yes	No	Don't know
8.	Have a dry mouth on waking in the morning?	Yes	No	Don't know
9.	Occasionally wet the bed?	Yes	No	Don't know
10.	Wake up feeling *un*refreshed in the morning?	Yes	No	Don't know
11.	Have a problem with sleepiness during the day?	Yes	No	Don't know
12.	Have a teacher or other supervisor comment that your child appears sleepy during the day?	Yes	No	Don't know
Other:				
13.	Is it hard to wake up your child in the morning?	Yes	No	Don't know
14.	Does your child wake up with headaches in the morning?	Yes	No	Don't know
15.	Did your child stop growing at a normal rate at any time since birth?	Yes	No	Don't know
16.	Is your child overweight?	Yes	No	Don't know
This child often …				
17.	Does not seem to listen when spoken to directly.	Yes	No	Don't know
18.	Has difficulty organizing tasks and activities.	Yes	No	Don't know
19.	Is easily distracted by extraneous stimuli.	Yes	No	Don't know
20.	Fidgets with hands or feet or squirms in seat.	Yes	No	Don't know
21.	Is 'on the go' or often acts as if 'driven by a motor'.	Yes	No	Don't know
22.	Interrupts or intrudes on others (e.g., interrupts conversations or games).	Yes	No	Don't know

Figure 8.7 — The Paediatric Sleep Questionnaire (PSQ), answered yes / no / don't know by the caregiver, covering three groups of items: while sleeping (snoring more than half the time, always snoring, loud snoring, heavy or loud breathing, trouble breathing or struggling to breathe), witnessed events (stopping breathing during the night), and daytime and behavioural items (mouth breathing during the day, dry mouth on waking, bed-wetting, waking unrefreshed, daytime sleepiness, teacher-reported daytime sleepiness, difficulty waking, morning headaches, poor growth, overweight), followed by attention and behaviour items (not listening, difficulty organising tasks, easy distraction, fidgeting, inability to remain seated, interrupting others).

*Why it is examinable: note how many items are directly observable by a dentist or reported by a parent in a dental chair, and note the substantial **behavioural** section — the questionnaire's structure embodies the central paediatric teaching point that children present with behaviour rather than sleepiness.*

ID#: _ _ _ _ _ _ Name Code: _ _ _ _ DOV: _ _ - _ _ _ - _ _ _ _ (d d m m m y y y y)

OSA-18 Quality of Life Survey

Evaluation of Sleep-Disordered Breathing

Instructions: For each question below, please fill in the circle that best describes how often each symptom or problem has occurred during the past *4 weeks* (or since the last survey, if sooner). Thank you.

	None of the time	Hardly any of the time	A little of the time	Some of the time	A good bit of the time	Most of the time	All of the time
SLEEP DISTURBANCE							
During the past *4 weeks*, how often has your child had…							
…loud snoring?	①	②	③	④	⑤	⑥	⑦
…breath holding spells or pauses in breathing at night?	①	②	③	④	⑤	⑥	⑦
…choking or gasping sounds while asleep?	①	②	③	④	⑤	⑥	⑦
…restless sleep or frequent awakenings from sleep?	①	②	③	④	⑤	⑥	⑦
PHYSICAL SUFFERING							
During the past *4 weeks*, how often has your child had…							
…mouth breathing because of nasal obstruction?	①	②	③	④	⑤	⑥	⑦
…frequent colds or upper respiratory infections?	①	②	③	④	⑤	⑥	⑦
…nasal discharge or runny nose?	①	②	③	④	⑤	⑥	⑦
…difficulty in swallowing foods?	①	②	③	④	⑤	⑥	⑦
EMOTIONAL DISTRESS							
During the past *4 weeks*, how often has your child had…							
…mood swings or temper tantrums?	①	②	③	④	⑤	⑥	⑦
…aggressive or hyperactive behavior?	①	②	③	④	⑤	⑥	⑦
…discipline problems?	①	②	③	④	⑤	⑥	⑦
DAYTIME PROBLEMS							
During the past *4 weeks*, how often has your child had…							
…excessive daytime drowsiness or sleepiness?	①	②	③	④	⑤	⑥	⑦
…poor attention span or concentration?	①	②	③	④	⑤	⑥	⑦
…difficulty getting out of bed in the morning?	①	②	③	④	⑤	⑥	⑦
CAREGIVER CONCERNS							
During the past *4 weeks*, how often have the above problems…							
…caused you to worry about your child's general health?	①	②	③	④	⑤	⑥	⑦
…created concern that your child is not getting enough air?	①	②	③	④	⑤	⑥	⑦
…interfered with your ability to perform daily activities?	①	②	③	④	⑤	⑥	⑦
…made you frustrated?	①	②	③	④	⑤	⑥	⑦

OVERALL, HOW WOULD YOU RATE YOUR CHILD'S QUALITY OF LIFE AS A RESULT OF THE ABOVE PROBLEMS?

(Circle one number)

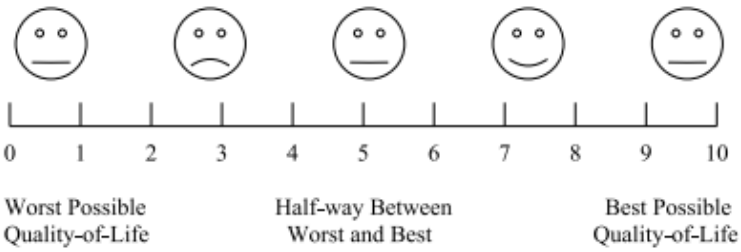
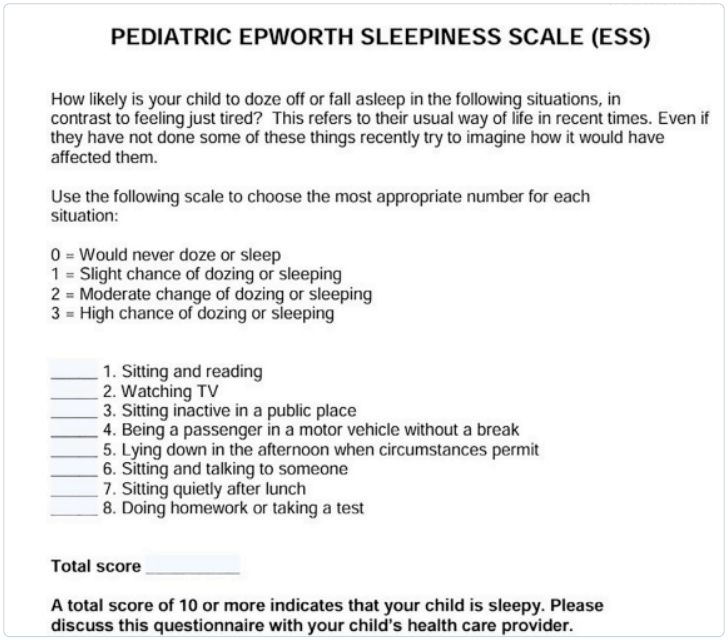


Fig. 2. OSA-18 Quality of Life Survey.

Figure 8.8 — The OSA-18 quality-of-life survey for the evaluation of sleep-disordered breathing, with its five domains: **sleep disturbance** (loud snoring, breath-holding or struggling to breathe, choking or gasping, restless sleep or frequent awakening), **physical suffering** (mouth breathing from nasal obstruction, frequent colds or upper respiratory infections, nasal discharge or runny nose, difficulty swallowing food), **emotional distress** (mood swings or temper tantrums, aggressive or hyperactive behaviour, discipline problems), **daytime problems** (excessive daytime drowsiness or sleepiness, poor attention span or concentration, difficulty getting out of bed in the morning) and **caregiver concerns** (worry over the child's general health, concern that the child is not getting enough air, interference with the caregiver's ability to perform daily activities, frustration), plus a global quality-of-life rating scale.

Why it is examinable: the OSA-18 is the standard measure of the burden of paediatric SDB on the child and family, complementary to the PSQ's symptom focus. Its inclusion of a caregiver-concerns domain is a reminder that paediatric SDB is a family diagnosis.



PEDIATRIC EPWORTH SLEEPINESS SCALE (ESS)

How likely is your child to doze off or fall asleep in the following situations, in contrast to feeling just tired? This refers to their usual way of life in recent times. Even if they have not done some of these things recently try to imagine how it would have affected them.

Use the following scale to choose the most appropriate number for each situation:

0 = Would never doze or sleep
1 = Slight chance of dozing or sleeping
2 = Moderate change of dozing or sleeping
3 = High chance of dozing or sleeping

_____ 1. Sitting and reading
_____ 2. Watching TV
_____ 3. Sitting inactive in a public place
_____ 4. Being a passenger in a motor vehicle without a break
_____ 5. Lying down in the afternoon when circumstances permit
_____ 6. Sitting and talking to someone
_____ 7. Sitting quietly after lunch
_____ 8. Doing homework or taking a test

Total score _____

A total score of 10 or more indicates that your child is sleepy. Please discuss this questionnaire with your child's health care provider.

Figure 8.9 — The paediatric Epworth Sleepiness Scale, rating the likelihood of the child dozing off in eight situations — sitting and reading, watching television, sitting inactive in a public place, being a passenger in a motor vehicle for an hour without a break, lying down in the afternoon when circumstances permit, sitting and talking to someone, sitting quietly after lunch, and doing homework or taking a test — each scored 0 to 3. **A total score of 10 or more indicates that the child is sleepy.**

Why it is examinable: note the adaptation of the adult instrument — "doing homework or taking a test" replaces the adult "in a car, stopped in traffic". The same caveat applies as in adults: it measures sleep propensity and cannot diagnose OSA, and children with significant OSA frequently score normally because they present with hyperactivity rather than sleepiness.

8.7 Treatment

Adenotonsillectomy — first line

Aspect	Detail
Status	**First-line treatment** and the most effective surgical therapy for paediatric patients, improving polysomnographic parameters, behaviour and quality of life significantly
Limitations	**Post-operative persistence of OSA after adenotonsillectomy is common.** Adenotonsillectomy alone may be insufficient because it does not address oropharyngeal obstruction secondary to tonsillar hyperplasia, especially in children with underlying illness — Down syndrome, craniofacial anomalies, neuromuscular disease
Risk factors for treatment failure	Obesity; baseline severity (high pre-operative AHI); age (<3 years or ≥7 years have a higher failure rate); nasal abnormality or craniofacial disproportion; family history of OSA; asthma. **Post-operative polysomnography is suggested for these high-risk patients even in the absence of persistent snoring** — although persistent OSA is less likely if there is no snoring post-operatively
Contraindication	Submucous cleft palate is a relative contraindication — a partial adenotonsillectomy may be performed instead
Post-operative management	Usually performed as an outpatient. For patients at high risk of post-operative complications, consider an overnight hospital stay — specifically where the SpO_2 nadir is <80%, the AHI ≥24/h, or the peak PCO_2 ≥60 mmHg (hypercapnia), because of the increased risk of post-operative respiratory compromise
Risk factors for post-operative complications	Age <3 years; severe OSA on polysomnography; cardiac complications; failure to thrive; obesity; upper respiratory tract infection — if acute URI on the day of surgery, reschedule or monitor closely post-operatively, but be aware that children with adenotonsillar hypertrophy exhibit chronic rhinorrhoea and nasal congestion even without viral infection; other medical comorbidity
Partial tonsillectomy	A portion of tonsil tissue is left covering the musculature of the tonsillar fossa. *Advantages:* reduced morbidity, less immediate post-operative pain, earlier return to normal activity and diet. *Dis-*

Aspect	Detail
	advantages: increased risk of tonsil regrowth (0.5–16%), increased intra-operative blood loss, risk of recurrent infection in the tonsillar remnant, and the need for long-term monitoring for OSA symptoms related to regrowth and recurrence
Other surgical options	Turbinate reduction, septoplasty; expansion sphincter pharyngoplasty for refractory cases; craniofacial procedures such as mandibular distraction osteogenesis for children with craniofacial anomalies; **tracheotomy as a last resort** when all other treatment fails — very effective but with much morbidity

CPAP

- Effective but **not first line where adenotonsillectomy is an option**, because of compliance problems. Adherence in children is around 50%, though respiratory therapist support improves compliance.
- **Indications:** children who do not respond adequately to surgery, or in whom surgery is contraindicated — which can include patient and family preference, for example religious belief against surgery or blood transfusion. Often used in children with neuromuscular or craniofacial disorders, or residual OSA after surgery.
- **Requires objective assessment of adherence.** Attempt to improve adherence by addressing adverse effects and using behaviour modification techniques; if the child remains non-compliant, use an alternative treatment.
- **CPAP pressure changes over time in children because of growth and development, so periodic reassessment is required.** Polysomnography is used for CPAP titration.

Medication

- **Intranasal corticosteroid.** Indicated **only for children with mild OSA (AHI <5/h)**, and not for moderate or severe disease. Effects: small reduction in AHI; improvement in symptoms of adenoidal hypertrophy including snoring and observed apnoea; demonstrated shrinkage of adenoidal tissue. Follow-up evaluation is needed as long-term effects are unknown.
- **Leukotriene receptor antagonists** may help in mild cases or where allergic rhinitis contributes to airway inflammation.

Orthodontic treatment

Intervention	Detail and evidence
Rapid maxillary expansion (RME)	*Indication:* children with maxillary constriction and dental malocclusion. *Mechanism:* increases the transverse diameter of the hard palate by reopening the mid-palatal suture, using a fixed appliance with an expansion screw anchored on selected teeth; after 3–4 months of expansion, normally mineralised structure is rebuilt. *Reported effect:* significant improvement in OSA signs, symptoms and polysomnographic parameters, with a reported mean reduction of about **4.8 events/h (−70%)**
Miniscrew-assisted rapid palatal expansion (MARPE)	Together with RME, has been shown to increase nasal cavity volume, improve nasal airflow and reduce upper airway resistance — effects that can contribute to alleviating mild-to-moderate SDB symptoms such as snoring and obstructive events, by promoting healthy airway development
Mandibular advancement	For class II malocclusion correction; together with maxillary expansion, one of the two orthodontic treatments in children that may change airway dimension. In children and adolescents with posterior crossbite or maxillary constriction and retrognathia, maxillary expansion and mandibular advancement produce upper airway dimensional change and AHI decrease
Role	Orthodontics may be used where adenotonsillectomy is not suitable, or in combination with surgery for better outcomes, particularly in patients with craniofacial abnormality or residual OSA. Long-term follow-up and comprehensive assessment including sleep studies are important

EXAM TRAP — THE HONEST EVIDENCE ON EXPANSION

Functional responses to orthopaedic change are **highly variable**. Expansion treatment does not always help: some patients do not improve and **some may get worse** after RME. In studies of bi-maxillary expansion (maxilla by RME, mandible by dental expansion), around **one third of patients became objectively worse** after expansion. There is also natural regression of symptoms in some children, which inflates uncontrolled before-and-after results. Expansion may therefore be considered **only for selected patients**, and — crucially — any intervention undertaken with the purpose of improving OSA needs the OSA to have been **diagnosed by a physician**. Presenting expansion as a reliable cure for paediatric OSA is the single easiest way to fail this section.

Positional therapy and other measures

- AHI is increased in the supine position, and lowest when children sleep **prone** — but position is difficult to maintain overnight in a child.
- **Weight management** for obese children.
- **Watchful waiting** with close monitoring may be appropriate for mild snoring or mild OSA.
- **Myofunctional therapy** as an adjunct.

Summary of treatment by indication

Treatment	Indications	Key points
Adenotonsillectomy	Moderate to severe OSA; enlarged tonsils and adenoids	Often first-line; reduces AHI significantly; risk of residual OSA
CPAP	Residual OSA; non-surgical candidates; co-morbidity	Effective but adherence challenging; mask fitting critical; re-titrate with growth
Orthodontics	Craniofacial abnormality; residual OSA	Improves airway via anatomical change; adjunct or alternative; response variable
Medical therapy	Mild OSA; allergic rhinitis	Nasal steroids and leukotriene inhibitors; supportive care only
Other surgery	Severe or persistent OSA	Turbinate reduction, septoplasty, advanced procedures for refractory cases; tracheotomy last resort

Effective treatment involves a multidisciplinary approach with paediatricians, ENT specialists, sleep medicine experts and orthodontists.

8.8 Differences between children and adults

	Children	Adults
Causes	Enlarged tonsils and adenoids are the leading cause; genetic predisposition; obesity in a minority (but increasingly in adolescents). Most commonly affects ages 3–6, when tonsillar tissue is proportionally largest	Obesity, age and male gender; lifestyle factors including smoking and alcohol. Often diagnosed as OSA or central sleep apnoea
Presentation	Snoring or noisy breathing; restless sleep or unusual sleeping positions; frequent bed-wetting; daytime fatigue and attention difficulties; **behavioural problems — hyperactivity, mood swings, attention difficulties that may mimic ADHD**; poor academic performance	Loud chronic snoring; episodes of stopped breathing; morning headache or dry mouth; daytime drowsiness; decreased libido and mood swings; impaired memory or concentration
Thresholds	AHI ≥1 is abnormal; mild 1–5, moderate 5–10, severe ≥10	AHI ≥5 with symptoms, or ≥15 regardless; mild 5–15, moderate 15–30, severe ≥30

	Children	Adults
Diagnosis	Thorough clinical evaluation of behaviour and sleep patterns; polysomnography; imaging (radiographs, endoscopy) may help identify anatomical obstruction. **End-tidal CO_2 is a standard PSG measurement**	Sleep studies in clinical or home settings; evaluation of coexisting cardiovascular disease or metabolic syndrome
Treatment	Surgical removal of enlarged tonsils and adenoids first; weight management if obesity contributes; positive airway pressure when surgery is ineffective; behaviour-focused therapy to improve sleep hygiene; orthodontics in selected cases	Lifestyle change including weight loss, alcohol avoidance and sleep position; CPAP; oral appliance therapy as a non-invasive alternative; surgery where necessary
Long-term outlook	Early treatment often resolves the condition entirely, with improvements in behaviour, academic performance and quality of life; ongoing monitoring may be needed to prevent recurrence. Where enlarged tonsils and adenoids shrink with age the symptoms may improve, but many children benefit substantially from early diagnosis and treatment	A chronic condition requiring continuous management, with increased risk of serious complications including stroke, hypertension and cardiovascular disease; long-term treatment adherence is essential
Age boundary	**12 years is the cut-off point for childhood SDB**; above this, adult criteria are generally applied	

8.9 SDB with underlying medical conditions and syndromes

Medical conditions in children

Obesity, asthma, allergic rhinitis, gastro-oesophageal reflux disease and neuromuscular disorders all contribute to airway obstruction, inflammation or decreased airway muscle tone, increasing the risk and severity of SDB. Children with neuromuscular diseases such as Duchenne muscular dystrophy and myasthenia gravis have weakened respiratory muscles, leading to hypoventilation and greater susceptibility to SDB.

Genetic syndromes

Children with Down syndrome, Prader–Willi syndrome, Crouzon syndrome and cleft lip and palate often have craniofacial abnormalities — midfacial hypoplasia, macroglossia, retrognathia — that narrow the airway and predispose to OSA. In Prader–Willi syndrome, hypersomnolence and varying degrees of airway obstruction contribute to SDB.

HIGH-YIELD — DOWN SYNDROME

- OSA is **highly prevalent**. Prevalence by AHI threshold: >1 in 69%; >1.5 in 76%; >2 in 75%; **>5 (moderate–severe) in 50%**; >10 in 34%. Compare 1–5% in the general paediatric population.
- Predisposing factors: **macroglossia, midface hypoplasia, overweight and poor muscle tone**.
- Moderate-to-severe OSA is more prevalent at younger ages.
- **All children with Down syndrome should be referred to a paediatric sleep laboratory for polysomnography by age 4** (AAPD recommendation).
- Clinical symptoms are **unreliable**: only a weak correlation exists between parental perception of sleep problems and polysomnographic results, and **53.8% of children with OSA had no symptoms**. This is the single strongest argument for objective screening in a syndromic population.

In adults and generally

Condition	Relationship
Cardiovascular disease	SDB is often a comorbidity of cardiovascular disease and may also be causal, with growing evidence linking it to heart failure, hypertension and stroke. Mechanisms include increased platelet aggregation, hypercoagulability and endothelial dysfunction, and contribution to left ventricular hypertrophy and heart failure. Guidelines now recommend considering SDB management in acute stroke and congestive heart failure
Obesity and diabetes	Strong link with obesity; SDB is a risk factor for and can exacerbate insulin resistance and type 2 diabetes, with higher SDB prevalence in the diabetic population
Chronic kidney disease	High prevalence of SDB in CKD; severe SDB increases the risk for CKD
Stroke	Contributes to risk through atherosclerosis, hypertension and cardiac arrhythmias
Pulmonary conditions	Sleep-related hypoventilation can be associated with restrictive pulmonary dysfunction, as in conditions such as Poland syndrome

Why the relationship matters: increased morbidity and mortality; reduced quality of life; healthcare burden and high cost; and — the therapeutic opportunity — treating SDB can positively affect the management of associated chronic disease and improve outcomes. In children, SDB with underlying conditions can lead to daytime behavioural problems, cognitive impairment, cardiovascular strain, growth delay and reduced quality of life. Early diagnosis and management are crucial, especially because **some signs of neuromuscular disease may present initially as sleep-disordered breathing**. Paediatricians and specialists must monitor at-risk children closely for snoring, mouth breathing, nocturnal gasping and behavioural problems.

8.10 Referral pathways

Pathway	Detail
Sleep physician or ENT	For children exhibiting multiple SDB risk factors or symptoms, refer to a paediatric sleep physician or ENT specialist for comprehensive evaluation, which may include polysomnography or further imaging. For children with enlarged adenoids and tonsils, surgical removal is the first-line treatment
Orthodontic intervention	Where craniofacial development suggests a narrow airway or malocclusion, consider early orthodontic intervention. **Any intervention with the purpose of improving OSA needs to be diagnosed by a physician.** Maxillary expansion, including RME and MARPE, increases nasal cavity volume, improves nasal airflow and reduces upper airway resistance, contributing to alleviation of mild-to-moderate SDB symptoms
Collaboration with paediatricians	Regular communication with the child's paediatrician ensures a coordinated approach to treatment and monitoring, especially where SDB overlaps with other health conditions
Parent and caregiver education	Ensure awareness of SDB symptoms and risks, the potential health impacts, and the importance of addressing these issues early. Provide guidance on promoting good sleep hygiene and avoiding environmental triggers

Diagnosis of paediatric SDB by a physician

- **Clinical evaluation:** detailed history focusing on habitual snoring, observed apnoeas, laboured or paradoxical breathing during sleep, daytime sleepiness, behavioural problems and learning difficulties. Physical examination attends to upper airway anatomy including tonsil and adenoid size, craniofacial features, and signs of obesity or comorbidity. Validated questionnaires aid screening but are not definitive.
- **Polysomnography** is the diagnostic **gold standard** — overnight, attended, in-laboratory. It records EEG, EOG, EMG, ECG, respiratory airflow through oronasal sensors, respiratory effort via chest and abdominal belts, oxygen saturation, body position and (in children) end-tidal CO_2, allowing correlation of respiratory events with sleep stages and arousals. An obstructive AHI of **≥1 event per hour is abnormal** in children.

- **Indications for PSG:** clinical symptoms of SDB, especially **before adenotonsillectomy in children under 2 years**, in obesity, in craniofacial or neuromuscular disorders, in Down syndrome, or where symptoms persist after surgery. Also for CPAP titration and evaluation of complex cases.
- **Other tools:** respiratory polygraphy records cardiorespiratory parameters but lacks sleep staging and is less valid-
  ated in children; nocturnal oximetry is a simple screening tool with good predictive value in high-risk children; **home sleep apnoea testing is still emerging in paediatrics and is not widely recommended as a standalone diagnostic tool**; daytime nap studies are less accurate; radiological imaging and biomarkers show promise but require further research.

EXAM TRAP — CBCT FOR AIRWAY ANALYSIS IN CHILDREN

The reliability and validity of CBCT airway analysis are both questionable. **Reliability:** the oropharynx is clearly imaged, but the most frequent sites of collapse are the nasopharynx and hypopharynx — and the adenoid site in children — so reliability is poor for the region that matters. **Validity:** upper airway dynamic function is very complex, and a healthy subject with a narrow airway can overcome anatomical risk by actively dilating the airway during inspiration. Cautions: poor inter-examiner reliability; normative values with large standard deviations; and, most fundamentally, **airways are not static**. Do not offer CBCT airway analysis as a diagnostic tool for paediatric OSA.

NUMBERS TO MEMORISE — TOPIC 8

- Paediatric OSA 1–5% (typically 2–3%); snoring 3–12%; peaks at 2–8 y and in adolescence; commonest at 3–6 y
- Paediatric AHI: normal <1 · mild 1–5 · moderate 5–10 · severe ≥10. Obstructive AHI ≥1 is abnormal
- Age cut-off for childhood SDB: 12 years
- Sleep duration: 4–12 mo 12–16 h · 1–2 y 11–14 h · 3–5 y 10–13 h · 6–12 y 9–12 h · 13–18 y 8–10 h
- <8 h in adolescence → ~3× increased risk of suicide attempt; caffeine consumers 2× more likely to have difficulty sleeping
- PSQ threshold >0.33 (>8 of 22 items); FAIREST-6: 2 mild / 4 moderate / 6 severe; paediatric ESS ≥10; tonsillar obstruction >50% warrants evaluation
- Brodsky grades 0–4: within pillars · <25% · 25–50% · 50–75% · >75%
- Adenotonsillectomy failure risk: age <3 or ≥7, obesity, high pre-op AHI, nasal/craniofacial abnormality, family history, asthma. Overnight admission if SpO_2 nadir <80%, AHI ≥24/h or peak PCO_2 ≥60 mmHg
- Partial tonsillectomy: regrowth risk 0.5–16%
- Intranasal steroid only for AHI <5; paediatric CPAP adherence ~50%
- RME: 3–4 months of expansion; reported mean AHI reduction ~4.8/h (−70%); **but ~⅓ objectively worse after bi-maxillary expansion**
- Down syndrome OSA prevalence: AHI >1 in 69% · >5 in 50% · >10 in 34%; PSG by age 4; 53.8% asymptomatic
- PSG indicated before adenotonsillectomy in children <2 years

SELF-TEST — TOPIC 8

1. A 5-year-old snores nightly, is hyperactive at school and wets the bed. What is your assessment and plan?

This is a classic paediatric SDB presentation — behavioural rather than sleepy, with secondary enuresis in a previously dry child. I would take a medical, dental and family history covering allergies, asthma, recurrent ear and throat infections, prematurity, syndromic conditions and family history of sleep apnoea; record height, weight and neck size and plot growth, looking for failure to thrive; administer the Paediatric Sleep Questionnaire (positive if >0.33, that is more than 8 of 22 items) and consider the FAIREST-6; and perform an oral and craniofacial examination assessing tonsil size by the Brodsky scale, modified Mallampati, palatal vault and arch width, jaw size and position, tongue posture and tie, and breathing habits. I would then refer to a paediatric sleep physician or ENT specialist for comprehensive evaluation and polysomnography, noting that an obstructive AHI ≥1/h is abnormal in a child and that adenotonsillectomy is first-line if adenotonsillar hypertrophy is confirmed. I would not perform CBCT airway analysis, and I would not commence orthodontic expansion for the purpose of treating OSA before a physician diagnosis.

2. Give the paediatric AHI thresholds and explain why they differ from adult values.

Normal <1, mild 1 to <5, moderate 5 to <10, severe ≥10 — approximately one third of the adult values of 5, 15 and 30. An obstructive AHI of ≥1 event per hour is abnormal in a child. The lower thresholds reflect that children normally have very stable breathing during sleep, so even a small number of obstructive events represents pathology, and that the consequences — on growth, neurocognitive development, behaviour and academic performance — occur at much lower event rates. The practical trap is that a child with an AHI of 6 has *moderate* disease, not mild. Twelve years is the cut-off above which adult criteria are generally applied.

3. What is first-line treatment for paediatric OSA, and what are the risk factors for failure?

Adenotonsillectomy — the most effective surgical therapy, improving polysomnographic parameters, behaviour and quality of life. However, post-operative persistence of OSA is common, and adenotonsillectomy alone may be insufficient because it does not address oropharyngeal obstruction secondary to tonsillar hyperplasia, especially in children with Down syndrome, craniofacial anomalies or neuromuscular disease. Risk factors for failure: obesity; high pre-operative AHI; age under 3 or 7 and over; nasal abnormality or craniofacial disproportion; family history of OSA; and asthma. Post-operative polysomnography is suggested for these high-risk children even without persistent snoring. Consider overnight admission where the SpO_2 nadir is below 80%, the AHI is 24/h or more, or peak PCO_2 is 60 mmHg or more.

4. A parent asks whether palatal expansion will cure their child's sleep apnoea. How do you answer?

Honestly and with the evidence. Rapid maxillary expansion increases the transverse diameter of the hard palate by reopening the mid-palatal suture, and both RME and MARPE increase nasal cavity volume, improve nasal airflow and reduce upper airway resistance; reported mean AHI reductions of around 4.8 events per hour (about 70%) have been published in children with maxillary constriction. **But** functional responses to orthopaedic change are highly variable: expansion does not always help, some children do not improve and some get worse, and in bi-maxillary expansion studies about one third of patients became objectively worse. There is also natural regression of symptoms in some children, which flatters uncontrolled results. Expansion should therefore be considered only for selected patients, as an adjunct or where adenotonsillectomy is unsuitable, and any intervention intended to improve OSA requires the diagnosis to have been made by a physician, with long-term follow-up including sleep studies.

5. Distinguish the Mallampati and Brodsky classifications.

The **modified Mallampati** classification grades *oral airway patency in relation to tongue position*: class I, soft palate, uvula, fauces and pillars fully visible; II, soft palate and most of the uvula; III, soft palate and base of uvula only; IV, hard palate only. It is performed with the patient upright, mouth wide open and tongue maximally protruded without phonation. The **Brodsky** scale grades *tonsil size as a percentage of the oropharyngeal width between the anterior pillars*: grade 0, tonsils within the pillars; 1, <25%; 2, 25–50%; 3, 50–75%; 4, >75% or meeting in the midline. Assess by depressing the tongue and asking the patient to say "ah". Obstruction of more than 50% warrants further evaluation — but tonsil size does not correlate with OSA severity, so the grading identifies a risk factor, not a severity measure.

6. Why should a child with Down syndrome be referred for polysomnography even if the parents report no sleep problems?

Because clinical symptoms are unreliable in this population. OSA prevalence in Down syndrome is very high — approximately 50% at an AHI above 5, and 69% at an AHI above 1, compared with 1–5% in the general paediatric population — driven by macroglossia, midface hypoplasia, overweight and poor muscle tone, with moderate-to-severe disease more prevalent at younger ages. Only a weak correlation exists between parental perception of sleep problems and polysomnographic findings, and 53.8% of affected children had no symptoms. Guidance therefore recommends that all children with Down syndrome be referred to a paediatric sleep laboratory for polysomnography by age 4, regardless of reported symptoms.

7. How do children with SDB differ from adults in presentation?

Children present with **behaviour rather than sleepiness**: hyperactivity, aggression, impulsivity, irritability, inattention and poor academic performance — a picture that closely mimics ADHD — together with restless sleep, unusual sleeping positions such as sitting up or hyperextending the neck, secondary bed-wetting, mouth breathing, morning headache and either excessive or poor weight gain. Adults present with loud chronic snoring, witnessed apnoeas, morning headache and dry mouth, daytime drowsiness, reduced libido and impaired memory. Causes also differ — adenotonsillar hypertrophy in children versus obesity, age and male sex in adults — as do thresholds, first-line treatment (adenotonsillectomy versus CPAP or an oral appliance), and prognosis: early treatment often resolves the condition entirely in children, whereas adult OSA is a chronic condition requiring lifelong management.

REFERENCES — TOPIC 8

1. Marcus CL, Brooks LJ, Draper KA, et al. Diagnosis and management of childhood obstructive sleep apnea syndrome. *Pediatrics*. 2012;130(3):576–584.
2. Kaditis AG, Alonso Alvarez ML, Boudewyns A, et al. Obstructive sleep disordered breathing in 2- to 18-year-old children: diagnosis and management. *Eur Respir J*. 2016;47(1):69–94.
3. Aurora RN, Zak RS, Karippot A, et al. Practice parameters for the respiratory indications for polysomnography in children. *Sleep*. 2011;34(3): 379–388.
4. Brodsky L. Modern assessment of tonsils and adenoids. *Pediatr Clin North Am*. 1989;36(6):1551–1569.
5. Chervin RD, Hedger K, Dillon JE, Pituch KJ. Pediatric Sleep Questionnaire (PSQ): validity and reliability of scales for sleep-disordered breathing, snoring, sleepiness, and behavioral problems. *Sleep Med*. 2000;1(1):21–32.
6. Franco RA, Rosenfeld RM, Rao M. Quality of life for children with obstructive sleep apnea (OSA-18). *Otolaryngol Head Neck Surg*. 2000;123(1):9–16.
7. Janssen KC, Phillipson S, O'Connor J, Johns MW. Validation of the Epworth Sleepiness Scale for children and adolescents using Rasch analysis. *Sleep Med*. 2017;33:30–35.
8. Zaghi S, Peterson C, Shamtoob S, et al. Assessment of nasal breathing using lip taping: a simple and effective screening tool. *Int J Otorhinolaryngol*. 2020;6(1):10–15.
9. Yoon A, Zaghi S, Weitzman R, et al. Toward a functional definition of ankyloglossia: validating current grading scales for lingual frenulum length and tongue mobility in 1052 subjects. *Sleep Breath*. 2017;21(3):767–775.
10. Paruthi S, Brooks LJ, D'Ambrosio C, et al. Recommended amount of sleep for pediatric populations: a consensus statement of the American Academy of Sleep Medicine. *J Clin Sleep Med*. 2016;12(6):785–786.
11. Owens J; Adolescent Sleep Working Group; Committee on Adolescence. Insufficient sleep in adolescents and young adults: an update on causes and consequences. *Pediatrics*. 2014;134(3):e921–e932.
12. Carskadon MA. Sleep in adolescents: the perfect storm. *Pediatr Clin North Am*. 2011;58(3):637–647.
13. Camacho M, Chang ET, Song SA, et al. Rapid maxillary expansion for pediatric obstructive sleep apnea: a systematic review and meta-analysis. *Laryngoscope*. 2017;127(7):1712–1719.
14. Villa MP, Rizzoli A, Miano S, Malagola C. Efficacy of rapid maxillary expansion in children with obstructive sleep apnea syndrome: 36 months of follow-up. *Sleep Breath*. 2011;15(2):179–184.
15. Pirelli P, Saponara M, Guilleminault C. Rapid maxillary expansion in children with obstructive sleep apnea syndrome. *Sleep*. 2004;27(4):761–766.
16. Bull MJ; Committee on Genetics. Health supervision for children with Down syndrome. *Pediatrics*. 2011;128(2):393–406.
17. Shott SR, Amin R, Chini B, Heubi C, Hotze S, Akers R. Obstructive sleep apnea: should all children with Down syndrome be tested? *Arch Otolaryngol Head Neck Surg*. 2006;132(4):432–436.
18. Bhattacharjee R, Kheirandish-Gozal L, Spruyt K, et al. Adenotonsillectomy outcomes in treatment of obstructive sleep apnea in children: a multicenter retrospective study. *Am J Respir Crit Care Med*. 2010;182(5):676–683.
19. Marcus CL, Moore RH, Rosen CL, et al. A randomized trial of adenotonsillectomy for childhood sleep apnea (CHAT). *N Engl J Med*. 2013;368(25):2366–2376.
20. Kuhle S, Hoffmann DU, Mitra S, Urschitz MS. Anti-inflammatory medications for obstructive sleep apnoea in children. *Cochrane Database Syst Rev*. 2020;1:CD007074.
21. Alsufyani NA, Flores-Mir C, Major PW. Three-dimensional segmentation of the upper airway using cone beam CT: a systematic review. *Dentomaxillofac Radiol*. 2012;41(4):276–284.

TOPIC 9

Best-practice principles in dental sleep medicine

BLUEPRINT

1. Diagnosis of SDB by a physician · 2. Record keeping: baseline sleep study data, clinical notes, treatment planning, study models, informed consent, interprofessional communication · 3. Item codes for private and Department of Veterans' Affairs billing · 4. Ethics in dental sleep medicine

Not clinically difficult but easy to fail, because the answers are matters of rule rather than judgement — and examiners can mark them unambiguously.

9.1 Diagnosis by a physician

- **In-laboratory PSG** — accurate with a low failure rate, attended by technical staff; expensive and technically complex. Also the standard for narcolepsy and central sleep apnoea.
- **Portable monitoring** — less expensive and quicker, and can be as accurate as PSG in selected populations. Patients must be at **high risk for OSA and must not have comorbid medical or comorbid sleep disorders**. Failure to meet these criteria in a patient at high risk of moderate–severe OSA should lead to in-laboratory PSG.

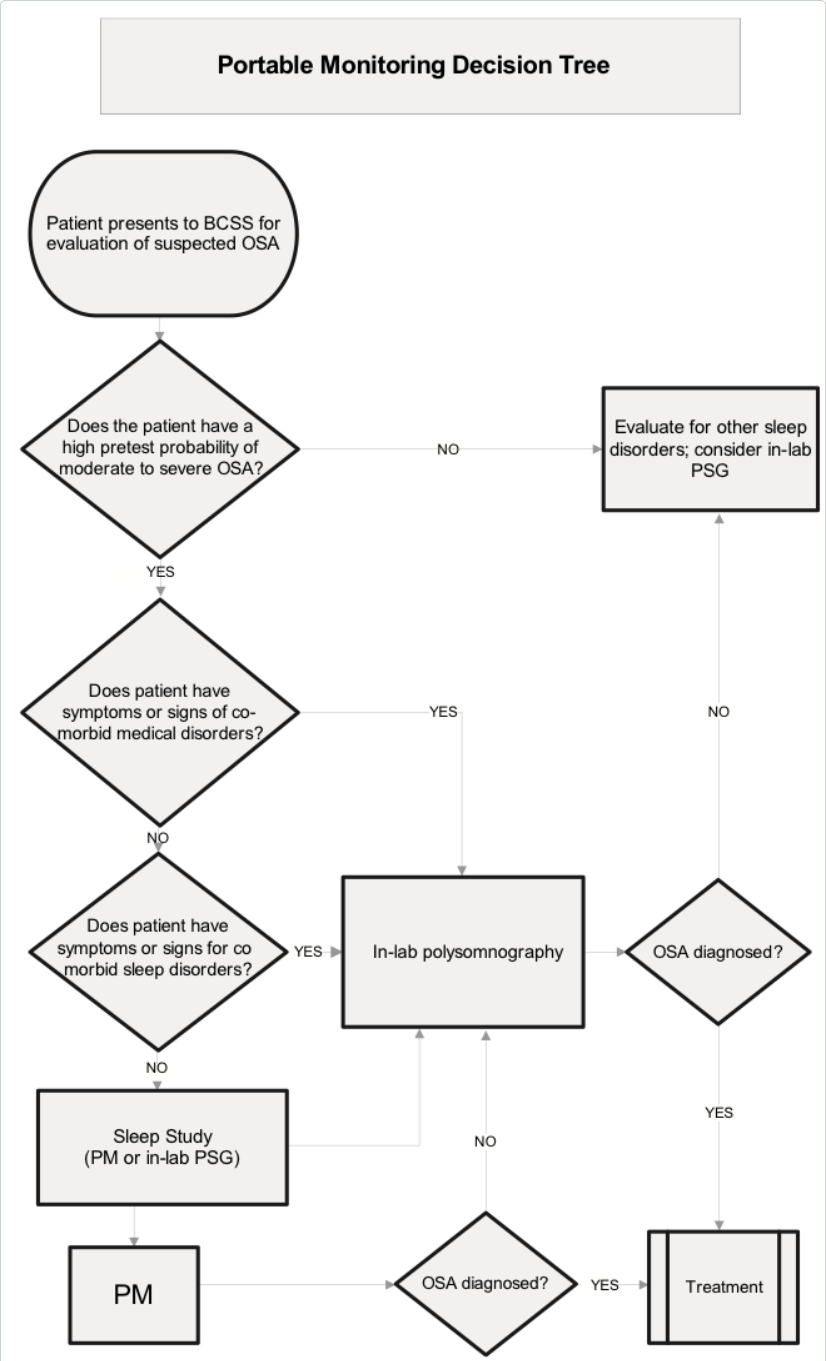


Figure 1—Flow chart depicting recommended pathway of patients considered for PM. Patients appropriate for PM should have moderate to high risk for OSA, have no comorbid medical conditions and no comorbid sleep disorders. Patients not considered appropriate for PM should have in-laboratory polysomnography. (BCSS = Board Certified Sleep Specialist or an individual who fulfills the eligibility criteria for the sleep medicine certification examination)

Fig 9.1 — Portable monitoring decision tree. Patients appropriate for a portable monitor should be at moderate-to-high risk for OSA, with no comorbid medical conditions and no comorbid sleep disorders. Those not appropriate should have in-laboratory polysomnography. Portable monitoring should be conducted under the supervision of a board-certified sleep specialist or an individual who fulfils the eligibility criteria for the sleep medicine certification examination. **Examinable because:** this defines the boundary of the dentist's involvement with objective testing. Where a dentist may lawfully order or administer a home sleep apnoea test, the resulting data are used **for titration purposes only** — never for diagnosis or formal follow-up assessment.

The portable monitoring guideline in detail

- **Supervision** — performed **only in conjunction with a comprehensive sleep evaluation**, by a board-certified sleep specialist or an individual meeting the eligibility criteria for the sleep medicine certification examination.
- **Indication** — *may* be used as an alternative to PSG in patients with a **high pre-test probability of moderate-to-severe OSA**. Limited to high-risk populations by clinical judgement informed by demographics, predisposing and precipitating factors, clinical

features and familial pattern. Relevant risk factors are **snoring, sleepiness, obesity and witnessed apnoea**, and their severity correlates with OSA severity. Only limited data in children and those over 65.

- **Contraindications** — should **not** be used in comorbidity, in other sleep disorders, or for **screening**. *Comorbidity degrades accuracy* — moderate-to-severe pulmonary disease, neuromuscular disease, congestive heart failure → use in-laboratory PSG. *Other sleep disorders* — central sleep apnoea, periodic limb movement disorder, insomnia, parasomnias, circadian rhythm disorders, narcolepsy — a portable monitor does not record the data needed for these diagnoses → in-laboratory PSG.
- **Special indications** — where in-laboratory PSG is impossible because of immobility, safety or critical illness; and to **monitor response to non-CPAP treatments** (oral appliances, upper airway surgery, weight loss) **where the diagnosis has already been made**.
- **Technology.** Must record **airflow, respiratory effort and blood oxygenation** at a minimum, and the biosensor types recommended for laboratory PSG are recommended for portable monitors. **Airflow** — **oronasal thermal sensor** to detect *apnoea* and **nasal pressure transducer** to detect *hypopnoea*; ideally **both**. Nasal pressure is superior for detecting flow but limited to nasal flow, leaving oral flow undetected and giving an unnoticed signal in a mouth breather — which is exactly why the oronasal sensor is also needed. **Effort** — ideally calibrated or uncalibrated **inductance plethysmography**, using chest and abdominal excursion to measure actual tidal volume; more accurate when assessed alongside another event such as desaturation. **Oxygenation** — **pulse oximetry** with appropriate signal averaging and motion-artefact accommodation: maximum signal averaging time **≤3 s at a heart rate of 80 bpm or more**; most studies use a finger probe. **Scoring** — the device must allow review of **raw data**, permitting manual scoring or manual editing of automated scoring by skilled personnel. **Paediatric addition** — **end-tidal CO_2** is a standard PSG measurement in children.
- **Diagnosis in children** — comprehensive clinical evaluation then objective study, with overnight PSG the gold standard (see 8.10). Obstructive AHI ≥1/h abnormal. PSG indicated for symptoms, especially before adenotonsillectomy under 2 years, in obesity, craniofacial or neuromuscular disorders, Down syndrome, or persistent post-operative symptoms; also for CPAP titration and complex cases. Respiratory polygraphy lacks staging and is less validated; nocturnal oximetry is a simple screen with good predictive value in high-risk children; home testing is emerging and not recommended standalone in paediatrics; nap studies are less accurate; imaging and biomarkers need further research.
- **Diagnosis in adults** — assess habitual loud snoring, witnessed apnoeas, choking or gasping, excessive sleepiness, fatigue, cognitive impairment and mood disorder; note obesity, large neck circumference, hypertension, cardiovascular disease. Screening questionnaires support suspicion but **cannot diagnose alone**. PSG quantifies the AHI; **AHI ≥5/h with symptoms, or ≥15/h regardless, confirms OSA**; PSG is also essential for grading severity and distinguishing central from obstructive events. HSAT may diagnose moderate–severe OSA where pre-test probability is high without significant comorbidity, monitoring fewer channels (airflow, effort, saturation, heart rate) — but PSG remains preferred where the diagnosis is uncertain or other disorders are suspected. Additional testing: MSLT for suspected narcolepsy or hypersomnia; acoustic pharyngometry and actigraphy as adjuncts; referral to ENT, cardiology or neurology as findings dictate.

THE FIVE-LINE SUMMARY OF THE DIAGNOSTIC PROCESS

1. Clinical suspicion from symptoms and risk factors initiates the process.
2. Overnight laboratory PSG is the definitive diagnostic tool.
3. Home testing may suit straightforward moderate-to-severe cases.
4. Diagnosis relies on the AHI threshold **combined with** the clinical presentation.
5. Complementary testing and specialist referral support diagnosis and management.

9.2 Record keeping

- **Baseline sleep study data** — the PSG or home test results with detailed sleep parameters, AHI, architecture and arousals, properly stored for treatment reference and follow-up comparison. **Without a baseline you cannot demonstrate treatment effect.**
- **Ongoing clinical notes** — patient assessments, symptoms, progress, adherence and response, including evaluations and observations from both physicians and qualified dentists. AADSM specifies **SOAP format** — subjective, objective, assessment, plan.
- **Treatment planning** — comprehensive plans setting out rationale, appliance choice, adjustments, titration and patient education, with any **shared decision-making** discussions about options and intolerances recorded.
- **Study models** — impressions and bite registrations used for fabrication and adjustment, retained as long as local regulation requires. Also pre-treatment panoramic and cephalometric radiographs and intraoral and extraoral photographs.
- **Informed consent** — complete signed forms explaining purpose, risks, benefits, alternatives and patient rights; clear, free of coercion, verifiable in the record; signed by the patient in the presence of the qualified dentist or team member, then countersigned and dated by the dentist.
- **Interprofessional communication** — records of communication between sleep physicians, dentists and others, ensuring coordinated care with shared assessment, adjustments and follow-up, documented **with patient permission**.
- **Delivery documentation** — written home care instructions, delivery form and warranty, signed and dated by patient and team member, one copy to the patient and one retained. **Side effects** — document presence, management and manner of resolution, and communicate to the medical provider and general dentist.

CODE OF PRACTICE IN FIVE LINES

1. Complete board certification requirements for dental sleep medicine, or obtain an equivalent diploma.
2. Maintain a file of patient history and examination results, which may include assessment of depression, mood and sleepiness.
3. Keep pre-treatment panoramic and cephalometric radiographs and dental casts.
4. Obtain written informed consent, including potential risks, **before** fitting.
5. Maintain regular written communication with the physician about plan, progress and follow-up, and keep that correspondence on file.

9.3 Item codes and billing

BEFORE YOU LEARN THIS SECTION

Item numbers, payer identifiers and fee schedules are **jurisdiction-specific and change frequently**. Do not memorise specific numbers unless your own syllabus supplies them. What is examinable is the **principle**: use the code matching the service actually provided and the patient's insurer; document well enough to survive audit; and keep records for different insurers and conditions separate. State the principle first, local detail second, flagged as local.

- **Private billing:** identify codes specific to the service and the insurer; maintain clear, **separate** consultation notes differentiating insurers and conditions; consult the payer's fee schedule or guidelines for correct item numbers.
- **Department of Veterans' Affairs:** use **DVA-specific codes**, which can differ from those for the general public insurer or private insurance; consult the relevant fee schedule; use the correct payer identifiers; and for veterans with both DVA and other insurance, clearly **separate documentation of unrelated conditions** when billing multiple insurers.
- **Both:** documentation is key — clear, accurate records support all claims, and **screening test scores must be recorded in the clinical record because they may be audited**. Recognise that codes differ between private, general public insurer, veterans' affairs and workers' compensation schemes, and familiarise yourself with each to prevent errors. Consult authoritative sources — the relevant government health department or insurer — for current guidance rather than memory or colleagues.
- **Ethical dimension:** billing and coding fall explicitly within the professionalism competencies of a qualified dentist. **Appliance selection must never be driven by what an insurer will reimburse.**

9.4 Ethics

- **Autonomy** — patients have the right to make informed decisions; in practice, genuine informed consent requiring clear understanding of purpose, risks, benefits and alternatives.
- **Non-maleficence** (do no harm) — use evidence-based practices, stay updated, and provide treatment **only within established competence**.
- **Beneficence** — promote well-being and provide high-quality care improving oral and overall health. **Justice** — fairness and equity for all patients. **Veracity** — be truthful, including about the limits of the evidence and the likelihood of side effects.
- **Informed consent** — disclose all material information: the nature of the disorder; the purpose of and alternatives to treatment; and the potential risks and consequences, including the risks of **no** treatment.
- **Scope of competence** — treat only within your qualifications and expertise; deliver care within the qualified dentist's competence.
- **Collaboration and referral** — work with other professionals, especially sleep physicians, for comprehensive care.
- **Screening versus diagnosis** — dentists are **not officially authorised to diagnose** sleep-related breathing disorders but can screen for symptoms, including tooth grinding and jaw pain, and recommend further testing.
- **Patient-centred care** — respect the diversity of patient populations and their unique needs and backgrounds. **Confidentiality** — a fundamental ethical and legal obligation, relevant to interprofessional communication (with patient permission) and to remote compliance monitoring, where loss of privacy is a recognised disadvantage. **Quality assurance** — adhere to high standards through ongoing processes. **Conflict of interest** — appliance choice belongs to the qualified dentist and the patient, not to insurers or manufacturers.

The competency framework for a "qualified dentist"

- **Clinical knowledge** — basic sleep physiology including SDB pathophysiology; the roles of dentist and team members; evidence-based therapeutic options and awareness of emergent treatments and technologies; the comorbid nature of OSA and TMD and the effects of appliance therapy on the TMJ, dental occlusion and related structures.
- **Patient care** — screening and risk assessment; comprehensive DSM-focused medical and dental examination and history, including relevant imaging; understanding objective sleep testing and complementary assessments; gathering and synthesising information to screen appropriately and establish a plan covering appliance selection, fitting and calibration, long-term care and management of potential side effects, taking patient values and preferences into account.
- **Learning and improvement** — evidence-based decision-making; continuing education; reflective practice and personal growth; identifying opportunities to standardise, simplify and improve quality.

- **Professionalism** — ethical principles including billing and coding; accountability and team training with adherence to applicable laws; SOAP documentation, record maintenance and protection of patient information; a collaborative interdisciplinary model; informed consent.
- **Interpersonal and communication skills** — effective patient communication; barrier and bias mitigation; interprofessional and team communication.
- Team members should receive appropriate training for their tasks. **Telehealth** may be used for screening and many follow-ups, expanding access for remote patients or those without a nearby qualified dentist — but the level of service should equal an in-person appointment, and the **comprehensive examination, impressions and construction bite registration should be done in person**.

RULES TO MEMORISE — TOPIC 9

- PM minimum: airflow + respiratory effort + blood oxygenation
- Airflow: oronasal thermal sensor (apnoea) + nasal pressure transducer (hypopnoea) — ideally both. Effort: calibrated or uncalibrated inductance plethysmography. Oximetry: averaging ≤3 s at HR ≥80 bpm, finger probe. Must allow raw data review
- PM only where high pre-test probability of moderate–severe OSA, no medical or sleep comorbidity, with comprehensive sleep evaluation supervised by a board-certified sleep specialist
- PM not for screening, not in comorbidity, not for other sleep disorders; limited data in children and over-65s; may monitor response to non-CPAP treatment where OSA is already diagnosed
- Adult diagnosis AHI ≥5 with symptoms or ≥15 regardless; paediatric obstructive AHI ≥1 abnormal; end-tidal CO_2 standard in children; PSG before adenotonsillectomy under 2 y

SELF-TEST — TOPIC 9

1. What must a portable monitor record, and with which sensors?

Minimum: **airflow, respiratory effort and blood oxygenation**, using the biosensor types recommended for laboratory PSG. Airflow — an oronasal thermal sensor for apnoea *and* a nasal pressure transducer for hypopnoea, ideally both, because nasal pressure is superior for detecting flow but records only nasal flow, leaving oral flow undetected in a mouth breather. Effort — calibrated or uncalibrated respiratory inductance plethysmography, using chest and abdominal excursion to measure tidal volume, more accurate alongside another event such as desaturation. Oxygenation — pulse oximetry with signal averaging ≤3 seconds at a heart rate of 80 bpm or above, usually a finger probe. The device must allow review of raw data so automated scoring can be manually edited by skilled personnel.

2. In whom is portable monitoring inappropriate?

Not for screening, not in significant medical comorbidity, and not where another sleep disorder is suspected. Comorbidity degrades accuracy — moderate-to-severe pulmonary disease, neuromuscular disease, congestive heart failure — and a portable monitor does not record what is needed to diagnose central sleep apnoea, periodic limb movement disorder, insomnia, parasomnias, circadian rhythm disorders or narcolepsy. All of these need in-laboratory PSG, as does a high-risk patient who fails the portable monitoring criteria. Data are also limited in children and over-65s. Two legitimate special indications remain: where in-laboratory study is impossible because of immobility, safety or critical illness; and to monitor response to a non-CPAP treatment, including an oral appliance, where the diagnosis has already been made.

3. List the records you must keep.

Baseline sleep study data with detailed parameters, AHI, architecture and arousals, stored for later comparison; ongoing SOAP-format notes recording assessments, symptoms, progress, adherence and response from both physician and dentist; documented treatment plans with rationale, appliance choice, adjustments, titration and education, plus shared decision-making discussions; study models, impressions and bite registrations, pre-treatment panoramic and cephalometric radiographs, and intraoral and extraoral photographs, retained as long as local regulation requires; signed informed consent covering purpose, risks, benefits, alternatives and patient rights, countersigned and dated; records of interprofessional communication kept with patient permission; the signed delivery form, written home care instructions and warranty; and documentation of side effects with their management and resolution.

4. Name the ethical principles with a dental sleep medicine example of each.

Autonomy — genuine informed consent covering purpose, risks, benefits and alternatives including no treatment. **Non-maleficence** — practising within established competence using evidence-based methods; for example declining to fit an appliance in untreated severe TMD or without a diagnosis. **Beneficence** — high-quality care improving oral and overall health. **Justice** — fairness and equity of access. **Veracity** — honesty about the likelihood of occlusal change and the limits of the evidence. Specific applications: the screening-versus-diagnosis boundary; collaboration and referral; patient-centred care respecting diversity; confidentiality in interprofessional communication and remote monitoring; quality assurance; and avoiding conflicts of interest — appliance choice must not be dictated by insurer requirements or commercial relationships.

TOPIC 9

Best-practice principles in dental sleep medicine

EXAMINER'S BLUEPRINT

1. Diagnosis of sleep-disordered breathing by a physician
2. Record keeping requirements: baseline sleep study data, ongoing clinical notes, treatment planning, study models, informed consent and interprofessional communication
3. Use of correct and appropriate item codes for billing patients privately and for the Department of Veterans' Affairs
4. Ethics in dental sleep medicine

Topic 9 is the professional-practice topic. It is not clinically difficult but it is easy to fail, because the answers are matters of rule rather than judgement — and examiners can mark them unambiguously. Learn the portable monitoring guideline in 9.1 properly; it is the most technically detailed part of the topic and the part most often examined in fine detail.

9.1 Diagnosis of SDB by a physician

In-laboratory polysomnography

- Accurate, with a low failure rate, and attended by technical staff — but expensive and technically complex.
- Also the standard for diagnosis in patients with other sleep disorders such as narcolepsy and central sleep apnoea.

Portable monitoring — the clinical guideline in detail

A portable monitor (PM) is an alternative diagnostic test: less expensive and quicker, and it can be as accurate as polysomnography in selected populations. Patients appropriate for a PM must be at high risk for OSA and must **not** have comorbid medical disorders or comorbid sleep disorders. Failure to meet these criteria in a patient at high risk for moderate-to-severe OSA should lead to in-laboratory polysomnography.

Element	Guideline content
Supervision	Portable monitoring for the diagnosis of OSA should be performed **only in conjunction with a comprehensive sleep evaluation**, by a board-certified sleep specialist or an individual who fulfils the eligibility criteria for the sleep medicine certification examination
Indication	A PM **may** be used as an alternative to PSG for the diagnosis of OSA in patients with a **high pre-test probability of moderate-to-severe OSA**. Use should be limited to high-risk populations, and this depends on clinical judgement informed by demographics, predisposing and precipitating factors, clinical features and familial pattern. The relevant risk factors are **snoring, sleepiness, obesity and witnessed apnoea**, and the severity of the risk factors correlates with the severity of the OSA. There are only limited data for the use of PMs in children and in those over 65
Contraindications	A PM should **not** be used in patients with comorbidities, in patients with other sleep disorders, or for screening. **Significant comorbid medical conditions may degrade PM accuracy** — moderate-to-severe pulmonary disease, neuromuscular disease, congestive heart failure — so use in-laboratory PSG. Where **other sleep disorders** are suspected — central sleep apnoea, periodic limb movement disorder, insomnia, parasomnias, circadian rhythm disorders, narcolepsy — a PM does not include the data needed to reach those diagnoses, so use in-laboratory PSG
Special indications	A PM may be indicated for the diagnosis of OSA in patients for whom in-laboratory PSG is not possible because of immobility, safety or critical illness. A PM may also be indicated to **monitor the response to non-CPAP treatments** for OSA — including oral appliances, upper airway surgery and weight loss — **where the diagnosis of OSA has already been made**

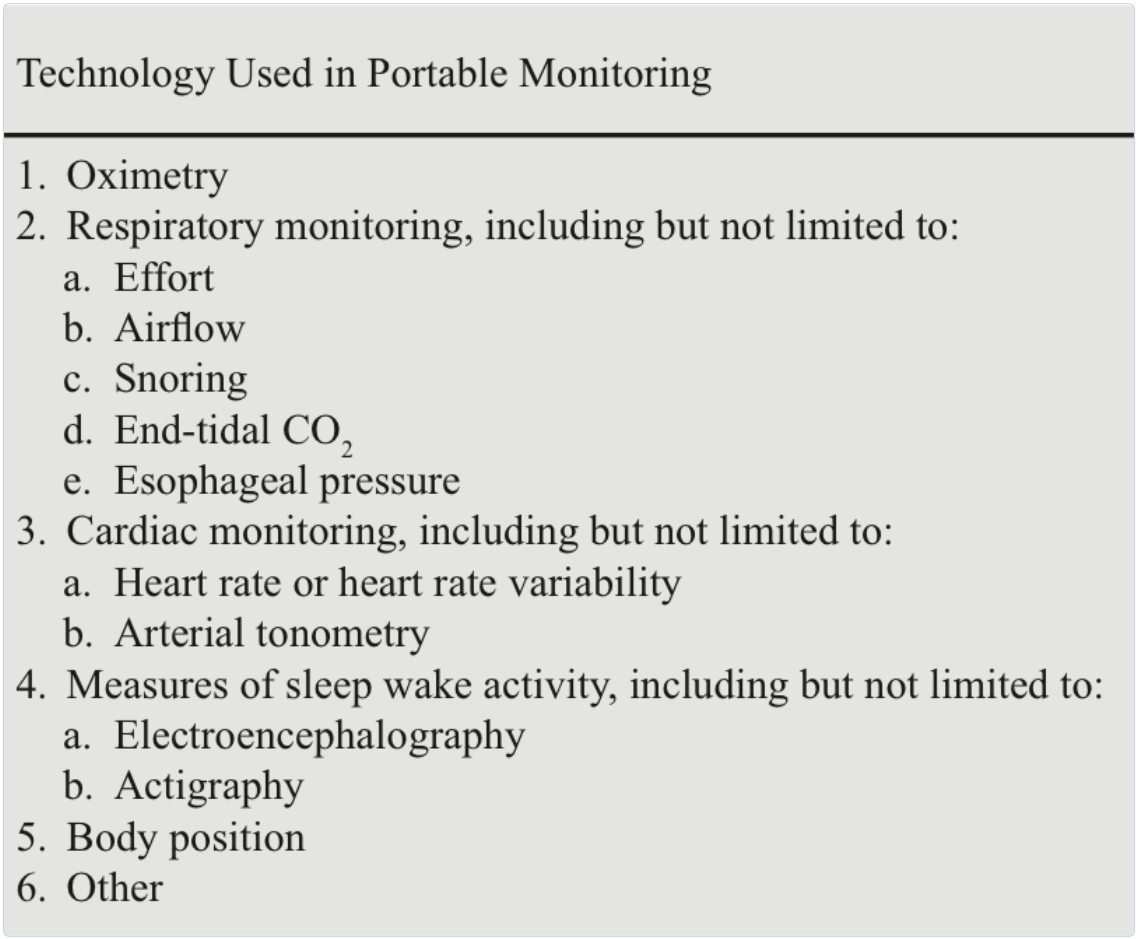


Figure 9.1 — Technology used in portable monitoring: (1) oximetry; (2) respiratory monitoring — effort, airflow, snoring, end-tidal CO_2, oesophageal pressure; (3) cardiac monitoring — heart rate or heart rate variability, arterial tonometry; (4) measures of sleep–wake activity — electroencephalography, actigraphy; (5) body position; and (6) other.

*Why it is examinable: examiners ask what a portable monitor must record as a minimum. The answer is **airflow, respiratory effort and blood oxygenation** — and the biosensor types recommended for laboratory PSG are the ones recommended for portable monitors.*

Technology requirements — the detail worth learning

Parameter	Recommended sensor and rationale
Minimum recording	A PM must record **airflow, respiratory effort and blood oxygenation** at a minimum
Airflow	**Oronasal thermal sensor** to detect *apnoea*, and **nasal pressure transducer** to detect *hypopnoea* — ideally a PM should use **both** sensor types. The nasal pressure device is superior to a thermistor for detecting flow but is limited to nasal flow, leaving oral flow undetected, giving an unnoticed signal in a mouth breather — which is exactly why the oronasal sensor is also needed
Respiratory effort	Ideally either calibrated or uncalibrated **inductance plethysmography**. Respiratory inductive plethysmography can use changes in chest and abdominal excursion during inspiration and expiration to measure actual tidal volume. To improve accuracy, effort should be assessed in conjunction with another event such as oxygen desaturation
Blood oxygenation	**Pulse oximetry** with an appropriate signal averaging time and accommodation for motion artefact: a maximum signal averaging time of **≤3 seconds at a heart rate of 80 bpm or more**. Most studies use a finger probe
Scoring	The device must allow review of **raw data**, permitting manual scoring or manual editing of automated scoring by skilled personnel with review of the raw signal
Paediatric addition	**End-tidal CO_2** is a standard PSG measurement in paediatric patients

Diagnosis in children

Diagnosis primarily involves comprehensive clinical evaluation followed by objective sleep studies, with overnight polysomnography the gold standard. Clinical evaluation begins with a detailed history focusing on habitual snoring, observed apnoeas, laboured or paradoxical breathing during sleep, daytime sleepiness, behavioural problems and learning difficulties, with physical examination attending to upper airway anatomy including tonsil and adenoid size, craniofacial features, and signs of obesity or comorbid conditions. Validated questionnaires aid screening but are not definitive. Attended in-laboratory PSG records EEG, EOG, EMG, ECG, respiratory airflow, respiratory effort, oxygen saturation and body position, allowing correlation of respiratory events with sleep stages and arousals; an obstructive AHI ≥1 event/hour is abnormal. PSG is indicated for children with clinical symptoms of SDB, especially before adenotonsillectomy in patients under 2 years, in obesity, in craniofacial or neuromuscular disorders, in Down syndrome, or where symptoms persist after surgery — and for CPAP titration and complex cases. Other tools: respiratory polygraphy (no sleep staging, less validated in children); nocturnal oximetry (a simple screening tool with good predictive value in high-risk children); home sleep apnoea testing (emerging, not recommended as a standalone diagnostic tool in paediatrics); daytime nap studies (less accurate); radiological imaging and biomarkers (promising, requiring further research).

Diagnosis in adults

Physicians first assess symptoms — habitual loud snoring, witnessed apnoeas, choking or gasping during sleep, excessive sleepiness, fatigue, cognitive impairment and mood disorder — and note risk factors on examination: obesity with high BMI, large neck circumference, hypertension, cardiovascular disease. Screening questionnaires such as the ESS support clinical suspicion but **cannot diagnose SDB alone**. Overnight polysomnography quantifies apnoeas and hypopnoeas per hour of sleep as the AHI; **an AHI ≥5/h with symptoms, or ≥15/h regardless of symptoms, confirms OSA**. PSG is also essential for assessing severity and distinguishing central from obstructive events. Home sleep apnoea testing may be used to diagnose moderate-to-severe OSA in patients with high pre-test probability and without significant comorbidity, monitoring fewer channels — airflow, respiratory effort, oxygen saturation, heart rate — but PSG remains preferred where the diagnosis is uncertain or other sleep disorders are suspected. Additional testing includes the MSLT where narcolepsy or hypersomnia is suspected, with acoustic pharyngometry and actigraphy as adjuncts, and referral to ENT, cardiology or neurology as findings dictate.

HIGH-YIELD — THE FIVE-LINE SUMMARY OF THE DIAGNOSTIC PROCESS

1. Clinical suspicion, based on symptoms and risk factors, initiates the diagnostic process.
2. Overnight laboratory polysomnography is the definitive diagnostic tool, measuring multiple physiological parameters during sleep.
3. Home sleep testing may be suitable for straightforward moderate-to-severe cases.
4. Diagnosis relies on the AHI threshold **combined with** the clinical presentation.
5. Complementary testing and specialist referral support diagnosis and management.

9.2 Record keeping

Record	Requirement
Baseline sleep study data	Documentation of the baseline sleep study results — polysomnography or home sleep apnoea test — with detailed sleep parameters, apnoea–hypopnoea index, sleep architecture and arousals. Proper storage of these baseline studies in the patient's record, for treatment reference and follow-up comparison. Without a baseline you cannot demonstrate treatment effect
Ongoing clinical notes	Detailed notes recording patient assessments, symptoms, treatment progress, adherence and response to therapy, including all relevant evaluations and observations from both physicians and qualified dentists. The AADSM refers to **SOAP-format** documentation — subjective, objective, assessment, plan

Record	Requirement
Treatment planning	Documentation of comprehensive treatment plans outlining rationale, appliance choice, adjustments, titration and patient education. Any **shared decision-making** discussions about therapy options and intolerances should be recorded
Study models	Retention of study models — dental impressions and bite registrations — used for fabrication and adjustment. These form part of the dental record for ongoing management, and should be retained for as long as local regulations require. Also retain pre-treatment panoramic and cephalometric radiographs, and intraoral and extraoral photographs
Informed consent	Complete and signed informed consent forms explaining treatment purpose, risks, benefits, alternatives and patient rights. Consent documentation must be clear, free of coercion, and verifiable in the record. Signed by the patient in the presence of the qualified dentist or team member, then countersigned and dated by the dentist and kept as part of the record of care
Interprofessional communication	Records must include communications between sleep physicians, dentists and other healthcare professionals involved, to ensure coordinated care with shared assessment, treatment adjustments and follow-up — documented with patient permission
Delivery documentation	A written copy of the home care instructions, the delivery form and the warranty information, signed and dated by the patient and a team member, with one copy given to the patient and one retained in the record
Side effect documentation	Where side effects are noted, document their presence, their management and the manner of their resolution, and communicate to the medical provider and general dentist

HIGH-YIELD — THE CODE OF PRACTICE, IN FIVE LINES

1. Complete board certification requirements for dental sleep medicine, or obtain an equivalent diploma.
2. Maintain a file containing patient history and examination results, which may include assessment of depression, mood and sleepiness.
3. Keep pre-treatment panoramic and cephalometric radiographs and dental casts.
4. Obtain written informed consent, including potential risks, before fitting the appliance.
5. Maintain regular written communication with the physician about treatment plan, progress and follow-up, and keep that correspondence on file.

9.3 Item codes and billing

A NOTE BEFORE YOU LEARN THIS SECTION

Item numbers, payer identifiers and fee schedules are **jurisdiction-specific and change frequently**. Do not memorise specific numbers for an examination unless your own syllabus supplies them. What is genuinely examinable is the **principle**: use the code that matches the service actually provided and the patient's insurer, document accurately enough to survive audit, and keep the records for different insurers and different conditions separate. State the principle first; offer local detail second, flagged as local.

Principles for private billing

- **Identify the correct codes.** Use item codes and descriptions specific to the service provided and to the patient's insurer.
- **Maintain detailed notes.** Keep clear, separate consultation notes differentiating between different insurers and different conditions.
- **Refer to payer guidelines.** Consult the payer's specific fee schedule or guidelines to ensure correct item numbers.

Principles for Department of Veterans' Affairs billing

- **Use DVA-specific codes.** Veterans' affairs schemes maintain their own set of codes, which can differ from those used for the general public health insurer or for private insurance.
- **Consult the relevant fee schedule** for the correct codes for therapy services and other treatments.
- **Use the correct payer identifiers** when submitting claims.
- **Separate conditions.** For veterans with both DVA and other insurance, clearly separate discussion and documentation of unrelated conditions when billing multiple insurers.

Best practice for both

- **Documentation is key.** Clear, accurate documentation supports all billing claims and ensures a smooth billing process — and screening test scores must be recorded in the clinical record because they may be subject to audit.
- **Understand that codes differ** between private, general public insurer, veterans' affairs and workers' compensation schemes. Familiarise yourself with the appropriate item number for each insurer to prevent errors and ensure claims are paid accurately.
- **Consult authoritative sources** — the relevant government health department or insurer website — for current guidance rather than relying on memory or on colleagues.
- **Ethical dimension:** billing and coding practices fall explicitly within the professionalism competencies expected of a qualified dentist in dental sleep medicine. Appliance selection must never be driven by what an insurer will reimburse.

9.4 Ethics in dental sleep medicine

The core principles

Principle	Application in dental sleep medicine
Autonomy	Patients have the right to make informed decisions about their treatment. In dental sleep medicine this means obtaining genuine informed consent, requiring clear understanding of the purpose, risks, benefits and alternatives to appliance therapy
Non-maleficence (do no harm)	Avoid causing harm: use evidence-based practices, stay updated on the latest research, and provide treatment **only within established competence**
Beneficence	Promote the patient's well-being and provide high-quality care that improves oral health and overall health
Justice	Fairness and equity in providing care to all patients
Veracity	Be truthful and honest with patients — including about the limits of the evidence and about the likelihood that side effects will occur

Specific ethical considerations

Consideration	Detail
Informed consent	Disclose all material information: the nature of the sleep-related breathing disorder; the purpose of, and alternatives to, the proposed treatment; and the potential risks and consequences of appliance therapy — including the risks of **no** treatment
Scope of competence	Treat patients only within the scope of your qualifications and expertise in managing sleep-related breathing disorders. Patient care should be delivered within the scope of the qualified dentist's competence
Collaboration and referral	Ethical practice often involves collaborating with other healthcare professionals, especially sleep physicians, to ensure comprehensive and appropriate care

Consideration	Detail
Screening versus diagnosis	Dentists are **not officially authorised to diagnose** sleep-related breathing disorders, but can screen for symptoms — including tooth grinding and jaw pain — and recommend that patients seek further testing
Patient-centred care	Deliver care in a way that respects the diversity of patient populations and considers their unique needs and backgrounds
Confidentiality	Maintaining the privacy of patient information is a fundamental ethical and legal obligation — relevant to interprofessional communication, which should occur with patient permission, and to remote compliance monitoring, where loss of privacy is a recognised disadvantage
Quality assurance	Adhering to high standards of care through ongoing quality assurance processes
Conflict of interest	Avoid conflicts of interest. Appliance choice should be the decision of the qualified dentist and the patient, and should not be determined solely by the requirements of governmental or private insurers, nor by commercial relationships with manufacturers

The competency framework for a qualified dentist

It is worth being able to describe what "qualified" means, because it defines the standard against which your practice is measured.

Domain	Competencies
Clinical knowledge	Basic sleep physiology, including the pathophysiology of sleep-related breathing disorders; the roles of the dentist and team members in managing SRBD; evidence-based therapeutic options and awareness of emergent treatments and technologies; the comorbid nature of OSA and temporomandibular disorders, and the effects of appliance therapy on the TMJ, dental occlusion and related structures
Patient care	Screening and risk assessment; comprehensive dental-sleep-medicine-focused medical and dental examination and history, including relevant imaging; understanding objective sleep testing and other complementary assessments; gathering and synthesising information to screen appropriately and establish a treatment plan covering appliance selection, device fitting and calibration, long-term care and management of potential side effects — taking patient values and preferences into account
Learning and improvement	Clinical practice and decision-making guided by evidence-based principles; continuing education; reflective practice and personal growth; identifying opportunities to standardise, simplify and improve quality of care
Professionalism	Ethical principles, including billing and coding practices; accountability and team training with adherence to applicable laws and regulations; SOAP note documentation, maintenance of treatment records and protection of patient information; a collaborative model of interdisciplinary care; informed consent
Interpersonal and communication skills	Effective patient communication; barrier and bias mitigation; interprofessional and team communication

Team members should also receive appropriate training to perform the tasks required of them. Note that **telehealth** may be used during screening and for many follow-up visits, expanding access for patients in remote areas or without a nearby qualified dentist — but the level of service in a telehealth appointment should be the same as in person, and the comprehensive examination, impressions and construction bite registration should be done **in person**.

NUMBERS AND RULES TO MEMORISE — TOPIC 9

- PM minimum recording: airflow + respiratory effort + blood oxygenation
- Airflow sensors: oronasal thermal sensor for apnoea, nasal pressure transducer for hypopnoea — ideally both
- Effort sensor: calibrated or uncalibrated inductance plethysmography
- Oximetry: signal averaging time ≤3 s at heart rate ≥80 bpm; finger probe
- PM must allow review of raw data with manual scoring or editing
- PM appropriate only in high pre-test probability of moderate–severe OSA, without medical or sleep comorbidity, and only with a comprehensive sleep evaluation supervised by a board-certified sleep specialist
- PM not for screening; not in comorbidity; not for other sleep disorders; limited data in children and those over 65
- PM may monitor response to non-CPAP treatment where OSA is already diagnosed
- Adult diagnosis: AHI ≥5 with symptoms or ≥15 regardless. Paediatric: obstructive AHI ≥1 abnormal; end-tidal CO_2 standard in children
- PSG indicated before adenotonsillectomy in children under 2 years

SELF-TEST — TOPIC 9

1. What must a portable monitor record as a minimum, and which sensors are recommended?

As a minimum, **airflow, respiratory effort and blood oxygenation**. The biosensor types recommended for in-laboratory PSG are the ones recommended for portable monitors: for airflow, an oronasal thermal sensor to detect apnoea *and* a nasal pressure transducer to detect hypopnoea — ideally both, because nasal pressure is superior for detecting flow but records only nasal flow, leaving oral flow undetected in a mouth breather; for effort, calibrated or uncalibrated respiratory inductance plethysmography, which uses chest and abdominal excursion to measure tidal volume, and which is more accurate when assessed alongside another event such as desaturation; and for oxygenation, pulse oximetry with a signal averaging time of no more than 3 seconds at a heart rate of 80 bpm or above, usually with a finger probe. The device must also allow review of raw data so that automated scoring can be manually edited by skilled personnel.

2. In whom is portable monitoring inappropriate?

Portable monitoring should not be used for screening, in patients with significant medical comorbidity, or where another sleep disorder is suspected. Significant comorbidity degrades accuracy — moderate-to-severe pulmonary disease, neuromuscular disease, congestive heart failure — and a portable monitor simply does not record the data needed to diagnose central sleep apnoea, periodic limb movement disorder, insomnia, parasomnias, circadian rhythm disorders or narcolepsy. In all of these situations in-laboratory polysomnography is required. Data are also limited for children and for patients over 65. Where a patient at high risk of moderate-to-severe OSA fails to meet the criteria for portable monitoring, they should have in-laboratory PSG. Two legitimate special indications remain: where in-laboratory study is impossible because of immobility, safety or critical illness, and to monitor response to a non-CPAP treatment — including an oral appliance — where the diagnosis has already been made.

3. List the records you must keep for a patient in oral appliance therapy.

Baseline sleep study data with detailed parameters, AHI, sleep architecture and arousals, properly stored for later comparison; ongoing clinical notes in SOAP format recording assessments, symptoms, progress, adherence and response, including observations from both physician and dentist; documented treatment plans setting out rationale, appliance choice, adjustments, titration and patient education, together with any shared decision-making discussions; study models, impressions and bite registrations, plus pre-treatment panoramic and cephalometric radiographs and intraoral and extraoral photographs, retained as long as local regulation requires; signed informed consent explaining purpose, risks, benefits, alternatives and patient rights, countersigned and dated; records of interprofessional communication with the sleep physician and others, kept with patient permission; the signed delivery form, written home care instructions and warranty; and documentation of any side effects with their management and resolution.

4. Name the ethical principles and give a dental sleep medicine example of each.

> **Autonomy** — obtaining genuine informed consent so the patient understands purpose, risks, benefits and alternatives, including no treatment. **Non-maleficence** — practising within established competence and using evidence-based methods, for example declining to fit an appliance in an untreated severe TMD or without a diagnosis. **Beneficence** — providing high-quality care that improves oral and overall health. **Justice** — fairness and equity of access. **Veracity** — honesty about the likelihood of occlusal change and about the limits of the evidence. Specific applications include the screening-versus-diagnosis boundary, collaboration and referral, patient-centred care respecting diversity, confidentiality in interprofessional communication and remote monitoring, quality assurance, and avoidance of conflicts of interest — appliance choice must not be dictated by insurer requirements or commercial relationships.

5. What does "qualified dentist" mean in dental sleep medicine?

> A dentist who has undertaken serious, recognised training in sleep medicine or sleep-related breathing disorders — diplomate certification in dental sleep medicine or the equivalent qualified-dentist designation — with focused emphasis on the proper protocol for diagnosis, treatment and follow-up. The competencies span five domains: clinical knowledge (sleep physiology and SDB pathophysiology, roles of the team, evidence-based and emerging therapies, the comorbidity of OSA and TMD and the effects of therapy on the joint, occlusion and related structures); patient care (screening and risk assessment, comprehensive examination including relevant imaging, understanding of objective testing, and treatment planning covering selection, fitting, calibration, long-term care and side-effect management, incorporating patient values); learning and improvement (evidence-based decision-making, continuing education, reflective practice, quality improvement); professionalism (ethics including billing and coding, accountability and team training, SOAP documentation and information protection, interdisciplinary collaboration, informed consent); and interpersonal and communication skills (patient communication, bias mitigation, interprofessional communication).

REFERENCES — TOPIC 9

1. Collop NA, Anderson WM, Boehlecke B, et al. Clinical guidelines for the use of unattended portable monitors in the diagnosis of obstructive sleep apnea in adult patients. *J Clin Sleep Med*. 2007;3(7):737–747.
2. Kapur VK, Auckley DH, Chowdhuri S, et al. Clinical practice guideline for diagnostic testing for adult obstructive sleep apnea: an AASM clinical practice guideline. *J Clin Sleep Med*. 2017;13(3):479–504.
3. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2).
4. Schwartz D, Adame M, Addy N, et al. American Academy of Dental Sleep Medicine position on the scope of practice for dentists ordering or administering home sleep apnea tests. *J Dent Sleep Med*. 2020;7(4).
5. White J, Richardson S, Levine M. Protocol for communicating with a patient's treating physician. American Academy of Dental Sleep Medicine.
6. Berry RB, Quan SF, Abreu AR, et al. *The AASM Manual for the Scoring of Sleep and Associated Events*. Version 3. AASM; 2023.
7. Aurora RN, Zak RS, Karippot A, et al. Practice parameters for the respiratory indications for polysomnography in children. *Sleep*. 2011;34(3): 379–388.
8. American Dental Association. Policy statement: the role of dentistry in the treatment of sleep-related breathing disorders. ADA; 2017.
9. Beauchamp TL, Childress JF. *Principles of Biomedical Ethics*. 8th ed. Oxford University Press; 2019.
10. American Dental Association. Principles of Ethics and Code of Professional Conduct.
11. Dental Board of Australia. Code of conduct for registered health practitioners.
12. Australian Government Department of Health. Medicare Benefits Schedule and Department of Veterans' Affairs fee schedules — current online editions.

TOPIC 10

Informed consent and management of oral appliance side effects

BLUEPRINT

1. Obtaining informed consent · 2. Evidence-based expectations of side effects · 3. Effect of protrusion on soft tissue, teeth and TMJs · 4. Occlusal guide and morning exercises · 5. Management of side effects · 6. Suspending or abandoning therapy

THE UNIFYING INSIGHT

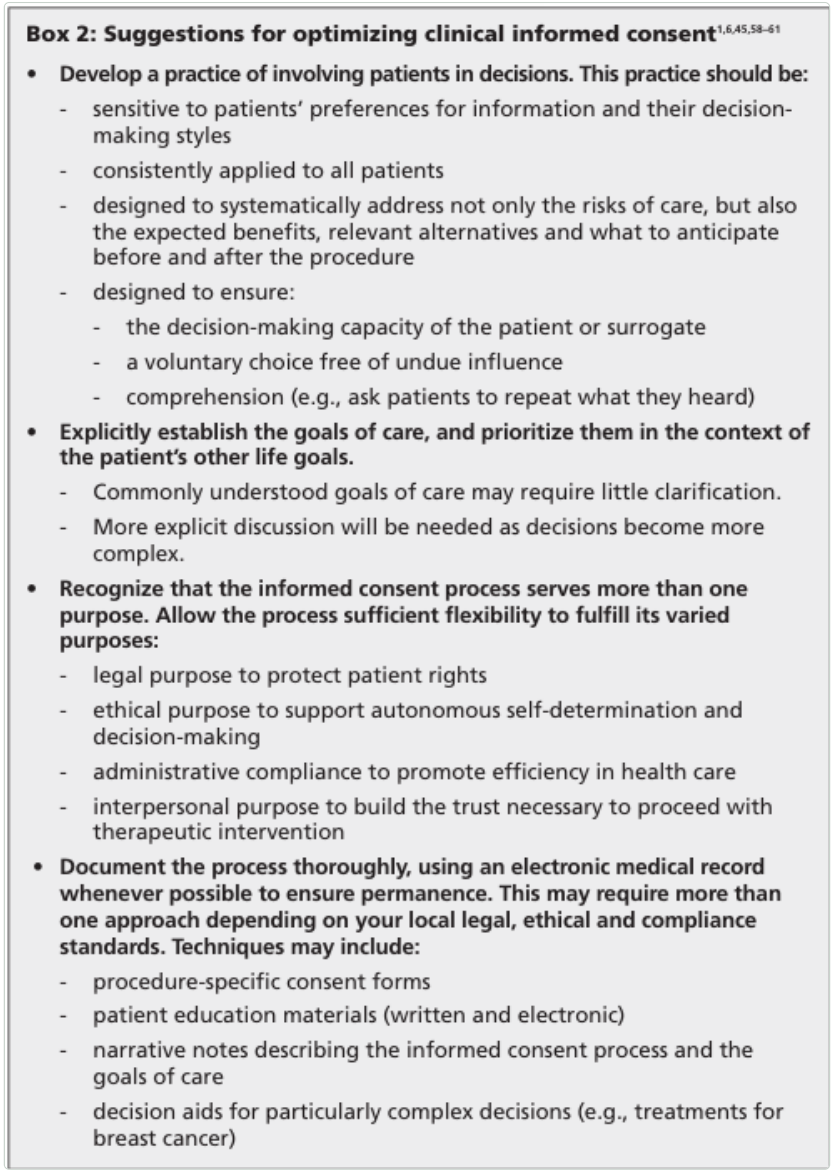
Occlusal change is not a complication of appliance therapy — it is an expected consequence of it, occurring in the great majority of long-term users. Your task is not to prevent it (you largely cannot) but to **predict, disclose, monitor, minimise and manage** it. Presenting bite change as a rare mishap loses marks; presenting it as a known, quantified, consented trade-off scores well.

10.1 Informed consent

DEFINITION

Simple consent entails that a patient — or surrogate — with decision-making capacity **freely authorises** a treatment plan aimed at a **mutually acknowledged treatment goal**. The authorisation is "**informed**" when the clinician discloses, and the patient understands, the diagnosis, the relevant options **including no treatment**, and the respective risks and benefits of each.

- **Four purposes of the consent process: legal** (protect patient rights; protection from assault; preventing unwanted procedures) · **ethical** (support autonomous self-determination and decision-making; support patient-defined goals) · **administrative compliance** (promote efficiency; provide safeguards ensuring nominal fulfilment of ethical and legal requirements) · **interpersonal** (build the trust necessary to proceed with therapeutic intervention). A signed form alone satisfies only the administrative purpose.
- **Three tests of valid consent: capacity**, **voluntariness** (free of undue influence) and **comprehension** — verified, not assumed, for example by asking patients to repeat back what they heard.



Box 2: Suggestions for optimizing clinical informed consent[1,6,45,58–61]

- **Develop a practice of involving patients in decisions. This practice should be:**
  - sensitive to patients' preferences for information and their decision-making styles
  - consistently applied to all patients
  - designed to systematically address not only the risks of care, but also the expected benefits, relevant alternatives and what to anticipate before and after the procedure
  - designed to ensure:
    - the decision-making capacity of the patient or surrogate
    - a voluntary choice free of undue influence
    - comprehension (e.g., ask patients to repeat what they heard)
- **Explicitly establish the goals of care, and prioritize them in the context of the patient's other life goals.**
  - Commonly understood goals of care may require little clarification.
  - More explicit discussion will be needed as decisions become more complex.
- **Recognize that the informed consent process serves more than one purpose. Allow the process sufficient flexibility to fulfill its varied purposes:**
  - legal purpose to protect patient rights
  - ethical purpose to support autonomous self-determination and decision-making
  - administrative compliance to promote efficiency in health care
  - interpersonal purpose to build the trust necessary to proceed with therapeutic intervention
- **Document the process thoroughly, using an electronic medical record whenever possible to ensure permanence. This may require more than one approach depending on your local legal, ethical and compliance standards. Techniques may include:**
  - procedure-specific consent forms
  - patient education materials (written and electronic)
  - narrative notes describing the informed consent process and the goals of care
  - decision aids for particularly complex decisions (e.g., treatments for breast cancer)

Fig 10.1 — Optimising clinical informed consent. Develop a practice of involving patients in decisions, sensitive to their preferences for information and decision-making styles, consistently applied, addressing not only risks but expected benefits, relevant alternatives and what to anticipate before and after the procedure, and designed to ensure capacity, a voluntary choice free of undue influence, and comprehension. Explicitly establish and prioritise the goals of care in the context of the patient's other life goals. Recognise the four purposes — legal, ethical, administrative and interpersonal — and allow the process flexibility to fulfil them. Document thoroughly using the electronic record, which may require more than one approach depending on local standards: procedure-specific consent forms, patient education materials (written and electronic), narrative notes describing the process and the goals of care, and decision aids for particularly complex decisions.

- **What must be disclosed:** the nature of the sleep-related breathing disorder and consequences if untreated; the purpose of treatment and the **alternatives** (PAP, positional therapy, weight loss, surgery, no treatment); the potential risks and consequences and the **expected longevity** of the appliance; **possible dental side effects before starting** — at minimum reduced overjet and overbite leading to suboptimal occlusion with fewer occlusal contact points, possible unilateral or bilateral crossbite in the (pre)molar region, and altered aesthetics; and the requirement for long-term follow-up.
- **Procedure:** explain the treatment **with long-term follow-up** and obtain consent **preferably in writing, before fitting**. The form should clearly indicate the **potential and probable side effects and the appliance's longevity**. Give the patient the opportunity to ask about the risks of treatment and educate them about the risks of **no** treatment. The patient signs in the presence of the qualified dentist or team member; the dentist **countersigns and dates**; the document is kept as part of the record. **Give the patient a copy.** Document thoroughly. **Signed, informed consent is mandatory** — and, given predictable occlusal change, it is your principal protection.

10.2 Evidence-based expectations of side effects

ONE-SENTENCE SUMMARY

Most side effects are **mild and transient**, disappearing during the first few months — but in approximately **7.5%** they lead to termination of therapy, most often because of **TMJ pain** (less commonly because of occlusal change). Side effects contribute to about **20–30%** of all discontinuations.

- **Short term (first weeks to months):** discomfort around teeth and jaws; **excessive salivation ~27%**; **dry mouth ~38%**; **tooth pain ~36%**; perception of tooth mobility; soft tissue, gingival and tongue irritation; difficulty chewing; abnormal occlusion on waking; jaw discomfort or stiffness and masticatory or myofascial pain (**more common on statins**); headache; tooth grinding; **allergic reaction <1%** (acrylic, nickel).
- **Long term (moderate–severe, ongoing): jaw pain ~34%**; TMJ restriction or noise; **bite change 14–26%**; damage to teeth or dental work; appliance issues. Significant and persistent TMJ problems are **rare**. Change in facial height and jaw relation noted as early as **6 months**.

OCCLUSAL CHANGE — THE QUANTITATIVE DETAIL

- **Direction of force: distally directed on the upper dentition, forwardly directed on the lower teeth** → dose-dependent shift with lower teeth moving anteriorly, upper teeth distally, and loss of posterior chewing contact area.
- **Magnitude:** overjet ~1.9 mm and overbite ~2.3 mm reduction (average ~2.5 mm described); ~⅓ exceed 1 mm; at 17 years median −1.1 mm overjet and −1.6 mm overbite.
- **Frequency:** ~**80%** of nightly users over 7 years experience change; **85%** detectable on models and cephalograms (some bites do not change); the proportion rose over the first 2 years to 13.3% of cases then remained relatively constant.
- **Time course:** overbite decreases *less* with time; overjet decreases *continuously at a constant rate*. Initial overjet does not affect the pattern but initial overbite does — with a smaller initial overbite the rate first decreases then increases after ~9 years, whereas a deep bite decreases gradually. Larger initial overjet → greater total reduction of both. Change increases progressively while treatment continues.
- **Other changes:** lower arch expansion — increased intercanine (continuously) and intermolar (rapidly then tapering) width, reduced lower crowding, opening of interdental spaces; **51% posterior open bite** (loss of contact on ≥2 posterior teeth); **62% anterior crossbite** of ≥1 tooth; increased maximal opening.
- **Anteroposterior:** tendency toward mesio-occlusion versus CPAP — class I → III, or class II → I/III; tendency to unilateral or bilateral (pre)molar crossbite, because with a mesial shift the broader part of the lower arch occludes with the narrower part of the upper. Significant decrease in (pre)molar occlusal contacts with **both** appliances and CPAP, attributed to the incisal guidance phenomenon — but **no tendency to anterior open bite**.
- **Predictors:** significant association between overbite change and amount of protrusion, and **initial AHI is positively related to the magnitude of overjet reduction** (more severe OSA needs more advancement). But degree of advancement is **not** related to magnitude of change in some analyses (any degree can cause change in predisposed people); change cannot be predicted from pre-treatment occlusal classification; and age, sex and pre-treatment BMI are unrelated. **Greatest vulnerability in the first 2 years**, after which change may partly reverse as a new occlusal equilibrium develops.
- **Mechanism:** change occurs because of a **failure to reposition the mandible during the day** after use, rather than immutable overnight change. Side effects are a product of **protruding the mandible to achieve a therapeutic effect rather than of specific design**.
- **CPAP is not innocent:** it also causes retroclination of the upper incisors from posteriorly directed forces of a tightly fitted nasal mask; long-term use of both produces significant dental change.

- **Causes — what it is *not*.** Change in mandibular length: unlikely. Change in skeletal pattern: unlikely — appliances cause **tooth movement, not craniofacial skeletal change**. Functional repositioning and neuromuscular adaptation: unlikely. **Tilting of teeth is the main cause** — retroclination of upper incisors, proclination of lower incisors, mesial tendency in the molar relationship, overall overjet reduction.

WHO IS VULNERABLE?

- **Edge-to-edge bite** (minimal overjet and overbite) — easily becomes class III
- **Younger patients** — more flexible bone, longer lifetime of wear
- **Previous orthodontic treatment**
- **Using >8 hours** — ask them to remove it on waking
- **≥80% of maximum protrusion**
- **Suboptimal result** — they keep advancing to chase residual symptoms
- **Severe OSA and obesity**
- **Using 7 nights a week** — consider 1–2 nights off
- **High aesthetic requirement**, narrow acceptance of change, or body dysmorphic features — these need the most careful consent

- **What patients actually experience.** Mostly changes are **minor and unnoticed**; only a minority notice. Most are unaware, untroubled, and **do not want to stop**. Where noticed: gaps and food impaction; occasional lisping; rarely difficulty chewing on

back teeth or tearing food with front teeth. Most do not lose posterior occlusion. Professional position: for most people the change is not annoying, not a risk to oral health, and does not require further dental treatment. **But** — the clinically consequential point — bite change **influences the mechanism of the device**, since a forward shift of the lower teeth relative to the upper progressively **reduces the effective advancement**, putting efficacy at risk.

10.3 Effect of protrusion on craniofacial structures

- **Teeth** — occlusal change (decreased overjet and overbite, proclined lower and retroclined upper incisors); tooth movement including **over-eruption of molars**; **biomechanical stress** — greater protrusion (e.g. 70% of maximum) induces significant stress particularly on the mandibular molars, increasing risk of root and bone resorption. Craniofacial changes are mainly related to significant tooth movement.
- **TMJ** — unnatural anterior positioning can cause pain, clicking and myofascial pain; reciprocal forces may alter position and function over time. But note the reassuring long-term data (Topic 4): no significant morphological alteration, condylar translation less than in maximum opening, no long-term impairment, and symptom intensity, joint noise and jaw fatigue may reduce.
- **Soft tissues** — hypersalivation or dry mouth; the intended airway enlargement by advancing the mandible and repositioning the tongue; and **soft tissue adaptation** — muscular activity adapts to the device, leading to superior and anterior repositioning of the hyoid.
- **Factors influencing severity** — risk increases with greater advancement, with levels **exceeding 50% of maximum** often correlating with significantly higher risk; underlying masticatory musculoskeletal disorders increase risk; appropriate planning and monitoring are crucial.
- **Corollary:** an appliance is already effective at a **less protrusive position** in mild and moderate OSA, and therefore causes less severe dental side effects. Minimising titration is a **side-effect prevention strategy**, not just a comfort measure.
- Although undesirable in some patients, effective treatment of a potentially life-threatening disease **may supersede** maintenance of a baseline occlusion — a sentence worth deploying, but only after documenting that the patient understood and accepted the trade-off.

10.4 Morning occlusal guide and exercises

> **WHAT A MORNING OCCLUSAL GUIDE IS AND DOES**
>
> A **morning occlusal guide** (morning repositioning device, "AM aligner") is used after waking to help realign the jaw and teeth after overnight advancement. The patient bites into it until the teeth re-establish habitual occlusion, which can reverse tooth movement and reposition the condyles. It repositions the mandible into its habitual pre-treatment position — and, just as importantly, **lets the patient know when occlusal change is occurring**: if they cannot fit into it with the back teeth in occlusion, bite change has occurred.

- **How it works:** realigns the bite toward original maximum intercuspation; exercises and stretches the muscles of mastication, tight after a night of wear; assists repositioning of the condyles to pre-treatment alignment, reducing strain; and **monitors progress** — if it does not fit, contact the practice.
- **Practical points: suggest it for every patient**. Two fabrication types — thermoacrylic beads, or a **prefabricated sheet** (AM aligner) placed in hot water, moulded around the teeth, with the patient biting in centric occlusion. **Should be custom made and recorded in centric occlusion. Check the occlusion with and without the aligner** to confirm the back teeth occlude with it seated. **Wear 15–20 minutes on waking** (some protocols ~10 minutes). Consider taking a centric occlusion record at the appliance impression appointment.
- **Morning exercises:** relieve masticatory muscle stiffness and accelerate repositioning of the mandible; may inhibit or minimise occlusal functional changes in predisposed patients. Perform slowly and gently — opening wide, moving the jaw left and right, resting the chin on a fist, curling the tongue back to find the retruded position. They increase contact between the teeth: good short-term, limited long-term evidence. Both guide and exercises are used immediately after removing the appliance, with duration and frequency set by the dentist. **Chewing gum** also helps, mostly with acute morning change. Recommended by the AADSM — but **not a cure-all**: some tooth or root change, though often clinically insignificant, may still occur.

10.5 Management of side effects

1. **Conservative management of occlusal change.** Temporary cessation or reduced use **plus remedial exercises every morning** after removal — with exercises, occlusal change resolves **within 2 weeks** of cessation in most cases. Use an occlusal guide. After satisfactory resolution, resume nightly treatment with an ongoing prophylactic exercise regimen and careful monitoring.
2. **It is reasonable to continue treatment in the presence of occlusal change**, provided it is regularly monitored (at least 6-monthly clinical reviews), is not associated with unacceptable symptoms, is not progressive, and there is **adequate posterior occlusal support**.
3. If conservative management fails — **especially with loss of posterior occlusion** — permanent cessation may be necessary.
4. **Correction** with restorations, crowns and/or orthodontics should be considered **only if the patient has discontinued treatment and can use another OSA treatment**, because change continues (more slowly) while the appliance is worn.

MINIMISING OCCLUSAL CHANGE

- **Minimise titration** — least advancement that relieves symptoms; allow a temporary increase during a cold, but remind the patient to titrate back
- **Reduce wearing time** where appropriate; remove on waking; consider 1–2 nights off weekly in vulnerable patients
- **Chewing gum** in the morning
- **Morning aligner for every patient**, custom, in centric occlusion, 15–20 min
- **Jaw exercises**
- **Cover all teeth** — do not leave molars uncovered; hard rather than flexible material; include incisor coverage
- Previous orthodontics: consider 4 nights appliance / 3 nights morning aligner, or appliance at night with a retainer by day
- **Review vulnerable patients early** and keep a list of them

- **Transient morning jaw pain** — muscles adjusting to the forward mandible. → jaw exercises, massage, heat; morning occlusal guide.
- **Bilateral jaw pain** — advancement beyond TMJ tolerance; vertical dimension outside tolerance (too open or closed); appliance too tight or loose. → retrude the mandible for a few weeks then gradually advance; modify vertical dimension; adjust retention; rest, ice, soft diet, anti-inflammatories; supportive exercise; decrease the titration rate.
- **Unilateral jaw pain** — prone (stomach) sleeping; appliance not seating; imbalance in appliance or midlines; uneven occlusal contact; asymmetrical protrusion. → self-care programme; **retrude the screw mechanism on the affected side by 0.5–1 mm**; adjust occlusion for bilateral contact; confirm seating; reset to a new bite registration.
- **Persistent TMJ pain** — asymmetric or excessive advancement; interocclusal distance. → correct asymmetric advancement; decrease titration rate; modify interocclusal distance; add **posterior stops** in some designs; passive jaw exercises and morning guides. TMD is usually acute and transient, resolving within about a week with massage, warm packs and adequate NSAIDs.
- **Other jaw pain** — bruxism while side-sleeping with pressure on the jaw; biting into food; trauma.
- **Soft tissue irritation** — screw mechanism or rough surface rubbing tongue or cheek; cheek or lip biting from bite change. → polish; discontinue until healed; change appliance; orthodontic wax.
- **Excessive salivation** — the appliance itself. → towel on the pillow; reassurance, resolving in 2–3 weeks.
- **Dry mouth** — poor nasal breathing → mouth breathing; vertical dimension too large making lip seal difficult. → improve nasal breathing (treat rhinitis, refer to ENT); reduce vertical dimension; elastic or chin strap; dry mouth gels; increase fluids, humidifier; chewing gum or xylitol lozenges.
- **Minor bite change** — mechanical forces causing gradual shifts. → jaw exercises, passive exercises, morning occlusal guide; adjust the appliance or settings.
- **The dentist's role:** monitoring and adjustment; palliative care during adaptation; patient education and reassurance that symptoms are likely temporary and manageable; **regular follow-up every 6 months to a year** with documentation of presence, management and resolution, communicated to the general dentist and medical provider. **Resolve issues early and promptly** — the strongest determinant of whether the patient stays in treatment.

10.6 Suspending or abandoning therapy

- **Occlusal change the patient finds unacceptable** — treatment should be discontinued **only if** the patient accepts another form of therapy from their attending physician, **or** understands the risks of discontinuation and their physician is informed. You cannot simply stop treating a patient with a potentially serious condition.
- **Other reasons:** non-compliance and long-term abandonment, even among initially compliant patients; persistent intolerable side effects (salivation, tooth discomfort, jaw pain); lack of efficacy (no effective reduction in AHI, snoring or daytime function); severe OSA where the appliance has failed and CPAP is preferred; developing contraindications (severe periodontal problems, insufficient teeth, active TMJ disorder, limited protrusion); and **failure of conservative management of occlusal change, particularly with loss of posterior occlusion**.
- **Factors in the decision:** patient-specific factors — overall health, tolerance for side effects, commitment; regular follow-up by qualified dentists to survey for dental side effects and occlusal change, essential for maximising adherence and effectiveness; and device and settings, since efficacy and side effects depend on type, protrusion and design — **modification may be preferable to abandonment**.
- **Alternatives:** CPAP is the primary alternative, particularly for severe OSA; combination therapy (appliance plus CPAP at lower pressure, or plus positional therapy or EPAP); surgery where other conservative treatments fail or are not tolerated; orthognathic surgery in a younger patient; and for those who cannot accept ongoing occlusal change, a half-and-half regimen alternating CPAP and appliance on different nights.
- **Where treatment stops:** notify the medical provider, and refer non-compliant patients back to consider alternatives, since untreated or subtherapeutic OSA carries long-term risk.
- **Treating established occlusal change:** only a small group need anything. **Leave it alone within the first 12–18 months** — most self-correct, which is why reviewing vulnerable patients early keeps this option open. If treatment is required: orthodontics; orthognathic surgery (very rare); optimising occlusion by equilibration and restoration (refer to a prosthodontist if the patient

simply wants to chew well); prosthodontic overlays. Definitive correction only once appliance therapy has stopped and the patient is on another OSA treatment.

COMMUNICATION

- **Signed, informed consent is mandatory** — and give the patient a copy.
- **Regular reviews** — at least 6-monthly for the first year, then annually.
- **Keep a list of vulnerable patients** and review them early.
- **Discuss with the patient and record the discussion** — the conversation is only protective if documented.

NUMBERS — TOPIC 10

- Discontinuation from side effects ~7.5% (commonest TMJ pain); 20–30% of all discontinuations; ~9.8% in some series
- Short term: salivation ~27% · dry mouth ~38% · tooth pain ~36% · allergy <1%. Long term: jaw pain ~34% · bite change 14–26%
- Occlusal change: OJ ~1.9 mm, OB ~2.3 mm (average ~2.5 mm); ~⅓ exceed 1 mm; at 17 y median −1.1 / −1.6 mm; ~80% of 7-year nightly users; 85% detectable on models and cephalograms; 13.3% by 2 years then constant
- 51% posterior open bite; 62% anterior crossbite of ≥1 tooth; facial height change from 6 months
- Greatest vulnerability first 2 years; resolves within ~2 weeks of cessation with exercises; leave alone 12–18 months
- Protrusion >50% of max = significantly higher risk; >80% = vulnerability factor
- Morning aligner 15–20 min on waking; review at least 6-monthly year 1 then annually

SELF-TEST — TOPIC 10

1. What must you disclose before fitting an appliance?

The nature of the disorder and consequences of leaving it untreated; the purpose of treatment; all alternatives including PAP, positional therapy, weight loss, surgery and **no treatment**; the potential and probable risks; the expected longevity of the appliance; and the need for long-term follow-up. On dental side effects specifically: reduced overjet and overbite leading to suboptimal occlusion with fewer contact points, possible unilateral or bilateral (pre)molar crossbite, and altered aesthetics. Consent **in writing before fitting**; the form states the potential and probable side effects and the appliance's longevity; the patient has the chance to ask questions; they sign in the presence of the dentist or team member; the dentist countersigns and dates; the patient gets a copy; and the process is documented in narrative notes alongside the form.

2. A patient asks whether the appliance will change their bite. Answer honestly, with figures.

Yes, most likely — an expected consequence rather than a rare complication. About 80% of nightly users over seven years experience occlusal change, and 85% show change detectable on models and cephalograms. Typically overjet reduces ~1.9 mm and overbite ~2.3 mm, with around one third exceeding 1 mm; at 17 years median changes are −1.1 mm overjet and −1.6 mm overbite. However, in most cases the change is minor and **unnoticed**: only a minority perceive it, most report no problems and do not wish to stop, and professional bodies consider that for most people it is not annoying, not a risk to oral health, and needs no further dental treatment. Where noticed: gaps and food impaction, occasional lisping, and rarely difficulty chewing on the back teeth. I would then explain the mitigation — morning aligner, minimal titration, jaw exercises, possible nights off — and the monitoring schedule, and document the conversation.

3. Which patients are most vulnerable to occlusal change?

Edge-to-edge bites with minimal overjet and overbite, which easily become class III; younger patients with more flexible bone and a longer expected duration of wear; those with previous orthodontic treatment; those wearing more than eight hours or seven nights a week; those advanced beyond about 80% of maximum protrusion; those with a suboptimal result, because they keep advancing chasing residual symptoms; and those with severe OSA and obesity. Additionally, patients with high aesthetic requirements, narrow acceptance of change, or body dysmorphic features need particularly careful consent, because a clinically trivial change may be subjectively intolerable. Keep a list of vulnerable patients and review them early — within 12–18 months, while spontaneous correction remains possible.

4. Describe the mechanism of occlusal change, and what it is *not*.

The appliance produces distally directed forces on the upper dentition and forwardly directed forces on the lower teeth. The main cause of the resulting change is **tilting of teeth** — retroclination of upper incisors, proclination of lower incisors, a mesial tendency in the molar relationship, and overall overjet reduction. It is **not** due to change in mandibular length, change in skeletal pattern, or functional repositioning and neuromuscular adaptation — appliances cause tooth movement, not craniofacial skeletal change. Importantly the change reflects a **failure to reposition the mandible during the day** after use rather than immutable overnight change, which is the rationale for morning aligners and exercises. And side effects are a product of protruding the mandible to achieve a therapeutic effect rather than of any specific design — while CPAP is not exempt, since a tightly fitted nasal mask also retroclines the upper incisors.

5. A patient on therapy three years has developed a posterior open bite on two teeth. Manage this.

> Establish the facts first: compare against baseline records, models and photographs; quantify the change; determine whether it is progressive, whether it is symptomatic, and critically whether **adequate posterior occlusal support** remains. Then conservative management: temporary cessation or reduced use with remedial exercises every morning after removal, since change resolves within about two weeks of cessation in most cases with exercises; plus a morning occlusal guide. On resolution, resume nightly treatment with a prophylactic exercise regimen and careful monitoring, minimising titration to the least effective advancement. It is reasonable to continue in the presence of change provided it is monitored at least 6-monthly, is not associated with unacceptable symptoms, is not progressive, and posterior support is adequate. If conservative management fails — particularly with loss of posterior occlusion — permanent cessation may be necessary, but only with the patient accepting an alternative therapy from their physician or understanding and accepting the risks of stopping, with the physician informed. Definitive occlusal correction should follow discontinuation and transfer to another OSA treatment, because change continues while the appliance is worn.

6. Unilateral jaw pain three weeks after starting. Causes and management?

> Unilateral pain points to asymmetry rather than excessive advancement. Causes: prone sleeping with pressure on one side; the appliance not seating; imbalance within the appliance or a midline discrepancy; uneven occlusal contact; asymmetrical protrusion. Management: a self-care programme; confirm proper seating; retrude the screw mechanism on the affected side by 0.5–1 mm; adjust the occlusion for bilateral contact; and if these fail, reset to a new bite registration. Contrast **bilateral** pain, which suggests advancement beyond TMJ tolerance, a vertical dimension outside tolerance, or an appliance too tight or too loose — managed by retruding for a few weeks then advancing gradually, modifying the vertical dimension, adjusting retention, reducing the titration rate, and rest, ice, soft diet and anti-inflammatories.

TOPIC 10

Informed consent and management of oral appliance side effects

EXAMINER'S BLUEPRINT

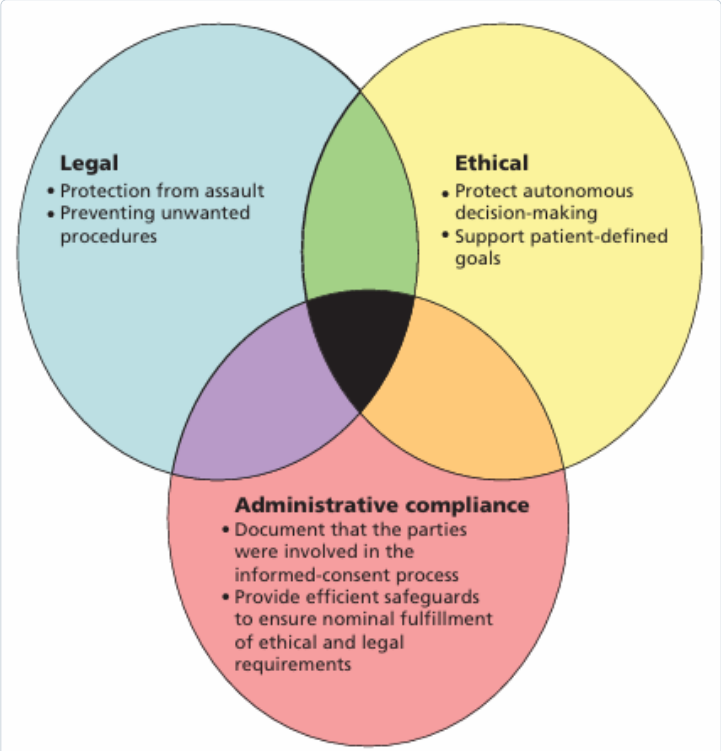
1. Obtaining informed consent for treatment
2. Evidence-based expectations of oral appliance side effects
3. Effect of mandibular protrusion on craniofacial structures including soft tissue, teeth and TMJs
4. Use of the occlusal guide and morning exercises for prevention and management of side effects
5. Management of side effects
6. Decision-making regarding suspending or abandoning oral appliance therapy in various clinical scenarios

Topic 10 is where the examination becomes medicolegal. The unifying insight is that **occlusal change is not a complication of appliance therapy — it is an expected consequence of it**, occurring in the great majority of long-term users. Your task is not to prevent it (you largely cannot) but to predict it, disclose it, monitor it, minimise it and manage it. Candidates who present bite change as a rare mishap lose marks; candidates who present it as a known, quantified, consented trade-off score well.

10.1 Obtaining informed consent

DEFINITION

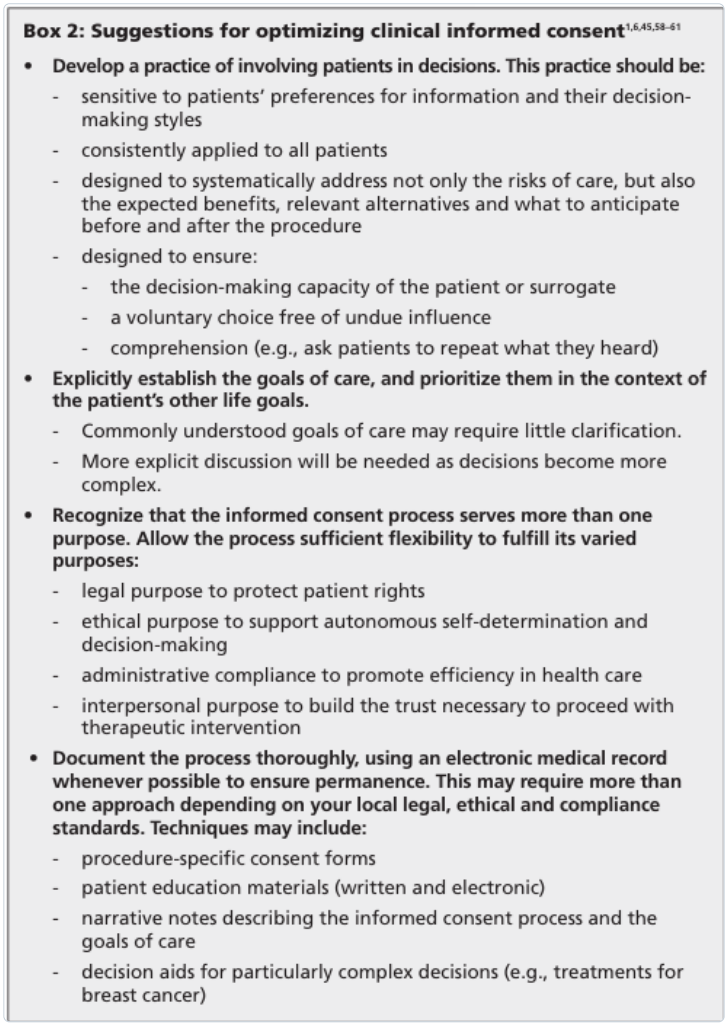
Simple consent entails that a patient — or surrogate — with decision-making capacity **freely authorises** a treatment plan aimed at a **mutually acknowledged treatment goal**. The authorisation is "**informed**" when the clinician discloses, and the patient understands, the diagnosis, the relevant options for treatment **including no treatment**, and the respective risks and benefits of each.



igure 1: Venn diagram showing the multiple overlapping purposes o nformed consent.

Figure 10.1 — The three overlapping purposes of informed consent. Legal — protection from assault, and preventing unwanted procedures. **Ethical** — protecting autonomous decision-making, and supporting patient-defined goals. **Administrative compliance** — documenting that the parties were involved in the informed-consent process, and providing efficient safeguards to ensure nominal fulfilment of ethical and legal requirements.

Why it is examinable: this figure distinguishes consent as a process from consent as a form. The administrative circle — mere nominal fulfilment — is the one that a signed form alone satisfies. Examiners want candidates who understand that all three purposes must be met.



Box 2: Suggestions for optimizing clinical informed consent[1,6,45,58–61]

- **Develop a practice of involving patients in decisions. This practice should be:**
  - sensitive to patients' preferences for information and their decision-making styles
  - consistently applied to all patients
  - designed to systematically address not only the risks of care, but also the expected benefits, relevant alternatives and what to anticipate before and after the procedure
  - designed to ensure:
    - the decision-making capacity of the patient or surrogate
    - a voluntary choice free of undue influence
    - comprehension (e.g., ask patients to repeat what they heard)
- **Explicitly establish the goals of care, and prioritize them in the context of the patient's other life goals.**
  - Commonly understood goals of care may require little clarification.
  - More explicit discussion will be needed as decisions become more complex.
- **Recognize that the informed consent process serves more than one purpose. Allow the process sufficient flexibility to fulfill its varied purposes:**
  - legal purpose to protect patient rights
  - ethical purpose to support autonomous self-determination and decision-making
  - administrative compliance to promote efficiency in health care
  - interpersonal purpose to build the trust necessary to proceed with therapeutic intervention
- **Document the process thoroughly, using an electronic medical record whenever possible to ensure permanence. This may require more than one approach depending on your local legal, ethical and compliance standards. Techniques may include:**
  - procedure-specific consent forms
  - patient education materials (written and electronic)
  - narrative notes describing the informed consent process and the goals of care
  - decision aids for particularly complex decisions (e.g., treatments for breast cancer)

Figure 10.2 — Suggestions for optimising clinical informed consent. Develop a practice of involving patients in decisions that is sensitive to patients' preferences for information and their decision-making styles, consistently applied to all patients, designed to substantively address not only the risks of care but also the expected benefits, relevant alternatives and what to anticipate before and after the procedure, and designed to ensure the decision-making capacity of the patient or surrogate, a voluntary choice free of undue influence, and comprehension — for example by asking patients to repeat what they heard. Explicitly establish the goals of care and prioritise them in the context of the patient's other life goals: commonly understood goals may require little clarification, while more explicit discussion is needed as decisions become more complex. Recognise that the informed-consent process serves more than legal purposes — legal (to protect patient rights), ethical (to support autonomous self-determination and decision-making), administrative compliance (to promote efficiency in health care) and interpersonal (to build the trust necessary to proceed with therapeutic intervention) — and allow the process sufficient flexibility to fulfil its varied purposes. Document the process thoroughly, using an electronic medical record wherever possible, and recognise that this may require more than one approach depending on local legal, ethical and compliance standards: procedure-specific consent forms, patient education materials (written and electronic), narrative notes describing the informed-consent process and the goals of care, and decision aids for particularly complex decisions.

*Why it is examinable: the four purposes — legal, ethical, administrative, **interpersonal** — and the three tests of valid consent: **capacity, voluntariness and comprehension**. Note the practical technique of asking patients to repeat back what they heard; naming it demonstrates that you understand comprehension must be verified, not assumed.*

What must be disclosed in dental sleep medicine

- The nature of the sleep-related breathing disorder and its consequences if untreated.
- The purpose of the proposed treatment and the **alternatives** — PAP therapy, positional therapy, weight loss, surgery, and no treatment.
- The potential risks and consequences of appliance therapy, and the **expected longevity** of the appliance.
- **Possible dental side effects, before starting therapy** — at minimum the chance of reduction in overjet and over-bite leading to a suboptimal occlusion, with fewer occlusal contact points, possible unilateral or bilateral crossbite in the (pre)molar region, and altered aesthetics.
- The requirement for long-term follow-up.

HIGH-YIELD — THE PROCEDURAL REQUIREMENTS

- Explain the proposed treatment, **with long-term follow-up**, and obtain the patient's consent **preferably in writing, before fitting the appliance**.
- The consent form should clearly indicate the **potential and probable side effects** of using the appliance **and its longevity**.
- Give the patient an opportunity to ask questions about the risks of treatment, and educate them about the risks of **no** treatment.
- The patient signs in the presence of the qualified dentist or another team member; the dentist then **countersigns and dates** the document, which is kept as part of the record of care.
- **Give the patient a copy.**
- Document the process thoroughly using the electronic record, procedure-specific consent forms, patient education materials and narrative notes describing the process and the goals of care.
- Signed, informed consent is **mandatory** — and, in the context of predictable occlusal change, it is your principal protection.

10.2 Evidence-based expectations of side effects

THE ONE-SENTENCE SUMMARY

Most side effects of appliance treatment are **mild and transient** and disappear during the first few months of treatment — but in approximately **7.5%** of patients they lead to termination of therapy, most often because of **TMJ pain** (it is less common to give up because of occlusal change). Side effects contribute to about 20–30% of patients discontinuing therapy overall.

Short-term side effects (first weeks to months)

Side effect	Reported frequency where available
Discomfort around the teeth and jaws	
Excessive salivation	~27%
Dry mouth	~38%
Pain in the teeth	~36%
Perception of tooth mobility	
Soft tissue irritation; gum irritation; tongue discomfort	
Difficulty chewing; abnormal occlusion on waking	
Jaw discomfort or stiffness; masticatory or myofascial pain	more common in patients on statin medication
Headache	
Tooth grinding	
Allergic reaction (e.g. acrylic, nickel)	<1%

Long-term side effects (moderate to severe and ongoing)

Side effect	Reported frequency
Jaw pain	~34%
TMJ restriction or noise	
Bite change	14–26%
Damage to teeth or dental work; appliance issues	

Significant and persistent TMJ problems are **rare**. Change in facial height and jaw relation has been noted as early as **6 months** into appliance use.

Occlusal change — the quantitative detail

THE NUMBERS ON BITE CHANGE

- **Direction of force:** appliances produce **distally directed forces on the upper dentition and forwardly directed forces on the lower teeth**. Over time there is a dose-dependent shift in occlusion, with lower teeth moving anteriorly, upper teeth moving distally, and a loss of chewing contact area posteriorly.
- **Magnitude:** reduced overjet ~1.9 mm and overbite ~2.3 mm; decreased overjet and overbite during the first 5 years; around **one third** of patients experience more than 1 mm of change. At 17 years, median overjet change −1.1 mm and median overbite change −1.6 mm. Average decrease of ~2.5 mm in overjet and overbite is described, which may be favourable or unfavourable.
- **Frequency:** approximately **80% of patients who use an appliance every night over 7 years** will experience occlusal change. **85%** had bite change detectable on study models and cephalograms — though some bites do not change. The proportion developing occlusal change increased over the first 2 years to 13.3% of cases and remained relatively constant thereafter.
- **Time course:** overbite decreases *less* with time; overjet decreases *continuously at a constant rate*. Initial overjet has no effect on the pattern, but initial overbite does: with a smaller initial overbite the rate of change first decreases then increases after ~9 years, whereas a deep bite decreases gradually with time. A larger initial overjet is associated with a greater total reduction in both overjet and overbite. Bite changes increase progressively as long as treatment continues.
- **Other occlusal changes:** lower arch expansion, with increased lower intercanine and intermolar distance and decreased lower crowding — intercanine width increases continuously while intermolar width increases rapidly then tapers off; opening of interdental spaces; **51% posterior open bite** (loss of occlusal contact on at least two posterior teeth); **62% anterior crossbite** of at least one tooth; increased maximal opening.
- **Anteroposterior relationship:** a tendency toward mesio-occlusion compared with CPAP — that is, a change from class I to class III, or class II to class I or III. Also a tendency to unilateral or bilateral crossbite in the (pre)molar region, because with a mesial shift the broader part of the lower arch occludes with the narrower part of the upper arch. There is a significant decrease in the number of occlusal contacts in the (pre)molar region with **both** appliances and CPAP, attributed to the incisal guidance phenomenon — but no tendency to anterior open bite.
- **Predictors:** a significant association exists between change in overbite and the amount of mandibular protrusion, and **initial AHI is positively related to the magnitude of overjet reduction** — because more severe OSA requires more advancement for symptom relief. But the **degree of advancement is not related to the magnitude of occlusal change** in some analyses (any degree of advancement can produce change in predisposed people), change cannot be predicted from the pre-treatment occlusal classification, and patient age, sex and pre-treatment BMI are not related to occlusal change. The greatest vulnerability is within the **first 2 years**, after which change may partly reverse as a new occlusal equilibrium develops.
- **Mechanism:** occlusal changes occur because of a **failure to reposition the mandible during the day** following use of the device, rather than as a result of immutable changes caused by the device overnight. Side effects are a product of protruding the mandible to achieve a therapeutic effect, **rather than of specific appliance design**.
- **CPAP is not innocent:** CPAP also causes retroclination of the upper incisors, from posteriorly directed forces on the anterior maxilla by a tightly fitted nasal mask, and long-term use of both CPAP and appliances produces significant dental change.

Causes of occlusal change — what it is *not*

Proposed cause	Verdict
Change in mandibular length	Unlikely
Change in skeletal pattern	Unlikely — appliances cause tooth movement but **not** craniofacial skeletal change
Functional repositioning of the jaw and neuromuscular adaptation	Unlikely
Tilting of teeth	**The main cause** — retroclination of upper incisors, proclination of lower incisors, a mesial tendency in the molar relationship, and overall reduction in overjet

HIGH-YIELD — WHO IS VULNERABLE?

- **Edge-to-edge bite** (minimal overjet and overbite) — easily becomes class III, so ensure the patient returns for review
- **Younger patients** — more flexible bone and a longer lifetime of wear (an anatomical rather than obesity phenotype)
- **History of orthodontic treatment**
- **Using the appliance more than 8 hours** — ask the patient to remove it on waking, especially those vulnerable to change
- **80% or more of maximum protrusion**
- **Patients with a suboptimal result** — because they keep advancing the mandible to reduce residual sleepiness or snoring
- **Severe OSA and obesity**
- **Using the appliance 7 days a week** — consider 1–2 nights off
- **High aesthetic requirement**, narrow acceptance of occlusal change, or body dysmorphic features — these patients need the most careful consent

What patients actually experience

- In most cases occlusal changes are **minor and not noticed** by the patient; only a minority notice bite change. Most patients are unaware, are not caused problems, and **do not want to stop using the appliance**.
- Where noticed, symptoms of bite change are: gaps and food impaction; occasional lisping when talking; and — rarely — difficulty chewing on the back teeth or tearing food with the front teeth.
- Most patients do not lose posterior occlusion.
- Position of professional bodies: for most people the change in bite is not annoying, is not a risk to oral health, and does not require further dental treatment.
- **Most importantly** — and this is the clinically consequential point — these bite changes influence the mechanism of the device, since a forward shift of the lower teeth relative to the upper results in a progressively **reduced degree of effective mandibular advancement**, putting the efficacy of treatment at risk.

10.3 Effect of protrusion on craniofacial structures

Structure	Effect
Teeth	**Occlusal change** — decreased overjet and overbite with proclination of the lower incisors and retroclination of the upper incisors. **Tooth movement** — prolonged use can produce over-eruption of molars and retroclination of upper incisors. **Biomechanical stress** — greater protrusion levels (e.g. 70% of maximum) induce significant stress on the teeth, particularly the mandibular molars, and increase the risk of root and bone resorption. Changes in craniofacial structures are mainly related to significant tooth movement
Temporomandibular joint	**Pain and discomfort** — unnatural anterior positioning can lead to TMJ pain, clicking and myofascial pain. **Position and function** — reciprocal forces may alter mandibular position and function over time. But note the reassuring long-term data from Topic 4: no significant TMJ morphological alteration, condylar translation less than in maximum opening, no impairment of the joint long term, and symptom intensity, joint noise and jaw fatigue may actually reduce
Soft tissues	

Structure	Effect
	Salivation — hypersalivation or dry mouth. **Airway** — the primary goal: enlargement of the upper airway by advancing the mandible and repositioning the tongue, preventing obstruction. **Soft tissue adaptation** — muscular activity within the airway adapts to the device, leading to superior and anterior repositioning of the hyoid bone
Factors influencing severity	**Degree of protrusion** — risk generally increases with greater advancement, with levels exceeding 50% of maximum often correlating with significantly higher risk. **Patient-specific factors** — underlying musculoskeletal disorders of the masticatory system may increase the risk of complications. **Monitoring** — appropriate planning and monitoring by a dental professional are crucial to prevent or manage these effects

A crucial clinical corollary: an appliance is already effective at a **less protrusive position** in mild and moderate OSA, and therefore causes less severe dental side effects in those patients. Minimising titration is therefore not merely a comfort measure — it is a side-effect prevention strategy.

Although these occlusal changes may be undesirable in certain patients, the effective treatment of a potentially life-threatening disease such as OSA **may supersede** the maintenance of a baseline occlusion. That sentence is worth being able to deploy — but only after you have documented that the patient understood and accepted the trade-off.

10.4 Morning occlusal guides and exercises

WHAT A MORNING OCCLUSAL GUIDE IS AND DOES

A **morning occlusal guide** (morning repositioning device, or "AM aligner") is a device used after waking to help realign the jaw and teeth after wearing a mandibular advancement device overnight. The patient bites into the guide for as long as it takes for the teeth to re-establish habitual occlusion, which can reverse tooth movement and reposition the condyles. It is used to reposition the mandible into its habitual pre-treatment position — and, just as importantly, it **lets the patient know when occlusal change is occurring**: if they cannot fit into the aligner with the back teeth in occlusion, bite change has occurred.

How it works

- **Realigns the bite** — custom-made to fit the patient's upper and lower teeth, guiding them back to their original maximum intercuspation.
- **Exercises the jaw muscles** — applying pressure exercises and stretches the muscles of mastication, which are tight after a night of wear.
- **Positions the condyles** — assists in repositioning the condyles to their pre-treatment alignment, reducing strain and discomfort.
- **Monitors progress** — the patient uses it each morning to check jaw alignment; if it does not fit, they should contact the practice.

Practical points

- **Suggest it for every patient** in whom appliance therapy is commenced.
- Two fabrication types: those using thermoacrylic beads, and those using a **prefabricated sheet** (AM aligner) placed in hot water, moulded around the teeth, with the patient asked to bite in centric occlusion.
- **Should be custom made and recorded in centric occlusion.**
- **Check the occlusion before and after**, with and without the morning aligner in place, to make sure the back teeth are in occlusion even with the aligner seated.
- **Wear for 15–20 minutes in the morning after waking** — some protocols specify around 10 minutes.
- Consider making one at the same appointment as the appliance impression by also taking a centric occlusion record (Topic 5.6).

Morning exercises

- Certain jaw exercises help relieve stiffness of the masticatory muscles and accelerate repositioning of the mandible to its normal position after a night of appliance use. They may also inhibit or minimise occlusal functional changes in predisposed patients.
- Perform slowly and gently: opening the mouth wide, moving the jaw left and right, resting the chin on a fist to let the jaw settle, and curling the tongue back to find the retruded position. These stretch the muscles of mastication and reduce discomfort.
- Exercises increase contact between the teeth — good in the short term, with limited long-term evidence.
- Both the guide and the exercises are used in the morning immediately after removing the appliance, with duration and frequency determined by the treating dentist.
- **Chewing gum** also helps, mostly with acute changes in the morning.
- These measures are recommended by the AADSM to prevent and manage common side effects such as changes in tooth position and jaw muscle pain — but they are **not a cure-all**: some tooth or root changes, though often clinically insignificant, may still occur.

10.5 Management of side effects

Conservative management of occlusal change

1. **Temporary cessation or reduction** in use of the device, **plus remedial exercises every morning** following removal. With exercises, occlusal changes resolve within **2 weeks** of treatment cessation in most cases.
2. **Use of an occlusal guide.**
3. Following satisfactory resolution, **nightly treatment can resume** with an ongoing exercise regimen as a prophylactic measure and careful monitoring.
4. **It is reasonable to continue treatment in the presence of occlusal change**, provided that it is regularly monitored (at least 6-monthly clinical reviews), is not associated with unacceptable symptoms, is not progressive, and there is **adequate posterior occlusal support**.
5. If conservative management fails — **especially if there is loss of posterior occlusion** — permanent treatment cessation may be necessary.
6. **Correction** of the occlusion with restorations, crowns and/or orthodontic treatment should be considered **only if the patient has discontinued treatment and is able to use another OSA treatment**, because the changes will continue — albeit at a slower pace — as long as the patient wears the device.

Minimising occlusal change

- **Minimise titration** — use the minimum advancement that relieves symptoms. Allow a temporary increase during a cold when snoring worsens, but remind the patient to titrate back
- **Reduce wearing time** where appropriate; ask the patient to remove the appliance on waking; consider 1–2 nights off per week in vulnerable patients
- **Chewing gum** in the morning
- **Morning aligner** for every patient, custom-made in centric occlusion, worn 15–20 minutes
- **Jaw exercises**
- **Cover all teeth** — do not leave molars uncovered; use a hard rather than flexible material for better protection; include incisor coverage
- For a patient with previous orthodontic treatment, consider alternating regimens: 4 nights with the appliance and 3 nights with the morning aligner, or an appliance at night and a retainer during the day, to reduce tooth movement
- **Review vulnerable patients early** and keep a list of them

Management of specific side effects

Side effect	Possible causes	Management
Transient morning jaw pain	Muscles adjusting to the appliance holding the mandible forward	Jaw exercises, muscle massage, heat application; a morning occlusal guide to return the jaw to its normal position
Bilateral jaw pain	Mandibular advancement beyond the tolerance of the TMJ; vertical dimension outside the tolerance of the jaw (too open or too closed); appliance too tight or too loose	Retrude the mandible for a few weeks until comfortable then gradually advance; modify the vertical dimension of the appliance; adjust retention; rest, ice, soft diet, anti-inflammatories; supportive therapy such as exercise; decrease the titration rate
Unilateral jaw pain	Stomach (prone) sleeping; appliance not seating; imbalance in the appliance or midlines; uneven occlusal contact; asymmetrical mandibular protrusion	Self-care programme; retrude the screw mechanism on the affected side by 0.5–1 mm; adjust the appliance occlusion to gain bilateral contact; confirm proper seating; reset to a new bite registration; correct asymmetric mandibular advancement
Persistent TMJ pain	Asymmetric advancement; excessive advancement; interocclusal distance	Correct asymmetric advancement; decrease the appliance titration rate; modify the interocclusal distance; add **posterior stops** to the occlusal surfaces in some designs; passive jaw exercises and morning occlusal guides. TMD is usually acute and transient, resolving within about a week with massage, warm packs and adequate NSAIDs
Other jaw pain	Bruxism while sleeping on the side with pressure on the jaw; biting into food; trauma	Address the specific cause
Soft tissue irritation	Screw mechanism or rough surface rubbing on the tongue or cheek; cheek or lip biting from bite change	Polish; discontinue use until healed; change the appliance; orthodontic wax
Excessive salivation	Presence of the appliance in the mouth	Towel on the pillow; reassurance — should resolve in 2–3 weeks
Dry mouth	Poor nasal breathing leading to mouth breathing; vertical dimension too large, making lip seal difficult	Improve nasal breathing (treat rhinitis, refer to ENT); reduce the vertical dimension; use an elastic or chin strap to keep the appliance together and the mouth closed; dry mouth gels and products; increase fluid intake, use a humidifier; chewing gum or xylitol lozenges to stimulate saliva
Minor bite change	Mechanical forces from the appliance causing gradual shifts in dental contact	Jaw exercises, passive jaw exercises, morning occlusal guide; adjust the appliance or its settings

The dentist's role and follow-up

- **Monitoring and adjustment** — a qualified dentist is essential for monitoring side effects and making necessary adjustments to the appliance or plan.
- **Palliative care** — supportive care to manage discomfort, especially during the initial adaptation period.
- **Patient education** — reassure that symptoms are likely temporary and manageable.
- **Regular follow-up** — every 6 months to a year, to evaluate for and manage side effects appropriately, with documentation of presence, management and resolution, and communication to the general dentist and medical provider.

- **Resolve issues and side effects early and promptly** — this is the single strongest determinant of whether the patient stays in treatment.

10.6 Suspending or abandoning therapy

Reason	Detail
Occlusal change the patient finds unacceptable	If the patient decides that occlusal changes are undesirable, treatment should be discontinued **only if** the patient accepts another form of therapy from their attending physician, **or** the patient understands the risks of discontinuing therapy and their physician is informed. You cannot simply stop treating a patient with a potentially serious condition
Non-compliance and abandonment	A major limitation of therapy is patient non-compliance and long-term abandonment, even among initially compliant patients
Persistent intolerable side effects	Excessive salivation, tooth discomfort or jaw pain that are not manageable or become intolerable
Lack of efficacy	Where therapy does not effectively reduce the AHI, snoring or improve daytime function
Severe OSA with appliance failure	Appliances are generally most suited to mild-to-moderate disease; in severe cases more effective treatments such as CPAP are preferred, especially where the appliance has failed
Contraindications developing	Severe periodontal problems, insufficient teeth to support the device, active TMJ disorder, or limited ability to protrude the mandible
Failure of conservative management of occlusal change	Particularly where there is **loss of posterior occlusion** — this may necessitate permanent cessation

Factors in the decision, and what comes next

- **Patient-specific factors** — overall health, tolerance for side effects and commitment to therapy.
- **Regular follow-up** — monitoring by qualified dentists is necessary to survey for dental side effects and occlusal change, and is essential for maximising adherence and effectiveness.
- **Device and settings** — efficacy and side effects depend on appliance type, degree of protrusion and other design settings, so modification may be preferable to abandonment.
- **Alternatives:** CPAP is the primary alternative, particularly for severe OSA, and can be considered when appliance therapy fails or is refused; combination therapy (appliance plus CPAP at lower pressure, or appliance plus positional therapy or EPAP); surgical intervention where other conservative treatments are unsuccessful or not tolerated; in a younger patient, orthognathic surgery may be considered; and for those who cannot accept ongoing occlusal change, a half-and-half regimen alternating CPAP and appliance on different nights.
- **Where treatment is stopped:** notify the medical provider, and refer non-compliant patients back to the medical provider to consider alternative therapies, since untreated or subtherapeutic OSA carries long-term health risk.

Treatment of established occlusal change

- Only a small group of patients need any treatment at all.
- **Leave it alone within the first 12–18 months** — most will self-correct. This is why it is important to review vulnerable patients early, so that this option remains available.
- If treatment is required: orthodontics; orthognathic surgery (very rare); optimising the occlusion by occlusal equilibration and restoration — refer to a prosthodontist if the patient simply wants to chew well; prosthodontic overlays.
- Remember the rule from 10.5: definitive correction should be undertaken only once appliance therapy has been discontinued and the patient is on another OSA treatment, because change continues while the appliance is worn.

HIGH-YIELD — COMMUNICATION

- **Signed, informed consent is mandatory** — and give the patient a copy.
- **Regular reviews**, at least 6-monthly for the first year then annually.
- **Keep a list of vulnerable patients** and review them early.
- **Discuss with the patient and record the discussion** — the conversation is only protective if it is documented.

NUMBERS TO MEMORISE — TOPIC 10

- Discontinuation from side effects ~7.5% (most often TMJ pain); overall discontinuation from side effects 20–30%; discontinuation ~9.8% in some series
- Short term: excessive salivation ~27% · dry mouth ~38% · tooth pain ~36% · allergy <1%
- Long term: jaw pain ~34% · bite change 14–26%
- Occlusal change: OJ ~1.9 mm and OB ~2.3 mm reduction (average ~2.5 mm described); ~⅓ exceed 1 mm; at 17 y median −1.1 mm OJ and −1.6 mm OB
- ~80% of nightly users over 7 years show occlusal change; 85% detectable on models and cephalograms; 13.3% by 2 years then constant
- 51% posterior open bite; 62% anterior crossbite of at least one tooth
- Facial height and jaw relation change noted as early as 6 months
- Greatest vulnerability in the first 2 years; occlusal change resolves within 2 weeks of cessation with exercises in most cases; leave alone for 12–18 months as most self-correct
- Morning aligner: worn 15–20 minutes on waking, custom-made in centric occlusion
- Protrusion >50% of maximum correlates with significantly higher side-effect risk; >80% of maximum is a vulnerability factor
- Review at least 6-monthly in year 1, then annually

SELF-TEST — TOPIC 10

1. What must you disclose before fitting an oral appliance?

The nature of the sleep-related breathing disorder and the consequences of leaving it untreated; the purpose of the proposed treatment; all relevant alternatives including PAP therapy, positional therapy, weight loss, surgery and **no treatment**; the potential and probable risks; the expected longevity of the appliance; and the requirement for long-term follow-up. Specifically on dental side effects, at minimum the chance of reduction in overjet and overbite leading to a suboptimal occlusion with fewer occlusal contact points, possible unilateral or bilateral crossbite in the (pre)molar region, and altered aesthetics. Consent should be obtained **in writing before fitting**, the form should state the potential and probable side effects and the appliance's longevity, the patient should have the opportunity to ask questions, they sign in the presence of the dentist or a team member, the dentist countersigns and dates, the patient is given a copy, and the process is documented in narrative notes alongside the form.

2. A patient asks whether the appliance will change their bite. Answer them honestly, with figures.

Yes, most likely — this is an expected consequence rather than a rare complication. Approximately 80% of patients using an appliance every night over seven years will experience occlusal change, and 85% show change detectable on study models and cephalograms. Typically overjet reduces by about 1.9 mm and overbite by about 2.3 mm, with around one third of patients exceeding 1 mm of change; at 17 years the median changes are −1.1 mm overjet and −1.6 mm overbite. However, in most cases the change is minor and **not noticed**: only a minority perceive it, most patients report no problems and do not wish to stop, and professional bodies consider that for most people the change is not annoying, not a risk to oral health, and does not require further dental treatment. Where noticed, symptoms are gaps and food impaction, occasional lisping, and rarely difficulty chewing on the back teeth. I would then explain the mitigation — a morning aligner, minimal titration, jaw exercises, possible nights off — and the monitoring schedule, and document that we had this conversation.

3. Which patients are most vulnerable to occlusal change?

Patients with an edge-to-edge bite and minimal overjet and overbite, who can easily become class III; younger patients, with more flexible bone and a longer expected duration of wear; those with a history of orthodontic treatment; those wearing the appliance more than eight hours or seven nights a week; those advanced beyond about 80% of maximum protrusion; those with a suboptimal therapeutic result, because they keep advancing the mandible chasing residual symptoms; and patients with severe OSA and obesity. Additionally, patients with high aesthetic requirements, narrow acceptance of occlusal change, or body dysmorphic features need particularly careful consent, because for them a clinically trivial change may be subjectively intolerable. I would keep a list of vulnerable patients and review them early — within the first 12–18 months, while spontaneous correction remains possible.

4. Describe the mechanism of occlusal change, and what it is *not*.

The appliance produces distally directed forces on the upper dentition and forwardly directed forces on the lower teeth. The main cause of the resulting change is **tilting of teeth** — retroclination of the upper incisors, proclination of the lower incisors, a mesial tendency in the molar relationship and an overall reduction in overjet. It is **not** due to a change in mandibular length, a change in skeletal pattern, or functional repositioning and neuromuscular adaptation — appliances cause tooth movement, not craniofacial skeletal change. A further important point is that the change reflects a **failure to reposition the mandible during the day** after use, rather than immutable overnight change, which is the rationale for morning aligners and exercises. Finally, side effects are a product of protruding the mandible to achieve a therapeutic effect rather than of any specific appliance design, and CPAP is not exempt: a tightly fitted nasal mask also produces retroclination of the upper incisors.

5. A patient on therapy for three years has developed a posterior open bite on two teeth. How do you manage this?

First establish the facts: compare against baseline records, models and photographs, quantify the change and determine whether it is progressive, whether it is symptomatic, and — critically — whether **adequate posterior occlusal support** remains. Then conservative management: temporary cessation or reduced use of the device together with remedial exercises every morning after removal, since occlusal changes resolve within about two weeks of cessation in most cases with exercises; and use of a morning occlusal guide. Following satisfactory resolution, nightly treatment can resume with an ongoing prophylactic exercise regimen and careful monitoring, minimising the titration to the least effective advancement. It is reasonable to continue treatment in the presence of occlusal change provided it is monitored at least 6-monthly, is not associated with unacceptable symptoms, is not progressive, and posterior support is adequate. If conservative management fails — particularly where posterior occlusion is being lost — permanent cessation may be necessary, but only with the patient accepting an alternative therapy from their physician or understanding and accepting the risks of stopping, with the physician informed. Definitive occlusal correction with restorations, crowns or orthodontics should be undertaken only after appliance therapy has been discontinued and the patient is on another OSA treatment, because the change continues while the appliance is worn.

6. A patient reports unilateral jaw pain three weeks after starting therapy. Work through the causes and management.

Unilateral pain points to asymmetry rather than to excessive advancement. Causes: prone (stomach) sleeping with pressure on one side of the jaw; the appliance not seating properly; imbalance within the appliance or a midline discrepancy; uneven occlusal contact; or asymmetrical mandibular protrusion. Management: institute a self-care programme; confirm the appliance is seating properly; retrude the screw mechanism on the affected side by 0.5–1 mm; adjust the appliance occlusion to gain bilateral contact; and if these fail, reset to a new bite registration. Contrast with **bilateral** pain, which suggests advancement beyond TMJ tolerance, a vertical dimension outside the jaw's tolerance, or an appliance that is too tight or too loose — managed by retruding the mandible for a few weeks then advancing gradually, modifying the vertical dimension, adjusting retention, reducing the titration rate, and supportive measures of rest, ice, soft diet and anti-inflammatories.

7. Describe the morning occlusal guide: purpose, fabrication and use.

> Its purpose is to reposition the mandible into its habitual pre-treatment position after a night of advancement — realigning the bite toward original maximum intercuspation, exercising and stretching the tight muscles of mastication, and assisting repositioning of the condyles to their pre-treatment alignment. It also serves as a **self-monitoring device**: if the patient cannot fit into the aligner with the back teeth in occlusion, bite change has occurred and they should contact the practice. Fabrication is either from thermoacrylic beads or from a prefabricated sheet placed in hot water and moulded around the teeth with the patient biting in centric occlusion; it **should be custom made and recorded in centric occlusion**, and it is convenient to take the centric record at the same appointment as the appliance impression. Check the occlusion with and without the aligner to confirm the back teeth occlude with it seated. It is worn for 15–20 minutes on waking, immediately after removing the appliance, and should be suggested for every patient. Alongside it, gentle jaw exercises — wide opening, lateral movements, resting the chin on a fist, curling the tongue back to find the retruded position — and chewing gum help with acute morning change. These measures are recommended by the AADSM but are not a cure-all; some tooth and root change, though usually clinically insignificant, may still occur.

REFERENCES — TOPIC 10

1. Sheats RD, Schell TG, Blanton AO, et al. Management of side effects of oral appliance therapy for sleep-disordered breathing. *J Dent Sleep Med*. 2017;4(4):111–125.
2. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2).
3. Ramar K, Dort LC, Katz SG, et al. Clinical practice guideline for the treatment of obstructive sleep apnea and snoring with oral appliance therapy: an update for 2015. *J Clin Sleep Med*. 2015;11(7):773–828.
4. Pliska BT, Nam H, Chen H, Lowe AA, Almeida FR. Obstructive sleep apnea and mandibular advancement splints: occlusal effects and progression of changes associated with a decade of treatment. *J Clin Sleep Med*. 2014;10(12):1285–1291.
5. Almeida FR, Lowe AA, Otsuka R, Fastlicht S, Farbood M, Tsuiki S. Long-term sequellae of oral appliance therapy in obstructive sleep apnea patients: part 2. Study-model analysis. *Am J Orthod Dentofacial Orthop*. 2006;129(2):205–213.
6. Almeida FR, Lowe AA, Sung JO, Tsuiki S, Otsuka R. Long-term sequellae of oral appliance therapy in obstructive sleep apnea patients: part 1. Cephalometric analysis. *Am J Orthod Dentofacial Orthop*. 2006;129(2):195–204.
7. Doff MHJ, Finnema KJ, Hoekema A, Wijkstra PJ, de Bont LGM, Stegenga B. Long-term oral appliance therapy in obstructive sleep apnea syndrome: a controlled study on dental side effects. *Clin Oral Investig*. 2013;17(2):475–482.
8. Doff MHJ, Veldhuis SKB, Hoekema A, et al. Long-term oral appliance therapy in obstructive sleep apnea syndrome: a controlled study on temporomandibular side effects. *Clin Oral Investig*. 2012;16(3):689–697.
9. Marklund M. Predictors of long-term orthodontic side effects from mandibular advancement devices in patients with snoring and obstructive sleep apnea. *Am J Orthod Dentofacial Orthop*. 2006;129(2):214–221.
10. Marklund M, Franklin KA, Persson M. Orthodontic side-effects of mandibular advancement devices during treatment of snoring and sleep apnoea. *Eur J Orthod*. 2001;23(2):135–144.
11. Ueda H, Almeida FR, Chen H, Lowe AA. Effect of 2 magnitudes of mandibular advancement on demand for oral appliance therapy. *Am J Orthod Dentofacial Orthop*. 2009;135(6):822.e1–7.
12. Cunali PA, Almeida FR, Santos CD, et al. Mandibular exercises improve mandibular advancement device therapy for obstructive sleep apnea. *Sleep Breath*. 2011;15(4):717–727.
13. Ueda H, Almeida FR, Lowe AA, Ruse ND. Changes in occlusal contact area during oral appliance therapy assessed on study models. *Angle Orthod*. 2008;78(5):866–872.
14. Tsuda H, Almeida FR, Tsuda T, Moritsuchi Y, Lowe AA. Craniofacial changes after 2 years of nasal continuous positive airway pressure use in patients with obstructive sleep apnea. *Chest*. 2010;138(4):870–874.
15. Spatz D, Bickley SR. Informed consent: a communication process. *Chest*. 2015;148(6):1352–1354.
16. Beauchamp TL, Childress JF. *Principles of Biomedical Ethics*. 8th ed. Oxford University Press; 2019.

FINAL SECTION

Mock examination — 60 questions

Sixty single-best-answer items sampling all ten competencies in proportion. Work through the whole paper in one sitting without notes, allowing about 60 minutes. Mark yourself against the answer key that follows, which gives the parent topic for every item so you can go back to the source rather than merely learn the answer. A score of 48 or above (80%) suggests you are on track for a strong pass; anything below 42 means returning to the topics where you dropped marks before attempting the paper again.

QUESTIONS 1–60

1. An apnoea is defined as an airflow reduction of at least:
(a) 30% for 10 s (b) 50% for 10 s (c) 90% for 10 s (d) 100% for 20 s

2. Under the AASM-recommended rule, a hypopnoea requires a ≥30% airflow reduction for ≥10 s *plus*:
(a) ≥3% desaturation only (b) ≥4% desaturation only (c) ≥3% desaturation and/or an arousal (d) an arousal only

3. Central sleep apnoea is considered the primary diagnosis when central events comprise at least:
(a) 25% (b) 50% (c) 75% (d) 90% of apnoeas

4. Which is *absent* during REM sleep?
(a) Sympathetic activity (b) Cerebral blood flow (c) Thermoregulation (d) Dreaming

5. Approximate proportion of total sleep spent in REM:
(a) 5% (b) 10–15% (c) 20–25% (d) 45–55%

6. Sleep spindles and K-complexes characterise:
(a) N1 (b) N2 (c) N3 (d) REM

7. Growth hormone is secreted predominantly during:
(a) REM (b) N1 (c) early slow-wave sleep (d) the final third of the night

8. In the PALM construct, "L" refers to:
(a) Lung volume (b) Loop gain (c) Lateral wall collapse (d) Latency to arousal

9. Approximate proportion of OSA patients with impairment of at least one non-anatomical trait:
(a) 20% (b) 40% (c) 70% (d) 95%

10. A 10% weight gain predicts an AHI increase of approximately:
(a) 10% (b) 26% (c) 32% (d) 50%

11. In the Wisconsin cohort (AHI >5, ages 30–60), prevalence in men was:
(a) 9% (b) 13% (c) 17% (d) 24%

12. Adults with two or more signs or symptoms of OSA had what increase in first-onset TMD incidence?
(a) 23% (b) 50% (c) 73% (d) 300%

13. Chronic insomnia disorder under ICSD-3 requires symptoms for at least:
(a) 1 week (b) 1 month (c) 3 months (d) 6 months

14. First-line treatment for chronic insomnia is:
(a) Zolpidem (b) CBT-I (c) Melatonin (d) Quetiapine

15. Which drug class most characteristically suppresses REM and may induce sleep bruxism?
(a) Benzodiazepines (b) SSRIs/SNRIs (c) Orexin antagonists (d) Antihistamines

16. Approximately what proportion of people with OSA do *not* snore?
(a) One tenth (b) One quarter (c) One third (d) One half

17. A type 3 sleep study:
(a) Includes full EEG staging (b) Uses 10–15 sensors (c) Monitors breathing without sleep assessment (d) Records oximetry only

18. A home (type 2) study typically:
(a) Overestimates AHI by 10% (b) Underestimates AHI by 10% (c) Is unaffected (d) Cannot measure AHI

19. Moderate adult OSA is:
(a) AHI 5–15 (b) AHI 15–30 (c) AHI 20–40 (d) AHI ≥30

20. Moderate paediatric OSA is:
(a) AHI 1–5 (b) AHI 5–10 (c) AHI 10–15 (d) AHI 15–30

21. Normal sleep efficiency in a healthy adult is at least:
(a) 65% (b) 75% (c) 85% (d) 95%

22. Which best describes a complete response to appliance therapy?
(a) ≥50% AHI reduction (b) AHI <10 (c) Symptom resolution and AHI <5 (d) Patient satisfaction

23. With effective treatment, total sleep time typically:
(a) Increases markedly (b) Decreases (c) Is unchanged, with redistribution to more REM (d) Becomes unmeasurable

24. The MSLT measures:
(a) Ability to stay awake (b) Tendency to fall asleep (c) Arousal threshold (d) Loop gain

25. MSLT findings supporting narcolepsy:
(a) MSL ≤8 min with ≥2 SOREMPs (b) MSL ≥15 min (c) 1 SOREMP (d) Reverse first-night effect

26. STOP-BANG threshold for increased OSA risk:
(a) ≥2 (b) ≥3 (c) ≥5 (d) ≥8

27. OSA50 validation performance for moderate–severe OSA:
(a) 94% sensitivity, 31% specificity (b) 31% sensitivity, 94% specificity (c) 70% both (d) 50% both

28. Which statement about the ESS is correct?
(a) It diagnoses OSA (b) It correlates strongly with AHI (c) It correlates poorly with AHI and must not be used alone (d) A score >5 is abnormal

29. Modified Mallampati class IV means visible:
(a) Soft palate, uvula, fauces, pillars (b) Soft palate and most of uvula (c) Soft palate and uvula base (d) Hard palate only

30. Brodsky grade 3 tonsils occupy:
(a) <25% (b) 25–50% (c) 50–75% (d) >75% of the space between the anterior pillars

31. Who described the tongue retaining device, and when?
(a) Robin, 1903 (b) Cartwright & Samelson, 1982 (c) Meier-Ewert, 1985 (d) Catlin, 19th century

32. An effective oral appliance must permit advancement increments of:
(a) ≤0.1 mm (b) ≤1 mm (c) ≤2 mm (d) ≤5 mm

33. An effective oral appliance requires a protrusive adjustment range of at least:
(a) 2 mm (b) 3 mm (c) 5 mm (d) 10 mm

34. A mandibular advancement device enlarges the airway predominantly:
(a) Anteroposteriorly at the tongue base (b) Laterally at the velopharynx (c) At the larynx (d) At the nasopharynx

35. An appliance reduces pharyngeal collapsibility but does *not* systematically change:
(a) Airway calibre (b) Tongue position (c) Loop gain and arousal threshold (d) Minimum SaO_2

36. CPAP titration in an adult should start at:
(a) 2 cmH_2O (b) 4 cmH_2O (c) 8 cmH_2O (d) 12 cmH_2O

37. BPAP should be considered when the required pressure exceeds:
(a) 10 (b) 15 (c) 20 (d) 25 cmH_2O

38. Long-term CPAP compliance at 12 months is approximately:
(a) 20–30% (b) 40–60% (c) 70–80% (d) >90%

39. Mean disease alleviation is:
(a) Efficacy + compliance (b) Efficacy × compliance (c) Efficacy ÷ compliance (d) AHI reduction alone

40. Reported MDA for oral appliances versus CPAP is approximately:
(a) 25% vs 50% (b) 51% vs 50% (c) 75% vs 42% (d) 90% vs 40%

41. Hypoglossal nerve stimulation is contraindicated by:
(a) BMI 30 (b) AHI 35 (c) Complete concentric collapse on DISE (d) Retrognathia

42. In the VOTE classification, "V" stands for:
(a) Vallecula (b) Velum (c) Vocal cords (d) Vestibule

43. Strictly defined, positional OSA means:
(a) Supine AHI > non-supine AHI (b) Non-supine AHI <5 (c) >50% of events supine (d) Any positional variation

44. Routine imaging before appliance therapy is:
(a) CBCT (b) Panoramic and lateral cephalometric (c) MRI (d) None required

45. Regarding sleep bruxism and OSA, the correct position is:
(a) SDB causes SB (b) SB causes SDB (c) They co-occur but causation is not established (d) They are unrelated

46. A flat-plane stabilisation splint in a patient with untreated OSA:
(a) Reduces AHI (b) Has no effect (c) May increase AHI and snoring (d) Is curative

47. Initial protrusive setting for a construction bite is usually:
(a) 25% (b) 50–70% (c) 80–90% (d) 100% of maximum protrusion

48. Increasing the vertical dimension of an appliance:
(a) Improves efficacy (b) Reduces efficacy (c) Has no effect (d) Eliminates side effects

49. Minimum protrusive capacity generally required for a MAD:
(a) 2 mm (b) 4 mm (c) 6 mm (d) 12 mm

50. Which material is preferred in a heavy bruxer, and why?
(a) Acrylic, for repairability (b) Nylon, for reduced breakage (c) EVA, for impact absorption (d) Alginate, for accuracy

51. The AADSM defines appliance compliance as wear for:
(a) ≥4 h per night (b) ≥6 h per night (c) ≥80% of the night, ≥5 nights per week (d) Every night without exception

52. Objective appliance adherence at 6 months is approximately:
(a) 40% (b) 67% (c) 80% (d) 95%

53. Appliance efficacy: proportion achieving AHI <5:
(a) 15% (b) 37% (c) 64% (d) 80%

54. Recommended follow-up after efficacy is established:
(a) Annually from the start (b) 6-monthly for year 1, then annually (c) Only if symptoms recur (d) Monthly indefinitely

55. Discontinuation of appliance therapy specifically due to side effects is approximately:
(a) 2% (b) 7.5% (c) 20% (d) 40%

56. The commonest side effect leading to cessation of appliance therapy is:

(a) Occlusal change (b) Dry mouth (c) TMJ pain (d) Excessive salivation

57. The principal cause of occlusal change with appliance therapy is:
(a) Increased mandibular length (b) Skeletal pattern change (c) Tilting of teeth (d) Condylar remodelling

58. A morning occlusal guide is worn for approximately:
(a) 2–3 min (b) 15–20 min (c) 2 h (d) All day

59. First-line treatment for paediatric OSA with adenotonsillar hypertrophy:
(a) CPAP (b) Rapid maxillary expansion (c) Adenotonsillectomy (d) Intranasal steroid

60. Children with Down syndrome should have polysomnography by age:
(a) 2 (b) 4 (c) 8 (d) Only if symptomatic

ANSWER KEY WITH CROSS-REFERENCES

Q	Ans	Topic	Explanation
1	**c**	2.2	≥90% airflow reduction for ≥10 s. Obstructive if effort persists, central if absent
2	**c**	2.2	≥3% desaturation *and/or* an arousal. The ≥4% rule became optional in Scoring Manual v3 (2023)
3	**b**	1.10	≥50% of apnoeas central in origin
4	**c**	1.2	Thermoregulation is suspended in REM — no shivering or sweating; temperature drifts toward ambient
5	**c**	1.2	REM 20–25%; NREM 75–80%
6	**b**	1.2	N2. Spindle density rises after learning a new task
7	**c**	1.1	Early slow-wave sleep, in the first few hours after sleep onset
8	**b**	1.3	P_{crit}, Arousal threshold, Loop gain, Muscle responsiveness
9	**c**	1.3	~70%; loop gain ~36%, low arousal threshold ~37%, poor muscle responsiveness ~36%
10	**c**	1.5	~32%. Conversely 10% loss predicts ~26% decrease
11	**d**	1.4	24% of men, 9% of women (Young *et al.* 1993)
12	**c**	1.7	73% greater incidence, independent of age, gender, ethnicity, obesity, smoking and autonomic parameters
13	**c**	1.9	≥3 nights per week for ≥3 months. Older texts say one month — that was ICSD-2
14	**b**	1.9	CBT-I, without the tolerance and adverse effects of pharmacotherapy
15	**b**	1.12	SSRIs/SNRIs — ↓REM, ↑REM latency, and may induce RLS, bruxism and nightmares
16	**c**	1.10	One third of OSA patients do not snore; one third of snorers have no OSA
17	**c**	2.5	Limited channel monitoring of breathing parameters without sleep assessment; 4–7 sensors
18	**b**	2.5	Underestimates by ~10% through loss of effort signals and use of recording rather than sleep time. A negative home study in a symptomatic patient does not exclude OSA
19	**b**	2.2	15 to <30 events/hour
20	**b**	2.2	5 to <10 — roughly one third of adult thresholds
21	**c**	2.1	≥85%, often >90% in young adults
22	**c**	2.4	Symptom resolution *and* treatment AHI <5/h
23	**c**	2.4	TST unchanged; REM rises and NREM falls — redistribution, not more sleep
24	**b**	2.6	Tendency to fall asleep. The MWT measures the ability to stay awake

Q	Ans	Topic	Explanation
25	**a**	2.6	But SOREMPs are also frequent in untreated OSA, so the finding is not specific
26	**b**	2.7	≥3; a higher cut-off reduces sensitivity but increases specificity
27	**a**	2.7	Highly sensitive, poorly specific — the general pattern of OSA screening tools
28	**c**	2.7	It measures sleep propensity; excessive daytime sleepiness has many other causes. Useful in combination, never alone
29	**d**	2.7	Only the hard palate visible. Perform upright with maximal tongue protrusion and no phonation
30	**c**	8.6	50–75%. Obstruction >50% warrants further evaluation, but tonsil size does not correlate with OSA severity
31	**b**	3.1	Cartwright & Samelson, 1982. Robin described the monobloc in 1903; Meier-Ewert an intraoral protraction device in 1985
32	**b**	3.1	1 mm or less
33	**c**	3.1	At least 5 mm, with reversibility and a verifiable setting
34	**b**	3.2	Laterally at the velopharyngeal level, through traction on soft tissue connections between pharynx and mandibular ramus
35	**c**	3.2	It improves passive anatomy only — which is why non-anatomical phenotypes are non-responders
36	**b**	3.4	4 cmH_2O, titrated to the lowest effective pressure up to ~20 cmH_2O
37	**c**	3.4	>20 cmH_2O, or hypercapnic SDB, or intolerance of high CPAP settings
38	**b**	3.4	40–60%, with the ≥4 h/night threshold and failure rates of 29–83% reported
39	**b**	3.10	Efficacy × compliance — the reconciliation of the CPAP/appliance paradox
40	**b**	3.10	~51% versus ~50%: lower efficacy offset by greater adherence
41	**c**	3.7	Complete concentric collapse on DISE, and grade 3–4 tonsils. Selection favours BMI <35 and AHI 20–50
42	**b**	3.5	Velum, Oropharynx, Tongue base, Epiglottis
43	**b**	3.6	Non-supine AHI <5 — a strict definition; supine-predominant is not the same thing
44	**b**	4.1	Panoramic and lateral cephalometric as a pre-treatment record. CBCT is not indicated
45	**c**	4.7	Co-occurrence is strongly supported; causation is weakly supported; treating SDB reducing SB is moderately supported. Refrain from implying causation to patients
46	**c**	4.7	Increases AHI and snoring — always screen for OSA before fitting a splint
47	**b**	5.6	50–70% of maximum (AADSM 2025: 25–75% is within the comfort and therapeutic range). Do not start at maximum
48	**b**	5.4	Increases pharyngeal collapsibility, reduces efficacy, and promotes mouth breathing with loss of lip seal
49	**c**	5.3	6 mm; protrusion <6 mm is a contraindication (normal range 10–12 mm)
50	**b**	5.4	Nylon — less breakage than acrylic in bruxers, though it stains readily and cannot be relined or repaired
51	**c**	6.6	≥80% of the night, ≥5 nights per week, with ≥7 h sleep. The ≥4 h figure is the CPAP research convention
52	**b**	6.6	~67%, falling from ~80% at 1 month, then stable through 1 year
53	**b**	6.5	~37% reach AHI <5; ~52% AHI <10; ~64% achieve ≥50% reduction

Q	Ans	Topic	Explanation
54	**b**	5.1	Every 6 months for the first year, then at least annually; physician reassessment approximately every 5 years
55	**b**	10.2	~7.5%; side effects contribute to 20–30% of all discontinuations
56	**c**	10.2	TMJ pain. It is less common to stop because of occlusal change
57	**c**	10.2	Tilting of teeth — retroclination of upper incisors, proclination of lower incisors, mesial molar tendency. Not skeletal change
58	**b**	10.4	15–20 minutes on waking, custom made and recorded in centric occlusion
59	**c**	8.7	Adenotonsillectomy — the most effective surgical therapy, though residual OSA is common
60	**b**	8.9	By age 4, regardless of symptoms: OSA prevalence ~50% at AHI >5, and 53.8% of affected children are asymptomatic

WHERE DID YOU LOSE MARKS?

Items you got wrong	Go back to
1–16	Topic 1 — sleep physiology and SDB pathophysiology. Focus on the PALM construct and the numbers table
17–30	Topic 2 — diagnostic testing. Re-learn the event definitions verbatim and the screening thresholds
31–43	Topic 3 — treatments. Focus on the effective-appliance definition, PAP modalities and mean disease alleviation
44–46	Topic 4 — history, examination and bruxism. Focus on imaging and the causation caveat
47–50, 54	Topic 5 — selection, design and bite registration. Focus on protrusion and vertical dimension
51–53	Topic 6 — titration, efficacy and compliance definitions
55–58	Topic 10 — side effects and consent. Focus on the occlusal change figures
59–60	Topic 8 — paediatrics. Focus on treatment hierarchy and syndromic screening

Abbreviations

Abbreviation	Expansion	Abbreviation	Expansion
AADSM	American Academy of Dental Sleep Medicine	**MAD / MAS / MAA**	Mandibular advancement device / splint / appliance
AASM	American Academy of Sleep Medicine	**MARPE**	Miniscrew-assisted rapid palatal expansion
ABDSM	American Board of Dental Sleep Medicine	**MDA**	Mean disease alleviation (efficacy × compliance)
ADA	American Dental Association / Australian Dental Association	**MMA**	Maxillomandibular advancement
AHI	Apnoea–hypopnoea index	**MPA**	Mandibular plane angle
APAP	Auto-adjusting positive airway pressure	**MSL**	Mean sleep latency
ASV	Adaptive servo-ventilation	**MSLT**	Multiple Sleep Latency Test
BPAP / BiPAP	Bilevel positive airway pressure	**MWT**	Maintenance of Wakefulness Test
CBCT	Cone-beam computed tomography	**NREM**	Non-rapid eye movement sleep
CBT / CBT-I	Cognitive behavioural therapy / for insomnia	**OA / OAT**	Oral appliance / oral appliance therapy
CFS	Chronic fatigue syndrome	**OB / OJ**	Overbite / overjet
COMISA	Comorbid insomnia and sleep apnoea	**ODI**	Oxygen desaturation index
CPAP	Continuous positive airway pressure	**OHS**	Obesity hypoventilation syndrome
CSA	Central sleep apnoea	**OSA**	Obstructive sleep apnoea
DISE	Drug-induced sleep endoscopy	**PAP**	Positive airway pressure
DSM	Dental sleep medicine	**P_{crit}**	Critical closing pressure
DVA	Department of Veterans' Affairs	**PLMD / PLMS**	Periodic limb movement disorder / of sleep
EBD / EBP	Evidence-based dentistry / practice	**PM**	Portable monitor
ECG / EEG / EMG / EOG	Electrocardiogram / electroencephalogram / electromyogram / electro-oculogram	**PMMA**	Polymethyl methacrylate
EPAP	Expiratory positive airway pressure	**PSG**	Polysomnography
ESS	Epworth Sleepiness Scale	**PSQ**	Paediatric Sleep Questionnaire
EUPF	Extended uvulopalatal flap	**PVS**	Polyvinyl siloxane (addition silicone)
EVA	Ethylene-vinyl acetate	**RBD**	REM sleep behaviour disorder
FM	Fibromyalgia	**RCMP**	Remotely controlled mandibular positioner
FOSQ	Functional Outcomes of Sleep Questionnaire	**RDI / REI**	Respiratory disturbance index / respiratory event index

Abbreviation	Expansion	Abbreviation	Expansion
FRC	Functional residual capacity	**RERA**	Respiratory effort-related arousal
GORD / GERD	Gastro-oesophageal reflux disease	**REM**	Rapid eye movement sleep
HNS / HGNS	Hypoglossal nerve stimulation	**RFA**	Radiofrequency ablation
HSAT / HST	Home sleep apnoea test / home sleep test	**RIP**	Respiratory inductance plethysmography
ICSD-3-TR	International Classification of Sleep Disorders, 3rd edition, Text Revision	**RLS**	Restless legs syndrome
LAUP	Laser-assisted uvulopalatoplasty	**RME**	Rapid maxillary expansion
MBS	Medicare Benefits Schedule	**SAQLI**	Calgary Sleep Apnea Quality of Life Index
SB	Sleep bruxism	**SDB**	Sleep-disordered breathing
SE	Sleep efficiency	**SOL**	Sleep onset latency
SOREMP	Sleep-onset REM period	**SRBD**	Sleep-related breathing disorder
SWS	Slow-wave sleep	**TMD / TMJ**	Temporomandibular disorder / joint
TORS	Transoral robotic surgery	**TRD**	Tongue retaining device
TST	Total sleep time	**UARS**	Upper airway resistance syndrome
UPPP	Uvulopalatopharyngoplasty	**VDO**	Vertical dimension of occlusion
VOTE	Velum, Oropharynx, Tongue base, Epiglottis	**WASO**	Wake after sleep onset

Core sources

The full reference list for each topic appears at the end of that topic. The documents below are the ones worth reading in the original, and worth being able to name in a viva.

Governing guidelines and standards

1. Ramar K, Dort LC, Katz SG, et al. Clinical practice guideline for the treatment of obstructive sleep apnea and snoring with oral appliance therapy: an update for 2015. An American Academy of Sleep Medicine and American Academy of Dental Sleep Medicine clinical practice guideline. *J Clin Sleep Med*. 2015;11(7):773–828. — the single most important document for this examination.
2. Levine M, Cantwell MK, Postol K, Schwartz DB. Dental sleep medicine standards for screening, treatment and management of sleep-related breathing disorders in adults using oral appliance therapy: an update. *J Dent Sleep Med*. 2025;12(2). — the current standards of practice; supersedes the 2018 and 2022 versions.
3. Mogell K, Blumenstock N, Mason E, Rohatgi R, Shah S, Schwartz D. Definition of an effective oral appliance for the treatment of obstructive sleep apnea and snoring: an update for 2019. *J Dent Sleep Med*. 2019;6(3).
4. Sheats R, Essick G, Grosdidier J, et al. Identifying the appropriate therapeutic position of an oral appliance. *J Dent Sleep Med*. 2020;7(4).
5. Sheats RD, Schell TG, Blanton AO, et al. Management of side effects of oral appliance therapy for sleep-disordered breathing. *J Dent Sleep Med*. 2017;4(4):111–125.
6. Radmand R, Chiang H, Di Giosia M, et al. Defining and measuring compliance with oral appliance therapy. *J Dent Sleep Med*. 2021;8(3).
7. Schwartz D, Adame M, Addy N, et al. AADSM position on the scope of practice for dentists ordering or administering home sleep apnea tests. *J Dent Sleep Med*. 2020;7(4).
8. Patil SP, Ayappa IA, Caples SM, Kimoff RJ, Patel SR, Harrod CG. Treatment of adult obstructive sleep apnea with positive airway pressure: an AASM clinical practice guideline. *J Clin Sleep Med*. 2019;15(2):335–343.
9. Kapur VK, Auckley DH, Chowdhuri S, et al. Clinical practice guideline for diagnostic testing for adult obstructive sleep apnea: an AASM clinical practice guideline. *J Clin Sleep Med*. 2017;13(3):479–504.

10. Collop NA, Anderson WM, Boehlecke B, et al. Clinical guidelines for the use of unattended portable monitors in the diagnosis of obstructive sleep apnea in adult patients. *J Clin Sleep Med*. 2007;3(7):737–747.
11. Berry RB, Quan SF, Abreu AR, et al. *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications*. Version 3. AASM; 2023.
12. American Academy of Sleep Medicine. *International Classification of Sleep Disorders*. 3rd ed, Text Revision (ICSD-3-TR). AASM; 2023.
13. Marcus CL, Brooks LJ, Draper KA, et al. Diagnosis and management of childhood obstructive sleep apnea syndrome. *Pediatrics*. 2012;130(3):576–584.
14. Kaditis AG, Alonso Alvarez ML, Boudewyns A, et al. Obstructive sleep disordered breathing in 2- to 18-year-old children: diagnosis and management. *Eur Respir J*. 2016;47(1):69–94.
15. Sateia MJ, Buysse DJ, Krystal AD, Neubauer DN, Heald JL. Clinical practice guideline for the pharmacologic treatment of chronic insomnia in adults: an AASM clinical practice guideline. *J Clin Sleep Med*. 2017;13(2):307–349.
16. Edinger JD, Arnedt JT, Bertisch SM, et al. Behavioral and psychological treatments for chronic insomnia disorder in adults: an AASM clinical practice guideline. *J Clin Sleep Med*. 2021;17(2):255–262.
17. Lobbezoo F, Ahlberg J, Raphael KG, et al. International consensus on the assessment of bruxism: report of a work in progress. *J Oral Rehabil*. 2018;45(11):837–844.
18. American Dental Association. Policy statement: the role of dentistry in the treatment of sleep-related breathing disorders. ADA; 2017.
19. Australasian Sleep Association. Position statement on the assessment and management of primary snoring in adults.

Landmark primary literature

1. Young T, Palta M, Dempsey J, Skatrud J, Weber S, Badr S. The occurrence of sleep-disordered breathing among middle-aged adults. *N Engl J Med*. 1993;328(17):1230–1235.
2. Peppard PE, Young T, Barnet JH, Palta M, Hagen EW, Hla KM. Increased prevalence of sleep-disordered breathing in adults. *Am J Epidemiol*. 2013;177(9):1006–1014.
3. Benjafield AV, Ayas NT, Eastwood PR, et al. Estimation of the global prevalence and burden of obstructive sleep apnoea. *Lancet Respir Med*. 2019;7(8):687–698.
4. Eckert DJ, White DP, Jordan AS, Malhotra A, Wellman A. Defining phenotypic causes of obstructive sleep apnea. *Am J Respir Crit Care Med*. 2013;188(8):996–1004.
5. Phillips CL, Grunstein RR, Darendeliler MA, et al. Health outcomes of CPAP versus oral appliance treatment for obstructive sleep apnea. *Am J Respir Crit Care Med*. 2013;187(8):879–887.
6. Trzepizur W, Cistulli PA, Glos M, et al. Health outcomes of CPAP versus mandibular advancement device for severe OSA: an individual participant data meta-analysis. *Sleep*. 2021;44(7):zsab015.
7. Vanderveken OM, Dieltjens M, Wouters K, De Backer WA, Van de Heyning PH, Braem MJ. Objective measurement of compliance during oral appliance therapy. *Thorax*. 2013;68(1):91–96.
8. Chan ASL, Sutherland K, Schwab RJ, et al. The effect of mandibular advancement on upper airway structure in obstructive sleep apnoea. *Thorax*. 2010;65(8):726–732.
9. Remmers J, Charkhandeh S, Grosse J, et al. Remotely controlled mandibular protrusion during sleep predicts therapeutic success with oral appliances. *Sleep*. 2013;36(10):1517–1525.
10. Pliska BT, Nam H, Chen H, Lowe AA, Almeida FR. Obstructive sleep apnea and mandibular advancement splints: occlusal effects and progression of changes associated with a decade of treatment. *J Clin Sleep Med*. 2014;10(12):1285–1291.
11. Sanders AE, Essick GK, Fillingim R, et al. Sleep apnea symptoms and risk of temporomandibular disorder: OPPERA cohort. *J Dent Res*. 2013;92(7 Suppl):70S–77S.
12. Strollo PJ, Soose RJ, Maurer JT, et al. Upper-airway stimulation for obstructive sleep apnea. *N Engl J Med*. 2014;370(2):139–149.
13. Marcus CL, Moore RH, Rosen CL, et al. A randomized trial of adenotonsillectomy for childhood sleep apnea. *N Engl J Med*. 2013;368(25): 2366–2376.
14. Malhotra A, Grunstein RR, Fietze I, et al. Tirzepatide for the treatment of obstructive sleep apnea and obesity. *N Engl J Med*. 2024;391(13): 1193–1205.
15. Chung F, Yegneswaran B, Liao P, et al. STOP questionnaire: a tool to screen patients for obstructive sleep apnea. *Anesthesiology*. 2008;108(5): 812–821.
16. Chai-Coetzer CL, Antic NA, Rowland LS, et al. A simplified model of screening questionnaire and home monitoring for obstructive sleep apnoea in primary care. *Thorax*. 2011;66(3):213–219.
17. Johns MW. A new method for measuring daytime sleepiness: the Epworth Sleepiness Scale. *Sleep*. 1991;14(6):540–545.
18. Sutherland K, Phillips CL, Cistulli PA. Efficacy versus effectiveness in the treatment of obstructive sleep apnea: CPAP and oral appliances. *J Dent Sleep Med*. 2015;2(4):175–181.

About this notebook. The content, structure and sub-topic numbering follow the candidate's own ten-topic syllabus notes and lecture material. All clinical content has been cross-checked against, and where necessary corrected and expanded from, the current primary

sources listed above — in particular the 2015 AASM/AADSM clinical practice guideline, the 2025 AADSM standards of practice, the 2023 AASM Scoring Manual version 3, and ICSD-3-TR. Where the source notes reflected superseded criteria — the four-stage NREM nomenclature, the one-month insomnia duration criterion, the 1990 fibromyalgia tender-point criteria, and the ≥4% hypopnoea desaturation rule — both the classic and the current position are given, with the discrepancy flagged in an exam-trap box. Figures are reproduced from the candidate's own source documents. Reimbursement and referral pathways are jurisdiction-specific and are flagged as such throughout; verify local rules against the current official schedule before relying on them clinically.